

Dendritic morphology as a dictionary for local credit assignment

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Abstract

Dendritic trees compartmentalize synaptic inputs, but which spatial distinctions a learning signal must preserve is unclear. We describe delivered credit as combinations of spatial profiles constrained by morphology. In image classifiers with one dendritic neuronal layer, preserving neuronal identity accounts for most of the feedback benefit. Contextual conflict favors branch selection, while hierarchical distractors favor ancestry grouping at a bandwidth predicted by the task. Targets with matched input-sensitivity spectra develop different learned credit spectra and tolerance of fixed profiles; a local inhibitory gate also supports learning of opposing branch tuning in a conductance tree. Serial compartments help when nuisance-factor information is distributed across modules. Reconstructed arbors compress modeled passive perturbation fields through ancestry profiles, and focal shunts regulate their gains with selectivity set by electrotonic state. Measured visual responses provide no evidence of preferential ancestry alignment in the sampled cells. The framework links task structure to useful credit distinctions while separating available delivery patterns from the mechanisms that generate their coefficients.

Introduction

Learning must link a synapse’s local activity to its contribution to a circuit outcome. Cable filtering, nonlinear integration and spatially organized inhibition make locations within a dendritic tree functionally distinct [1–7]. These differences expand computation. We ask when branching also helps distribute useful learning signals.

Credit assignment asks how changing a synapse would affect task error. Backpropagation (BP) answers this by calculating the loss gradient—the sensitivity of task error to each parameter [8]; related sensitivity calculations apply to recurrent and steady-state networks [9, 10]. For a point neuron, each weight gradient combines a local sensitivity, called eligibility, with a neuron-specific learning signal. Hebbian plasticity captures the association between local pre- and postsynaptic variables [11], while three-factor rules combine that eligibility with an outcome-dependent signal [12]. We call a rule local when a synapse combines its own eligibility with an available learning signal, whether or not that signal is computed locally.

Biologically motivated learning theories differ in how they obtain such signals. Feedback alignment communicates errors through separate pathways [13, 14]; equilibrium, predictive and prospective methods infer target states or errors from dynamics [15–17]; eligibility propagation combines temporal traces with neuron-specific signals [18]. One number shared across neurons carries less task information than separately assigned neuronal signals, although each synapse still has its own eligibility [19–21]. Recent causal evidence for distinct instructive signals across neuronal populations motivates asking what additional distinctions are useful within a neuron’s arbor [22].

A *route dictionary* describes the spatial profiles available for delivering learning signals. A route identifies the locations reached; its profile specifies their relative amplitudes, and a trial-dependent coefficient scales that profile. Each dictionary column stores one profile. A shared profile broadcasts throughout an arbor; combining several profiles can distinguish subtrees. We use *basis* for linearly independent profiles and *dictionary* when profiles may be redundant. Morphology constrains the available routes; the task and teaching mechanism determine useful coefficients.

Conductance changes the gains of anatomical delivery routes. Excitatory and inhibitory contacts open nonnegative conductances whose reversal potentials determine current; inhibition can lower input resistance and attenuate other inputs even when its own current is small [23–27]. Rescaling a delivery channel preserves its spatial profile, whereas altered cable transport can reshape how local voltage affects somatic output and task error.

Dendritic models already connect nonlinear compartments and structural plasticity to memory capacity [28], and branch-specific plasticity can organize nonlinear feature combinations [29]. Successive computations in artificial trees and detailed cortical models also establish substantial forward capacity [30, 31]. Dendritic learning theories supply mechanisms for somatic prediction, segregated teaching, interneuron-mediated errors, bursts and prospective dynamics [32–38]; cable-learning models optimize reconstructed neurons using synaptic sensitivities [39]. We ask which spatial distinctions these mechanisms must preserve within an arbor.

The conductance-tree gradient factorization and LocalCA rule were introduced in our predecessor preprint [40]. This Article extends those foundations with controlled task-to-credit comparisons, matched-spectrum and conductance learning experiments, and anatomical transport and response tests.

We test a progression from neuronal identity to branch selection, ancestry grouping and example-dependent credit across interacting branches. Controlled learning tasks establish useful distinctions; forward-compatibility tests separate learning gaps from representational limits. Reconstructed arbors then test anatomical delivery capacity and conductance-dependent gains, while measured responses bound the biological interpretation.

Results

Exact gradients separate local sensitivity from downstream credit

For a point neuron $y_u = f_u(\sum_i w_i x_i + b_u)$, the chain rule gives $\partial \mathcal{L} / \partial w_i = x_i f'_u \delta_u$, where x_i and w_i are synaptic input and weight, b_u is bias, and $\delta_u = \partial \mathcal{L} / \partial y_u$ is the neuronal error. The derivative f'_u is evaluated at the current input sum. Local eligibility $x_i f'_u$ and task-derived credit therefore supply different information (Fig. 1A).

The rooted-tree model has soma $n = 0$, compartment voltage V_n , and nonnegative synaptic conductances g_i . A contact with reversal potential E_i^{rev} and nonnegative activity x_i contributes current $g_i x_i (E_i^{\text{rev}} - V_n)$. In the directed artificial trees, child outputs drive their parents and a terminal-to-soma pass computes the steady state. Total input resistance R_n^{tot} is the reciprocal of the summed leak, excitatory, inhibitory and child-coupling conductances (Methods). Differentiating the current balance yields

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n)}_{\text{local eligibility}} \underbrace{\delta_{n,u}^V}_{\partial \mathcal{L} / \partial V_n}. \quad (1)$$

Excitatory (E) and inhibitory (I) contacts share this factorization (Fig. 1A, filled blue and filled carmine contacts on the arbor). Eligibility tells how conductance changes local voltage; compartment credit tells how that voltage changes task loss.

Voltage error denotes a loss derivative. Each compartment has one directed path to the soma, giving

$$\delta_{n,u}^V = \delta_{0,u}^V \tilde{\alpha}_n, \quad \tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}. \quad (2)$$

Here $f'_i(V_i)$ is the child compartment’s output derivative, $g_{i \rightarrow k}^{\text{den}}$ is its coupling conductance to parent k , and $\delta_{0,u}^V = f'_0(V_0) \delta_u$ is the somatic voltage error. The path multiplier $\tilde{\alpha}_n$ converts one somatic error into a spatial field. The conductance-only component of this product is attenuated by additional conductance on a path’s ancestors; branch nonlinearities and forward voltages can change as well (Supplementary Section S1 and Fig. S1).

Reciprocal cables instead require a joint adjoint calculation of how local currents affect task loss (Methods). These mathematical sensitivities do not establish a biological error-transmission mechanism.

Local credit assignment (LocalCA) retains eligibility and supplies an approximate compartment signal, denoted by a hat (Fig. 1B shows a strict scalar, one signal per neuron and the exact path, each on a generic tree):

$$\widehat{\nabla}_{g_i} \mathcal{L} = x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n) \widehat{\delta}_{n,u}^V, \quad (3)$$

$$\widehat{\delta}_u^V = A_u \mathbf{c}_u. \quad (4)$$

Columns of $A_u \in \mathbb{R}^{N_u \times K}$ are fixed spatial delivery profiles over N_u routed nonsomatic compartments; $\mathbf{c}_u$ contains their example-dependent coefficients. To distinguish coefficient generation from delivery gain, we can write $\mathbf{c}_u = \Gamma_u \mathbf{c}_u^{\text{source}}$, where diagonal Γ_u scales each profile and $\mathbf{c}_u^{\text{source}}$ is the supplied signal (Fig. 1A, gain dial). Fixed nonzero gains leave

the dictionary span and its oracle capture unchanged while altering conditioning and reliability-weighted learning (Supplementary Figs. S4–S5). Broadcast, subtree and compartment profiles give increasingly detailed spatial addresses (Fig. 1C). This is a delivery model, distinct from the forward cable equations and the mechanism estimating coefficients.

At $K = 1$, the all-ones column $A_u = \mathbf{1}$ broadcasts one coefficient through the arbor; at $K = N_u$, the identity matrix $A_u = I$ can reproduce the full nonsomatic field. Nested subtrees provide intermediate dictionaries. Profile count, rank (the number of independent profiles) and coefficient information are distinct resources. Exact path transport combines one somatic scalar with state-dependent local derivatives; it does not require an independent external error for every compartment.

Capture is the fraction of squared learning-signal magnitude that a dictionary can reproduce. For a nonzero field δ ,

$$\mathcal{C}(A; \delta) = \frac{\|P_A \delta\|^2}{\|\delta\|^2}, \quad P_A = AA^\dagger.$$

Here P_A gives the least-squares reconstruction, $A^\dagger$ is the Moore–Penrose pseudoinverse and $\|\cdot\|$ the Euclidean norm. Capture measures field energy rather than decision accuracy. Averaging these per-field ratios weights fields equally; $\text{tr}(P_A C) / \text{tr}(C)$, with $C = \mathbb{E}[\delta \delta^\top]$, is the fraction of total ensemble energy captured, weighting fields by their energy. The uncentered second-moment matrix C has trace $\text{tr}(C)$, the sum of its diagonal entries. These capture measures can differ, and both assume optimal projection coefficients rather than an implemented encoder.

The field coordinate also matters. In nonlinear image models, we route errors of compartment outputs and retain the local output derivative in eligibility; voltage-error capture is reported separately. Identity-output conductance trees use voltage errors directly. Supplementary Table S2 compares the coordinates, coefficient sources and state dependence. Learning experiments test whether this available capacity improves performance.

Projection can also filter update noise outside a dictionary’s span. Retaining most of the useful gradient while admitting less noise can improve a one-step guarantee (Supplementary Fig. S3G). If the span contains the entire gradient, its mean is preserved. A one-step smoothness bound formalizes this balance between retained descent and update variance (Methods; Supplementary Fig. S3G). Its initialization utility did not select the best final morphology: in a separate 12,800-fit prospective experiment, it lost to a fixed maximum-budget tree in all twenty seed averages in both selection arms (Supplementary Fig. S34). We therefore use the bound to interpret local filtering, not to select a final tree.

One signal per neuron captures most of the image-task learning benefit

Image classification first tests whether within-arbor resolution helps when neuronal errors are available. All image networks used one dendritic layer and a trainable readout, which supplied exact neuronal output errors before the primary feedback interventions. The MNIST comparison used 128 directed $[3, 3]$ trees, each with three proximal and nine terminal compartments, receiving flattened images. Shunting and raw-additive architectures shared the graph, initial contact construction, nonlinearities and training protocol, with architecture-specific initialization. The additive control removed reversal-potential driving forces and the conductance denominator, retaining signed current summation without shunting.

Six rules separated neuronal identity, spatial resolution and learning in the dendritic core (Fig. 1B–D): a strict scalar, per-neuron feedback (neuron-specific), a projected $K = 1$ rule, projected $K = 3$ subtree profiles, the exact path and a decoder-only reference. The strict scalar shared one signal across neurons; per-neuron feedback broadcast each neuron’s own output error through its arbor. The projected $K = 1$ rule averaged the exact activation errors over all twelve nonsomatic compartments, while the projected $K = 3$ profiles averaged them within each proximal compartment and its three children. Per-neuron feedback uses the same all-ones profile as the projected $K = 1$ rule and differs only in the coefficient source (Fig. 1C). Both used the same exact field and preserved its mean at identical parameters, while allowing different norms. The exact path retained every compartment error. A decoder-only reference trained the readout while freezing the core. Three development seeds selected each rule’s rate from three common multipliers; ten fresh paired seeds per architecture evaluated selected and original common rates over 180 epochs (Methods; Supplementary Fig. S7A,B).

At selected rates, per-neuron feedback exceeded the strict scalar by 9.23 percentage points in shunting trees and 7.46 points in additive trees, reaching 97.12% and 97.11% accuracy (Fig. 1D,E). Decoder-only learning reached 80.96% and 89.57%, respectively (Fig. 1D). Increasing the projected dictionary from one to three profiles changed accuracy by only 0.049 points in shunting trees (95% paired interval, -0.048 to 0.141) and 0.036 points in additive trees (-0.056 to

0.122; point estimates in Fig. 1D, note; intervals in Source Data). The common-rate comparison gave similarly small differences (Supplementary Fig. S7C). The exact path added 0.010 points over per-neuron feedback in shunting trees (5 of 10 seeds positive) and 0.182 points in additive trees (0.114–0.252; positive in all ten seeds; Fig. 1F); against the projected $K = 3$ rule it was 0.09 and 0.04 points lower (Fig. 1D, note). Both the exact path and one projected profile reached about 97.3% in additive trees. Most learning benefit therefore came from preserving neuronal coordinates; the smaller additive improvement was attainable without separating subtrees.

The richer dictionary nevertheless captured more of the trained activation-error field in the same cohort. Moving from one to three profiles raised mean capture from 0.622 to 0.655 in shunting trees and from 0.764 to 0.859 in additive trees, and the twelve-profile identity dictionary reproduces the field exactly by construction (Fig. 1G). Additive capture nevertheless fell over training at both budgets (0.851 to 0.764 at $K = 1$; 0.893 to 0.859 at $K = 3$), and the two shunting values at initialization were indistinguishable (0.5322 and 0.5323; open markers in Fig. 1G). These ratios average over example–neuron fields and use the coordinates delivered during learning. Thus improved reconstruction did not yield a comparable accuracy benefit. Earlier voltage-credit diagnostics and recipient-permutation controls support spatial field structure and the importance of neuronal identity, respectively (Supplementary Section S2 and Figs. S8A–C).

With fixed random soma-level feedback (DFA), per-neuron feedback improved over the strict scalar by 11.03 points in shunting trees and 10.80 points in additive trees (fifteen paired seeds; Fig. 1E; Supplementary Fig. S6D), while exact paths added at most 0.09 points (Fig. 1F; Supplementary Fig. S6E). This control changes weight transport but retains the single dendritic layer and the same output-error dimensionality. Fashion-MNIST also favored per-neuron feedback over the matched-width scalar fallback, by 4.46 and 3.78 points, with exact paths within 0.14 points of it (ten paired seeds; Fig. 1E,F; Supplementary Fig. S6C). On flattened CIFAR-10, exact paths reduced accuracy relative to per-neuron feedback by 0.86 points (0.55–1.17), while remaining equivalent to matched backpropagation within the prespecified one-point margin (−0.115 points, −0.457 to 0.226; two one-sided tests $P = 1.6 \times 10^{-5}$; twenty paired seeds; Fig. 1F, diamond in the band; Supplementary Fig. S6A,B). These four cohorts differ in feedback source, scalar construction and tree (Methods).

Thus finer delivery need not improve learning even when the error field has spatial structure. Further depth and task-generator controls are in Supplementary Figs. S6 and S9A–D, which cover input coverage, depth and the historical noise generators; the serial-depth half is Supplementary Fig. S9E,F. We next introduce tasks that make specific spatial distinctions useful.

Branch-selective information prevents updates from inactive distractors

Branch selection is useful when otherwise valid local activity comes from irrelevant inputs. We tested this with one logistic output and $B \in \{2, 4, 8\}$ parallel branch weight vectors and no internal forward compartments or conductance nonlinearity (Fig. 2A). Context selected the branch determining the output, while every branch received an image and formed an ungated local eligibility.

Inputs were average-pooled, standardized Fashion-MNIST images from classes 0 and 6. The selected image determined the binary target. At conflict dose $\chi = 0$, all branches received that image; increasing χ independently replaced nonselected inputs with opposite-class images. For example i , let b_i^* be the selected branch, $\mathbf{x}_{i,b}$ its input vector and $y_i \in \{0, 1\}$ the target. The branch weights $\mathbf{w}_b$ give the pre-probability score (logit) and output error

$$z_i = \mathbf{w}_{b_i^*}^\top \mathbf{x}_{i,b_i^*}, \quad \delta_i = \sigma(z_i) - y_i. \quad (5)$$

Here $\sigma(z) = 1/(1 + e^{-z})$ is the logistic function. Unselected branches have zero exact derivative despite nonzero eligibility. Conflict therefore challenges a rule that ignores selection by exposing it to irrelevant, opposite-class eligibilities.

For N examples, the branch-specific and neuron-shared gradient estimates were

$$\mathbf{d}_b^{\text{selected}} = N^{-1} \sum_i \delta_i \mathbf{1}[b_i^* = b] \mathbf{x}_{i,b}, \quad \mathbf{d}_b^{\text{shared}} = N^{-1} \sum_i \delta_i B^{-1} \mathbf{x}_{i,b}. \quad (6)$$

The indicator $\mathbf{1}[\cdot]$ is one when its condition holds, zero otherwise; $1/B$ matches scale without supplying branch information. A feedback selector and a branch-local eligibility gate give the same update (Fig. 2B), testing access to selection rather than a need for independently computed external errors. We call these the branch-specific and neuron-shared rules. A deranged route instead delivers the same signal to the neighbouring branch at matched rank and sparsity (Fig. 2B, bottom row).

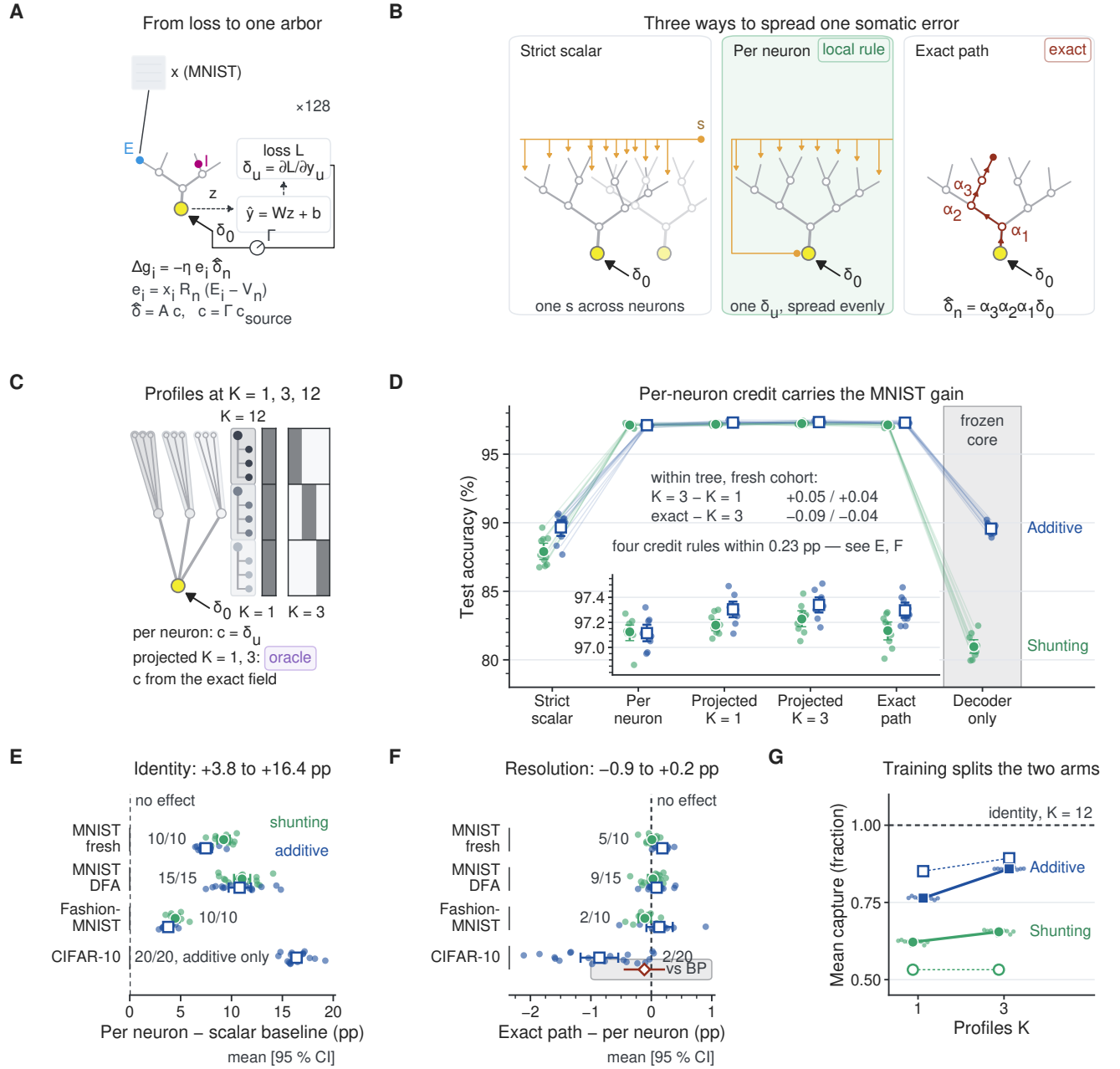


Figure 1: Neuronal identity, not finer spatial address, carries the image-learning benefit. **A**, MNIST input (leader) reaches an excitatory contact (blue) on one of 128 trees (the $\times 128$ tag); an inhibitory contact (carmine) is also drawn, on a schematic arbor, not C's [3,3] morphology; readout $\hat{y}$ and loss $\mathcal{L}$ return δ_u into the soma as δ_0 (ink arrow) through the diagonal gain Γ (dial), $c = \Gamma c_{\text{source}}$; no data. **B**, Three deliveries of one somatic error on a generic depth-three tree: a strict scalar s shared with a ghosted neighbour, one signal per neuron, and the exact path $\hat{\delta}_n = \alpha_3 \alpha_2 \alpha_1 \delta_0$; schematic, no data. **C**, Dictionaries over the twelve nonsomatic sites: the tinted strip is the $K = 12$ identity, beside $K = 1$ all-ones and $K = 3$ proximal subtrees; columns are indicator supports averaged by the projected rules (oracle badge); schematic, no data. **D**, Test accuracy for six rules at selected rates; circles shunting, squares additive, dots seeds paired by hairlines; band, decoder only (frozen core); note, the fresh-cohort within-tree contrasts; means with 95% seed-bootstrap intervals over ten fresh paired seeds per architecture, smaller than the symbol for the credit rules. Inset, those four credit-rule columns at the same x positions and marks, rescaled until their intervals and ten seeds resolve. **E**, Per neuron minus each cohort's scalar baseline (the strict scalar, except the matched-width fallback for Fashion-MNIST), shunting on the row and additive open below it, each row tagged with shunting positive seeds out of the total ($n = 10, 15, 10$ and 20 seeds; CIFAR-10 is additive only); means with paired 95% seed-bootstrap intervals in percentage points, several narrower than the symbol and hidden by it. **F**, Exact path minus per neuron in the same rows, marks and interval type; in the CIFAR-10 row the carmine diamond and interval are exact path minus backpropagation, inside the band of the prespecified ± 1 percentage-point margin. **G**, Mean activation-error capture of D's trained exact-path fields by C's dictionaries at $K = 1, 3$; filled, trained, with its ten seeds; open, initialization ($n = 10$, seeds not drawn); reference line at 1, the $K = 12$ identity, which reproduces the field exactly. All 95% seed-bootstrap intervals are at most 0.007 capture units and are not drawn. pp, percentage points. Source Data: [source_data/curated_publication/figure_01_plotted.csv](#).

With centered, symmetric class evidence, the shared-mode prediction crosses zero at $\chi_c = B/[2(B-1)]$: 1, 2/3 and 4/7 for two, four and eight branches (Fig. 2C; Methods). The measured initial signed utility of the neuron-shared update tracked this line to within 0.018 at all twenty-four sampled cells, although its bootstrap interval excluded the prediction in eleven of them (Fig. 2C). This is a statement about initialization: the trained chance crossing is a separate endpoint, displaced from the boundary because update alignment is lost during learning (Methods; Supplementary Fig. S10B–D). It nonetheless followed the predicted order $B = 8 < 4 < 2$ in 20 of 20 seeds, and five two-branch seeds never reached chance within the sweep (Fig. 2D). Selection also protects what has already been learned: in a separate two-stream interference task, neuron-shared credit erased a previously acquired context, losing 55.7 accuracy points [53.8, 57.6], while branch-specific subtree routes lost none in all ten seeds (Fig. 2E; Supplementary Fig. S10A).

At zero conflict, the branch-specific and neuron-shared rules achieved similar accuracy. Their difference increased with conflict, reaching 34.9–58.2 percentage points at full conflict across the tested branch counts (20 paired seeds; Fig. 2F–H). In every seed, the advantage of correct routing increased with conflict. A deranged route sent every signal to the wrong branch at matched rank and sparsity. It was harmful even at zero conflict, reducing accuracy by 12–25 points relative to branch-specific selection, whereas the neuron-shared rule was indistinguishable from branch-specific selection there (Fig. 2F–H, $\chi = 0$); at high conflict the deranged rule settled below chance, near 22% correct at every branch count, as did the neuron-shared rule at $B = 4$ and $B = 8$, while the two-branch neuron-shared rule reached 45.1%. BP and the gated point matched branch-specific selection exactly in every seed, as required by their algebraic equivalence (Fig. 2B; Source Data; Supplementary Section S3). Branch-count comparisons do not hold capacity fixed.

The neuron-shared rule reached chance at lower conflict as branch count increased, consistent with the predicted shift (Fig. 2D). Credit reaching inactive branches can therefore harm learning, and credit reaching the wrong branch harms it at every dose.

Ancestry provides a modest advantage at an intermediate feedback budget

Between shared and branch-resolved feedback, nested subtree routes offer an intermediate dictionary. The task contained eight independent, eight-dimensional terminal weight blocks at the leaves of a balanced binary feedback tree. Only the context-selected block determined the logit. Internal tree nodes defined feedback supports and did not perform serial forward computation (Fig. 3A).

Distractors opposed the selected stream’s class signal increasingly with distance through the feedback hierarchy, deliberately favoring grouping that matches the task-to-route assignment. At each budget $K \in \{1, 2, 4, 8\}$, the routing field had K independent patterns and unit-length rows. Ancestry routes grouped the terminals into the whole tree, halves, sibling pairs or individual terminals (Fig. 3B). Permuting task streams across leaves changed the task-to-route assignment without changing the streams or forward parameter budget. Here bandwidth counts independent feedback patterns, not physical stages or independent scalar losses; other single-pattern controls need not be uniform. The complete generator and update are given in Methods.

The generator predicts the budget at which ancestry can help. The selected stream has class coefficient one; its sibling, same-half nonsiblings and other-half streams have coefficients -0.15 , -0.45 and -0.75 . The unnormalized sum of the class signal within the selected ancestry group gives -3.05 , -0.05 , 0.85 and 1 at $K = 1, 2, 4, 8$, respectively. Coarse pooling reverses or nearly cancels the signal; sibling pairs first preserve a positive intermediate-group signal, and full resolution removes grouping differences. The implemented unit-row normalization divides these sums by the square root of group size, preserving their signs. This budget follows from the design, not an independently discovered optimum (Fig. 3B, card footers).

Controls preserved rank while changing recipient assignment, support geometry or sparsity: cyclic derangement, depth-interleaving, random-sparse and dense random fields, whose delivered support of the cued stream’s credit is shown in Fig. 3B.

Increasing bandwidth improved correct routing, while derangement caused large losses (Fig. 3C). At $K = 1$ and $K = 2$ both fell below chance, as the design predicts: pooling reverses the class signal within the cued group (Fig. 3B). Matched-bandwidth partitions had coincident field capture, so capture does not explain the ordering (Supplementary Fig. S11A). Degree- and depth-preserving rewiring removed 23.15 and 5.02 points at $K = 2$ and $K = 4$ but left fully shared and fully resolved ancestry routes exactly tied (Fig. 3D).

The ancestry-specific effect was much smaller. Ancestry performed worse than the best matched control at low budgets, exceeded it by 1.27 percentage points at $K = 4$, and tied at full resolution. The paired $K = 4$ interval was

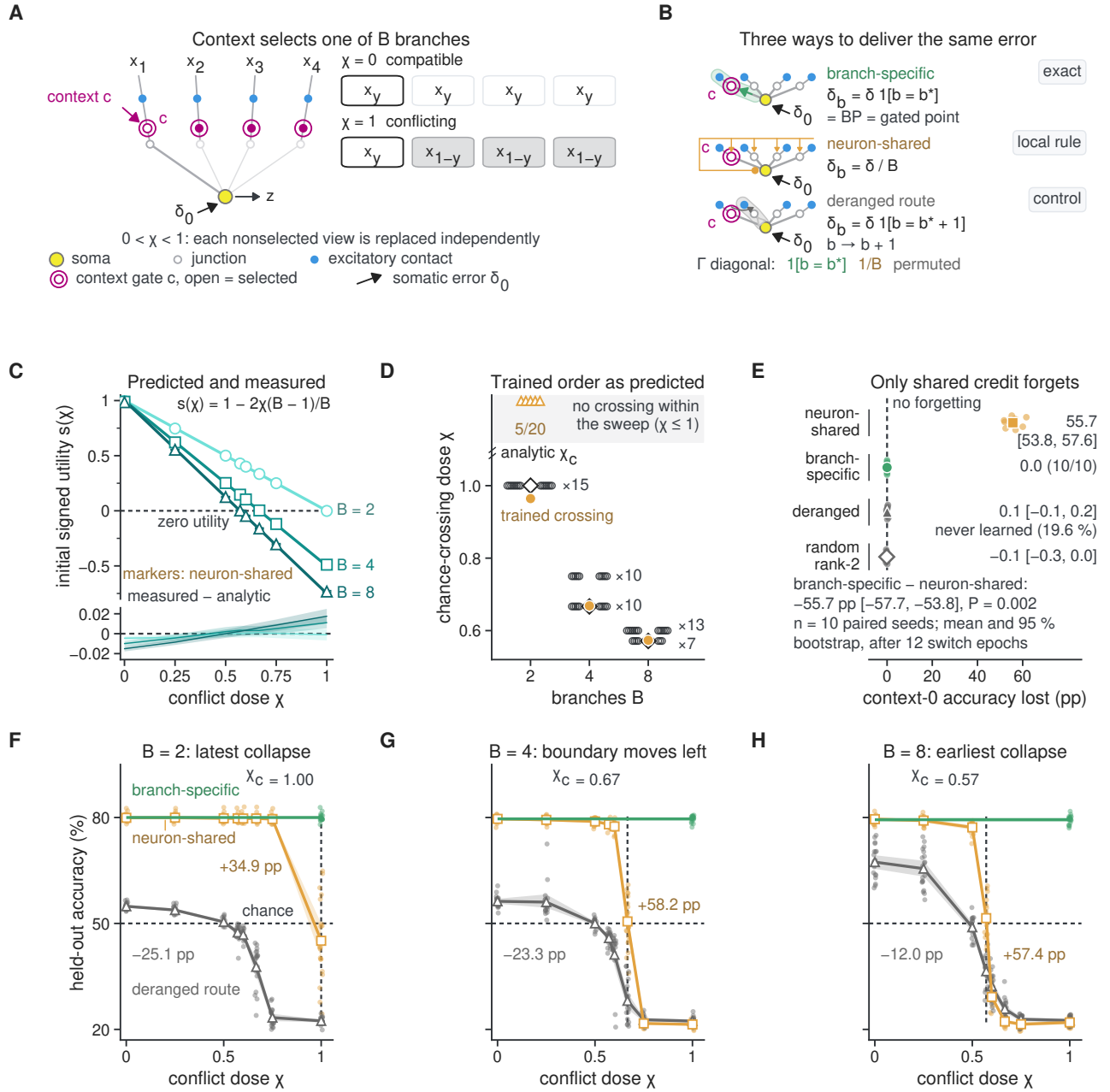


Figure 2: Credit reaching an unselected branch is harmless until its input conflicts. **A**, Context-gated Fashion-MNIST task; schematic, no data. Carmine ring tagged c , the context gate on the selected branch, drawn open; closed gates filled, segments elsewhere faded; ink arrow, somatic error δ_0 . Tile rows, conflict dose $\chi = 0$ and $\chi = 1$. **B**, The same δ_0 delivered three ways on three identical fans; schematic, no data: branch-specific $\delta \mathbf{1}[b = b^*]$ (green capsule, identical to BP and to a gated point), neuron-shared δ/B (amber bus from the soma, one drop per junction) and a deranged route $\delta \mathbf{1}[b = b^* + 1]$ (grey capsule on the neighbouring branch). Branch selection is a diagonal gain Γ (Eq. 4). **C**, Analytic $s(\chi) = 1 - 2\chi(B-1)/B$ (lines; the teal ramp is ordinal in B) against the neuron-shared rule's measured initial signed utility (open markers). Lower strip, the same three series as residuals (measured - analytic; hairline, mean; ribbon, 95% interval) on a ± 0.026 scale. $n = 20$ seeds per cell, mean and 95% seed-bootstrap interval at initialization. **D**, Analytic $\chi_c = B/[2(B-1)]$ (open ink-edged diamonds) against the trained mean neuron-shared curve's first chance crossing (amber) and each seed's first at-or-below-chance dose ($\times n$, seeds at that level); trained crossings 0.964, 0.668 and 0.573 (exact two-sided sign test on the predicted order, $P = 1.9 \times 10^{-6}$). The strip above the axis break holds five $B = 2$ non-crossing seeds; $n = 20$ seeds per B , sweep levels without intervals, endpoint state. **E**, Context-0 accuracy lost after a context switch in the two-stream task, for four routes: neuron-shared (amber square), branch-specific (green circle), deranged (grey filled triangle) and a random dense rank-2 field (grey open diamond), matched in rank but not a routing rule. The deranged route never learned context 0 (held-out 19.6%), so its loss is undefined. Depth-shared equalled neuron-shared in every seed and is not drawn. $n = 10$ paired training seeds, mean and 95% seed-bootstrap interval. **F-H**, Held-out accuracy against χ at $B = 2, 4$ and 8 for **B**'s three deliveries; dashed vertical, χ_c ; dashed horizontal, chance; faint dots, individual seeds. The branch-specific arm is χ -invariant by construction and is drawn as one line, band and seed fan. Grey tags, the deranged - branch-specific difference at $\chi = 0$; amber tags, branch-specific - neuron-shared at $\chi = 1$. $n = 20$ paired seeds, mean and 95% seed-bootstrap band at the endpoint state. Source Data: [source_data/curated_publication/figure_02_plotted.csv](#).

0.59–1.95 points, with 15/20 positive differences; a paired Wilcoxon signed-rank test gave Holm-adjusted $P = 0.0101$ across four budgets. Ancestry also exceeded the dense rank-4, random-sparse and depth-interleaved controls by 2.31, 3.77 and 7.11 points, and the deranged control by 61.06 points, so the small best-of-four contrast reflects a maximum taken over controls rather than a weak effect (Fig. 3E).

A separate twenty-seed cohort tested how supplied context cues could set the coefficients (Fig. 3F). An encoder trained on 256 calibration cues with noise SD 0.5 and their four-route activation targets improved task accuracy over a frozen profile by 7.16 points, but remained 54.47 points below oracle routing. Exploratory delivery of only its most probable route reduced this noisy-cue gap to 16.26 points. With noise-free cues the correct route was identified at every calibration size, and the leakage cost then depended on calibration alone: soft coefficients lost 60.90 points with sixteen cues but only 0.31 points with 256, whereas hard selection matched the oracle in all twenty seeds. Leakage into incompatible routes therefore mattered when calibration was small, not whenever the readout was soft. Hard selection retained a deficit under noisy cues, and delaying either readout across independent contexts removed its benefit (Supplementary Fig. S11B,C and Section S3).

Cross-branch interactions change the spatial credit that learning uses

Branch interactions can change credit’s sign and magnitude between examples. We isolated this effect using targets with identical input-sensitivity spectra on the same compatible tree.

Eight independent inputs x_i were equally likely to be -1 or $+1$. Each branch computed $u = a + bu_L + cu_R + du_Lu_R$ with four trainable coefficients. This multi-affine unit is affine in either child while the other is fixed. Pairwise targets summed four disjoint products, $f_M = \frac{1}{2} \sum_{(i,j) \in M} s_{ij}x_ix_j$; quartic targets combined complementary quartets, $f_A = \frac{1}{2}(s_A \prod_{i \in A} x_i + s_{\bar{A}} \prod_{i \notin A} x_i)$. Here M partitions inputs into pairs, A is one quartet, $\bar{A}$ its complement, and signs s were independently sampled. Within each seed, targets shared a compatible balanced tree, input assignment, initial weights and examples (Fig. 4A). Both have input-sensitivity second moment $\mathbb{E}[\nabla_x f \nabla_x f^T] = I_8/4$, giving identical strengths of independent sensitivity directions.

A scalar branch output cannot transmit several independently required input combinations to its parent. This communication constraint gives an exact compatibility criterion: at each subtree boundary, the matrix of interaction coefficients between inside and outside input products must have rank at most one after removing the constant inside row. The condition is necessary and sufficient across all boundaries; discarded singular values bound representation error (Methods; Supplementary Section S4).

Logical rules make this forward constraint concrete. All fifteen binary trees over four inputs can represent AND, OR and parity exactly, but mixed rules can require particular input groupings. The rule $(a \text{ AND } b) \text{ OR } (c \text{ AND } d)$ needs only two serial stages, whereas $a \text{ AND } [b \text{ OR } (c \text{ AND } d)]$ needs three in this scalar model, despite containing the same two AND gates and one OR gate. These representability statements are independent of training (Supplementary Figs. S14–S15). They identify possible morphologies before testing credit-dependent learning.

All credit rules used the same exact local eligibilities and scalar output error. We compared exact paths with unit broadcast and calibrated broadcast, whose six-site profile is calibrated from each student’s mean path derivatives over 256 unlabeled initialization examples (Fig. 4B). Three development seeds selected learning rates with equal budgets; twenty fresh paired seeds tested the frozen choices. A separate nested target summed products over prefixes of two, four, six and eight inputs on its own compatible tree, which requires depth four and does not share the $I_8/4$ input spectrum (Supplementary Fig. S12F).

Calibrated broadcast learned pairwise targets but remained inaccurate on quartic and nested targets (Fig. 4C–E). At the primary checkpoint (1,024 updates), mean normalized mean squared errors (NMSE) for exact path and calibrated broadcast were 0.0317 and 0.0236 on pairwise targets, 0.1516 and 0.9813 on quartic targets, and 0.0230 and 0.7690 on nested targets. NMSE divides squared prediction error by clean target variance. The predefined quartic-minus-pairwise difference in calibrated-broadcast-minus-exact gaps was 0.838 (95% paired interval, 0.719–0.931), positive in all twenty seeds; a common learning rate gave 0.843 (0.727–0.935). Exact quartic learning was heterogeneous at this checkpoint: seventeen fits were near the noise floor and three remained stalled, and those three resolve one at a time between 1,024 and 4,096 updates (Fig. 4D). The lower pairwise mean for calibrated broadcast likewise reflected one slow exact fit; exact path performed better in fifteen of twenty seeds (Supplementary Figs. S16–S17).

A balanced continuation of every algebraic trajectory to 16,384 updates tested whether these gaps persisted without



Figure 3: Ancestry routes help only where the tree's partition matches the task, and only when the coefficients are supplied. **A**, Eight streams enter the terminals b_1 – b_8 of a balanced tree; context c selects b_3 , whose block sets the logit z , δ_0 enters the soma, and the $K = 4$ capsule delivers it to b_3 and sibling b_4 . Junctions define feedback supports; the tinted band gives each tier's coefficient. **B**, Ancestry dictionaries A ($8 \times K$): K columns of $8/K$ terminals, the channel carrying b_3 's credit green and the others grey; $K = 8$ is the identity, one column of eight unit channels labelled $A = I$. Raw class-signal sums are $-3.05, -0.05, +0.85, +1.00$ (row normalization preserves the signs); beside them, b_3 's support under the four matched controls at $K = 4$ (one draw). **C**, Held-out accuracy across budgets for the ancestry route, the deranged route and the best matched control (the per-seed maximum over the four, a ceiling); 20 paired seeds, per cent, means with 95% seed-bootstrap intervals, smaller than the markers where no bar is visible. **D**, Task-matched versus degree- and depth-matched rewired tree under ancestry feedback: $+23.15$ [21.03, 25.32] and $+5.02$ [4.19, 5.94] points at $K = 2, 4$, exact ties at $K = 1, 8$; 20 paired seeds. Bracketed intervals are paired and reported; the drawn bars are per-condition and smaller than the markers. **E**, Ancestry minus each control at $K = 4$; the top row is the per-seed best of the four below it plus the off-scale deranged control, so its seeds repeat theirs; Holm-adjusted P across four budgets (top row) or four controls; 20 paired seeds, percentage points, means with 95% seed-bootstrap intervals; the seed fan is clipped at 12 points and the seed beyond it is named on the panel. **F**, Coefficient source across the calibration $\times$ cue-noise grid at zero delay: soft solid with filled marks, exploratory hard dashed with open marks at three noise levels (covered at noise 0, where the hard series meet), labelled by calibration size, against the oracle ceiling and frozen-profile floor. At 256 cues and noise 0.5, soft is -54.47 [$-55.45, -53.54$] below oracle and $+7.16$ [6.26, 8.07] above the floor; 20 fresh seeds, means with 95% seed-bootstrap bands. Teal ramps in **A** and **F** are ordinal within this figure alone; **C**–**F** report epoch 80 and **A**, **B** are schematics. Source Data: source_data/curated_publication/figure_03_plotted.csv.

retuning. All twenty selected-rate Adam exact quartic fits reached the noise floor by 4,096 updates, and the pairwise mean lag disappeared. At 16,384 updates, the quartic-minus-pairwise calibrated-broadcast deficit was 0.952 (95% paired interval, 0.883–1.012) at endpoint states and 0.790 (0.691–0.880) at validation-selected states, positive in all twenty seeds at all four budgets (Fig. 4F). Calibrated-broadcast arms contacted coefficient bounds; exact-path arms did not. These are the same previously observed seeds and bounded dynamics, with common-rate, per-seed and optimizer controls (selected-rate Adam trajectories in Supplementary Fig. S16A,C; Supplementary Figs. S16–S17 and Table S8).

The trained fields reveal a distinction missed by the input spectrum. Let $\mathbf{q}(x)$ contain the six nonsomatic derivatives of root output with respect to branch output; exact branch credit is scalar output error times $\mathbf{q}$. At the 1,024-step checkpoint, allowing an oracle amplitude per example, the best rank-one profile captured 99.7% of exact-trained path-field energy for pairwise targets, but 46.1% for quartic and 44.8% for nested targets. The separation persisted at 16,384 updates: capture was approximately 100%, 39.6% and 40.5%, respectively (Source Data; **G** draws the 1,024-update fit). Reaching 95% of the field energy took one fitted direction for pairwise targets and about five for quartic and nested targets. Loss-error weighting, eligibility weighting, site-normalized spectra and replay of 600 earlier fits preserved the original separation (Supplementary Section S4). Uniform-profile capture did not (0.122, 0.169 and 0.176; Fig. 4G, leftmost category): a nearly one-dimensional field can have a signed, nonuniform direction. Positive fixed-profile scaling largely cancels in Adam, explaining calibrated/sign-only agreement; it does not explain the slower pairwise learning with changing exact paths.

The learned credit spectra help explain these input-matched targets' different tolerance of calibrated broadcast. The results favor example-dependent spatial credit without establishing that all six exact channels are necessary.

Additional controls delimit the result. Morphology also matters within a fixed rule: shuffling which input enters which leaf, at fixed tree shape and parameter count, raised exact-path NMSE by 0.575 (95% paired interval, 0.557–0.592; positive in all twenty seeds) pooled over the three task families (Fig. 4H and Supplementary Fig. S13C). Calibrated-broadcast Adam learned a compatible positive-conductance teacher accurately; broadcast learned aligned XOR-of-AND, whereas parity favored exact credit under the tested optimizer (Supplementary Figs. S14–S15 and S18), and the 3,600-fit family-by-optimizer-by-profile screen behind these comparisons is in Supplementary Fig. S13. Noisy-query interaction estimation also yielded useful trees, retained as supporting structure-design evidence in Supplementary Fig. S36. Interaction structure therefore motivates specific credit comparisons rather than a universal monotonic resolution law.

Context-dependent branch tuning makes spatial credit useful in conductance trees

Conductance trees provide a biophysical model for testing this distinction. All rules learned a seven-compartment teacher with monotonic branch responses: extended Adam training kept broadcast NMSE below 0.0006 despite a varying path field (Supplementary Fig. S19). A separate construction tested context-selected branches with aligned or opposing feature tuning (Fig. 5A).

Four terminal compartments fed two proximal compartments and a soma, using the directed E/I conductance balance (Methods). Both subtrees received the same positive feature pairs, $\exp(z_j)$ and $\exp(-z_j)$, for two independent latent inputs z_j . A binary context increased inhibitory conductance in one proximal compartment. Teacher excitation and inhibition produced increasing feature responses in both subtrees for the aligned task, but increasing responses in one and decreasing responses in the other for the opposed task. Teacher and student shared the circuit, ensuring representability. Students shared the teacher's nominal proximal and coupling gains, with independent Gaussian perturbations of SD 0.4 in log conductance; terminal gains started near one. All 24 conductances were plastic, and paired conditions shared initial weights and example streams.

At fixed weights, binary context generates at most two exact path profiles. Their gains are positive: context changes the relative weighting of the two distal subtrees, while proximal credit is context invariant. The nominal parent's conductance denominator changes by approximately eightfold. Thus branch selection is the relevant distinction, connecting this task to Fig. 2A,B.

To supply selection without oracle coefficients, a local rule delivered the somatic error to terminal parameters only when their parent was uninhibited, while retaining ungated error at proximal compartments and the soma. It used the supplied inhibitory context and exact local eligibilities, without evaluating exact paths or projecting an exact field. Two fixed distal profiles with oracle amplitudes and unit proximal credit supplied a separate capacity control; the four deliveries of the somatic error are drawn together in Fig. 5B. An additional proximal profile in the earlier three-profile control therefore addresses proximal updates, rather than another context distinction.

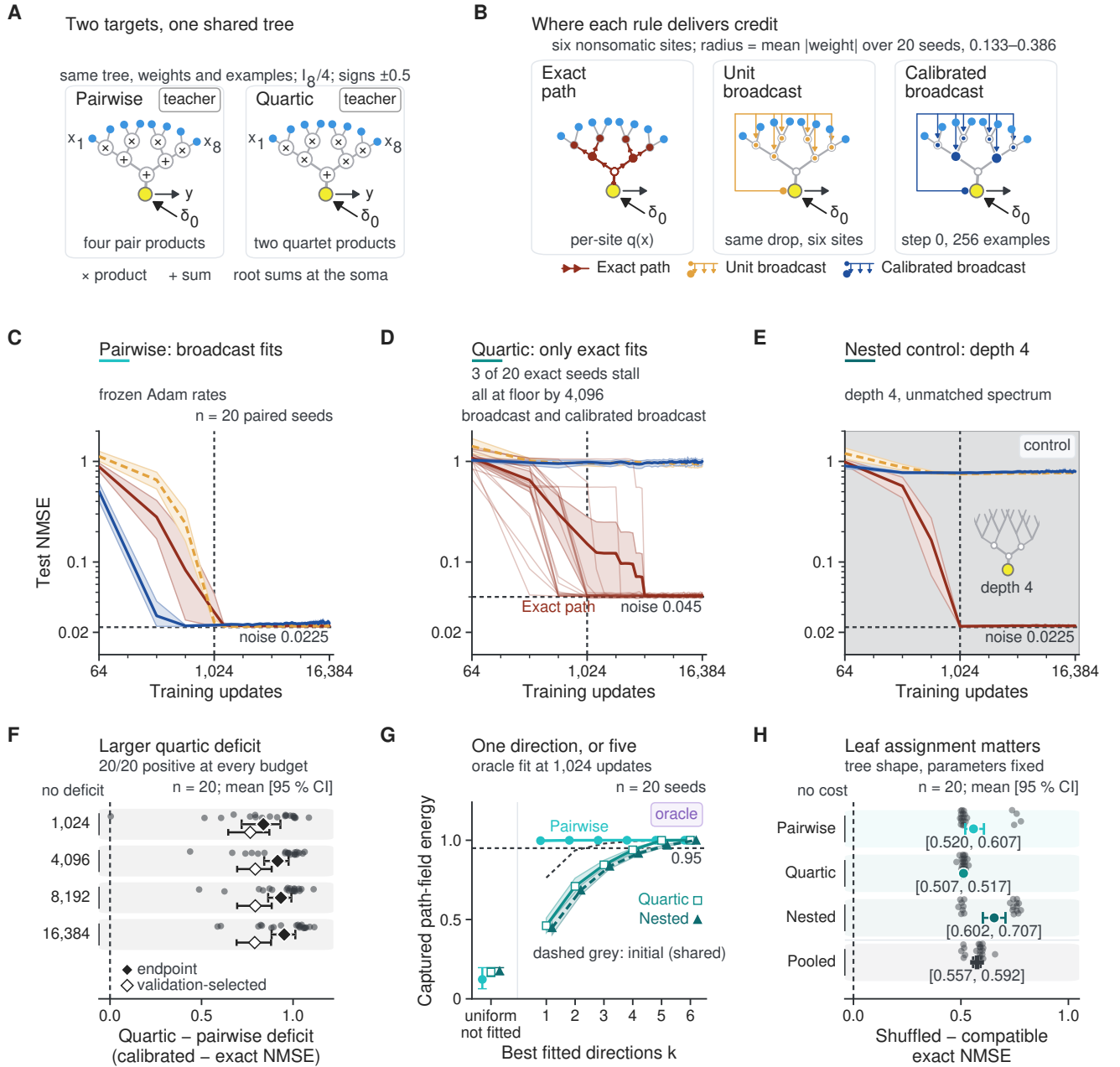


Figure 4: Input-spectrum-matched targets develop different credit geometry and different tolerance of calibrated broadcast credit. **A**, Pairwise and quartic targets on one shared balanced tree: soma lowest, inputs x_1 – x_8 (outer two labelled), each card badged teacher; output y and somatic error δ_0 are the student's, only the junction badges differing. Schematic, no data. **B**, The three credit rules over the six nonsomatic sites: exact path (per-site $q(x)$), unit broadcast (amber bus) and calibrated broadcast (blue bus; drop-dot radii are the mean absolute calibrated weight, 0.133–0.386, signs mixed). Schematic, except those radii. **C**, Held-out NMSE against updates on the pairwise target at the frozen rule-specific rates; label-noise floor dashed; the vertical dashed line in **C**–**E** is the 1,024-update primary checkpoint used in **G** and **H** ($n = 20$ paired seeds, 95% percentile bootstrap bands, fixed checkpoints). **D**, As **C** for the quartic target, all twenty exact trajectories drawn ($n = 20$ seeds, 95% bands, fixed checkpoints). **E**, A depth-four nested control on its own tree, not sharing $I_8/4$ ($n = 20$ seeds, 95% bands, fixed checkpoints); **C**–**E** share one axis, but their NMSE denominators are the clean target variances (1.0, 0.5, 1.0), so floor and curve heights are not comparable between panels; the teal ramp orders targets within this figure only. **F**, Quartic minus pairwise difference in calibrated-broadcast minus exact NMSE at four budgets; dots the paired seeds, filled diamonds endpoint and open diamonds validation-selected states ($n = 20$ seeds, 95% paired percentile intervals). **G**, Cumulative exact-trained path-field energy captured by the best k oracle directions, the uniform profile a separate unfitted category; the series are dodged in k against overprinting. Grey dashed, the step-0 initial spectrum shared by the pairwise and quartic trees; horizontal dashed, the 0.95 threshold; the nested mean is dashed ($n = 20$ seeds, 95% percentile intervals, at that checkpoint). **H**, Exact-path NMSE after shuffling leaf input assignments at fixed shape and parameter count; circles, one task family each; dark plus, pooled over the three families (0.575 [0.557, 0.592]). Grey dots the paired seeds, whose pairwise and nested distributions are bimodal, so those means are not typical seeds; each row prints its own interval under its marker ($n = 20$ seeds per target, 95% paired percentile intervals, endpoint state). Source Data: source_data/curated_publication/figure_04_plotted.csv.

Twenty new paired seeds tested the rule under a protocol frozen after exploratory results. All nine rules ran at three common rates through 16,384 updates; the primary rate and 4,096-update window were fixed before these outcomes. On opposed targets, mean held-out NMSE at validation-selected states within the primary window was 2.18×10^{-5} for exact credit, 2.33×10^{-5} for the local distal gate, 2.28×10^{-5} for the two-profile oracle with unit proximal credit, and 0.487 for broadcast; the fixed-checkpoint broadcast trajectory reaches 0.522 at the 4,096-update tick of Fig. 5D. On aligned targets, the corresponding exact, local-gate and broadcast errors were 7.53×10^{-6} , 1.06×10^{-5} and 5.53×10^{-4} . The opposed-minus-aligned difference in broadcast-minus-gate error was 0.487 (95% paired interval, 0.360–0.615), positive in all twenty seeds. It remained 0.468 (0.332–0.604) within the longer window (Fig. 5E).

The controls identify what the gate must preserve. Reversing its distal selection gave opposed-task NMSE 0.949, with four of twenty fits never improving on their initialization. Applying it at proximal compartments as well gave 0.123 and prevented both proximal inhibitory gains from learning in every seed: their eligibility is nonzero only during inhibition, precisely when that gate closes, and the validation-selected state came within 256 updates, its fixed checkpoint reaching 0.553 (Fig. 5F). The proper local rule changed all 24 parameters and succeeded across the three tested rates. The continuous, shunt-proportional shape of the same local distal gate reached NMSE 2.32×10^{-5} under the same primary rate and window, within 1.4×10^{-6} (-1.7 to 4.8×10^{-6}) of the exact path (Fig. 5F). Both gates were tested with binary inhibitory contexts; continuously varying or noisy context remains untested.

The gate coarsens the exact rule’s strong context-dependent gain ratio. At matched weights, broadcast mixes opposing context-conditioned updates despite a nonnegative per-example inner product with the exact gradient in every seed: in an earlier twenty-seed cohort the fraction of the distal gradient surviving context averaging was 0.64 for exact credit at both initial and broadcast-trained weights, but fell from 0.33 to 0.17 for the calibrated broadcast (Fig. 5G). This supplies a local implementation given inhibitory context and somatic error, while leaving their biological origin open. The comparisons remain window- and parameter-bounded. Exact-path and local-distal-gate fits reached no conductance bound, whereas eighteen of twenty opposed broadcast fits had done so by 4,096 updates and all twenty by 16,384; the earlier wide-bound and rate controls remain in Supplementary Fig. S20.

Task organization sets the benefit; budget sets the ranking

The preceding comparisons hold a compatible tree fixed. Varying serial structure instead tests whether successive compartments help remove unwanted multiplicative changes in input strength. Physical depth D_p counts forward stages, whereas feedback resolution counts distinctions in the backward field.

Each network contained 64 somata and eight nonsomatic compartments per soma. D1 [8], D2 [2,3] and D3 [2,1,2] distributed those compartments across one to three stages in the three- and four-tier comparisons while preserving contacts and trainable resources. The two-tier D2 tree used [4,1], also with eight compartments. Branching vectors are ordered from soma to tips. A resource-identical grouped-point control connected all eight modules directly to the soma; raw-additive trees removed divisive conductance normalization. Flexible point networks provided additional, approximately parameter-matched comparisons (the ceiling of Fig. 6D).

Distal excitation carried class information multiplied by nuisance gains. Inhibitory sensors reported those gains with sensor fidelity α : zero retained the sensor distribution but removed trial-specific information, and one reported the factor exactly. With inhibitory reversal zero, the sensors entered the conductance denominator and could support successive divisions [24, 41]. Nested factors acted over fine, coarse and global input groups; flat factors had equal-resolution supports; locally cancellable factors placed the needed ratio within each module (Fig. 6B). Both nested and flat targets used commutative products; they differed in where factor information was available.

Task-family controls showed what serial organization contributed: at full sensor fidelity, exact-BP serial trees exceeded the resource-identical grouped-point control by 30.32 percentage points for nested gains (30.02–30.62) and 22.20 points for flat gains (21.73–22.65), and by exactly zero for locally available ratios, where all ten pairs were numerically identical. Removing trial-specific sensor information eliminated these benefits (Fig. 6C). Within the nested three-tier task, aligned exact-BP accuracy rose from 61.0% at D1 to 67.5% at D2 and 91.8% at D3, whereas the resource-identical grouped-point control stayed at 61.0% at every depth, reversed sensor placement stayed between 60.6 and 61.1%, and raw-additive trees fell to 51.9%; a parameter-matched point network reached 99.0% (Fig. 6D). Adding a fourth stage did not help: in the four-tier task D4 minus D3 was -1.46 points (-1.74 to -1.16), negative in all ten seeds, and the two-tier task was tested only at D1 and D2 (Supplementary Figs. S22B,C and S24B). Serial structure was useful when the task required integrating factor information distributed across modules.

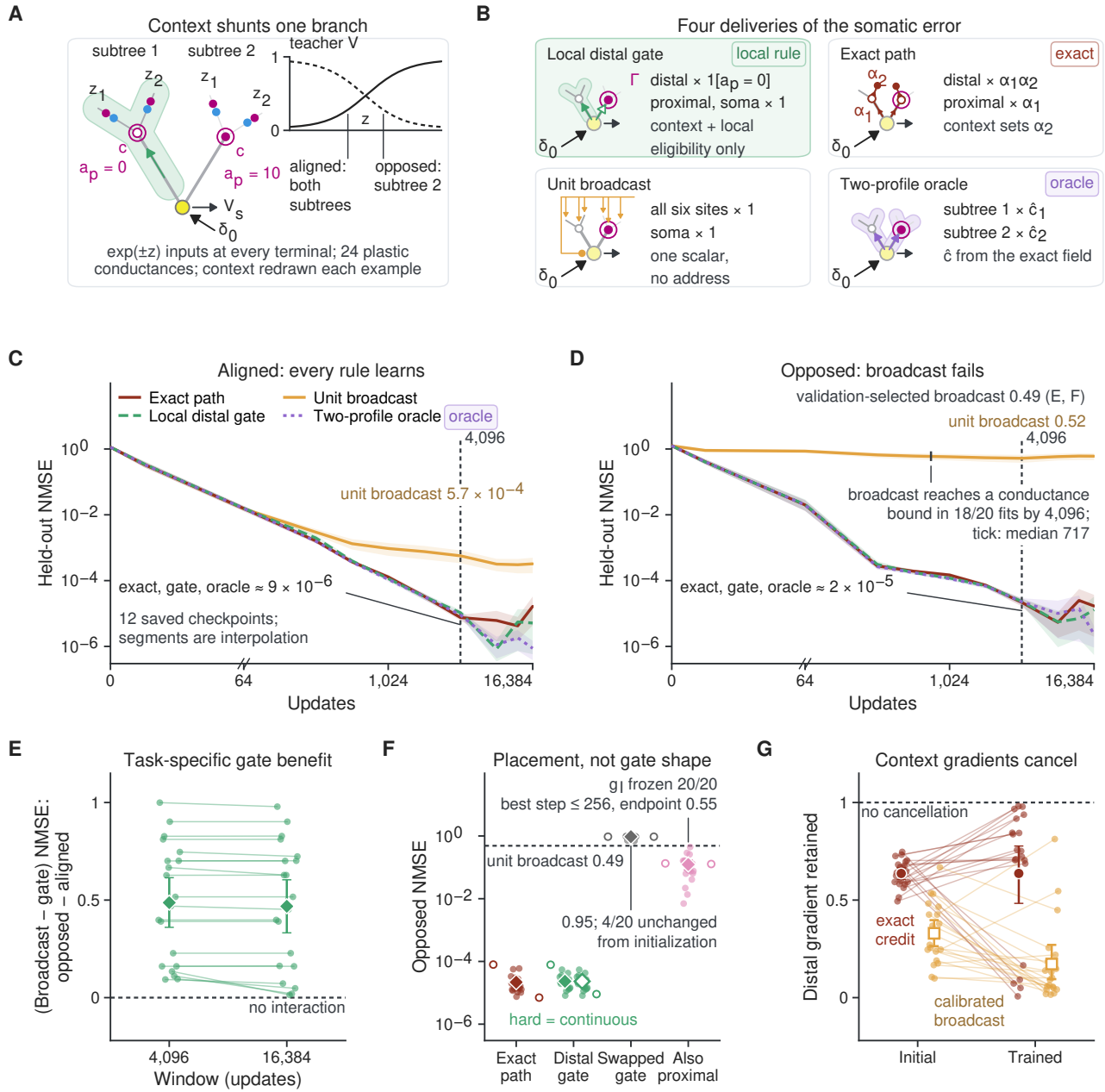


Figure 5: A local inhibitory gate delivers branch-selective credit that matches exact path credit in a conductance tree. **A**, Seven-compartment circuit: four terminals driven by $\exp(\pm z_j)$ feed two junctions and a soma that δ_0 enters; binary context raises one junction's inhibitory conductance ($a_p = 10$; closed, descendants attenuated). Inset, nominal teacher voltage against the latent feature, aligned solid and opposed dashed. Blue, excitatory contacts; magenta, inhibitory contacts and the context gate. Schematic, except the inset. **B**, Four deliveries of δ_0 on identical trees: local distal gate ($\propto \mathbb{I}[a_p = 0]$ at terminals, unit above), exact path (α factors), unit broadcast and two-profile oracle. The card draws the hard gate; both gates restrict Γ (Eq. 4) to $\{0, 1\}$ and $(0, 1]$, and both appear in **F**. Schematic, no data. **C, D**, Held-out NMSE for aligned and opposed targets; means of 20 paired seeds with 95% pointwise whole-seed bootstrap bands; rule key in **C** (exact path solid, gate dashed, oracle dotted, unit broadcast solid); tick in **D**, the median first bound contact at 717 updates. Exact path, gate and oracle coincide within the line width from 256 updates to the 4,096-update tick, and separate beyond it. **E**, Opposed-minus-aligned difference in broadcast-minus-gate NMSE at both windows, hairlines joining seeds; dashed rule, no interaction; $n = 20$ paired seeds, 95% paired whole-seed bootstrap intervals at validation-selected states, prespecified sign-flip $P = 3.8 \times 10^{-6}$ (Holm). **F**, Opposed NMSE within the 4,096-update window by gate shape and placement; at the distal-gate tick the filled and open diamonds are the hard gate $\mathbb{I}[a_p = 0]$ and the continuous gate $1/(1 + g_p^I a_p)$, both with filled seed clouds; small open circles, each column's hard gate at Adam 0.01 (left) and 0.1 (right); dashed rule, unit broadcast. At the swapped gate spread and intervals are narrower than the markers; at the two shapes only the intervals are. $n = 20$ seeds per condition and rate, 95% whole-seed bootstrap intervals, validation-selected states. **G**, Earlier cohort: distal gradient retained after context averaging at initial and broadcast-trained weights, hairlines joining each seed; dashed rule at 1, no cancellation. Exact-credit means coincide at the two states but the trained distribution splits, five of twenty seeds below 0.17. $n = 20$ seeds, 95% whole-seed bootstrap intervals. Source Data: source_data/curated_publication/figure_05_plotted.csv.

Credit comparisons within this useful forward architecture depended on training duration. Ten paired seeds compared D1 exact BP with five D3 credit/optimization conditions for up to 600 epochs, retaining the original rates, parameter groups and thirty-epoch validation patience. Because optimizer and random states were not archived, these are restarts of the original configurations and seeds rather than checkpoint continuations, which is why the 180-epoch D3-minus-D1 gain reads 30.82 points here and 30.86 in the original-budget fits (Supplementary Figs. S22–S24 and Table S9). At validation-selected states from the same trajectories, exact-BP D3 exceeded D1 by 30.82 points at 180 epochs and 38.17 at 600 (Fig. 6E). Eight of ten D1 references stopped early, whereas all fifty D3 fits reached the cap with validation loss still falling (Fig. 6F). The forward benefit is therefore budget-specific, with unequal optimization maturity across depths.

Within D3, the LocalCA exact-path-minus-shared-soma accuracy gap changed from +10.86 points at 180 epochs to −1.52 at 600 (95% paired interval, −1.83 to −1.23), with this budget-indexed sign change in all ten seeds. The mean gap remained negative from epoch 315, reached −2.94 points at epoch 486, and moved back towards zero by 600; the pointwise interval straddles zero intermittently between epochs 300 and 325, so the crossing epoch is not resolved (Fig. 6G). Exact-path LocalCA nevertheless retained lower mean cross-entropy over epochs 180–600, its advantage decaying from −0.073 to −0.021 nats and ending at 0.254 versus 0.276, negative in nine of ten seeds at 600 (Fig. 6H). Accuracy counts threshold crossings, whereas cross-entropy also reflects predicted probabilities. Under the BP recipe, the 600-epoch exact-minus-broadcast accuracy gap was 0.48 points (0.40–0.55), positive in all ten seeds (Fig. 6E). Thus serial organization and spatial credit answer distinct questions: distributed factor information supports the forward architecture, while the eventual ranking of the tested credit rules remains unresolved (Supplementary Figs. S22–S24 and Table S9).

Reconstructed arbors provide spatial routes beyond a shared broadcast

The learning tasks motivate asking what delivery patterns real arbors supply. MICrONS links reconstructed anatomy and connectivity to visual responses [42]. Observed skeletons, soma positions and mapped contact locations and areas define the geometry; conductances, voltages and gradient fields are modeled.

Each inhibitory-bearing segment supplied a candidate route over its input-bearing descendants (Fig. 7A). These supports form dictionary columns, supplemented by a constant column for common broadcast; in the median-sized initial cell the seven selected routes nest and cover 32 of its 70 input-bearing sites (Fig. 7B). For a site-by-route matrix A_K , diagonal excitatory-area weights W and weighted least-squares projector $P_{A_K, W}$, capture of a field $\mathbf{t}$ was

$$\mathcal{C}(A_K; \mathbf{t}) = \frac{\|P_{A_K, W} \mathbf{t}\|_W^2}{\|\mathbf{t}\|_W^2}, \quad \|\mathbf{v}\|_W^2 = \mathbf{v}^T W \mathbf{v}. \quad (7)$$

For an ensemble of perturbations, numerator and denominator sum over field columns. The field is the modeled gradient change after one small focal shunt, and capture is the fraction of its weighted energy inside the dictionary span (Fig. 7C). A common-constrained SVD dictionary provided a ceiling at the same column budget. Controls used random anatomical routes, cable-distance bins, shuffled routes preserving nonzero counts, and surrogate trees preserving degree and depth while changing ancestry.

Small focal shunts in reciprocal cables generated gradient-change fields without inspecting the selected dictionary. These fields nevertheless favor ancestry by construction under a somatic loss, as derived below. The test asks how a small dictionary compresses the tree’s own passive responses. In the initial eight-cell cohort, broadcast captured 31.4% of field energy, near the 32.6% rank-one ceiling (Source Data). Every dictionary therefore contained one common broadcast and $K - 1$ spatial profiles, with the protocol fixed before evaluation in 47 disjoint cells from the same mouse and a reconstructed second mouse.

Varying K separates the common signal from the additional spatial information each dictionary can capture (Fig. 7D). At eight columns in the disjoint 47-cell cohort a shared broadcast captured 0.203 of total energy and broadcast-plus-ancestry 0.619, against 0.578 for degree–depth matched surrogate trees, 0.511 for depth bins and 0.436 for shuffled routes (Fig. 7D); the spatial share beyond that broadcast is 0.416 for ancestry against 0.376, 0.309 and 0.233 for the three controls (Fig. 7E). Cell by cell, ancestry exceeded its own surrogate mean in 38 of 47 cells, but in 11 cells at least half of the 200 surrogate trees matched or exceeded the actual tree (Fig. 7F). The paired ancestry-minus-surrogate advantage was 0.041 of total energy (95% cell-bootstrap interval, 0.020–0.062) and 5.08 percentage points of post-broadcast energy (2.44–7.68); against depth bins it was 0.108 (0.077–0.135) and 13.51 points (9.86–16.84), positive in 45 cells, with



Figure 6: Task organization sets the forward benefit of serial dendrites, and the training budget sets the credit-rule ranking. **A**, Eight nonsomatic compartments per soma in one stage (D1 [8]) or three (D3 [2, 1, 2]): blue, the eight excitatory class-bearing contacts; carmine, inhibitory sensors, all tiers in D1 and one per stage in D3; δ_0 , somatic error; footer, the resource-identical grouped point. D2 and D4 keep that inventory. Schematic, no data. **B**, Generative model: nested gains act on 4, 2 and 1 blocks of the stream, flat gains on eight blocks, local-ratio inputs inside every module; sensors report tier ℓ at fidelity α . Schematic, no data. **C**, Serial-minus-grouped-point test accuracy at fixed D3 under exact BP, by task family and fidelity (colours ordinal here); the local-ratio value at $\alpha = 1$ is an exact tie, not a sampled null (LocalCA counterpart, Supplementary Fig. S22D). **D**, Three-tier task: accuracy against serial depth for three credit arms, a raw-additive mechanism control, two resource-matched controls (grouped and reversed, dodged in depth, agreeing within 0.5 points) and the point-network ceiling and its interval. Intervals are smaller than the markers except for the additive arm, whose means vary by 1.6 points across re-runs; the four-tier cohort is separate and not drawn. **E**, Validation-selected accuracy for six 600-epoch restarts of the same seeds (restarts, not continuations); dotted, each rule's autograd-broadcast variant, the local one overprinting shared soma; thin grey, the D1 reference, lightened once fewer than eight of its ten fits run; bands exceed the stroke only before epoch 100. Paired labels name arms whose 600-epoch ends differ by 0.48 and 0.02 points. **F**, Best validation loss for the same arms; open markers, D1 stopping epochs. **G,H**, Paired exact-path-minus-shared-soma differences in accuracy and cross-entropy; the band in **G** is the window where the crossing is unresolved (epochs 300–330); the dashed rule in **E–H** is the 180-epoch budget. Points are means over ten paired seeds; bars and shading are descriptive 95% seed-bootstrap intervals, of each arm's mean in **D–F** and of the paired difference elsewhere; accuracy endpoints are validation-selected states. Source Data: source_data/curated_publication/figure_06_plotted.csv.

shuffled and random routes at 23.06 and 27.17 points (Fig. 7G). Actual ancestry improves average compression within this passive response family, while coarse geometry explains much of the capacity.

Capture and delivery cost are separate constraints (Fig. 7G, right columns and footer). The eight-column ancestry dictionary used 18.75% of the entries of a fully dense dictionary on average—12.5% for the always-dense broadcast column and 6.25% for the seven route columns—and its shuffled-routes control preserved exactly the same nonzero count at much lower capture. Equal column count does not ensure equal rank, and nonzero entries are a delivery-cost proxy rather than feedback synapses or metabolic expenditure.

The ancestry-over-surrogate-over-depth ordering held in the initial eight-cell cohort and in the Pinky second-mouse cohort, where eight of ten qualifying cells supported the eight-column comparison (Fig. 7H); the rank of the random-route control, by contrast, exchanged with depth bins between cohorts. Pinky’s 9–13 input-bearing sites put ancestry near the residual-capture ceiling, so lower-budget comparisons and exclusions remain separate from the larger arbors. Each cohort derives from a single mouse. Geometry, labeling and compression sensitivities, including direct-label coverage and direct-versus-proxy agreement, accompany the cohort controls in Supplementary Figs. S25–S28, with the wiring-density reconciliation in Supplementary Fig. S25A, the coarse-energy controls in Supplementary Fig. S27B,C and the preprocessing audit in Supplementary Fig. S28A; there, no inhibitory-targeting endpoint persists after joint three-dimensional matching (Supplementary Fig. S26B–D), and the initial cohort’s anatomy-specific coarse-energy excess does not replicate in the disjoint 47-cell cohort (Supplementary Fig. S27D).

Focal shunting sets ancestry-defined route gains with selectivity set by electrotonic state

A focal shunt provides an exact link between anatomical routes and conductance-dependent gains. We compared a focal inhibitory shunt with a current injection matching its baseline focal current in reconstructed reciprocal cables. A separately solved somatic current restored baseline somatic voltage after either intervention (Fig. 8A). The endpoint was the change in the modeled excitatory-conductance gradient at open excitatory contacts.

Let G be the passive conductance matrix and $\mathbf{b}$ the current-source vector, so that baseline voltages satisfy $G\mathbf{V} = \mathbf{b}$. With $\mathbf{e}_k$ the unit vector at compartment k , adding inhibitory conductance κ_k at compartment k changes both G and its source: $G' = G + \kappa_k \mathbf{e}_k \mathbf{e}_k^\top$, $\mathbf{b}' = \mathbf{b} + \kappa_k E_1 \mathbf{e}_k$. The matched injection supplies baseline current $\kappa_k (E_1 - V_k)$ without changing G . The adjoint $\mathbf{q}$ gives loss sensitivity to current injected at each compartment. For a fixed somatic loss derivative, it changes by

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa_k q_k}{1 + \kappa_k (G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (8)$$

The column $G^{-1} \mathbf{e}_k$ gives the voltage response to unit current at k . On a passive tree with a somatic loss, the resulting adjoint change has an exact ancestry structure: all descendants of the shunt share the attenuation factor $[1 + \kappa_k (G^{-1})_{kk}]^{-1}$, while each sister block along the path to the soma shares its own, and the soma-side remainder a fifth factor (Fig. 8B; Supplementary Section S8). These disjoint blocks form an ancestry partition. Weighting their indicators by the baseline soma-to-site transfer gives a dictionary on which the shunt acts through diagonal gains, providing a physical instance of Γ_u , drawn as the block indicator matrix and its gain column in Fig. 8B. The diagonal form applies in this partition basis; it need not hold for an arbitrary selected set of overlapping routes.

In the small-perturbation fields used for Fig. 7, the driving-force term was substantial (0.99–3.50 times the full field’s weighted squared energy across the initial eight-cell cohort) and opposed the adjoint term (weighted correlations -0.979 to -0.913), yet 97.1–99.9% of full-field energy remained within each focal ancestry partition (Supplementary Section S8).

We compared descendants $D(k)$ with depth-matched off-route sites $C(k)$; Supplementary Section S8 gives a kernel criterion for stronger absolute adjoint changes below a shunt, with the within-cell control in Supplementary Fig. S29B and the contact-label and reversal-potential controls in Supplementary Fig. S29E,F. The analyzed gradient was sensitivity to open excitatory conductance, $\gamma_i = q_i (E_E - V_i)$. The local driving force $E_E - V_i$ can change differently within each block, so the complete synaptic gradient need not share a single block gain. With primes denoting post-intervention values and ϵ_γ a numerical floor defined in Methods, we measured absolute log change, $m_i = |\log[(|\gamma'_i| + \epsilon_\gamma)/(|\gamma_i| + \epsilon_\gamma)]|$, and localization $\Lambda_k = \text{median}_{D(k)} m_i - \text{median}_{C(k)} m_i$. Positive localization means a larger gradient change below the shunt.

In the permissive normalized model, shunts changed descendant gradients more strongly than depth-matched off-route gradients, with increasing localization at higher dose (Fig. 8C,D). Current injection changed descendant gradients

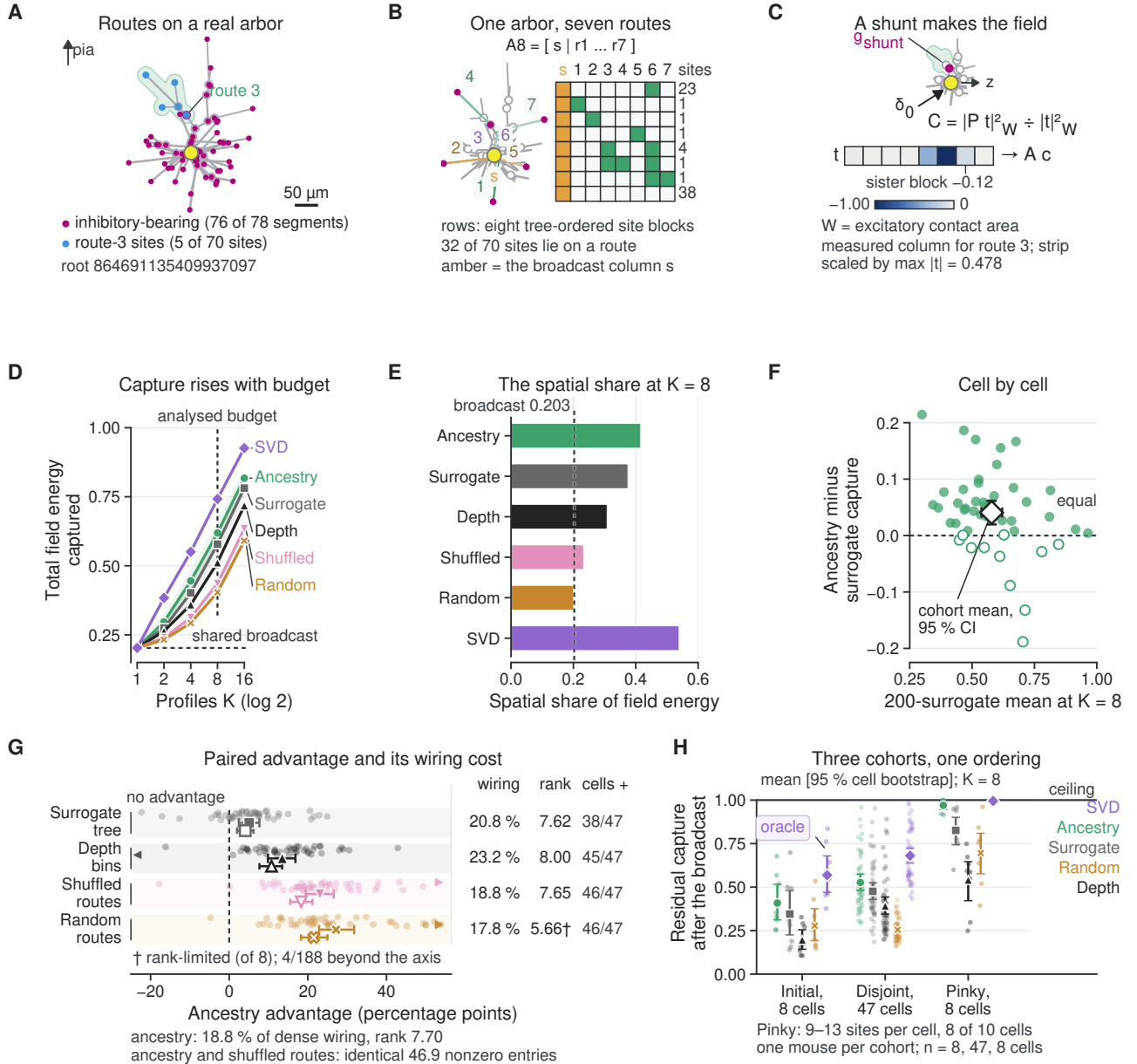


Figure 7: Ancestry routes on reconstructed arbors compress a cell's own focal-shunt response fields better than four budget-matched controls, at a fifth of dense wiring, and the advantage is modest and heterogeneous. **A**, Median-sized arbor of the initial eight-cell cohort (root 864691135409937097, 78 segments): its 76 inhibitory-bearing segments and route 3 over the five sites it addresses; schematic, no data; scale bar, 50 μm . **B**, The same arbor's seven $K = 8$ routes and the matrix A_8 , collapsed to eight tree-ordered site blocks; supports nest; schematic, no data (entries measured). **C**, A focal shunt on route 3 attenuates its descendants and makes the field t (block means); capture is the W -weighted energy of $P_{A_k, W} t$ as a fraction of that of t ; schematic, no data (field column measured). **D**, Total captured energy against budget K ; cell means with unpaired 95% cell-bootstrap bands for ancestry and surrogates; dashed floor, the shared broadcast 0.203; $n = 47$ cells, 46 at $K = 16$. **E**, The spatial share of that energy at $K = 8$, i.e. capture beyond the shared broadcast, for the six dictionaries; dashed line, the shared broadcast 0.203, shown for scale and removed from every bar; cell means, no interval; $n = 47$ cells. **F**, Each cell's ancestry capture minus its own 200-surrogate mean, against that surrogate mean; dashed line, zero difference; open green circles, the 11 cells whose surrogates match the actual tree at least half the time; open black diamond, the cohort mean, with 95% cell-bootstrap intervals on both coordinates; $n = 47$ cells at $K = 8$. **G**, Paired ancestry-minus-control advantages at $K = 8$ on post-broadcast (filled) and total (open) scales, with all 47 within-cell differences; means and 95% cell-bootstrap intervals; right: wiring, rank, positive cells. **H**, Residual capture after the broadcast in three cohorts from two mice (the initial and disjoint cohorts are one MICrONS mouse, Pinky a second); means with 95% cell-bootstrap intervals at $K = 8$; $n = 8, 47$ and 8 cells. Typed-contact and joint 3D matching controls are in Supplementary Figs. S25B and S26B–D. Fields are modeled passive responses: compression capacity, not observed teaching. Source Data: source_data/curated_publication/figure_07_plotted.csv.

roughly threefold less. Its unrelated-site changes were zero by construction under the soma-restoring control, and its depth-matched off-route changes were at most a few times 10^{-4} ; this is not a general claim that current injection cannot change a gradient. Holding the post-shunt voltage but replacing its adjoint with the baseline adjoint removed most localization, supporting a transport contribution beyond the voltage change alone (Fig. 8E). Adding the post-shunt adjoint on top of the substituted driving force ($q'd' - d'$) added 0.079 log units of localization, positive in all eight cells, while substituting the driving force alone left 0.026 against 0.104 for the full shunt.

Physical calibration constrained the spatial selectivity of the gradient change. At axial resistivity $R_a = 150 \Omega \text{cm}$ and specific membrane resistance $R_m = 15,000 \Omega \text{cm}^2$, mean shunt-minus-current localization was small and opposite-signed in the initial eight cells and a disjoint 45-cell cohort from the same mouse: -0.0019 and $+0.0024$, respectively. Signed changes showed attenuation of both descendant and off-route gradients at high resistance; lowering resistance primarily increased descendant attenuation (Fig. 8F), whereas the matched injection enhanced them ($+0.0099$ at $R_m = 15,000$, $+0.043$ at $R_m = 300$, initial eight cells). This increased the localization contrast by more than an order of magnitude (Fig. 8G). Distributed background conductance produced the same qualitative recovery at fixed high membrane resistance (Fig. 8H; -0.0004 , $+0.121$ and $+0.259$ at background leak zero, one and four times baseline, positive in eight of eight cells at one and four), consistent with a dependence on electrotonic state [25, 43].

Adding weak voltage-dependent channel currents produced a dose response close to the passive response at $R_m = 1,000 \Omega \text{cm}^2$ with one baseline-leak unit of distributed background conductance (Supplementary Section S8; Fig. S30D,E). This checks the local linearization around solved steady states. The baseline Jacobian included gate derivatives but remained fixed across doses, so the check does not establish robustness to strongly active dendrites or channel kinetics.

The partition identity and electrical calibration connect anatomical dictionaries to conductance-dependent route gains [26]. Their selectivity depends on cable state. These imposed perturbations establish modeled transport, not endogenous error production or plasticity.

Measured responses bound the anatomical alignment proposal

Measured responses test whether functional relationships follow the ancestry structure suggested by modeled capacity and transport. Seven MICrONS targets had complete arbors and imaged partners: 69 partners in the selected-scan dataset and thirteen eligible scans in the scan-complete analysis. These recordings sampled stimulus responses rather than learning, so no credit was delivered (Fig. 9A).

Partners with more shared dendritic ancestry did not have reliably more similar condition-mean responses after controlling Euclidean separation and soma-to-contact path-length differences (statistic, Fig. 9A; estimate, Fig. 9B). Mean partial rank correlations were -0.069 for selected scans and -0.048 for all eligible scans, the same seven targets re-analyzed, and both target-bootstrap intervals included zero (Fig. 9B,C). The null did not depend on the measure: raw shared-path (-0.025), tree-distance (-0.017) and same-major-branch (-0.003) associations were also centered on zero (Fig. 9C; Supplementary Fig. S31). This provides no evidence of preferential ancestry alignment in the sampled partners, at a different scale from population-level like-to-like wiring [44].

To establish the sensitivity of this small sample, response-level simulations preserved observed partners, shared inputs, repeated stimuli and seven-target inference, with noise calibrated to measured repeat reliabilities. Under the specified Gaussian ancestry-signal model, 80% detection required a mean simulated target-level partial rank correlation of approximately 0.25 (0.249 by interpolation, or $\lambda = 0.65$ once the Monte Carlo interval is respected), and the observed estimate and its whole interval lie below that point (Fig. 9D). With perfect reliability the same threshold falls only to 0.248, so the limit is the seven-target sample rather than measurement noise. The observed repeat reliabilities used for attenuation vary across recordings (Fig. 9D, inset). This conditional threshold leaves weaker effects, different signal geometries and unobserved partners unresolved (Methods; Supplementary Section S9).

A separate passive-tree model predicted each target's responses from its recorded partners. Its selected ancestry routes reached only 48.0% of mapped inputs, and in six of the thirteen scans every route reached a single input (Fig. 9E,F), so restricting the transfer profile to these supports tests transfer geometry more than example-dependent credit. Held-out prediction, input-removal and reconstruction controls are grouped in SI (Supplementary Figs. S31–S32), where the ancestry-minus-ridge difference (0.029; 95% target-bootstrap interval, -0.013 to 0.074) is matched by random anatomical routes (0.031; -0.009 to 0.068). This model diagnostic and the measured ancestry-similarity test constrain different parts of the anatomical proposal.

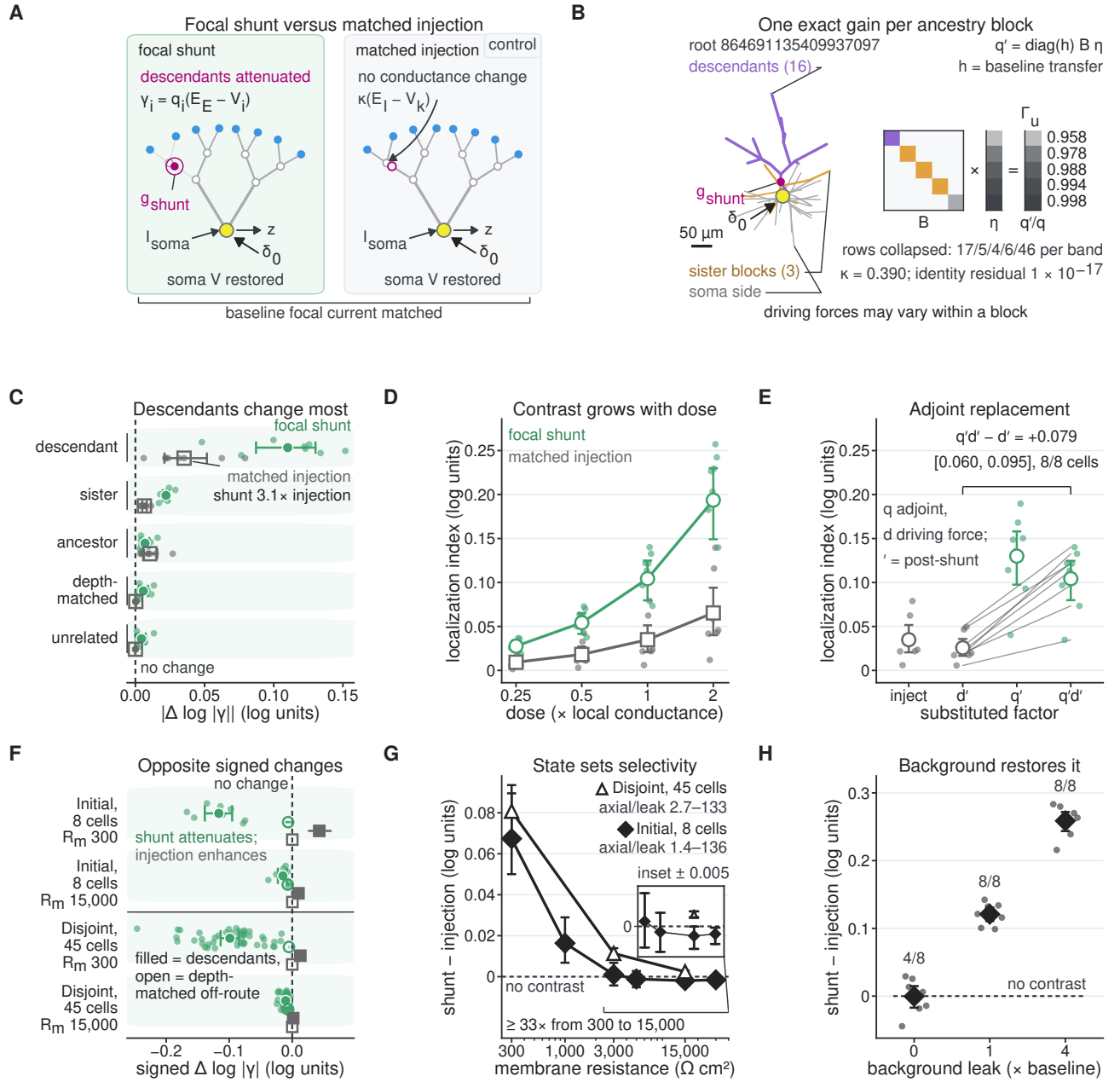


Figure 8: In a passive model, focal shunts apply ancestry-defined gains whose spatial selectivity is set by electrotonic state. **A**, Two state-matched interventions: a focal inhibitory shunt (g_{shunt} , ringed contact; descendants faded) and a matched current injection $\kappa(E_I - V_k)$ (open contact); I_{soma} restores baseline somatic voltage. Blue, excitatory contacts. Schematic. **B**, Exact ancestry partition of the median reconstructed cell (as in Fig. 7A) shunted at segment 4227: the site, sixteen descendants, three sister blocks, soma side. Block indicators times one gain per block ($\eta = 0.958\text{--}0.998$, $\kappa = 0.390$) give the adjoint ratio exactly: a physical instance of Γ_u (Eq. 4). Scale bar, 50 μm . Anatomical schematic; blocks and gains are derived (Source Data). **C**, Absolute log change in gradient magnitude by relation to the site (unit dose, permissive normalized passive parameters); depth-matched injection changes average 2.7×10^{-4} , unrelated-site changes are zero by construction. **D**, Localization index (descendant minus depth-matched off-route change) against dose normalized to local leak plus synaptic conductance. **E**, Factor substitution in the exact gradient: injection, driving force (d'), adjoint (q'), both ($q'd'$). Green, conditions carrying the post-shunt adjoint q' ; grey, the baseline adjoint q . Thin lines join the same cell across d' and $q'd'$; the bracket is their paired difference. Adding the post-shunt adjoint on top of the substituted driving force ($q'd' - d'$) adds 0.079 log units [0.060, 0.095]. **F**, Signed log change at two membrane resistances; green circles, focal shunt (attenuates); grey squares, matched injection (enhances); filled, descendants; open, depth-matched off-route sites. **G**, Shunt-minus-injection localization against membrane resistance; filled diamonds, the initial eight-cell cohort; open triangles, the disjoint 45-cell cohort. Inset, the bracketed window at ± 0.005 (the 45-cell $R_m = 3,000$ point is outside it): opposite cohort signs at $R_m = 15,000$. **H**, Distributed background conductance restores that contrast at $R_m = 15,000 \Omega \text{ cm}^2$, normalized dose one. **C–H**, unit, a reconstructed cell (sites averaged within it); symbols and whiskers, cohort means and 95% cell-bootstrap intervals; dots, cells (**F**, shunt descendants only; none in **G**); endpoint, the post-intervention steady state. Fill is panel-local and keyed in **F** and **G**; in **C** it repeats colour, and **E** draws every mean open. $n = 8$ cells (101 focal sites) from one MICrONS mouse; **F** and **G** add the disjoint 45-cell cohort (235 sites). Source Data: source_data/curated_publication/figure_08_plotted.csv.

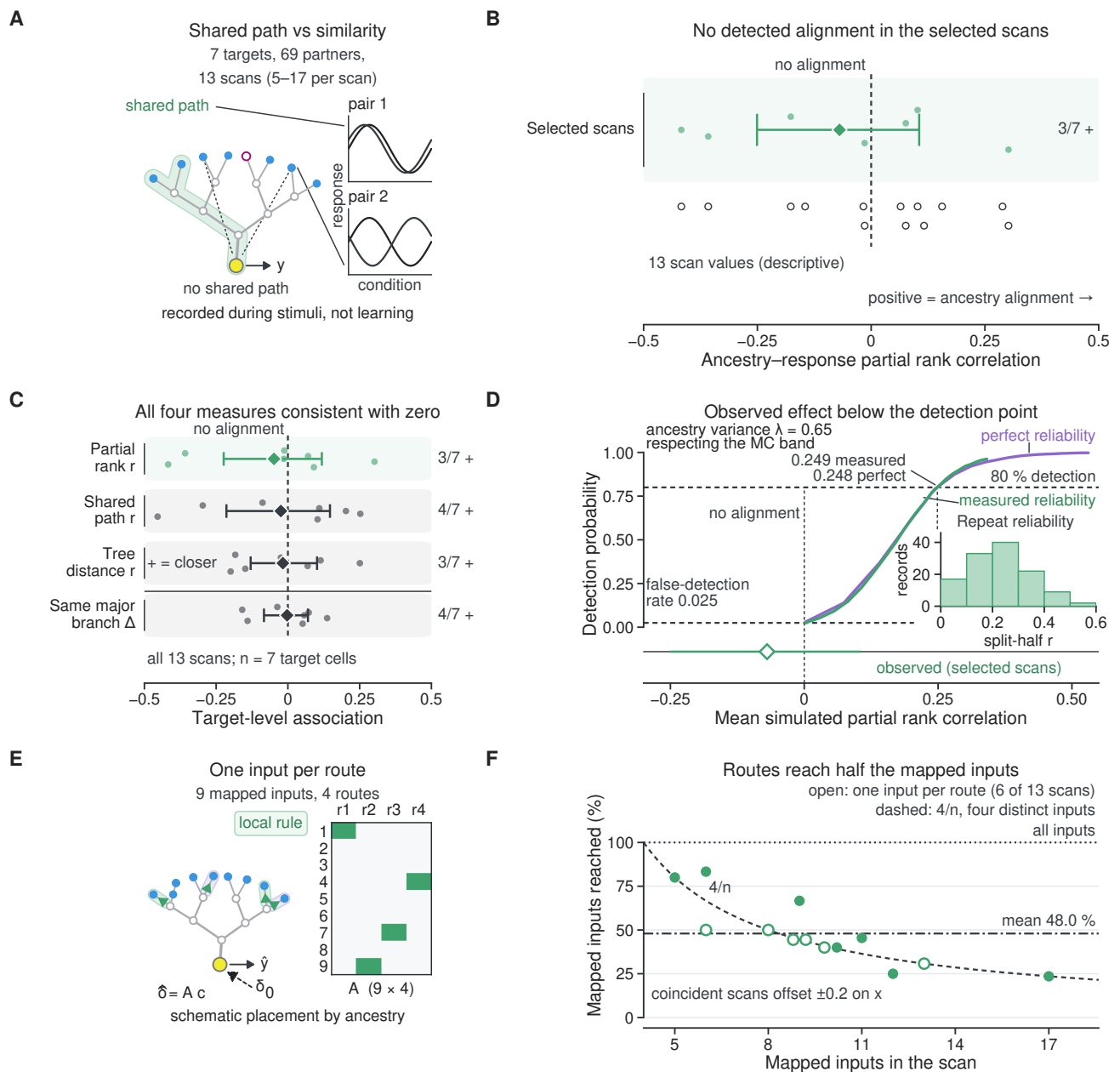


Figure 9: Recorded partner responses show no dendritic-ancestry alignment, and this sample detects only effects above a partial rank correlation of about 0.25. **A**, The statistic: two contact pairs on one arbor, one sharing a soma-to-ancestor path (green patch), the other linked to the soma by dashed hairlines; filled blue discs, excitatory contacts; open magenta ring, an inhibitory partner; insets, their response comparison. No data: these are stimulus recordings, with no δ_0 . **B**, Ancestry–response partial rank correlation, selected-scan cohort; dots, the seven target values; open circles, the thirteen all-eligible-scan values (same targets, stacked where they would overlap); diamond, mean with 95% target-bootstrap interval ($n = 7$): $-0.069 [-0.251, 0.106]$. **C**, Same inference, four topology measures, all thirteen scans; means and 95% target-bootstrap intervals ($n = 7$ target cells). Top row, the all-eligible-scan partial rank correlation, $-0.048 [-0.224, 0.118]$. Tree distance is sign-flipped, so positive means closer. Below the rule, same-major-branch Δ is a difference in mean similarity, not a correlation: its narrower interval reflects smaller units, not precision. **D**, Detection probability against the frozen model’s mean simulated partial rank correlation; shaded 95% Wilson Monte Carlo intervals (half-width ≤ 0.016); the measured-reliability intercept is its false-detection rate, 0.025 (0.030 for perfect reliability). The rug locates the observed estimate ($n = 7$ target cells, 95% target bootstrap), not power. Inset, split-half repeat reliability, 123 of the 125 partner–scan records (from 102 partners), the two with negative r omitted. **E**, Representative scan (median mapped-input count, no outcome selection): nine mapped inputs placed by ancestry beside the realized 9×4 support, $\hat{\delta} = A c$ imposed rather than observed (schematic; dashed δ_0). Route-capsule tints aid separability only and the *local rule* badge applies to the whole card; neither encodes a per-route property. **F**, Mapped inputs reached per scan: open, the six scans whose four routes each reached one input; dashed, 4/n, coverage when four routes reach four distinct inputs. Not a ceiling: two scans fall below it (repeated inputs), three above (multi-input routes). Coincident scans are offset ± 0.2 horizontally; $n = 13$ scans from seven targets, descriptive. No measured positive control exists. Source Data: `source_data/curated_publication/figure_09_plotted.csv`, `source_data/functional_topology_all_scans/`, `source_data/credit_first_figures/figure_08_sources.json` and `figure_08_support.csv`.

Discussion

Learning benefits from spatial selection when a common signal mixes incompatible synaptic changes. Dendritic morphology supplies candidate delivery patterns, teaching mechanisms supply their coefficients, and conductance regulates their gains. In the single-layer image classifiers, preserving neuronal identity supplies most of the benefit, including with random between-neuron feedback. Contextual conflict favors branch selection, supplied through feedback or local eligibility gating. Hierarchical distractors favor ancestry at a bandwidth predicted by the generator's coefficients. These results distinguish useful information from its physical carrier.

The interaction comparison explains why input rank alone is insufficient: pairwise and quartic targets share that spectrum and a compatible tree, yet learn different credit fields with different tolerance of fixed profiles. Spatial direction matters alongside dimension: even a nearly one-dimensional field can be poorly captured by uniform broadcast. These learned-state properties do not rescue the failed initialization selector. Intermediate dictionaries, adaptive profiles and alternative dynamics remain plausible solutions.

Conductance trees connect this distinction to computation: simple credit learns compatible monotonic teachers, whereas opposed tuning under contextual inhibition benefits from selection. A local inhibitory gate supplies that selection without oracle path coefficients, provided proximal credit remains ungated. The task supplies context and somatic error. Forward compatibility is a separate constraint: the scalar multi-affine cut criterion connects input grouping to hierarchical tensor descriptions and supports structure estimation from noisy examples [45–47]. Serial composition likewise helps remove distributed nuisance factors, but the LocalCA accuracy advantage of exact credit changes sign between the 180- and 600-epoch budgets while exact credit retains lower cross-entropy. The gap is still moving at the cap; forward depth and backward resolution need separate interpretation.

Obtaining coefficients remains a limitation beyond this explicit context gate. The separate encoder experiment exposes two obstacles: identifying a route from noisy cues and, when calibration is small, preventing coefficient leakage into incompatible routes even after identification is correct (Fig. 3F). With 256 noise-free cues the soft readout came within 0.31 points of the oracle, so leakage is a calibration cost rather than an intrinsic property of a soft readout. Exploratory hard selection removes it at every calibration size, but leaves a substantial noisy-cue deficit. Exact paths and projected coefficients elsewhere remain information-access controls. An adequate span alone does not ensure useful delivery.

Reconstructed arbors compress their modeled passive response fields beyond a common broadcast. Degree–depth surrogates explain much of this capacity, and the target fields themselves favor ancestry through passive transport under a somatic loss. Testing alignment beyond that construction requires measured fields or a model with different error sources or active dynamics; changing the passive perturbation family alone would not suffice. Focal shunting supplies a biophysical link by reweighting the baseline-weighted ancestry partition. Driving-force changes partly cancel adjoint changes, while cable state sets the spatial selectivity of the resulting gradient. This concerns sensitivity transport, distinct from inhibition that computes errors or regulates forward gain and load [48–51].

The small recorded cohort provided no evidence of preferential ancestry alignment; the offline learning diagnostic chiefly tests an almost fixed transfer profile. A direct experiment would compare plasticity below a focal inhibitory site with electrotonically matched off-route sites while matching somatic voltage, first-order current and outcome signal. Changing conductance state would test the predicted separation between shared and ancestry-selective effects [22, 52, 53].

The framework complements somatic prediction, segregated teaching, interneuron-mediated errors and burst plasticity [32, 34–36], alongside cable-learning and engineered gating models [39, 54–57]. Spikes, temporal eligibility, recurrence and channel kinetics may change the available profiles and coefficient estimation [58, 59]. The organizing question remains which task-relevant distinctions the arbor and learning mechanism can jointly preserve.

Methods

The Supplementary Information's contents and cohort index (Table S11) locate complete derivations and protocols by scientific question. Resolved configurations, seeds, checkpoints, exclusions and source hashes accompany the numerical tables. Diagnostics reusing a cohort retain its original independent units.

Conductance models and delivery coordinates

In a directed tree, $[b_1, \dots, b_D]$ lists branching factors from soma to tips, giving $\sum_{d=1}^D \prod_{r=1}^d b_r$ nonsomatic compartments. A child c supplies output $a_c = f_c(V_c)$. With leak g_n^L , aggregate open excitatory/inhibitory conductances $G_n^X = \sum_{i \in \mathcal{S}_n^X} g_i^X x_i^X$ for contact class X , and child couplings $g_{c \rightarrow n}^{\text{den}}$, the steady voltage is

$$V_n = R_n^{\text{tot}} \left(g_n^L E_L + G_n^E E_E + G_n^I E_I + \sum_c g_{c \rightarrow n}^{\text{den}} a_c \right), \quad (R_n^{\text{tot}})^{-1} = g_n^L + G_n^E + G_n^I + \sum_c g_{c \rightarrow n}^{\text{den}}. \quad (9)$$

Here $x_i \geq 0$ is contact activity, $g_i \geq 0$ its conductance, and E_X its reversal potential. A terminal-to-soma pass evaluates the directed steady state. Artificial conductance trees use dimensionless leak one and reversals $E_L = E_I = 0$, $E_E = 1$. Monotone parameter transforms enforce nonnegative conductances. Branch-output nonlinearities, termed reactivations in the implementation, are identity functions when disabled. Both excitatory and inhibitory contacts learn in the regular-tree experiments.

For a reciprocal cable residual $\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = 0$, the nonsingular voltage Jacobian J_V gives the adjoint

$$J_V^T \mathbf{q} = \nabla_{\mathbf{V}} \mathcal{L}, \quad \partial \mathcal{L} / \partial g_i = -\mathbf{q}^T \partial \mathbf{F} / \partial g_i. \quad (10)$$

With passive current balance, the synaptic gradient is $x_i(E_i^{\text{rev}} - V_n)q_n$. This established sensitivity calculation supplies the reference for restricted credit [9, 10, 33, 39]. For output $a_n = f_n(V_n)$, activation error $\delta_n^a = \partial \mathcal{L} / \partial a_n$ gives voltage error $\delta_n^V = f'_n(V_n) \delta_n^a$. Image LocalCA replaces δ_n^a while retaining f'_n and exact local gate derivatives in eligibility. Fixed activation-space profiles can therefore vary in voltage space; exact supplied fields recover exact selected gradients.

The raw-additive control retains the graph but sums signed inputs without the conductance denominator or reversal-potential driving force. Historical normalized-additive controls additionally standardize voltage across stage coordinates within each example; Supplementary Section S2 defines those variants. The optional four- and five-factor rules add stage-level preconditioners and are described in Supplementary Section S1.

In $A\Gamma$, nonzero diagonal gains preserve the dictionary span and oracle capture while changing coefficients and conditioning (Supplementary Fig. S5B–D). This describes a passive shunt in its baseline-weighted ancestry partition, not arbitrary overlapping dictionaries. Stage count, feedback rank, external error dimension and state-dependent path resolution remain distinct (Supplementary Tables S1–S2).

Image feedback ladders and checkpoint diagnostics

The trained decoder supplies somatic activation errors $\delta_{i,u} = \partial \mathcal{L} / \partial a_{i,0,u}$ for example i and neuron u . Strict scalar feedback uses their mean, or their largest-magnitude signed coordinate when the mean’s magnitude is at most 10^{-6} , throughout the core including somatic gates. Per-neuron feedback preserves each coordinate; the exact path additionally applies local path derivatives. The historical matched-width fallback retained neuronal identity at the 128-wide somatic stage, affecting 640 of 2,413,312 core parameters. A separately trained strict-scalar control tests that exception (Supplementary Fig. S2, Supplementary Section S2 and Source Data indexed in Table S12).

The new MNIST ladder uses 128-soma [3, 3] shunting and raw-additive networks, with 40 excitatory and 20 inhibitory contacts per nonsomatic compartment. Within each architecture, rules share initial contact construction, initialization policy, decoder and data streams. Pixels are divided by 255 and flattened without further normalization or augmentation. Each seed divides the 60,000 training images into 48,000 training and 12,000 validation examples; all 10,000 official test images remain held out. Training uses 180 epochs, batches of 256 and the validation-selected state (minimum validation cross-entropy), without early stopping. Adam uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ and no weight decay. Original rates are 0.0015 for input synapses and decoder, 0.0007 for branch couplings and 0.0005 for reactivation gates.

The projected $K = 1, 3$ rules average exact activation errors over the supports in Fig. 1C, preserving the field mean but not necessarily its norm. Both retain exact somatic errors and local derivatives. Decoder-only learning freezes the initialized core, verified by its hash. Three development seeds compare rate multipliers 0.3, 1 and 3 for every rule and architecture, giving 108 fits. Validation alone selects rates. Ten fresh paired seeds per architecture and rule give 120 selected-rate and 120 original-common-rate outcomes from 190 unique fits; 50 fits serve both policies. The projected- $K = 1$ addendum was frozen before fresh outcomes. Complete protocols, clipping and excluded implementation checks are in Supplementary Section S2.

Supplementary Section S2 gives the original fifteen-seed MNIST [60] ladder, ten-seed Fashion-MNIST [61] and twenty-seed raw-additive CIFAR-10 [62] protocols; CIFAR uses a 20-soma [3, 3, 3, 3] tree. Original gradient diagnostics appear in Fig. S8A–C. The fifteen-seed direct-feedback-alignment factorial replaces the decoder derivative by a fixed Gaussian matrix scaled by inverse square root of output dimension, paired with exact-readout references. Recipient permutations preserve coordinate values and bandwidth.

Common-checkpoint diagnostics compare rules at identical weights and states. Exact path transport is checked against automatic differentiation. New capture measurements use 2,048 held-out images at initial and exact-trained states; activation- and voltage-space metrics are kept separate. Gradient cosine, error magnitude by depth and mean-subtracted path energy are distinct summaries; none alone certifies learning performance, and their joint behaviour at 120 valid trained checkpoints is in Supplementary Fig. S3E. Historical depth and noise cohorts, including their task-identity and conductance-validity records, are specified in Supplementary Section S2.

Contextual conflict and ancestry routing

The branch-conflict task uses Eqs. 5–6, with independent opposite-class replacements of nonselected Fashion-MNIST inputs at probability χ . Twenty paired seeds cross $B = 2, 4, 8$ with eight doses. Gated-point and BP calculations receive equivalent updates; A deranged route changes recipients at fixed rank and support. A separate two-stream interference cohort (ten paired seeds 4100–4109, $K = 2$ subtree routes, two contexts) trains on context 0, switches to context 1 for twelve epochs and re-tests context 0 on 1,024 held-out examples; the retained loss is reported as context-switch forgetting (source_data/trained_subtree_address/).

Under centered, symmetric class evidence, shared-mode compatibility is $s(\chi) = 1 - 2\chi(B - 1)/B$, with zero at $\chi_c = B/[2(B - 1)]$. Trained crossings are evaluated separately by mean-curve interpolation and each seed’s first sampled at-or-below-chance dose, retaining non-crossings. Checkpoint re-execution reproduces endpoints and compares exact/shared fields at both common and own trained states (Supplementary Fig. S10B–D). The measured counterpart of $s(\chi)$, reported as the initial signed utility, is the Source Data column mean_initial_signed_utility: the initialization-state inner product of the neuron-shared update with the exact update, signed and normalized so that the analytic $s(\chi)$ is its prediction. It is distinct from the nonnegative capture U reported elsewhere.

The ancestry factorial contains eight eight-dimensional blocks. Context c_i and binary label sign s_i are balanced, and inputs are $\mathbf{x}_{i,b} = a(b, c_i)s_i\mathbf{t}_b + \boldsymbol{\epsilon}_{i,b}$, with seed-specific unit teachers $\mathbf{t}_b$ and Gaussian noise $\boldsymbol{\epsilon}_{i,b} \sim \mathcal{N}(0, 1.1^2 I_8)$. The selected coefficient is one; sibling, same-half nonsibling and opposite-half coefficients are $-0.15, -0.45, -0.75$. A leaf permutation changes the task-to-route assignment without changing the streams. For budget K , ancestry feedback has context-by-block entries $\Phi_{cb}^{(K)} = \sqrt{K/8} \mathbf{1}[g_K(\pi(c)) = g_K(b)]$, where g_K indexes contiguous groups and π maps streams to leaves. Thus every row has unit Euclidean norm.

The update is $\Delta \mathbf{w}_b = -(8\eta/N) \sum_i \delta_i \Phi_{cib}^{(K)} \mathbf{x}_{i,\pi^{-1}(b)}$, with N training examples and rate η . Each seed supplies 2,048 training, 2,048 test and 1,024 switch examples. Full-batch descent uses rate 0.3, eighty training epochs and twenty switch epochs. Controls include deranged ancestry, depth-interleaved and random-sparse supports, dense rank- K fields and a training-informed PCA reference. The best matched control is the per-seed post-training maximum over the four non-PCA controls. Integer test counts preserve ties in the named Wilcoxon comparison; Holm correction covers four budgets. Alternative mean-sign-flip and historical BH analyses remain explicitly labeled in the Source Data indexed in Supplementary Table S12. The separate noisy-context encoder and cue-delay experiments are fully specified in Supplementary Section S3.

Update utility and prospective selection

Let the exact gradient be $\boldsymbol{\mu}$ and the delivered update estimator have mean $\mathbf{m}$ and covariance Σ_{route} at the current weights. For an L -smooth loss and a nonnegative step η , expected loss after the update is bounded above by $\mathcal{L} - \eta \boldsymbol{\mu}^\top \mathbf{m} + (L\eta^2/2)(\|\mathbf{m}\|^2 + \text{tr} \Sigma_{\text{route}})$. Optimizing this expression gives utility $U = [\boldsymbol{\mu}^\top \mathbf{m}]_+^2 / [2L(\|\mathbf{m}\|^2 + \text{tr} \Sigma_{\text{route}})]$, with zero for a zero denominator. Utility certifies one-step progress under the stated moments; it is not a long-horizon ordering.

Synthetic quadratic screens use sixteen Haar coordinates and a designed signal/noise crossover, without a forward neuron (Supplementary Fig. S3). Retrospective factorial utility uses actual full-batch updates and a global logistic-curvature bound; maximum bandwidth matches its selection success. Supplementary Figs. S34–S35 retain the failed

12,800-fit prospective selector, its loss to fixed maximum-budget and rank-only baselines in every seed, and the 25,600-fit dynamics extension with evolving noise, observed-context-count controls, sealed choices; the computational-cost accounting is indexed in Supplementary Table S12.

Interaction compatibility, fixed profiles and learned spectra

For a subtree leaf set S , C_S arranges the target’s Fourier interaction coefficients by products inside and outside S ; C_S° removes the constant inside row. Scalar multi-affine trees represent the target exactly if and only if every centered cut has rank at most one. If $\mathcal{S}(T)$ contains all proper cuts of tree T and singular values are ordered from largest to smallest,

$$\text{MSE}(f, \hat{f}_T) \geq B(T; f) := \max_{S \in \mathcal{S}(T)} \sum_{j \geq 2} \sigma_j(C_S^\circ)^2. \quad (11)$$

Overlapping cut tails cannot be added as a lower bound. Supplementary Section S4 gives the constructive proof, minimum-depth algorithm and its relation to hierarchical tensor representations. The twelve-tree census with its candidate-selection outcome, the noisy-query estimator, end-to-end learning pipeline, conductance grouping controls and Boolean truth tables remain in Supplementary Figs. S12 (selection in panel C), S14–S15, S18 and S36. The theorem’s assumptions do not extend automatically to general conductance ancestors.

Three development seeds select rates with equal budgets; twenty fresh seed blocks pair matching and quartic conditions across trees, assignments, initialization, inputs, additive noise and minibatch order. Exact, unit, calibrated and sign-only profiles share local eligibilities; calibration averages path derivatives over 256 unlabeled initialization examples. Selected/common rates under SGD and Adam yield 432 development and 1,200 fresh fits. Coefficients stay within $[-2, 2]$, with bound contacts recorded. A balanced extension replays all seven original checkpoints through 1,024 steps, then continues all 720 unique selected/common-rate algebraic trajectories to 16,384. Terminal and validation-selected extensions describe the same seeds (Supplementary Section S4). Noisy held-out NMSE divides by clean target variance: one for matching and nested, one-half for quartic. All have noise variance 0.0225. The primary Adam contrast subtracts the pairwise calibrated-minus-exact gap from its quartic counterpart.

For evaluation-example-by-site path matrix Q , the best rank-one profile is the leading right singular vector. Its capture is $\sigma_1(Q)^2 / \sum_j \sigma_j(Q)^2$, an oracle reconstruction with example-dependent amplitudes. Additional diagnostics evaluate uniform, calibrated and ancestry profiles, loss-residual weighting, eligibility-weighted updates and root-mean-square normalization of each site. Participation-ratio effective rank is $(\text{tr}C)^2 / \text{tr}(C^2)$ for uncentered $C = Q^T Q / N$. At fixed weights and inputs, changing labels leaves the path field unchanged; eighty relabeling controls verify this. Independent checkpoint analysis and faithful replay of 600 earlier fits provide numerical and trajectory checks. Complete definitions, pairing audits, optimizer effects and the positive-conductance boundary are in Supplementary Sections S4–S5.

Conductance trees with context-dependent feature tuning

The seven-compartment model uses identity branch outputs in Eq. 9. Two independent $z_j \sim \mathcal{U}[-2, 2]$ generate positive inputs $\exp(\pm z_j)$ for both subtrees. Every terminal has two excitatory and two inhibitory gains; binary context inhibits one proximal compartment. All 24 conductances are positive and parameterized by logarithms. Students start from terminal gains one, proximal inhibitory gains 15, daughter couplings 20 and 0.2 per subtree, and soma couplings four. These proximal and coupling values match the teacher’s nominal scaffold, not its realized weights. Independent Gaussian log perturbations have SD 0.4 in students and 0.15 in teachers. Full generators and eligibility derivatives are given in Supplementary Section S5.

The local-gate follow-up used twenty new paired seeds and nine rules at Adam rates 0.01, 0.03 and 0.1, giving 1,080 trajectories. Each task supplied 2,048 training, 1,024 validation, 4,096 test and 512 unlabeled calibration examples. The primary rate 0.03 was fixed from earlier work, not selected on these seeds. Minibatches contained 128 examples; logarithmic conductances stayed within $[-7, 7]$. Minimum validation NMSE selected states within fixed 4,096- and 16,384-update windows. The somatic error was $(V_s - y) / \text{Var}_{\text{train}}(y)$; NMSE uses clean target variance. At a terminal with parent p , the hard gate multiplied this error by $\mathbf{1}[a_p = 0]$, where proximal inhibitory activity a_p was zero or ten. Proximal and somatic multipliers remained one. The continuous gate used $1/(1 + g_p^1 a_p)$. Both combined these signals with the same exact local eligibilities and accessed no exact path derivatives.

Controls used exact credit, unit and initially calibrated broadcast, two weighted distal oracle profiles with unit proximal credit, three oracle profiles including a proximal column, reversed distal gating, and gating at proximal sites as well. Two primary paired sign-flip tests—opposed broadcast-minus-gate error and its opposed-minus-aligned difference—were Holm-adjusted together. Other comparisons and 95% whole-seed bootstrap intervals were descriptive. The protocol followed exploratory results but preceded execution on new seeds. Supplementary Section S5 and Figs. S19–S21 give full rules, development, wider-bound and local-gate controls; the context-cancellation diagnostic is in Fig. 5G.

Serial depth, sensor fidelity and optimization

Physical-depth networks contain 64 somata and eight nonsomatic modules each. D1–D4 reallocate these modules across branching vectors [8], [2, 3], [2, 1, 2] and [1, 1, 2, 2] in the three- and four-tier comparisons; the two-tier D2 tree uses [4, 1]. Contacts and resources are matched; serial composition changes. Grouped-point controls connect every module directly to the soma. Raw-additive controls retain the graph without division, while flexible MLP controls are approximately parameter matched.

Excitatory blocks carry class evidence multiplied by lognormal nuisance factors. Inhibitory blocks sense those factors with sensor fidelity α : zero removes trial-specific information while preserving the sensor distribution, and one reports the factor. The generator floors noisy input rates before multiplying them by gains. Nested, flat and local-ratio families distinguish distributed hierarchical support from generic factor cancellation and locally available ratios. Full generators, floors, sensor-placement controls and contact inventories are defined in Supplementary Section S6, and the calibration fixing this study’s operating point is in Supplementary Fig. S23.

Original cohorts use ten paired seeds, a 180-epoch cap, thirty-epoch patience and validation-selected states. BP and LocalCA differ in rates and parameter groups; exact-path LocalCA still computes exact selected gradients. Source checks of one hundred original trajectories verified configurations and final metrics: all fifty D3 fits across five credit/optimizer arms reached the cap with falling validation loss. Sixty source-consistent restarts extend these fifty D3 fits and ten exact-BP D1 references to 600 epochs, selecting the validation-selected state (minimum validation loss) within each budget. All D3 fits and two D1 references reach the larger cap; eight D1 references stop early. D3 selected states lie near the cap while validation loss continues falling. Supplementary Section S6 retains complete histories, paired accuracy/loss contrasts and source-concordance discrepancies.

Anatomical dictionaries and common-mode controls

Observed MICrONS skeletons and mapped contacts define soma-rooted branch-to-branch segments. Initial eight and disjoint 47-cell cohorts come from the same mouse; Pinky is a second mouse, with ten of twelve selected cells meeting the inherited typed-contact criterion. Each subtree route contains the input-bearing descendants of its origin. Contact labels are direct where available and otherwise proxy-based as specified in Supplementary Section S7. Compression preserves path length but only approximates resistance; direct-label, mapping and geometry sensitivities are retained.

The modeled passive perturbation field consists of reciprocal passive-cable gradient changes induced by small focal shunts. Every corrected dictionary receives one constant column and $K - 1$ spatial columns. Ancestry selection uses inhibitory fraction times excitatory-area-weighted descendant fraction without field outcomes. Depth bins are reparameterized with the same constant while preserving their span; random routes, independent site shuffles and degree–depth surrogate trees provide controls. The weighted common component is removed before computing residual-energy capture. A common-constrained SVD oracle supplies the rank-budget ceiling. Numerical rank, coverage and nonzero density are reported separately; equal column count alone does not match those resources.

The initial eight cells provided post hoc development for common-mode correction; the protocol preceded evaluation of new v661 and Pinky outcomes. Two hundred randomized controls are averaged within each cell. At eight columns, all 47 v661 cells and eight Pinky cells qualify; other eligible budgets are retained without substituting for missing eight-column outcomes. Paired cell-bootstrap intervals and Holm-adjusted comparisons describe within-cohort consistency, not animal-population replication. The older route-generated positive controls use separate fields and endpoint definitions (Supplementary Section S7).

Focal shunting and the electrotonic boundary

Up to sixteen focal sites per cell span depth quartiles and require at least three excitatory descendants and three comparison sites. Normalized E/I scales are 0.35 with reversals one and -0.2 . Shunt doses are 0.25, 0.5, one and two times local membrane-plus-synaptic conductance. A matched injection supplies first-order current $\kappa_k(E_I - V_k)$ without changing the conductance matrix. A separately solved somatic current restores baseline voltage and is held fixed during differentiation. The squared-loss target is one normalized unit below baseline; finite differences check the adjoint gradients.

Localization subtracts depth-matched off-route from descendant medians of $|\log[(|\gamma'_i| + \varepsilon)/(|\gamma_i| + \varepsilon)]|$, with $\varepsilon = 10^{-15} \max(1, \max_i |\gamma_i|)$. Because this is unsigned, signed gradient changes are reported separately. Adjoint and driving-force substitutions in $\gamma_i = q_i(E_E - V_i)$ distinguish transport from local voltage effects. Sites and relation-reassignment templates are nested within cells. The passive ancestry-partition identity and its dictionary qualification are derived in Supplementary Section S8.

Physical calibration uses cylindrical axial conductance $\pi r^2/(R_a \ell)$ and leak $2\pi r \ell/R_m$, expressed in nS, with spherical soma area $4\pi r^2$. Here r and ℓ are segment radius and length, R_a axial resistivity and R_m specific membrane resistance. Sweeps include $R_a = 100, 150, 250 \Omega \text{cm}$ and $R_m = 300\text{--}30,000 \Omega \text{cm}^2$, fixed and locally normalized doses, and background conductance. The disjoint calibration cohort contains 235 eligible sites in 45 cells. Geometric fallbacks and dose definitions are specified in Supplementary Section S8.

The weak-channel check uses $R_a = 150$, $R_m = 1,000$ in those units and background conductance equal to leak. Median maximal Na, K, Ca and HCN conductances are 0.06, 0.10, 0.025 and 0.05 times leak; NMDA is 0.15 times excitatory conductance. Sixty-four draws per cell passed the predefined steady-state checks. Responses use rank-one updates of the fixed baseline Jacobian, including voltage-gate derivatives, followed by somatic state matching. They are local linearizations rather than newly solved nonlinear equilibria at each dose (Supplementary Fig. S30D,E).

Measured-response prediction and geometric diagnostics

The seven-target cohort includes thirteen eligible scans. Trial responses are the mean of the supplied ROI fluorescence traces in `RoiResponseSeries` over each stimulus interval; our analysis did not deconvolve the traces or apply a reliability correction. Repeated responses are averaged by stimulus condition for the ancestry-similarity analysis; partial rank correlations control Euclidean separation and soma-to-contact path-length differences. Offline learning holds out stimulus identities. Training-only percentile transforms produce nonnegative presynaptic proxies, and each partner enters at its dominant mapped contact segment. A linear somatic readout predicts the standardized target response. Only input-conductance credit differs across rules; readout and bias gradients remain exact.

Four-route dictionaries are selected without target responses and receive the projected baseline transfer profile multiplied by the scalar readout error. Supplementary Section S9 defines delivery and reconstruction. Fig. S31 reconstructs supports from saved segment identifiers, illustrating the median-input-count scan with identifier-based tie breaking; all scan supports are available. At common exact-trained states, field and eligibility-weighted update reconstruction compare the fixed profile, per-trial field projection and the best within-span update. These are oracle diagnostics rather than newly trained rules. Reproducibility comparisons report endpoint discrepancies across 130 exact-rule re-executions.

Ridge regularization uses training data within each outer split and the same measured predictors. Input removal, predictor noise and reliability masks are separate sensitivities; none uses held-out information for selection. Ten split/initialization replicates are averaged within scans and then within seven targets. Imposed route-span-alignment quadratic fields and the six-animal external signed-credit reanalysis are separate datasets and tests, described in Supplementary Sections S7 and S9 and Supplementary Fig. S33, respectively.

Statistics and computational provenance

Confirmatory choices precede outcome inspection but are not described as public preregistration. Pilots, post hoc analyses and same-seed reruns are identified separately. Effect sizes accompany experiment-specific 95% intervals, or adjusted 97.5% intervals where prespecified. Tests are two sided unless a directional protocol states otherwise. Named multiplicity families and exact tie treatment are retained in Source Data; null results imply equivalence only when a margin is tested.

For the sensitivity simulation, a positive-semidefinite shared-path covariance generated joint tuning across the 102 observed presynaptic roots. We mixed ancestry-correlated and independent reliable tuning, then added noise calibrated to each recording’s split-half reliability and actual repeat counts. Four thousand paired datasets at each of 21 signal fractions were analyzed with the original partial-rank statistic and an exact seven-target signed-rank test. Detection required two-sided $P \leq 0.05$ and a positive mean effect; Wilson intervals describe Monte Carlo uncertainty. The model and analysis were frozen before simulated outcomes. Supplementary Section S9 gives the covariance, reliability calibration, dependence assumptions and exact test resolution; the released pair-level inputs permit reproduction.

Independent units are paired training seeds, reconstructed cells, postsynaptic targets or external animals. Sites, contacts, scans, splits, checkpoints and randomized draws remain nested. Seed counts reflect the experiment’s purpose; no prospective parametric power calculation was performed. Source hashes certify retained inputs and implementations, not unavailable historical runtime states. The matched credit and new conductance studies recorded Python 3.10.13, NumPy 2.2.6 and pandas 2.3.3. The new MNIST controls used PyTorch 2.9.1/CUDA 12.8 and torchvision 0.24.1 with NumPy 2.2.6; the depth extension used Python 3.10 and PyTorch 2.9.1. The isolated reviewer environment was validated with NumPy 1.26.4, SciPy 1.15.3, PyTorch 2.9.1, torchvision 0.24.1 and Matplotlib 3.10.9. Cohort-specific runtime records distinguish executed experiments from this installation check. Supplementary Section S10 consolidates statistical families and release records; the question-specific sections retain their exclusions and numerical checks.

OpenAI Codex assisted with code, statistical and consistency audits, figure preparation, literature checks and language editing. The authors remain responsible for design, interpretation, numerical accuracy, citations and the final manuscript.

Ethics and data reuse

No new experiments involving animals or human participants were performed. All biological analyses reused public MICrONS, DANDI, and published animal source data. The source studies describe the original data collection and institutional approvals [22, 42, 63].

Data availability

MICrONS anatomy and connectivity are available through the public CAFE datastack minnie65_public. The primary cohort used materialization 1822, skeleton-service reconstructions, and the synapses_pni_2 table. The disjoint sensitivity cohort used the public v661 static release. Secondary target-structure labels used only in the full-contact analysis came from the in-development synapse_target_predictions_ssa_v2 table. These labels are distributed only as derived data and remain subject to the source-data access terms. All principal structural findings were repeated without them.

The independent-animal replication used the MICrONS phase-1 layer-2/3 Pinky v185 fixed meshes, soma-valence table, and dense synapse table archived at Zenodo DOI 10.5281/zenodo.3710459 [64]. File sizes, Zenodo MD5 checksums, and local SHA-256 checksums are recorded in the supplied manifest. Volumetric imagery is archived in BossDB under DOI 10.60533/BOSS-2021-T0SY, and co-registered functional data are available as DANDI:000402. The supplied manifest records the exact asset and table identifiers, and the source publication describes the full data release [42].

Public benchmark datasets are available from their original repositories [60–62]. The inhibitory-census tables and SWCs are available with the source record for Schneider-Mizell et al. [63]. The Francioni et al. workbook is available through the Source Data link associated with the Nature article [22]; its checksum and the sheet names used here are recorded in the manifest. Derived cell and exclusion manifests, numerical source data underlying every figure, and non-restricted summaries are supplied with this manuscript. The measured-response release includes the analyzed partner-pair tables, contact geometry, recording identifiers and repeated-stimulus response arrays needed for the sensitivity analysis. Authentication tokens and externally hosted raw caches are not redistributed.

Code availability

The accompanying reviewer software archive contains the complete training implementation, exact-gradient and perturbation analyses, versioned configurations, tests, figure scripts, and byte-identified snapshots of the critical training components under the MIT License. A separate Source Data archive contains the numerical data underlying every display item. These archives are available to editors and reviewers with the submission. The same version will be released publicly under the MIT License in a versioned repository and permanent archival record upon publication.

References

- [1] Wilfrid Rall. Theory of physiological properties of dendrites. *Annals of the New York Academy of Sciences*, 96(4): 1071–1092, 1962. doi: 10.1111/j.1749-6632.1962.tb54120.x.
- [2] Christof Koch. *Biophysics of Computation: Information Processing in Single Neurons*. Oxford University Press, 1999. doi: 10.1093/oso/9780195104912.001.0001.
- [3] Panayiota Poirazi, Terrence Brannon, and Bartlett W. Mel. Pyramidal neuron as two-layer neural network. *Neuron*, 37(6):989–999, 2003. doi: 10.1016/S0896-6273(03)00149-1.
- [4] Michael London and Michael Häusser. Dendritic computation. *Annual Review of Neuroscience*, 28:503–532, 2005. doi: 10.1146/annurev.neuro.28.061604.135703.
- [5] Nelson Spruston. Pyramidal neurons: dendritic structure and synaptic integration. *Nature Reviews Neuroscience*, 9: 206–221, 2008. doi: 10.1038/nrn2286.
- [6] Tiago Branco and Michael Häusser. The single dendritic branch as a fundamental functional unit in the nervous system. *Current Opinion in Neurobiology*, 20(4):494–502, 2010. doi: 10.1016/j.conb.2010.07.009.
- [7] Matthew Larkum. A cellular mechanism for cortical associations: an organizing principle for the cerebral cortex. *Trends in Neurosciences*, 36(3):141–151, 2013. doi: 10.1016/j.tins.2012.11.006.
- [8] David E. Rumelhart, Geoffrey E. Hinton, and Ronald J. Williams. Learning representations by back-propagating errors. *Nature*, 323:533–536, 1986. doi: 10.1038/323533a0.
- [9] Luis B. Almeida. A learning rule for asynchronous perceptrons with feedback in a combinatorial environment. In *Proceedings of the First IEEE International Conference on Neural Networks*, volume 2, pages 609–618, 1987.
- [10] Fernando J. Pineda. Generalization of back-propagation to recurrent neural networks. *Physical Review Letters*, 59 (19):2229–2232, 1987. doi: 10.1103/PhysRevLett.59.2229.
- [11] Donald O. Hebb. *The Organization of Behavior: A Neuropsychological Theory*. Wiley, New York, 1949.
- [12] Nicolas Frémaux and Wulfram Gerstner. Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9:85, 2016. doi: 10.3389/fncir.2015.00085.
- [13] Timothy P. Lillicrap, Daniel Cownden, Douglas B. Tweed, and Colin J. Akerman. Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7:13276, 2016. doi: 10.1038/ncomms13276.
- [14] Arild Nøkland. Direct feedback alignment provides learning in deep neural networks. In *Advances in Neural Information Processing Systems*, volume 29, pages 1037–1045, 2016.
- [15] Benjamin Scellier and Yoshua Bengio. Equilibrium propagation: bridging the gap between energy-based models and backpropagation. *Frontiers in Computational Neuroscience*, 11:24, 2017. doi: 10.3389/fncom.2017.00024.
- [16] James C. R. Whittington and Rafal Bogacz. An approximation of the error backpropagation algorithm in a predictive coding network with local Hebbian synaptic plasticity. *Neural Computation*, 29(5):1229–1262, 2017. doi: 10.1162/NECO_a_00949.
- [17] Yuhang Song, Beren Millidge, Tommaso Salvatori, Thomas Lukasiewicz, Zhenghua Xu, and Rafal Bogacz. Inferring neural activity before plasticity as a foundation for learning beyond backpropagation. *Nature Neuroscience*, 27: 348–358, 2024. doi: 10.1038/s41593-023-01514-1.
- [18] Guillaume Bellec, Franz Scherr, Anand Subramoney, Elias Hajek, Darjan Salaj, Robert Legenstein, and Wolfgang Maass. A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 11: 3625, 2020. doi: 10.1038/s41467-020-17236-y.

- [19] Justin Werfel, Xiaohui Xie, and H. Sebastian Seung. Learning curves for stochastic gradient descent in linear feedforward networks. *Neural Computation*, 17(12):2699–2718, 2005. doi: 10.1162/089976605774320539.
- [20] James C. R. Whittington and Rafal Bogacz. Theories of error back-propagation in the brain. *Trends in Cognitive Sciences*, 23(3):235–250, 2019. doi: 10.1016/j.tics.2018.12.005.
- [21] Timothy P. Lillicrap, Adam Santoro, Luke Marris, Colin J. Akerman, and Geoffrey Hinton. Backpropagation and the brain. *Nature Reviews Neuroscience*, 21:335–346, 2020. doi: 10.1038/s41583-020-0277-3.
- [22] Valerio Francioni, Vincent D. Tang, Enrique H. S. Toloza, Zilan Ding, Norma J. Brown, and Mark T. Harnett. Vectorized instructive signals in cortical dendrites. *Nature*, 652:1254–1263, 2026. doi: 10.1038/s41586-026-10190-7.
- [23] C. Koch, T. Poggio, and V. Torre. Nonlinear interactions in a dendritic tree: localization, timing, and role in information processing. *Proceedings of the National Academy of Sciences USA*, 80(9):2799–2802, 1983. doi: 10.1073/pnas.80.9.2799.
- [24] Gary R. Holt and Christof Koch. Shunting inhibition does not have a divisive effect on firing rates. *Neural Computation*, 9(5):1001–1013, 1997. doi: 10.1162/neco.1997.9.5.1001.
- [25] Frances S. Chance, L. F. Abbott, and Alex D. Reyes. Gain modulation from background synaptic input. *Neuron*, 35(4):773–782, 2002. doi: 10.1016/S0896-6273(02)00820-6.
- [26] Albert Gidon and Idan Segev. Principles governing the operation of synaptic inhibition in dendrites. *Neuron*, 75(2):330–341, 2012. doi: 10.1016/j.neuron.2012.05.015.
- [27] Matthew Lovett-Barron, Gergely F. Turi, Patrick Kaifosh, Peter H. Lee, Frédéric Bolze, Xiao-Hua Sun, Jean-François Nicoud, Boris V. Zemelman, Scott M. Sternson, and Attila Losonczy. Regulation of neuronal input transformations by tunable dendritic inhibition. *Nature Neuroscience*, 15:423–430, 2012. doi: 10.1038/nn.3024.
- [28] Panayiota Poirazi and Bartlett W. Mel. Impact of active dendrites and structural plasticity on the memory capacity of neural tissue. *Neuron*, 29(3):779–796, 2001. doi: 10.1016/S0896-6273(01)00252-5.
- [29] Robert Legenstein and Wolfgang Maass. Branch-specific plasticity enables self-organization of nonlinear computation in single neurons. *Journal of Neuroscience*, 31(30):10787–10802, 2011. doi: 10.1523/JNEUROSCI.5684-10.2011.
- [30] Ilenna Simone Jones and Konrad Paul Kording. Might a single neuron solve interesting machine learning problems through successive computations on its dendritic tree? *Neural Computation*, 33(6):1554–1571, 2021. doi: 10.1162/neco_a_01390.
- [31] David Beniaguev, Idan Segev, and Michael London. Single cortical neurons as deep artificial neural networks. *Neuron*, 109(17):2727–2739.e3, 2021. doi: 10.1016/j.neuron.2021.07.002.
- [32] Robert Urbanczik and Walter Senn. Learning by the dendritic prediction of somatic spiking. *Neuron*, 81(3):521–528, 2014. doi: 10.1016/j.neuron.2013.11.030.
- [33] Mathieu Schiess, Robert Urbanczik, and Walter Senn. Somato-dendritic synaptic plasticity and error-backpropagation in active dendrites. *PLOS Computational Biology*, 12(2):e1004638, 2016. doi: 10.1371/journal.pcbi.1004638.
- [34] Jordan Guerguiev, Timothy P. Lillicrap, and Blake A. Richards. Towards deep learning with segregated dendrites. *eLife*, 6:e22901, 2017. doi: 10.7554/eLife.22901.
- [35] João Sacramento, Rui Ponte Costa, Yoshua Bengio, and Walter Senn. Dendritic cortical microcircuits approximate the backpropagation algorithm. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [36] Alexandre Payeur, Jordan Guerguiev, Friedemann Zenke, Blake A. Richards, and Richard Naud. Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience*, 24:1010–1019, 2021. doi: 10.1038/s41593-021-00857-x.
- [37] Benjamin Ellenberger, Paul Haider, Federico Benitez, Jakob Jordan, Kevin Max, Ismael Jaras, Laura Kriener, and Mihai A. Petrovici. Backpropagation through space, time and the brain. *Nature Communications*, 17:66, 2026. doi: 10.1038/s41467-025-66666-z.
- [38] Blake A. Richards and Timothy P. Lillicrap. Dendritic solutions to the credit assignment problem. *Current Opinion in Neurobiology*, 54:28–36, 2019. doi: 10.1016/j.conb.2018.08.003.
- [39] Brendan A. Bicknell and Michael Häusser. A synaptic learning rule for exploiting nonlinear dendritic computation.

Neuron, 109(24):4001–4017.e10, 2021. doi: 10.1016/j.neuron.2021.09.044.

[40] Houman Safaai, Maceo Richards, and Bernardo L. Sabatini. Shunting inhibition and dendritic branching shape local credit assignment. *arXiv preprint arXiv:2607.03556*, 2026. URL <https://arxiv.org/abs/2607.03556>.

[41] Matteo Carandini and David J. Heeger. Normalization as a canonical neural computation. *Nature Reviews Neuroscience*, 13:51–62, 2012. doi: 10.1038/nrn3136.

[42] MICrONS Consortium. Functional connectomics spanning multiple areas of mouse visual cortex. *Nature*, 640: 435–447, 2025. doi: 10.1038/s41586-025-08790-w.

[43] Alain Destexhe, Michael Rudolph, and Denis Paré. The high-conductance state of neocortical neurons in vivo. *Nature Reviews Neuroscience*, 4:739–751, 2003. doi: 10.1038/nrn1198.

[44] Zhuokun Ding, Paul G. Fahey, Stelios Papadopoulos, Eric Y. Wang, Brendan Celii, Christos Papadopoulos, Andersen Chang, Alexander B. Kunin, et al. Functional connectomics reveals general wiring rule in mouse visual cortex. *Nature*, 640:459–469, 2025. doi: 10.1038/s41586-025-08840-3.

[45] Wolfgang Hackbusch and Stefan Kühn. A new scheme for the tensor representation. *Journal of Fourier Analysis and Applications*, 15:706–722, 2009. doi: 10.1007/s00041-009-9094-9.

[46] Lars Grasedyck. Hierarchical singular value decomposition of tensors. *SIAM Journal on Matrix Analysis and Applications*, 31(4):2029–2054, 2010. doi: 10.1137/090764189.

[47] Nadav Cohen, Or Sharir, and Amnon Shashua. On the expressive power of deep learning: A tensor analysis. In *Proceedings of the 29th Conference on Learning Theory*, volume 49 of *Proceedings of Machine Learning Research*, pages 698–728, 2016. URL <https://proceedings.mlr.press/v49/cohen16.html>.

[48] Julian Rossbroich and Friedemann Zenke. Dis-inhibitory neuronal circuits can control the sign of synaptic plasticity. In *Advances in Neural Information Processing Systems*, volume 36, pages 64059–64082. Curran Associates, Inc., 2023. doi: 10.52202/075280-2799.

[49] Alessandro R. Galloni, Ajay Peddada, Yash Chennawar, and Aaron D. Milstein. Cellular and subcellular specialization enables biology-constrained deep learning. *Cell Reports*, 45(4):117159, 2026. doi: 10.1016/j.celrep.2026.117159.

[50] Will Greedy, Heng Wei Zhu, Alexia Duriez, Joseph Pemberton, Patrick T. McCarthy, Kevin Kermani Nejad, and Rui Ponte Costa. Cell-type-specific cortical feedback coordinates hierarchical credit assignment. *bioRxiv*, 2026. doi: 10.64898/2026.06.16.732595.

[51] Houman Safaai, Maceo Richards, Naeem Khoshnevis, and Bernardo L. Sabatini. When branch-local shunting helps: A gain-load-alignment principle for dendritic E/I networks. *arXiv preprint arXiv:2607.24990*, 2026. URL <https://arxiv.org/abs/2607.24990>.

[52] Joseph Cichon and Wen-Biao Gan. Branch-specific dendritic ca^{2+} spikes cause persistent synaptic plasticity. *Nature*, 520:180–185, 2015. doi: 10.1038/nature14251.

[53] Erik B. Bloss, Mark S. Cembrowski, Bill Karsh, Jennifer Colonell, Richard D. Fetter, and Nelson Spruston. Structured dendritic inhibition supports branch-selective integration in CA1 pyramidal cells. *Neuron*, 89(5): 1016–1030, 2016. doi: 10.1016/j.neuron.2016.01.029.

[54] Spyridon Chavlis and Panayiota Poirazi. Dendrites endow artificial neural networks with accurate, robust and parameter-efficient learning. *Nature Communications*, 16:943, 2025. doi: 10.1038/s41467-025-56297-9.

[55] Joel Veness, Tor Lattimore, David Budden, Avishkar Bhoopchand, Christopher Mattern, Agnieszka Grabska-Barwinska, Eren Sezener, Jianan Wang, Peter Toth, Simon Schmitt, and Marcus Hutter. Gated linear networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 10015–10023, 2021. doi: 10.1609/aaai.v35i11.17202.

[56] Abhiram Iyer, Karan Grewal, Akash Velu, Lucas Oliveira Souza, Jeremy Forest, and Subutai Ahmad. Avoiding catastrophe: active dendrites enable multi-task learning in dynamic environments. *Frontiers in Neurorobotics*, 16: 846219, 2022. doi: 10.3389/fnbot.2022.846219.

[57] Changze Lv, Jingwen Xu, Yiyang Lu, Xiaohua Wang, Zhenghua Wang, Zhibo Xu, Di Yu, Xin Du, Xiaoqing Zheng, and Xuanjing Huang. Dendritic localized learning: Toward biologically plausible algorithm. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 41682–41700. PMLR, 2025. URL <https://proceedings.mlr.press/v267/lv25c.html>.

- [58] Ken C. Bittner, Aaron D. Milstein, Christine Grienberger, Sandro Romani, and Jeffrey C. Magee. Behavioral time scale synaptic plasticity underlies CA1 place fields. *Science*, 357(6355):1033–1036, 2017. doi: 10.1126/science.aan3846.
- [59] Jeffrey C. Magee and Christine Grienberger. Synaptic plasticity forms and functions. *Annual Review of Neuroscience*, 43:95–117, 2020. doi: 10.1146/annurev-neuro-090919-022842.
- [60] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998. doi: 10.1109/5.726791.
- [61] Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms. arXiv preprint arXiv:1708.07747, 2017.
- [62] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- [63] Casey M. Schneider-Mizell, Agnes L. Bodor, Derrick Brittain, JoAnn Buchanan, Daniel J. Bumbarger, Leila Elabbady, Clare Gamlin, Daniel Kapner, Sam Kinn, Gayathri Mahalingam, Sharmishta Seshamani, Shelby Suckow, Marc Takeno, Russel Torres, Wenjing Yin, Sven Dorkenwald, J. Alexander Bae, Manuel A. Castro, Akhilesh Halageri, Zhen Jia, Chris Jordan, Nico Kemnitz, Kisuk Lee, Kai Li, Ran Lu, Thomas Macrina, Eric Mitchell, Shanka Subhra Mondal, Shang Mu, Barak Nehoran, Sergiy Popovych, William Silversmith, Nicholas L. Turner, William Wong, Jingpeng Wu, Jacob Reimer, Andreas S. Tolia, H. Sebastian Seung, R. Clay Reid, Forrest Collman, and Nuno Maçarico da Costa. Inhibitory specificity from a connectomic census of mouse visual cortex. *Nature*, 640:448–458, 2025. doi: 10.1038/s41586-024-07780-8.
- [64] MICrONS Consortium. MICrONS Layer 2/3 Data Tables, 2020. URL <https://doi.org/10.5281/zenodo.3710459>.

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Author contributions

H.S. developed the computational framework, performed the analyses, and wrote the initial manuscript. M.R. contributed to the theory of local credit assignment, its implementation, and its interpretation. B.L.S. supervised the work and contributed to the biological framing, study design, and manuscript. All authors reviewed and revised the manuscript.

Competing interests

The authors declare no competing interests.

Supplementary Information for *Dendritic morphology as a dictionary for local credit assignment*

Houman Safaai, Maceo Richards and Bernardo L. Sabatini

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12 **How to use this supplement**

13 The notes follow the scientific questions in the main paper. They distinguish three roles for a tree: serial forward
14 computation, spatial delivery of learning signals, and a reconstructed cable model. A spatial dictionary is a set of
15 delivery profiles; the coefficients that combine them are a separate information resource. Exact transported errors, best-fit
16 projections and supplied reliability moments are identified as oracles when a restricted learner cannot calculate them.
17 Each experiment specifies its independent unit. Synapses, sites, scans, data splits and channel draws remain nested
18 within cells, targets or training seeds. All scientific outcomes and exclusions are retained in Source Data, including
19 comparisons summarized numerically rather than drawn again. Tables S11 and S12 locate the cohorts and their complete
20 records. Negative, small and budget-dependent results are reported beside the claims they qualify.

Main figure	Scientific question	Supplementary support
1	Exact local rules and image credit	Sections S1–S2; Figs. S1–S9
2–3	Branch conflict and ancestry	Section S3; Figs. S10–S11
4	Interactions and credit resolution	Section S4; Figs. S12–S17
5	Local conductance gating	Section S5; Figs. S18–S21
21 6	Serial computation and budgets	Section S6; Figs. S22–S24
7	Anatomical dictionaries	Section S7; Figs. S25–S28
8	Conductance and route gain	Section S8; Figs. S29–S30
9	Measured-response evidence	Section S9; Figs. S31–S33
Design exten- sion	What can be selected from data	Section S10; Figs. S34–S36

S1 Exact gradients and local learning rules

1.1 Exact gradients in a directed conductance tree

To determine which factors a local learning rule can compute, we first derive the exact conductance gradient, following the factorization and LocalCA rule introduced in our predecessor preprint [1]. This requires the forward steady state, the local synaptic sensitivity and transport of the somatic loss derivative through the tree.

1.2 Steady-state compartment voltage

Consider a rooted dendritic tree with soma node 0. Let V_n and C_n be the voltage and capacitance of compartment n , and let $\mathcal{C}_n$ be its set of children. A non-somatic compartment n receives excitatory contacts $\mathcal{S}_n^E$ and inhibitory contacts $\mathcal{S}_n^I$, with $\mathcal{S}_n = \mathcal{S}_n^E \cup \mathcal{S}_n^I$. A contact of type $X \in \{E, I\}$ has nonnegative maximal conductance g_i^X , nonnegative presynaptic activity x_i^X , open conductance $g_i^X x_i^X$, and reversal E_X . We use $E_i^{\text{rev}} = E_E$ on $\mathcal{S}_n^E$ and $E_i^{\text{rev}} = E_I$ on $\mathcal{S}_n^I$; when contact type is not at issue, we write the generic g_i^{syn} , x_i , and E_i^{rev} . Define the aggregate open conductances

$$G_n^E = \sum_{i \in \mathcal{S}_n^E} g_i^E x_i^E, \quad G_n^I = \sum_{j \in \mathcal{S}_n^I} g_j^I x_j^I.$$

The compartment also receives child-to-parent coupling conductances $g_{c \rightarrow n}^{\text{den}}$, and has leak g_n^L with reversal E_L . A child transmits $a_c = f_c(V_c)$, where f_c is the branch output nonlinearity (named “reactivation” in the implementation) and is the identity when that mechanism is disabled. Before taking the steady state, the directed compartmental dynamics are

$$C_n \dot{V}_n = g_n^L (E_L - V_n) + G_n^E (E_E - V_n) + G_n^I (E_I - V_n) + \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} (a_c - V_n). \quad (1)$$

The total child coupling is $G_n^{\text{child}} = \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}}$ and its voltage-weighted contribution is $J_n^{\text{child}} = \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} a_c$. Setting $\dot{V}_n = 0$ and collecting the coefficients of V_n then gives

$$g_n^{\text{tot}} = g_n^L + G_n^E + G_n^I + G_n^{\text{child}}, \quad (2)$$

$$V_n = R_n^{\text{tot}} (g_n^L E_L + G_n^E E_E + G_n^I E_I + J_n^{\text{child}}), \quad R_n^{\text{tot}} = (g_n^{\text{tot}})^{-1}. \quad (3)$$

Here g_n^{tot} and R_n^{tot} are the total conductance and its reciprocal, respectively. The soma obeys the same current balance with $n = 0$, including its child couplings and any explicitly modeled somatic inputs, but has no parent term. The controlled regular-tree architectures used here have no external somatic input synapses unless a control is explicitly described otherwise. For nonnegative conductances and activities, Eq. 3 is a conductance-weighted average of leak, synaptic reversal potentials and child activities. This boundedness is useful for numerical stability, but is not required for the chain-rule factorization below. Both E and I increase g_n^{tot} ; they differ through the driving forces $E_E - V_n$ and $E_I - V_n$, not through a signed conductance. When $E_I < V_n < E_E$, the former is depolarizing and the latter hyperpolarizing. At $V_n = E_I$, an inhibitory conductance has zero net current but still reduces R_n^{tot} , producing a pure shunt at that operating point.

The directed artificial-tree simulations use the dimensionless convention $g_n^L = 1$, $E_L = E_I = 0$, and $E_E = 1$, so inhibition is denominator-only. The reconstructed focal-cable analysis uses $E_E = 1$ and $E_I = -0.2$, so its inhibitory perturbation is both hyperpolarizing and shunting. Neither convention is a fit in millivolts.

1.3 Local sensitivities and synaptic eligibility

Writing the numerator of Eq. 3 as $b_n = g_n^{\text{tot}} V_n$, quotient differentiation gives

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = \frac{x_i E_i^{\text{rev}} g_n^{\text{tot}} - x_i b_n}{(g_n^{\text{tot}})^2} = x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n), \quad (4)$$

$$\frac{\partial V_n}{\partial V_c} = f'_c(V_c) g_{c \rightarrow n}^{\text{den}} R_n^{\text{tot}}, \quad (5)$$

$$\frac{\partial V_n}{\partial g_{c \rightarrow n}^{\text{den}}} = R_n^{\text{tot}} (a_c - V_n). \quad (6)$$

48 For a differentiable task loss $\mathcal{L}$, the exact synaptic conductance gradient therefore factors as

$$\frac{\partial \mathcal{L}}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n)}_{\text{local eligibility}} \underbrace{\frac{\partial \mathcal{L}}{\partial V_n}}_{\text{compartment error}}. \quad (7)$$

49 The eligibility contains quantities available at the synapse or its host compartment. All remaining task dependence
 50 lies in the compartment error. In particular, excitatory and inhibitory contacts use $x_i^E R_n^{\text{tot}} (E_E - V_n)$ and $x_j^I R_n^{\text{tot}} (E_I - V_n)$,
 51 respectively. Near $V_n = E_I$, the inhibitory contact's own gradient can be small even though its conductance still reduces
 52 R_n^{tot} and changes the sensitivity of other inputs.

53 1.4 Transport of the somatic error

54 Suppose the loss depends on the tree through its somatic output. Let $p(n)$ be the unique parent of non-somatic node n ,
 55 and define the somatic voltage error $\delta_0^V = \partial \mathcal{L} / \partial V_0$. The chain rule gives

$$\frac{\partial \mathcal{L}}{\partial V_n} = \frac{\partial \mathcal{L}}{\partial V_{p(n)}} f'_n(V_n) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}. \quad (8)$$

Because the graph is a tree, iterating Eq. 8 along the unique path from n to the soma yields

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad (9)$$

$$\frac{\partial \mathcal{L}}{\partial V_n} = \tilde{\alpha}_n \delta_0^V. \quad (10)$$

56 The effective path gain $\tilde{\alpha}_n$ includes every local transfer derivative. With identity reactivation, it reduces to the conductance-
 57 stage gain α_n^{cond} obtained by removing the f'_i factors. Equation 10 is an exact derivative, not a separately fitted model
 58 parameter.

59 1.5 Raw-parameter transforms

60 Trainable raw parameters θ are transformed to nonnegative conductances by a monotone map $g = \psi(\theta)$. The optimizer
 61 receives

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial \mathcal{L}}{\partial g} \psi'(\theta). \quad (11)$$

62 Where $\psi'(\theta) > 0$, this transform preserves the sign of the conductance gradient. Active synapse masks are applied after
 63 forming the conductance-space gradient. The exact-gradient diagnostics compare gradients in the same parameter space
 64 and include the transform derivative.

65 1.6 General implicit-adjoint form

66 The preceding path product is a corollary of a more general steady-state identity. Let $\mathbf{V}$, $\mathbf{g}$ and $\mathbf{x}$ collect compartment
 67 voltages, conductances and inputs. Suppose their steady state is defined by the residual equations $\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = \mathbf{0}$, with
 68 nonsingular voltage Jacobian $J_V = \partial \mathbf{F} / \partial \mathbf{V}$. Define the adjoint $\mathbf{q}$ by

$$J_V^T \mathbf{q} = \nabla_{\mathbf{V}} \mathcal{L}. \quad (12)$$

69 The implicit-function theorem gives

$$\frac{\partial \mathcal{L}}{\partial g_i} = -\mathbf{q}^T \frac{\partial \mathbf{F}}{\partial g_i}. \quad (13)$$

70 For a passive residual $\mathbf{F} = \mathbf{G}\mathbf{V} - \mathbf{b}$, let $\mathbf{G}$ be the conductance matrix, $\mathbf{b}$ the vector of imposed current contributions and
 71 $\mathbf{e}_n$ the coordinate vector for compartment n . A conductance synapse at n has $\partial \mathbf{F} / \partial g_i = x_i (V_n - E_i^{\text{rev}}) \mathbf{e}_n$, and therefore

$$\frac{\partial \mathcal{L}}{\partial g_i} = x_i (E_i^{\text{rev}} - V_n) q_n. \quad (14)$$

Both contact classes add positive diagonal loading to G , while their different reversals enter $\mathbf{b}$ and the local driving force in Eq. 14. Equations 12–14 recover the directed path product when J_V is triangular. For the reciprocal passive cable, $J_V = G$ is symmetric positive definite, so $G^{-\top} = G^{-1}$. A general nonsymmetric or recurrent operator instead requires the inverse transpose. Active steady states enter through their Jacobian whenever the required differentiability and nonsingularity conditions hold; no passive symmetry is assumed by the general identity.

1.7 Conductance-specific control of path gain

Holding coupling conductances and the forward state fixed, add effective inhibitory conductance $\Delta G_k^I \geq 0$ at compartment k . This is an activity-weighted aggregate conductance, not an individual maximal synaptic conductance. For a compartment n whose path crosses k ,

$$\frac{\alpha_n^{\text{cond}}(\Delta G^I)}{\alpha_n^{\text{cond}}(0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}}{g_k^{\text{tot}} + \Delta G_k^I}, \quad \frac{\partial \log \alpha_n^{\text{cond}}}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n), \end{cases} \quad (15)$$

where $\mathcal{A}(n) = \{k : (i \rightarrow k) \in \text{path}(n \rightarrow 0)\}$ contains the parent endpoints of the path edges: the strict ancestors of n , including the soma if its conductance is varied. Equation 15 proves path-selective attenuation. It does not prove that attenuation improves optimization. No reversal potential appears in Eq. 15 because this calculation retains only the total-conductance denominator while holding the forward state fixed. Any positive conductance has the same direct denominator effect. It is specifically a shunt when E_I is near the local voltage, so that its accompanying current is small; the full forward and gradient effects retain E_I .

Let $z_n = \log \alpha_n^{\text{cond}}$ and let β_n be the accumulated log-gain attenuation along path n . The perturbed log gain is $z'_n = z_n - \beta_n$. Across any specified ensemble of paths,

$$\text{Var}(z') = \text{Var}(z) + \text{Var}(\beta) - 2 \text{Cov}(z, \beta). \quad (16)$$

Shunting narrows this distribution only if $\text{Cov}(z, \beta) > \text{Var}(\beta)/2$. Path-gain compression is therefore an operating-point-dependent empirical outcome rather than a theorem.

1.8 Local rules and feedback objects

The exact compartment error derived above is generally unavailable to a local synapse. We therefore retain the exact local eligibility and ask what restricted teaching field can replace the transported error.

1.9 Three-factor update

The three-factor rule retains the exact local eligibility and substitutes an available voltage-space teaching field $\hat{\delta}_n$ for the exact compartment error:

$$\hat{\nabla}_{g_i^{\text{syn}}} \mathcal{L} = \left\langle x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n) \hat{\delta}_n \right\rangle_B, \quad \hat{\nabla}_{g_{c \rightarrow n}^{\text{den}}} \mathcal{L} = \left\langle R_n^{\text{tot}} (a_c - V_n) \hat{\delta}_n \right\rangle_B, \quad (17)$$

where B is the number of examples in a minibatch and $\langle \cdot \rangle_B$ denotes their average. Excitatory and inhibitory contacts use the same expression; their reversal potentials set the sign and magnitude of the driving force.

The four- and five-factor rules multiply Eq. 17 by slowly varying, bounded stage-level scalar reliability estimates. For stage ℓ , mini-batch t , and example b , the implementation averages the recorded layer voltage and corresponding somatic population activity over their feature axes, giving $\bar{V}_{\ell,b}^{(t)}$ and $\bar{V}_{0,b}^{(t)}$. The runs reported here used an uncentered dot product,

$$s_{\ell,t} = \frac{1}{B} \sum_{b=1}^B \bar{V}_{\ell,b}^{(t)} \bar{V}_{0,b}^{(t)}, \quad (18)$$

$$m_{\ell,1} = s_{\ell,1}, \quad m_{\ell,t} = 0.9m_{\ell,t-1} + 0.1s_{\ell,t} \quad (t > 1), \quad (19)$$

$$r_{\ell,t}^{\text{4F}} = \text{clip}_{[0.1, 2.0]}(m_{\ell,t}). \quad (20)$$

Thus $r_{\ell,t}^{\text{4F}}$ is one scalar for a recorded stage, not a separate branch-wise signal, and is not a centered covariance or a Pearson correlation. Recorded voltages are detached from automatic differentiation, and the same minibatch supplies $s_{\ell,t}$

101 and the local update. The positive clamp prevents a negative or very small dot product from reversing or eliminating the
 102 update.

The five-factor rule retains this multiplier and adds a conditional variance-ratio factor. Let $Y_\ell^{(t)}$ be the recorded stage voltage and $X_\ell^{(t)}$ its source-resolved parent proxy, each centered across mini-batch t . Let $\mathcal{E}_{0.1}[\cdot]$ denote an exponential moving average (EMA) with update rate 0.1. Define

$$v_{Y,\ell,t} = \mathcal{E}_{0.1}[(Y_\ell^{(t)})^2], \quad v_{X,\ell,t} = \mathcal{E}_{0.1}[(X_\ell^{(t)})^2], \quad c_{YX,\ell,t} = \mathcal{E}_{0.1}[\langle Y_\ell^{(t)} X_\ell^{(t)} \rangle], \quad (21)$$

$$\beta_{\ell,t} = \frac{c_{YX,\ell,t}}{v_{X,\ell,t} + 10^{-3}}, \quad v_{\text{res},\ell,t} = \max\{v_{Y,\ell,t} - \beta_{\ell,t} c_{YX,\ell,t}, 10^{-12}\}, \quad (22)$$

$$\phi_{\ell,t} = \text{clip}_{[0.25, 4.0]} \left(\frac{v_{Y,\ell,t}}{v_{\text{res},\ell,t}} \right), \quad \widehat{\nabla}^{5F} = r_{\ell,t}^{4F} \phi_{\ell,t} \widehat{\nabla}^{3F}. \quad (23)$$

103 If $v_{X,\ell,t} \leq 10^{-12}$, the implementation sets $\phi_{\ell,t} = 1$. These factors are empirical preconditioners. They are neither needed
 104 for the exact factorization nor extra task-error channels. Under exact transported error, the three-factor rule recovers the
 105 exact dendritic parameter gradient in the stated steady-state model. This separates feedback approximation from the
 106 validity of the local eligibility term; agreement of trained endpoints with backpropagation is tested empirically rather
 107 than assumed from the identity.

108 1.10 Feedback fields

109 For neuron u with N_u routed dendritic compartments, indexed $n = 1, \dots, N_u$ while the soma retains index $n = 0$, let
 110 $\delta_u^V = (\delta_{1,u}^V, \dots, \delta_{N_u,u}^V)^\top$ collect the exact compartment errors and let $\widehat{\delta}_u^V$ collect the values available to the local rule.
 111 Let $\mathbf{c}_u = (c_{u,1}, \dots, c_{u,K})^\top$ contain the signed values carried by K feedback channels. Let $A_u \in \mathbb{R}^{N_u \times K}$ be their fixed
 112 compartmental address matrix: column k is the spatial field produced by unit amplitude on channel k , and $(A_u)_{nk}$ is its
 113 delivery weight at compartment n . Linear superposition of the channel fields gives

$$\widehat{\delta}_u^V = A_u \mathbf{c}_u, \quad \widehat{\delta}_{n,u}^V = \sum_{k=1}^K (A_u)_{nk} c_{u,k}. \quad (24)$$

114 Equation 24 specifies the allowed feedback routes; it is a modeling assumption separate from the membrane current
 115 balance. The address matrix specifies where signals can be delivered; the coefficient vector specifies the values delivered
 116 on the current example. The model is exact for $A_u = I$ and $\mathbf{c}_u = \delta_u^V$. It becomes a neuron-wide broadcast for $K = 1$ and
 117 $A_u = \mathbf{1}_{N_u}$. A binary subtree column is one on descendants of its branch point and zero elsewhere. The equation leaves
 118 coefficient estimation open: projection analyses use best-fitting oracle coefficients to measure representational capacity,
 119 whereas learning experiments use the feedback rule specified for that task.

120 Here K counts fixed spatial delivery profiles, not an intrinsic number of independent external errors. Conditional
 121 on the forward state, a directed single-output tree has the scalar somatic error multiplied by its state-dependent path
 122 profile (Eq. 10). Exact path transport uses that scalar together with the sample-specific derivatives and is tested as a
 123 separate transport resource. A context gate can likewise reside in the local eligibility. The dimension of a field ensemble
 124 therefore measures the demand on a fixed dictionary across states; it does not exclude scalar error delivery combined
 125 with additional local transport or gating.

126 Table S2 defines the feedback objects without conflating bandwidth with routing. The strict scalar field used for
 127 the plotted MNIST ladder applies the scalar reduction defined below at every local-rule stage. The matched-width
 128 scalar-fallback field reuses a somatic vector only at a stage of the same width; at wider stages it uses the same robust
 129 scalar reduction as the strict scalar condition. For an N -dimensional somatic error vector δ_b in example b , the code first
 130 computes $\mu_b = N^{-1} \sum_u \delta_{b,u}$. With $\varepsilon = 10^{-6}$,

$$s_b = \begin{cases} \mu_b, & |\mu_b| > \varepsilon, \\ \delta_{b,u^*}, & u^* = \arg \max_u |\delta_{b,u}|, \quad |\mu_b| \leq \varepsilon. \end{cases} \quad (25)$$

131 For a one-dimensional error, the coordinate is retained. Selecting the largest-magnitude coordinate when the mean nearly
 132 cancels prevents the reduction from producing a near-zero teaching signal. The replacement count was not recorded in
 133 the initial runs and was therefore measured directly in the same-seed computational rerun. The matched-width field does
 134 not repeat each neuron's coordinate over that neuron's descendants. The neuron-specific feedback field does exactly that,

using one communicated coordinate per neuron and the tree as a fixed coordinate-to-arbor map. Exact path transport supplies the entire path derivative and is an information upper bound.

In the regular [3, 3] networks, the matched-width scalar-fallback field retained the 128-dimensional somatic vector only at the 128-wide stage. It reduced the field to one scalar at the 384- and 1,152-wide stages, so all 2,408,448 candidate feedforward synaptic parameters (92,160 active contacts) received strict scalar feedback. The exception comprised 384 final branch-to-soma couplings and 256 somatic gain and threshold parameters: 640 of 2,413,312 core parameters (0.027%). We therefore trained a separate 15-seed strict-scalar condition in each architecture, which supplies the scalar rung of the original fifteen-seed ladder; that ladder is retained in Source Data rather than in a figure, and Supplementary Fig. S7 shows the fresh ten-seed ladder. The fresh six-rule ladder also uses strict scalar feedback. Despite its small parameter count, this near-somatic exception affected learning. Strict scalar reduced mean accuracy relative to matched-width scalar-fallback feedback by 4.859 percentage points in shunting networks (95% paired-seed interval, 3.994–5.669; fallback higher in 15/15 seeds) and by 2.274 points in raw-additive networks (1.796–2.699; fallback higher in 14/15 seeds, one tie). Both intervals lay well outside the prespecified ± 0.5 -point practical-equivalence margin. These results show that a small near-somatic parameter group can exert disproportionate control over learning; they do not imply that the remaining neurons collapse under strict scalar feedback, because their local eligibility traces remain distinct. The paired comparison is retained in the Source Data indexed in Table S12.

1.11 Conditional gradient alignment

For one tree and one example, suppose the correct somatic error is available but path gain is omitted. Write local eligibility as e_i and let $n(i)$ be the host compartment. Then

$$g_i^{\text{exact}} = e_i \tilde{\alpha}_{n(i)} \delta_0^V, \quad g_i^{\text{shared}} = e_i \delta_0^V. \quad (26)$$

Let g^{exact} and g^{shared} collect these synaptic components, and let CV_w denote the weighted coefficient of variation defined below. For nonnegative path gains and nonzero gradient vectors,

$$\cos(g^{\text{shared}}, g^{\text{exact}}) = \frac{1}{\sqrt{1 + \text{CV}_w(\tilde{\alpha})^2}}, \quad w_i = (e_i \delta_0^V)^2. \quad (27)$$

Here

$$\bar{\alpha}_w = \frac{\sum_i w_i \tilde{\alpha}_{n(i)}}{\sum_i w_i}, \quad \text{CV}_w(\tilde{\alpha})^2 = \frac{\sum_i w_i [\tilde{\alpha}_{n(i)} - \bar{\alpha}_w]^2}{\bar{\alpha}_w^2 \sum_i w_i}, \quad (28)$$

so CV_w is the eligibility-energy-weighted coefficient of variation of the path gain. This identity explains why path-gain dispersion can matter after the correct somatic coordinate has been supplied. It does not solve the separate problem of supplying the correct coordinate across network layers.

1.12 Topology-matched credit capacity

A restricted field is useful only if its spatial supports approximate the required compartment-level credit. We therefore ask how much of that field can be represented by K ancestry-defined routes, and how this compares with a point-neuron partition at the same bandwidth.

1.13 Ancestry dictionary

Let A_T be the ancestry matrix of tree T , with one row per synaptic coordinate and one column per candidate route. Let β contain the state-dependent gains of these routes. A low-bandwidth teaching coefficient vector $\mathbf{c} \in \mathbb{R}^K$ generates a synaptic field

$$\hat{\mathbf{g}} = \Phi_{T,K} \mathbf{c}, \quad \Phi_{T,K} = D_e A_T \text{diag}(\beta) S_K, \quad (29)$$

where the diagonal matrix D_e holds local eligibility or anatomical weights and S_K assigns the K communicated channels to route sites. In this dictionary model, topology fixes binary support and β rescales route columns. Nonzero column rescaling preserves their span at a fixed selected route set, although it can affect selection, conditioning and coefficient dynamics. The focal cable interventions studied below are different: a physical shunt changes spatial transfer and local eligibility, and need not be a diagonal gain on these fixed columns. The compartment-level matrix A_u in Eq. 24 is the general address map. Here A_T is its anatomy-specific lift to synapse coordinates, and $\Phi_{T,K}$ is the resulting synaptic-update dictionary after eligibility, route gain and channel selection are included.

175 1.14 Address dimension and the point-neuron resource comparison

176 Separate the synapse-local eligibility from the task-dependent target credit field and let $\mathbf{t}_u \in \mathbb{R}^{N_u}$ denote the latter for
 177 neuron u . For this comparison, Φ acts on the chosen task-credit coordinates; eligibility-weighted update dictionaries use
 178 the same projection construction in their own coordinates. Let W be a positive diagonal coordinate-weight matrix, and let
 179 $\dagger$ denote the Moore–Penrose pseudoinverse. Define

$$\|\mathbf{v}\|_W^2 = \mathbf{v}^\top W \mathbf{v}, \quad P_{\Phi, W} = \Phi(\Phi^\top W \Phi)^\dagger \Phi^\top W. \quad (30)$$

180 This is the W -orthogonal projector onto $\text{col}(\Phi)$. Equivalently, with $\bar{\Phi} = W^{1/2}\Phi$, $\bar{P}_\Phi = \bar{\Phi}\bar{\Phi}^\dagger = W^{1/2}P_{\Phi, W}W^{-1/2}$ is the
 181 ordinary Euclidean projector in the transformed coordinates. For a prespecified nested family of dictionaries $\Phi_{u, K}$, a
 182 residual tolerance $0 \leq \varepsilon < 1$ and positive expected target energy, define

$$d_{\varepsilon, u} = \min \left\{ K : \frac{\mathbb{E}\|P_{\Phi_{u, K}, W}\mathbf{t}_u\|_W^2}{\mathbb{E}\|\mathbf{t}_u\|_W^2} \geq 1 - \varepsilon \right\}. \quad (31)$$

183 We set $d_{\varepsilon, u} = +\infty$ if no tested dictionary attains the threshold. This definition concerns the task-dependent coefficient
 184 after eligibility has been factored out; heterogeneous presynaptic activity alone is therefore not counted as an extra
 185 teaching coordinate. It is conditional on the model, state ensemble, coordinate metric and dictionary family. It specifies a
 186 resolution at a stated residual tolerance, rather than an intrinsic task complexity or a unique optimal morphology.

187 For any fixed $\Phi \in \mathbb{R}^{N_u \times K}$, all fields generated as $\Phi \mathbf{c}$ lie in $\text{col}(\Phi)$ and hence occupy at most $\text{rank}(\Phi) \leq K$ dimensions.
 188 Conversely, the weighted least-squares field in that space is $P_{\Phi, W}\mathbf{t}_u$, and its expected weighted residual energy is

$$\mathbb{E}\|(I - P_{\Phi, W})\mathbf{t}_u\|_W^2 = \text{tr}[(I - \bar{P}_\Phi)\bar{\Sigma}_t], \quad \bar{\Sigma}_t = \mathbb{E}[W^{1/2}\mathbf{t}_u\mathbf{t}_u^\top W^{1/2}]. \quad (32)$$

189 Equations 31–32 prove the fixed-dictionary dimension statement used in the main text; $\bar{\Sigma}_t$ is an uncentered second
 190 moment. The ordinary point-neuron local rule has $\Phi = \mathbf{1}$ and therefore one task-dependent coefficient per neuron.
 191 A K -route dendritic dictionary can use up to K structured fields inside that neuron. A point-neuron network is not
 192 mathematically forbidden from matching them, but doing so requires K separately routed teaching coefficients, K gated
 193 parameter groups, or an equivalent expansion into additional units. Accordingly, the comparison is about where address
 194 resources are represented, not about an absolute expressive separation between all possible dendritic and point-neuron
 195 networks.

196 1.15 Field capture

197 We now measure how much of a target field lies in a dictionary’s span. Let $\mathbf{g}$ denote that field in the coordinates under
 198 consideration, and use the positive weighting from Eq. 30. Set $\bar{\mathbf{g}} = W^{1/2}\mathbf{g}$, $\bar{\Phi}_{T, K} = W^{1/2}\Phi_{T, K}$, and let $\bar{P}_\Phi$ be the ordinary
 199 Euclidean orthogonal projector onto the columns of $\bar{\Phi}_{T, K}$. For a nonzero target field, capture is

$$\mathcal{C}_K(\mathbf{g}; \Phi, W) = \frac{\|\bar{P}_\Phi \bar{\mathbf{g}}\|_2^2}{\|\bar{\mathbf{g}}\|_2^2} = 1 - \frac{\|\bar{\mathbf{g}} - \bar{P}_\Phi \bar{\mathbf{g}}\|_2^2}{\|\bar{\mathbf{g}}\|_2^2}. \quad (33)$$

200 The unweighted case has $W = I$. The MICrONS structural analysis uses summed mapped excitatory synaptic area on
 201 the diagonal of W . For the transformed uncentered second moment $\bar{\Sigma}_g = \mathbb{E}[\bar{\mathbf{g}}\bar{\mathbf{g}}^\top]$, aggregate energy capture for a fixed
 202 projector is

$$\mathcal{C}_K(\bar{\Sigma}_g; \Phi, W) = \frac{\text{tr}(\bar{P}_\Phi \bar{\Sigma}_g)}{\text{tr}(\bar{\Sigma}_g)}. \quad (34)$$

203 This is the ratio of expected projected energy to expected total energy. It is generally different from the mean per-field
 204 fraction $\mathbb{E}[\mathcal{C}_K(\mathbf{g}; \Phi, W)]$, which weights nonzero fields equally. The two coincide for a fixed field norm, or when the
 205 second moment is formed after normalizing each field. The MNIST dictionary gallery averages per-field fractions; the
 206 cable-response operator analysis aggregates energy over its columns. The second moment equals a centered covariance
 207 only for mean-zero fields. Equation 34 states the topology–task matching condition: a sparse anatomical basis is useful
 208 only to the extent that task credit lies in its span.

Proposition 1 (Ky–Fan credit-capacity bound). Let $\bar{\Sigma}_g \in \mathbb{R}^{p \times p}$ be positive semidefinite, with eigenvalues satisfying $\lambda_1 \geq \dots \geq \lambda_p \geq 0$ and $\text{tr}(\bar{\Sigma}_g) > 0$. For every rank- K orthogonal projector P ,

$$\frac{\text{tr}(P\bar{\Sigma}_g)}{\text{tr}(\bar{\Sigma}_g)} \leq \frac{\sum_{j=1}^K \lambda_j}{\sum_{j=1}^p \lambda_j}. \quad (35)$$

Equality is attained by any rank- K projector whose range is a Ky–Fan-maximizing invariant subspace: it contains every eigenspace with eigenvalue strictly greater than λ_K and may choose the remaining directions within the eigenspace at λ_K . The difference between the right side and anatomical capture is the rank-matched spectral regret.

Proof. Ky–Fan’s maximum principle [2] gives $\max_{\substack{P=P^T=P^2 \\ \text{rank}(P)=K}} \text{tr}(P\bar{\Sigma}_g) = \sum_{j=1}^K \lambda_j$. Division by the fixed total spectral energy proves Eq. 35. $\square$

The spectral screen interpolates between two task-field second moments C_0 and C_1 with equal trace, using $C(\rho) = (1 - \rho)C_0 + \rho C_1$ for $0 \leq \rho \leq 1$. Thus, for two fixed projectors P and Q , the normalized capture difference has the sign of

$$\Delta(\rho) = \text{tr}[(P - Q)C_0] + \rho \text{tr}[(P - Q)(C_1 - C_0)], \quad (36)$$

so an interior crossover is determined exactly by its two endpoints. This affine result distinguishes alignment from rank: increasing K raises the oracle bound, whereas increasing ρ determines whether the anatomical subspace approaches it.

1.16 One-step descent bound

When the target field is a true loss gradient, capture also determines a one-step descent guarantee in the corresponding metric. Let $\mathbf{w}$ be the parameter vector and $\mathbf{g} = \nabla_{\mathbf{w}} \mathcal{L}$. To express the weighted calculation as Euclidean descent, define

$$\bar{\mathbf{w}} = W^{-1/2} \mathbf{w}, \quad \mathbf{w} = W^{1/2} \bar{\mathbf{w}}, \quad (37)$$

and define $\bar{\mathcal{L}}(\bar{\mathbf{w}}) = \mathcal{L}(W^{1/2} \bar{\mathbf{w}})$. Then $\nabla_{\bar{\mathbf{w}}} \bar{\mathcal{L}} = W^{1/2} \mathbf{g} = \bar{\mathbf{g}}$. If $\bar{\mathcal{L}}$ has L_W -Lipschitz gradients and η is the step size, the descent lemma gives

$$\bar{\mathcal{L}}(\bar{\mathbf{w}} - \eta \bar{P}_{\Phi} \bar{\mathbf{g}}) \leq \bar{\mathcal{L}}(\bar{\mathbf{w}}) - \eta \bar{\mathbf{g}}^T \bar{P}_{\Phi} \bar{\mathbf{g}} + \frac{L_W \eta^2}{2} \|\bar{P}_{\Phi} \bar{\mathbf{g}}\|_2^2. \quad (38)$$

Orthogonal projection implies $\bar{\mathbf{g}}^T \bar{P}_{\Phi} \bar{\mathbf{g}} = \|\bar{P}_{\Phi} \bar{\mathbf{g}}\|_2^2$, hence

$$\bar{\mathcal{L}}(\bar{\mathbf{w}} - \eta \bar{P}_{\Phi} \bar{\mathbf{g}}) \leq \bar{\mathcal{L}}(\bar{\mathbf{w}}) - \eta \left(1 - \frac{L_W \eta}{2}\right) \|\bar{P}_{\Phi} \bar{\mathbf{g}}\|_2^2. \quad (39)$$

At $\eta = 1/L_W$, the ratio between the projected- and full-gradient guarantees is Eq. 33. In the original coordinates, the projected step is

$$\mathbf{w}^+ = \mathbf{w} - \eta W^{1/2} \bar{P}_{\Phi} W^{1/2} \mathbf{g}, \quad (40)$$

which is a particular preconditioned update rather than an ordinary Euclidean gradient step in $\mathbf{w}$. The implementation was checked on 10,000 random positive-definite quadratic losses: there were no bound violations, minimum numerical slack was 2.67×10^{-7} , and the capture identity agreed to 1.89×10^{-15} . This numerical test checks the Euclidean projector algebra, which also applies after an explicit coordinate transformation; it does not validate synaptic-area weighting as a physical parameter metric. The result is a one-step guarantee only when the tested field is an exact gradient in the stated coordinates. It is not a convergence or generalization theorem and does not show that a biological circuit can infer the oracle coefficients.

1.17 Signal–noise theory of restricted credit routing

Capture alone favors increasing route resolution, but a broader operator also admits more stochastic error. We next formalize this signal–noise trade-off and ask when restricting the communicated credit improves expected progress.

1.18 Expected descent from routed-update moments

Let $\mathcal{L} : \mathbb{R}^p \rightarrow \mathbb{R}$ have L -Lipschitz gradient, with $L > 0$, on a neighborhood containing every update segment under consideration. At a fixed parameter state, write $\mathbf{g} = \nabla \mathcal{L}(\mathbf{w})$ and let $\mathbf{d}$ be a routed update estimator with finite second moment. Define $\mathbf{m} = \mathbb{E}[\mathbf{d}]$ and $V = \mathbb{E}[(\mathbf{d} - \mathbf{m})(\mathbf{d} - \mathbf{m})^\top]$. The expectation is over the stated sampling or feedback noise, conditional on $\mathbf{w}$. The update is

$$\mathbf{w}^+ = \mathbf{w} - \eta \mathbf{d}. \quad (41)$$

Proposition 2 (Routed-update moment bound). *For Eq. 41,*

$$\mathbb{E}\mathcal{L}(\mathbf{w}^+) \leq \mathcal{L}(\mathbf{w}) - \eta \mathbf{g}^\top \mathbf{m} + \frac{L\eta^2}{2} [\|\mathbf{m}\|_2^2 + \text{tr}(V)]. \quad (42)$$

Write $a = \mathbf{g}^\top \mathbf{m}$ and $b = \|\mathbf{m}\|_2^2 + \text{tr}(V)$. For $b > 0$, minimizing the bound over nonnegative steps gives $\eta^* = [a]_+ / (Lb)$.

Provided the stated smoothness condition holds along the update segments at this step, the certified decrease is

$$U = \frac{[\mathbf{g}^\top \mathbf{m}]_+^2}{2L\{\|\mathbf{m}\|_2^2 + \text{tr}(V)\}}, \quad [a]_+ = \max(a, 0). \quad (43)$$

For $b = 0$, $\mathbf{d} = 0$ almost surely and we set $U = 0$. A zero score means that this bound certifies no positive one-step decrease at the evaluated state; it does not establish failure of every later update or learning rule.

Proof. Apply the descent lemma [3] to $-\eta \mathbf{d}$, take expectations and use $\mathbb{E}\|\mathbf{d}\|_2^2 = \|\mathbf{m}\|_2^2 + \text{tr}(V)$. Minimizing the resulting scalar quadratic over $\eta \geq 0$ gives the claim. $\square$

A fixed linear routing operator gives a useful special case. If $\mathbf{d} = M(\mathbf{g} + \boldsymbol{\xi})$ with deterministic $M \in \mathbb{R}^{p \times p}$, $\mathbb{E}\boldsymbol{\xi} = 0$ and $\text{Cov}(\boldsymbol{\xi}) = \Sigma$, then $\mathbf{m} = M\mathbf{g}$ and $V = M\Sigma M^\top$. Equations 42–43 then recover the credit-operator expressions, denoted $U(M)$ below. The identity, orthogonal projectors and fixed block-gain screens use this specialization.

The more general moment form is needed for sample-dependent eligibilities and gates. In the context-gated task, changing an unselected distractor leaves the exact parameter gradient unchanged but changes a shared-feedback update through that distractor’s eligibility. A fixed matrix of the exact parameter gradient therefore cannot represent every routed update. The output-error-to-update map remains linear conditional on the complete forward state; that conditional map is a different object. The factorial reanalysis measures routed means and second moments directly and does not require a fixed map from exact parameter gradients to routed updates.

At fixed infinitesimal update norm, $-\mathbf{g}$ is the steepest Euclidean descent direction by Cauchy–Schwarz. Consequently, no routed local operator provides a uniform first-order improvement over exact full-batch backpropagation on the same parameters and objective. Equation 43 instead compares restricted or noisy estimators and exposes when rejecting stochastic energy compensates for discarded signal.

Corollary 1 (Projected stochastic crossover). *Consider the identity-curvature quadratic $\mathcal{L}(\mathbf{w}) = \frac{1}{2}\|\mathbf{w} - \mathbf{w}^*\|_2^2$, initialize at $\mathbf{w} = 0$, set $\mathbf{g} = -\mathbf{w}^*$, use unit step size, and let P be an orthogonal projector. The expected post-update losses of the unprojected and projected unbiased gradients are*

$$\mathbb{E}\mathcal{L}[-(\mathbf{g} + \boldsymbol{\xi})] = \frac{1}{2} \text{tr}(\Sigma), \quad (44)$$

$$\mathbb{E}\mathcal{L}[-P(\mathbf{g} + \boldsymbol{\xi})] = \frac{1}{2} [\|(I - P)\mathbf{g}\|_2^2 + \text{tr}(P\Sigma P)]. \quad (45)$$

The projected estimator is better precisely when

$$\text{tr}[(I - P)\Sigma] > \|(I - P)\mathbf{g}\|_2^2, \quad (46)$$

that is, rejected noise exceeds discarded signal. A local estimator using the same P with additional independent, zero-mean coefficient noise cannot achieve a lower expected loss than backpropagation followed by P .

For completeness, at a common step $0 < \eta < 2$ the unprojected-minus-projected expected loss is

$$\frac{\eta}{2} \{ \eta \text{tr}[(I - P)\Sigma] - (2 - \eta) \|(I - P)\mathbf{g}\|_2^2 \}. \quad (47)$$

Hence projection is better exactly when $\text{tr}[(I - P)\Sigma] > [(2 - \eta)/\eta] \|(I - P)\mathbf{g}\|_2^2$; the unit-step result above is the special case $\eta = 1$. Smaller common steps therefore require proportionally more rejected noise to compensate for the discarded signal.

1.19 Interior optimum along the nested route ladder

The scalar, neuron-specific and correctly routed ancestry fields in the controlled hierarchy analyses span nested subspaces. Their orthogonal projectors form the ladder $P_{\text{scalar}} \subset P_{\text{neuron}} \subset P_{\text{ancestry},K} \subset I$. Along such a ladder the capture of Eq. 33 is monotone by construction, but the guaranteed utility is not. For an orthogonal projector with retained signal energy $S = \mathbf{g}^\top P \mathbf{g} = \|P \mathbf{g}\|_2^2$ and admitted noise $N = \text{tr}(P \Sigma P)$, the optimal-step utility of the credit-operator bound is

$$U(S, N) = \frac{S^2}{2L(S + N)}. \quad (48)$$

Let one rung up the ladder add marginal signal $\Delta S \geq 0$ and marginal noise $\Delta N \geq 0$. Substituting into Eq. 48 and clearing denominators gives the exact marginal condition

$$U(S + \Delta S, N + \Delta N) > U(S, N) \iff \Delta S [(S + N)(2S + \Delta S) - S^2] > S^2 \Delta N. \quad (49)$$

The bracket is positive, so for small marginal additions the condition reduces to a threshold on the marginal signal-to-noise ratio, $\Delta S / \Delta N > S / (S + 2N)$: the utility keeps rising while each added route scale contributes proportionally more signal than the running threshold, and falls once the added scales admit mostly noise. When the signal spectrum decays faster than the admitted noise across ladder scales, U can attain an interior maximum. On a finite tested ladder, the marginal condition must become unfavorable before the last rung; spectral decay alone does not guarantee that crossing or a unique optimum. The displayed marginal condition assumes $S > 0$; if $S = 0$, a rung with positive retained signal has positive utility whenever its total second moment is finite and nonzero. Equations 48 and 49, together with the interior-maximum construction, are verified numerically in the released test suite (`tests/test_laminar_optimality.py`). Random, deranged and independently gained route families elsewhere in the paper need not lie on this same nested ladder.

Two experiments evaluate this optimum empirically (Table S12; analysis code is included in the release). In the hierarchy-depth routing experiment the initialization utility of the aligned tree, computed per seed and task depth, placed its argmax at the trained loss minimum in 135 of 200 seed–task pairs, and the group-mean argmax matched the trained optimum exactly at $H_c = 1, 2, 3$; at the full-rank boundary $H_c = 4$ the one-step bound was conservative and favoured $D_r = 2$, reflecting its single-step horizon.

The trained subtree factorial provides a second, retrospective test. Its full-batch moment score uses the global logistic curvature bound and the same four-control comparator as the accuracy contrast: deranged, depth-interleaved, random-sparse and random-rank routes. The mean ancestry-minus-best-control utility was positive at $K = 4$, negative at $K = 1, 2$ and zero at $K = 8$. Its contrast-maximizing budget was $K = 4$ in all 20 seeds, matching the observed accuracy-contrast maximum in 13 of 20. Mean accuracy-contrast selection regret was 1.94 percentage points. Adding the hypothetical minibatch variance made the mean $K = 4$ utility contrast negative, reduced optimum agreement to 8 of 20 and gave mean regret 2.75 points. Thus the score matched to the deterministic training recipe recovers the mean intermediate-bandwidth ancestry peak, while its seed-level selection remains imperfect. This retrospective result is distinct from the raw within-family performance optimum at $K = 8$ and from prospective morphology selection. The comparator definitions, scores and prediction errors are supplied in `source_data/review_evidence_reanalysis/`.

1.20 Laminar route families and hierarchical task second moments

The distinction between a point neuron and a dendritic tree can be stated as a constraint on the feasible credit operators at fixed rank. A point neuron addresses its synapses with the single shared column **1**: a rank-one operator with uniform support. An ancestry dictionary uses structured columns whose supports form a *laminar* family: any two supports are nested or disjoint. An unconstrained rank- K feedback matrix is the comparison class. For the balanced binary tree used in the factorial, the span of the K -route ancestry dictionary equals the span of the K coarsest vectors of the orthonormal tree-Haar basis $U = [\mathbf{h}_1, \dots, \mathbf{h}_n]$ (constant, then sibling-subtree differences, coarse to fine) for every admissible budget $K \in \{1, 2, 4, \dots, n\}$. Here n is the number of leaf coordinates. Let $P_{A,K}$ denote the orthogonal projector onto the selected ancestry span.

Proposition 3 (Laminar capture of a hierarchical second moment). *Let the task-credit second moment be $C = U \Lambda U^\top$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ ordered so that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$ along the coarse-to-fine Haar order, with $\text{tr}(C) > 0$. Then for every admissible budget K the ancestry dictionary attains the Ky–Fan bound,*

$$\frac{\text{tr}(P_{A,K} C)}{\text{tr}(C)} = \frac{\sum_{i \leq K} \lambda_i}{\text{tr}(C)} = \max_{\substack{P = P^\top = P^2 \\ \text{rank}(P) = K}} \frac{\text{tr}(P C)}{\text{tr}(C)}. \quad (50)$$

313 For a general second moment C with eigenpairs $(\lambda_i, \mathbf{v}_i)$ ordered by decreasing eigenvalue, the shortfall from the Ky–Fan
 314 bound decomposes exactly as

$$\sum_{i \leq K} \lambda_i - \text{tr}(P_{A,K}C) = \sum_{i \leq K} \lambda_i \|(I - P_{A,K})\mathbf{v}_i\|_2^2 - \sum_{j > K} \lambda_j \|P_{A,K}\mathbf{v}_j\|_2^2, \quad (51)$$

315 the leading-eigenspace energy outside the laminar span minus the trailing energy that the span retains. It is therefore
 316 bounded above by the former term. The shortfall is zero exactly when $P_{A,K}$ is Ky–Fan maximizing; when $K < n$ and
 317 $\lambda_K > \lambda_{K+1}$, this is equivalent to the ancestry span being the unique leading- K eigenspace.

318 *Proof.* The ancestry span equals the span of the K coarsest Haar vectors, which under the stated ordering are the
 319 top- K eigenvectors of C ; a rank- K orthogonal projector maximizes $\text{tr}(PC)$ exactly on a top- K eigenspace (Ky–Fan).
 320 For general C , expanding $\text{tr}(P_{A,K}C) = \sum_j \lambda_j \|P_{A,K}\mathbf{v}_j\|_2^2$ and $\|P_{A,K}\mathbf{v}_i\|_2^2 = 1 - \|(I - P_{A,K})\mathbf{v}_i\|_2^2$ for $i \leq K$ gives the stated
 321 decomposition. Ky–Fan’s equality condition, including arbitrary choices inside a degenerate cutoff eigenspace, gives the
 322 last statement. $\square$

323 This ordered hierarchical second moment is one sufficient class for which ancestry attains the rank- K spectral
 324 optimum. At a fixed K , the equality condition is the leading-eigenspace condition in the proposition; it does not
 325 require the entire second moment to have a Haar eigenbasis and does not identify a unique physical tree. Hierarchical
 326 organization without the stated coarse-to-fine spectral ordering is not sufficient. The spectral-alignment experiment
 327 realizes both regimes: at full alignment and $K = 4$ ancestry attained the rank-matched Ky–Fan bound to numerical
 328 precision, whereas at random alignment its advantage over sampled random routes vanished (main text). The attain-
 329 ment, the shortfall identity and the failure case with a fine-scale-dominant second moment are verified numerically in
 330 `tests/test_laminar_optimality.py`.

331 1.21 Point-neuron partition residual and route-coefficient error

332 Let t_i be the exact task-dependent target remaining after local eligibility has been factored out, let $w_i > 0$ be coordinate
 333 weights, and let $\pi(i)$ assign synapse i to a group that must share one teaching coefficient. The best allowed field minimizes
 334 $\sum_i w_i [t_i - c_{\pi(i)}]^2$.

335 **Proposition 4** (Shared-coefficient residual). *For every group G , the optimum is its weighted mean*

$$c_G^* = \frac{\sum_{i \in G} w_i t_i}{\sum_{i \in G} w_i}, \quad (52)$$

336 and the irreducible residual is

$$\mathcal{R}(\pi) = \sum_G \sum_{i \in G} w_i (t_i - c_G^*)^2. \quad (53)$$

337 *Refining a partition cannot increase $\mathcal{R}$; it decreases the residual only when an old group contains coefficient variation*
 338 *resolved by the new groups.*

339 *Proof.* Differentiate the separable weighted least-squares objective with respect to each c_G . The refinement statement
 340 follows because the old solution remains feasible when two or more refined coefficients are constrained to be equal. $\square$

341 The ordinary point-neuron rule has one group per neuron. Dendritic ancestry offers a particular nested sequence of
 342 refinements, but is helpful only if the task coefficient varies across those subtrees. An explicitly grouped point model can
 343 implement the same partition, which is why the resource-matched point controls in the main article coincide with the
 344 dendritic implementation.

345 For a general dictionary Φ and positive diagonal W , let $\mathbf{c}^* = \arg \min_{\mathbf{c}} \|\mathbf{t} - \Phi \mathbf{c}\|_W^2$ and let $\hat{\mathbf{c}}$ be the coefficients actually
 346 supplied by a local circuit. Weighted projection orthogonality yields the exact decomposition

$$\|\mathbf{t} - \Phi \hat{\mathbf{c}}\|_W^2 = \underbrace{\|\mathbf{t} - \Phi \mathbf{c}^*\|_W^2}_{\text{address / representation error}} + \underbrace{\|\Phi(\hat{\mathbf{c}} - \mathbf{c}^*)\|_W^2}_{\text{coefficient-estimation error}}. \quad (54)$$

347 If the coefficient error is unbiased with covariance Ω_c , the expected second term is $\text{tr}(\Phi^\top W \Phi \Omega_c)$. The Gram matrix
 348 $G_\Phi = \Phi^\top W \Phi$ therefore controls how coefficient error is expressed. Estimating $\mathbf{c}^*$ is sensitive to noise along directions
 349 with small positive Gram eigenvalues. Correlated route columns can likewise make individual coefficients poorly
 350 identifiable even when their combined field is stable. Address rank alone is not a complete learning metric.

1.22 Static route gains preserve span but can change conditioning

Proposition 5 (Route-wise static-gain invariance). *Let D_β be an invertible diagonal matrix of fixed nonzero route gains. Then*

$$\text{col}(\Phi D_\beta) = \text{col}(\Phi), \quad P_{\Phi D_\beta, W} = P_{\Phi, W}. \quad (55)$$

Thus route-wise static scalar gains cannot change address rank, projection residual or spectral capture. They can change the Gram matrix from $\Phi^\top W \Phi$ to $D_\beta^\top \Phi^\top W \Phi D_\beta$ and hence alter conditioning and coefficient dynamics.

This proposition concerns nonzero scalar rescaling of existing route columns. Spatially varying row gains, state-dependent gating and gains that reach zero can change the effective span or active support; those cases are not covered by Eq. 55. Physical shunts also change the cable Green’s function. For a somatic adjoint source on a passive tree, that change has an exact diagonal-gain representation in the baseline-weighted ancestry partition associated with the shunt site (Corollary 2). Its partition indicators are differences of nested subtree indicators. The diagonal form does not extend to an arbitrary selected overlapping dictionary, and the local synaptic driving force can still vary within each block.

1.23 Finite-time coefficient learning has a bias–variance crossover

Equal span need not give equal learning trajectories. Let $\mathbf{f}_t = \Phi \mathbf{c}_t$ be the field produced by coefficients $\mathbf{c}_t$ after update t , and let $\lambda_{\max} > 0$ be the largest eigenvalue of $\Phi \Phi^\top$. Normalizing ordinary coefficient descent by this eigenvalue gives $H_\Phi = \Phi \Phi^\top / \lambda_{\max}$. For normalized step size η and noisy rate-based target estimates $\mathbf{g} + \boldsymbol{\xi}_t$, with $\mathbb{E} \boldsymbol{\xi}_t = 0$ and $\text{Cov}(\boldsymbol{\xi}_t) = \sigma^2 I/n$, where n is the effective sample size and σ^2 the single-observation noise scale, the field iteration is

$$\mathbf{f}_{t+1} = \mathbf{f}_t - \eta H_\Phi (\mathbf{f}_t - \mathbf{g} - \boldsymbol{\xi}_t). \quad (56)$$

Proposition 6 (Exact finite-time coefficient risk). *Assume $\mathbf{f}_0 = 0$, $\mathbf{g} \in \text{col}(\Phi)$ and let $(\mu_j, \mathbf{u}_j)$ be the positive eigenpairs of H_Φ , with $\theta_j = \mathbf{u}_j^\top \mathbf{g}$. For independent isotropic noise and a stable step satisfying $|1 - \eta \mu_j| < 1$ for every positive mode, the expected population loss after T updates is*

$$\mathbb{E} \frac{\|\mathbf{f}_T - \mathbf{g}\|_2^2}{2} = \frac{1}{2} \sum_j \left[(1 - \eta \mu_j)^{2T} \theta_j^2 + \frac{\sigma^2 (\eta \mu_j)^2 [1 - (1 - \eta \mu_j)^{2T}]}{1 - (1 - \eta \mu_j)^2} \right]. \quad (57)$$

For Gram-preconditioned descent, $H_\Phi = P_\Phi$ and all positive $\mu_j = 1$; therefore every dictionary with the same projector has the same field trajectory under common observations.

Proof. Project Eq. 56 onto each eigenvector. The deterministic error contracts by $1 - \eta \mu_j$ per step. Independent innovations have variance $(\eta \mu_j)^2 \sigma^2 / n$ and accumulate as a finite geometric series. Summing half the modal second moments gives Eq. 57. Gram preconditioning replaces $\Phi^\top \Phi$ by its pseudoinverse on the positive subspace, so the induced field operator is P_Φ . $\square$

Equation 57 makes the low-data boundary explicit. Small μ_j retain more finite-time bias but admit less estimator variance. For two operators with the same span, let ΔB be the difference in the bias term and ΔV the difference in the coefficient of the noise term. Their risk difference is $\Delta B + (\sigma^2/n) \Delta V$. If $\Delta B > 0$ and $\Delta V < 0$, the ordering reverses at $n^* = -\sigma^2 \Delta V / \Delta B$: slow coordinates can help below n^* by implicit noise filtering, whereas well-conditioned coordinates help above it by reducing bias. This is a finite-horizon regularization effect, not new address capacity and not an unconditional advantage of ill-conditioning.

1.24 Branch reliability shrinkage

Partition the parameter coordinates into disjoint branch blocks b . Let $\mathbf{g}_b$ be the clean gradient in block b , Σ_b its noise covariance, and I_b the identity on that block. Define signal and noise energies $S_b = \|\mathbf{g}_b\|_2^2$ and $N_b = \text{tr}(\Sigma_b)$. With attenuation a_b in each block, $M = \text{blockdiag}(a_b I_b)$ and a fixed step $\eta = c/L$ for $c > 0$, Eq. 42 guarantees a loss decrease of

$$\mathcal{G}_c(\mathbf{a}) = \frac{c}{L} \sum_b \left[a_b S_b - \frac{c a_b^2}{2} (S_b + N_b) \right]. \quad (58)$$

Proposition 7 (Fixed-step reliability-optimal attenuation). *For each block with $S_b + N_b > 0$, Eq. 58 is maximized independently at*

$$a_b^* = \min \left\{ 1, \frac{S_b}{c(S_b + N_b)} \right\}, \quad (59)$$

for a physical attenuator constrained to $0 \leq a_b \leq 1$. A block with $S_b = N_b = 0$ contributes zero for every gain and can be omitted from the ratios below. If at least one block has positive energy, the best common gain is $a_g^ = \min\{1, \sum_b S_b / [c \sum_b (S_b + N_b)]\}$, and the branch-specific guarantee is no smaller because its feasible set contains every common gain. At $c = 1$, Eq. 59 reduces to the reliability factor $S_b / (S_b + N_b)$; in that case $\mathcal{G}_{\text{branch}} = (2L)^{-1} \sum_b S_b^2 / (S_b + N_b)$ and $\mathcal{G}_{\text{global}} = (2L)^{-1} (\sum_b S_b)^2 / \sum_b (S_b + N_b)$, with equality when reliability is constant across positive-energy blocks. If all blocks have zero energy, every common gain is optimal and both guarantees are zero.*

Proof. Differentiate each concave quadratic in Eq. 58 and project its unconstrained maximizer onto $[0, 1]$. Restricting all coordinates to a common value gives a_g^* . The branchwise optimum cannot be worse than this restricted solution. For $c = 1$, the closed-form comparison also follows from Cauchy–Schwarz applied to $S_b / \sqrt{S_b + N_b}$ and $\sqrt{S_b + N_b}$. $\square$

This optimum assumes access to S_b and N_b . A point implementation provided with the same branch partition and multiplicative factors produces the same update. Its biological relevance depends on whether local conductance can estimate or realize those factors.

1.25 Why depth can help or hurt

Let $P_1 \preceq P_2 \preceq \dots \preceq P_{D_r}$ be nested tree-route projectors. Increasing depth reduces the address bias $\|(I - P_{D_r})\mathbf{g}\|^2$ but increases admitted noise $\text{tr}(P_{D_r} \Sigma P_{D_r})$ whenever the new modes carry stochastic energy. These competing terms can therefore produce an interior optimum when task signal is concentrated above one hierarchy level and noise continues into finer levels; its location depends on their magnitudes, as specified by Eq. 46. This is the controlled role isolated by the hierarchy-depth routing experiment. It is distinct from nonlinear forward composition, which can make deeper dendritic subunits representationally useful even when the backward address span is unchanged.

Depth can also impose a transport cost. In an idealized ensemble with equal local eligibility, let the route gain be $A = \prod_{\ell=1}^{D_p} G_\ell$ and assume independent Gaussian log gains, $\log G_\ell \sim \mathcal{N}(\mu_\ell, \sigma_\ell^2)$. The cosine between a shared coefficient and the exact gain-weighted field in the large-route limit is

$$\frac{\mathbb{E}A}{\sqrt{\mathbb{E}A^2}} = \exp \left(-\frac{1}{2} \sum_{\ell=1}^{D_p} \sigma_\ell^2 \right). \quad (60)$$

Additional levels can therefore add address resolution while simultaneously making uncompensated shared feedback less aligned. The balance depends on task hierarchy, gain variability, noise and the available coefficient estimator; “more depth” has no unconditional sign.

1.26 Controlled signal–noise experiment

The signal–noise experiment used 16 Euclidean coordinates represented in the orthonormal Haar basis of a balanced binary tree, ordered from the global mode (level 0) to individual sibling contrasts (level 4). The configuration, 50 analysis seeds 7000–7049, two preliminary validation seeds and numerical tolerances were specified before the analysis outcomes were inspected. The preliminary runs verified basis orthogonality, projector idempotence, static-gain span invariance and exact agreement with the explicit point-gain control. All planned observations were completed: 3,750 spectral, 1,800 depth, 3,750 projection and 1,500 reliability rows.

In the spectral screen, the task-gradient second moment assigned total energy $(0.15, 0.35, 0.30, 0.20, 0)$ to tree levels 0–4. An alignment coefficient $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$ mixed this second moment with an independently rotated endpoint having the same eigenvalues. Intermediate mixtures do not preserve the spectrum and jointly vary alignment and concentration. Ancestry, random rank-matched and dense principal-eigenspace projectors were compared at $K \in \{1, 2, 4, 8, 16\}$. The dense value was also calculated as the Ky–Fan sum of the leading K eigenvalues divided by total second-moment energy. Because the endpoint second moments had equal trace, ancestry-minus-random capture varied affinely with alignment. Endpoint differences therefore gave the exact per-seed crossover; seeds already favoring ancestry at zero alignment were assigned threshold zero. In these unweighted coordinates, let $\bar{\Sigma}_g$ denote the uncentered task-gradient second moment; energy-weighted capture was $\text{tr}(P \bar{\Sigma}_g) / \text{tr}(\bar{\Sigma}_g)$.

The same-span diagnostic compared raw nested indicators, their tree-Haar basis and indicators multiplied by fixed nonzero lognormal column gains. Rank, coherence, positive-spectrum Gram condition number and capture were calculated directly.

In the depth screen, a unit-norm target was sampled from modes at levels 0 through H_c , with $H_c \in \{1, 2, 3, 4\}$. The gradient-noise standard deviation at level ℓ was $0.65[1 + 1.5 \max(\ell - H_c, 0)]$. Starting from zero, 80 common-noise updates of size 0.12 used the projector onto levels 0 through $D_r \in \{1, 2, 3, 4\}$. A fixed leaf permutation provided the rewired control, and the identity provided stochastic backpropagation.

The projection screen set retained signal and retained noise independently, with $r \in \{0.25, 0.5, 0.75, 0.9, 1\}$ and $n \in \{0.1, 0.25, 0.5, 0.75, 1\}$. It compared an unprojected stochastic gradient, backpropagation followed by the same projection, and a routed estimator with an added total coefficient-noise fraction of 0.05. For a unit step on the identity-curvature quadratic, expected losses were calculated analytically and stochastic losses were recorded from paired draws.

In the reliability screen, eight disjoint blocks had signal and noise energies proportional to $\exp(hx_b)$ and $\exp(-hx_b)$, respectively, with equally spaced $x_b \in [-1, 1]$ and heterogeneity $h \in \{0, 0.5, 1, 1.5, 2\}$. We compared no attenuation, the best common gain, $S_b/(S_b + N_b)$, seed-shuffled gains, reversed gains and a point-gain control receiving the same branch factors. Intervals used 20,000 resamples of the 50 paired seeds, and paired tests were two-sided Wilcoxon signed-rank tests.

The resolved protocol is `configs/credit_phase_theory/confirmatory.json`. Source tables are in `source_data/credit_phase_theory/`. All full-rank projectors agreed to 4.44×10^{-16} , route-wise static gain rescaling preserved the projector to 1.33×10^{-15} , and the explicit point-gain control matched reliability-aligned attenuation exactly. This fixed-state quadratic experiment validates the signal-noise prediction but does not join oracle reliability to learned positive conductances.

1.27 Initialization score in the trained subtree factorial

The existing factorial used full-batch gradient descent. Its secondary stochastic diagnostic reconstructed the frozen training data, initialization and routes, then estimated routed-update variance from 64 hypothetical minibatches of 64 examples per seed. The full-data routed mean and that variance supplied Eq. 43. This variance was not noise experienced by the full-batch optimizer and cannot by itself explain the trained contrasts. The original released score used the largest logistic Hessian eigenvalue at initialization as its curvature scale. The corrected analysis uses the global logistic bound

$$L_{\text{glob}} = \max_c \lambda_{\text{max}} \left(\frac{8}{4N} X_c^T X_c \right),$$

where rows of X_c contain the selected-stream features for examples in context c , N is the total training-set size, and eight is the gradient context-rescaling factor. This bound is 1.00083–1.02125 times the initialization Hessian maximum across seeds. We scored the deterministic full-batch update with $V = 0$, the hypothetical minibatch estimator with its estimated V , and the deterministic bound at the actual training step $\eta = 0.3$.

Across 540 tree fits, deterministic utility correlated with norm-matched one-step progress at 0.9509 and final accuracy at 0.9242; hypothetical-minibatch values were 0.9373 and 0.9157. Intervals use 5,000 whole-seed resamples. These are retrospective associations within the factorial, not independent prospective validation.

To characterize the task-coordinate spectrum, the analysis used the uncentered second moment of $\delta_i \mathbf{e}_{c_i}$, where δ_i is the output error on example i and $\mathbf{e}_{c_i}$ is the coordinate vector for its context c_i . Eight balanced contexts and logits near zero make this spectrum nearly isotropic, giving the reported participation rank near eight. This value characterizes the chosen coefficient representation at initialization; it is not an independent estimate of task complexity or a morphology law. The mean within-family utility and accuracy maxima occur at $K = 8$ for all five nondegenerate route families. Across their 25 implementation-family cells, there are nine distinct nonconstant mean utility curves: four implementations coincide, whereas leaf rewiring changes four of the five family curves; the principal component analysis (PCA) oracle curve is unchanged. In the dendritic-tree implementation, deterministic utility selected an accuracy-maximizing bandwidth in 95 of 100 seed-family curves, the same success count as always choosing the largest K ; the hypothetical-minibatch score matched in 97 of 100. The 20 derangement curves had zero optimized-step utility throughout and were reported as degenerate. Repeated equivalent parameterizations do not provide independent tests of endpoint ordering. The distinct interior-optimum and ancestry-minus-control analyses, including their prediction failures, are reported in Section 1.19. The corrected scores, per-family selection errors and regret are provided in `source_data/review_evidence_reanalysis/`.

479 1.28 State-matched positive-conductance reliability experiment

480 We next tested whether positive conductances could realize the attenuation prescribed by Eq. 59 in a separate trained
 481 branch model (Table S12). Here b indexes branches, i examples and d inputs within a branch. Every input rate x_{bid} was
 482 nonnegative, and each excitatory conductance was parameterized by $g_{bd} = \text{softplus}(q_{bd})$, with trainable raw parameter
 483 q_{bd} . With leak conductance g^L , define total unshunted conductance and voltage on example i

$$G_{bi} = g^L + \sum_d x_{bid} g_{bd}, \quad V_{bi} = \frac{\sum_d x_{bid} g_{bd}}{G_{bi}}. \quad (61)$$

484 A fixed shunt $\kappa_b \geq 0$ had reversal zero. To isolate its effect on input resistance from its effect on the forward feature, an
 485 oracle current $\kappa_b V_{bi}$ was added on each example. The perturbed state is therefore

$$V'_{bi} = \frac{\sum_d x_{bid} g_{bd} + \kappa_b V_{bi}}{G_{bi} + \kappa_b} = V_{bi}, \quad \frac{\partial V'_{bi}}{\partial q_{bd}} = \frac{x_{bid}(1 - V_{bi})}{G_{bi} + \kappa_b} \text{sigmoid}(q_{bd}). \quad (62)$$

486 Relative to the unshunted eligibility, the example-specific physical input-resistance factor is $G_{bi}/(G_{bi} + \kappa_b)$. The com-
 487 pensating current was recomputed from the pre-shunt state and treated as a clamp, not differentiated as an independently
 488 learned pathway. Consequently this manipulation deliberately removes forward feature suppression while retaining
 489 conductance attenuation in the local update.

490 The binary task used eight branches, four inputs per branch, 512 training examples and 2,048 held-out examples.
 491 Input rates were generated from a positive baseline plus a class-dependent branch pattern and clipped at zero. Initial
 492 full-data gradients supplied $S_b = \|\mathbf{g}_b\|_2^2$. Coefficient-noise variance was then scaled so that

$$N_b = S_b \exp(-2hx_b), \quad x_b \in [-1, 1], \quad (63)$$

493 where x_b is the evenly spaced branch-order coordinate and $h \in \{0, 0.5, 1, 1.5, 2\}$ controls reliability heterogeneity. The
 494 statistical reliability was $r_b = S_b/(S_b + N_b)$, lower-bounded at 0.03 for numerical stability. Because the common step was
 495 $\eta = c/L$ with $c = 0.5$, the corrected fixed-step target was $a_b^* = \min\{1, r_b/c\}$ and the initial shunt was $\kappa_b = \bar{G}_b[(a_b^*)^{-1} - 1]$,
 496 where $\bar{G}_b$ is the training-example mean of G_{bi} at initialization. Shunts remained fixed during the 40 updates; the excitatory
 497 conductance parameters were trained, and the same coefficient-noise realizations were used across matched methods
 498 within seed.

499 Nine conditions separated the relevant alternatives: exact clean backpropagation, noisy credit without a shunt, a state-
 500 matched unattenuated-eligibility control, the step-consistent best single global shunt, reliability-aligned, seed-shuffled
 501 and anti-aligned shunts, an explicit point-gain control receiving the identical branch factors, and aligned shunting with
 502 clean credit. The point-gain control is a resource control, not a biological assertion: because it receives the same branch
 503 partition and factors, its update must equal Eq. 62. The clean-backpropagation condition retains the unattenuated exact
 504 gradient and is the optimization reference.

505 An initial implementation using r_b despite $c = 0.5$ was excluded; the corrected protocol preceded two excluded checks
 506 and fifty paired seeds 8100–8149 (2,250 outcomes). State mismatch was below 3×10^{-16} ; independently computed
 507 point-gain and unattenuated controls agreed to 1.4×10^{-14} , and fixed-compensating-current finite differences to relative
 508 error 3.60×10^{-8} .

509 At heterogeneity two, aligned-minus-best-global one-step test-loss reduction was 0.001817 (95% paired interval,
 510 0.001618–0.002020; 49/50 positive). Aligned attenuation also improved immediate progress over unshunted learning.
 511 After forty updates it remained better than global, shuffled and reversed attenuation, but its loss difference from no shunt
 512 was 0.000425 (95% interval, -0.002456 to 0.003071). Clean BP retained the lowest mean loss. Thus the state-clamped
 513 oracle factors improve relative placement without establishing sustained superiority to unshunted learning; compact
 514 endpoints and all seeds are indexed below.

515 1.29 Adaptive local reliability estimation

516 The preceding experiment supplied the reliability moments. We next asked whether they could be estimated from noisy
 517 local observations, specifying the adaptive-gain experiment before examining its outcomes (Table S12). At the current
 518 parameter state, branch b received two conditionally independent noisy local gradient observations, $\hat{\mathbf{g}}_{1b}$ and $\hat{\mathbf{g}}_{2b}$, on the
 519 same examples. It formed

$$\hat{S}_b^{\text{obs}} = \langle \hat{\mathbf{g}}_{1b}, \hat{\mathbf{g}}_{2b} \rangle, \quad \hat{N}_b^{\text{obs}} = \frac{1}{2} \|\hat{\mathbf{g}}_{1b} - \hat{\mathbf{g}}_{2b}\|_2^2. \quad (64)$$

For independent zero-mean credit noise, these estimate the squared clean branch gradient and the noise energy of one observation. Eight paired observations initialized the moments; exponential averages with decay 0.9 were then updated before each of 40 training steps. Negative finite-sample signal estimates and moments below 10^{-12} were clipped only when the gain was constructed. The step-consistent gain followed Eq. 59, was bounded to $[0.03, 1]$, and was converted to a shunt using the branch’s current mean total conductance.

The task and state clamp are unchanged. At heterogeneity 0/1/2, seven rules compare clean BP, noisy no-shunt, fixed initial oracle, adaptive local/global/permuted shunts and the independently computed adaptive point gain. Update and probe noise are paired. Two implementation-check seeds are excluded; fifty paired seeds 8200–8249 supply 1,050 outcomes. State, finite-difference and point-gain checks pass at the stated tolerances.

At heterogeneity two, adaptive local attenuation lowers final loss relative to global and shuffled estimates, yet exceeds no-shunt loss by 0.004250 (95% interval, 0.002499–0.005975) and fixed-oracle loss by 0.003449 (0.002577–0.004305). Only 26/50 immediate local-versus-shuffled differences favored local, failing the predefined direction criterion. Final gain ordering correlates with the initial oracle at 0.756, with mean absolute error 0.205. The supplied paired probes, optimizer step fraction and exact state clamp remain necessary assumptions; this is not a fully autonomous inhibitory-plasticity mechanism (Supplementary Fig. S4).

1.30 Same-span noisy coefficient-learning experiment

To isolate coefficient learning from address capacity, we compared three dictionaries with the same span in a rate-based factorial (Table S12). The 16-leaf balanced-tree Haar basis supplied the common rank-eight span through depth three. The tested dictionaries were the eight orthonormal Haar modes, all 15 nested indicators through depth three, and the same nested indicators scaled by positive gains evenly spaced on the log scale from $\exp(-3)$ to $\exp(3)$. The dictionaries’ projectors agreed to 9.44×10^{-16} and their positive-spectrum Gram condition numbers were 1, 15 and 388.52 (Supplementary Fig. S5).

Each independent seed drew a unit-norm target in the common span. At every update the estimator observed the target plus isotropic Gaussian noise with standard deviation $2/\sqrt{n_{\text{eff}}}$, where $n_{\text{eff}} \in \{4, 16, 64, 256\}$. The same 80 observation vectors were used for every matched parameterization. Vanilla coefficient descent used $0.2/\lambda_{\max}(\Phi^T\Phi)$; the oracle coordinate control used $0.2(\Phi^T\Phi)^+$. Population field loss was recorded after 1, 5, 20 and 80 updates. The target address residual was below 1.45×10^{-30} in every condition, so all reported loss is within-span coefficient-estimation error.

The predictions were specified before two implementation-check seeds, which were excluded from inference, and 50 paired seeds (9100–9149). The results contradicted the prediction that Haar coordinates would always reduce finite-time error. At $n_{\text{eff}} = 4$, raw-minus-Haar final loss was -0.2774 (95% paired interval -0.3260 to -0.2313) and scaled-minus-Haar was -0.1906 (-0.2500 to -0.1335): slower modes filtered high estimator noise. At $n_{\text{eff}} = 256$, the effects reversed to 0.02555 (0.02171–0.02952) and 0.17091 (0.14576–0.19686), positive in 50/50 seeds for both contrasts. Equation 57, derived after these outcomes to explain the reversal, predicted median crossovers at effective sample sizes 38.53 and 8.25. The Gram-preconditioned field trajectories agreed across dictionaries to 1.15×10^{-15} . Thus static parameterization can alter finite-data learning without altering the attainable credit field.

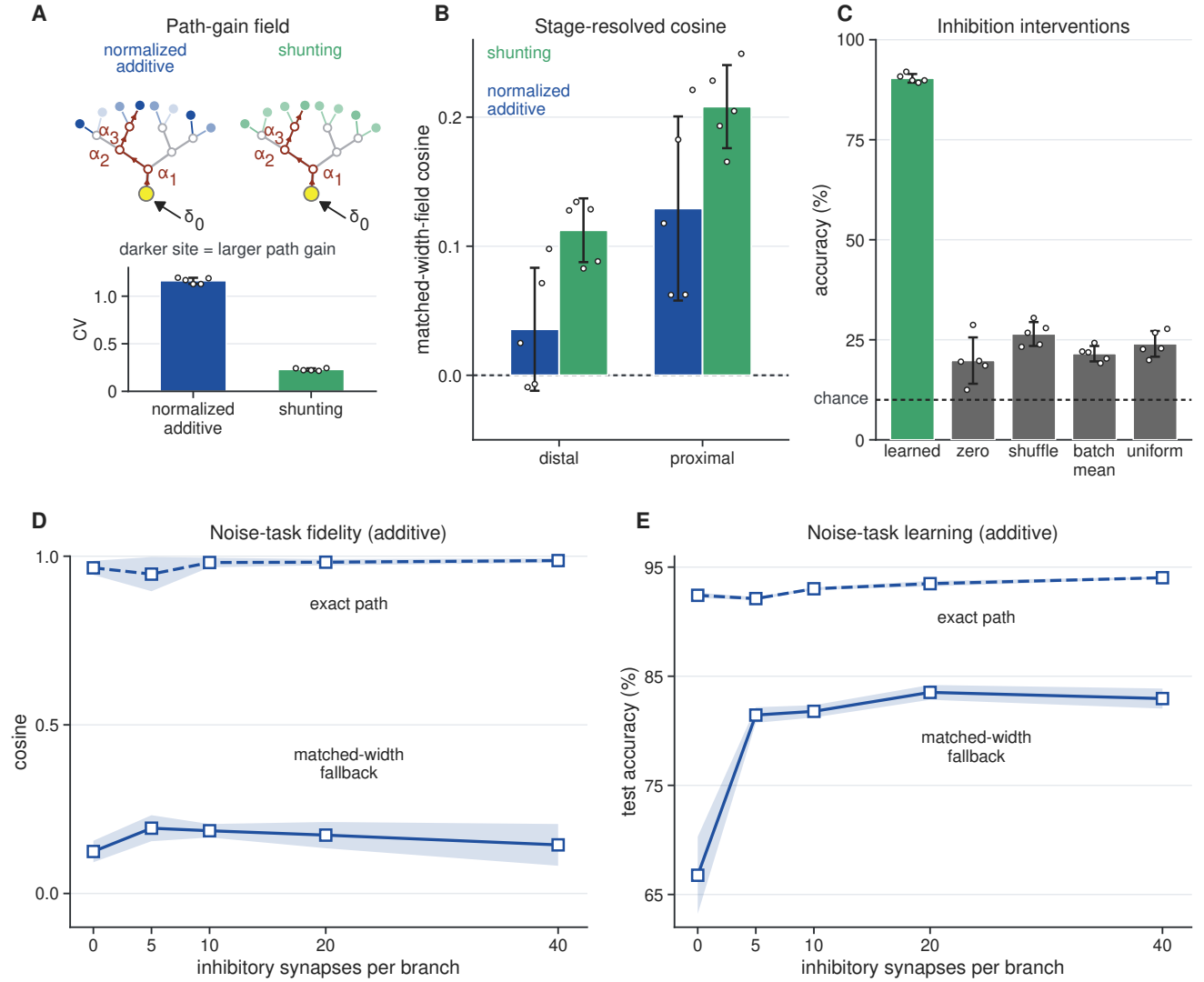


Figure S1: Mechanistic chain from path gains to learning in regular-tree models. MNIST panels A–C use two-stage [4,4] directed trees; the historical noise-resilience panels D,E use two-stage [3,3] additive trees. **A**, Conductance-stage path-gain field in shunting and normalized-additive models at five inhibitory synapses per branch; the inset shows paired-seed coefficients of variation (CV; standard deviation divided by mean). The schematic is drawn on the shared balanced tree of main Fig. 1B rather than the [4,4] geometry: one root-to-leaf route carries the serial path gains $\alpha_1, \alpha_2, \alpha_3$, distal sites are tinted by their illustrative path gain (darker, larger) in each model’s colour, and δ_0 is the somatic error. This loading quantity omits branch derivatives and is distinct from the trained exact-error profiles in Supplementary Fig. S8B,C. **B**, Stage-resolved cosine of matched-width scalar fallback versus exact MNIST errors; the width-matched somatic stage is excluded because its cosine is one by construction. **C**, MNIST post-training inhibition interventions: zero, sample shuffle, per-branch batch mean and a uniform matched mean. The dashed rule marks ten-class chance. These also change forward computation and do not isolate backward shunting. **D,E**, Feedback fidelity and learning on the additive noise task; solid and dashed lines denote matched-width scalar-fallback and exact transport. Bars show standard deviations across checkpoints in **B** and retained runs in **C**; shaded bands in **D,E** show standard deviations across retained runs (0.6–0.9 percentage points at doses 5–40 in **E**, and below 0.02 for the exact-path series in **D**, narrower than the marker); white points identify checkpoints in **B** and runs in **C**. Historical noise-task shunting aggregates with unresolved execution lineage are excluded and retained in Source Data only. The retained additive aggregates do not resolve their normalization setting. The state-matched focal analysis in main Fig. 8 provides the conductance-specific transport comparison. This chain is the figure support for Supplementary Section S1; the exact-path field it feeds is drawn at the main-text type scale in main Fig. 1.

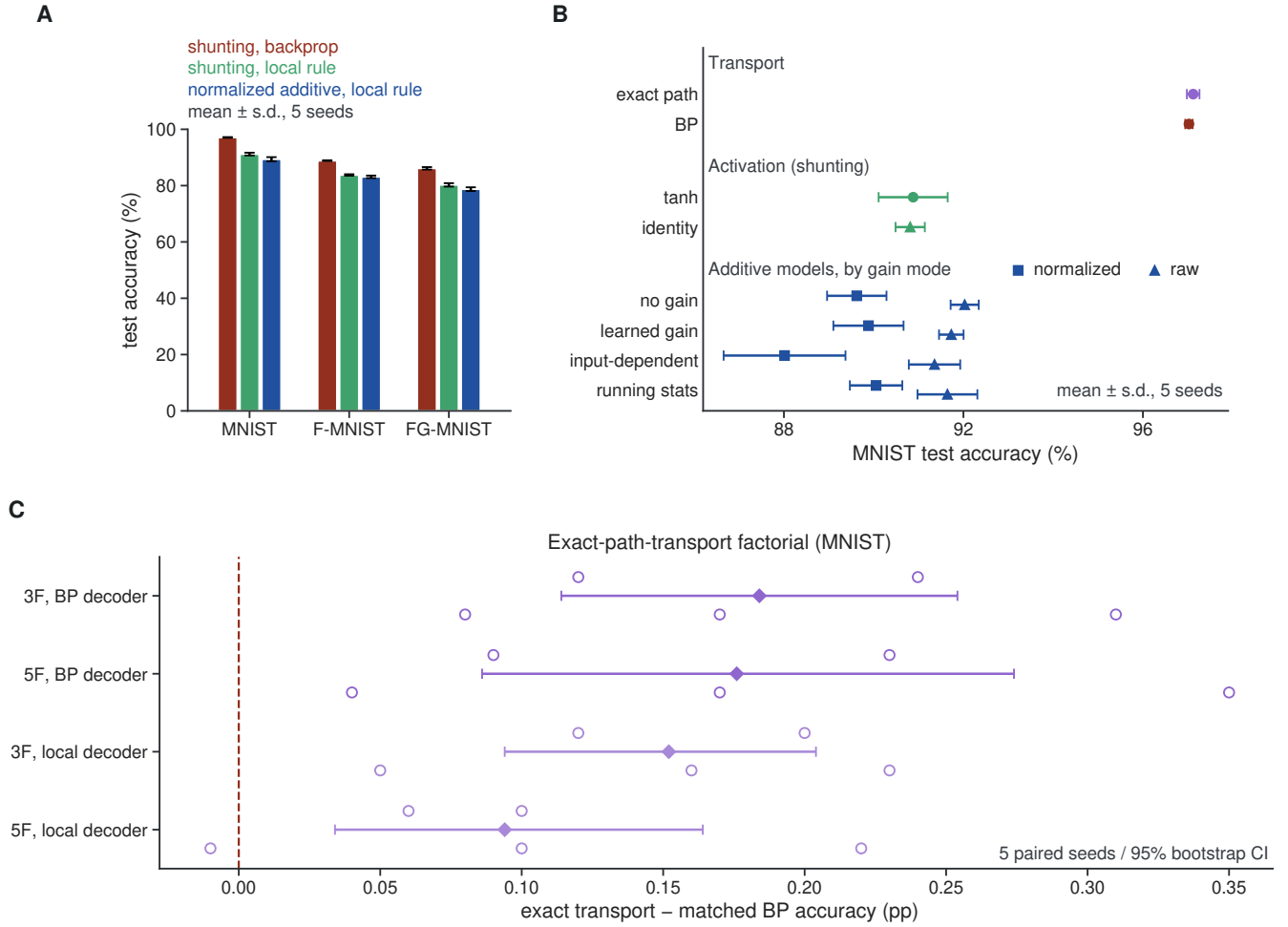


Figure S2: Local eligibility and exact transport reproduce the learning gradient. **A**, Test accuracy on MNIST, Fashion-MNIST (F-MNIST) and figure-ground MNIST (FG-MNIST) for the shunting architecture trained by backpropagation (red-brown) or by the local rule (green), and for the normalized-additive architecture trained by the local rule (blue); bars are means $\pm$ 1 s.d. across five seeds, and the backpropagation intervals are narrower than the line weight. **B**, Transport, activation and additive-model implementation controls on MNIST; points are means $\pm$ 1 s.d. across five seeds, and the backpropagation interval is smaller than its marker. Colour names the architecture as in **A** (red-brown backpropagation, green shunting, blue additive, violet exact path transport); within a family the house marker (circle for shunting, square for additive) is the configuration drawn in **A** and a triangle is the ablated variant (identity activation without tanh reactivation; raw additive without normalization). The additive group draws the complete normalized/raw $\times$ gain-mode factorial, one gain mode per row. The MNIST backpropagation and normalized-additive conditions in **A** and **B** come from different run cohorts (the matched ceiling-refresh and local-rule tuning cohorts in **A**; the exact-transport and gain-normalization revision cohorts in **B**): the backpropagation means differ by 0.09 percentage points, and the normalized-additive bar in **A** lies within the range of the four normalized rows in **B**. **C**, Exact-transport-minus-backpropagation accuracy across three-/five-factor and local/backpropagated decoder choices; open circles are five paired seeds, diamonds are condition means and bars are 95% paired-seed bootstrap intervals; dark violet denotes a backpropagated decoder and lighter violet a local decoder. Exact transport supplies an oracle compartment error. The local three-factor eligibility is sufficient when this error is supplied. Normalized- and raw-additive models are distinct controls; no noise-task shunting aggregate is used here.

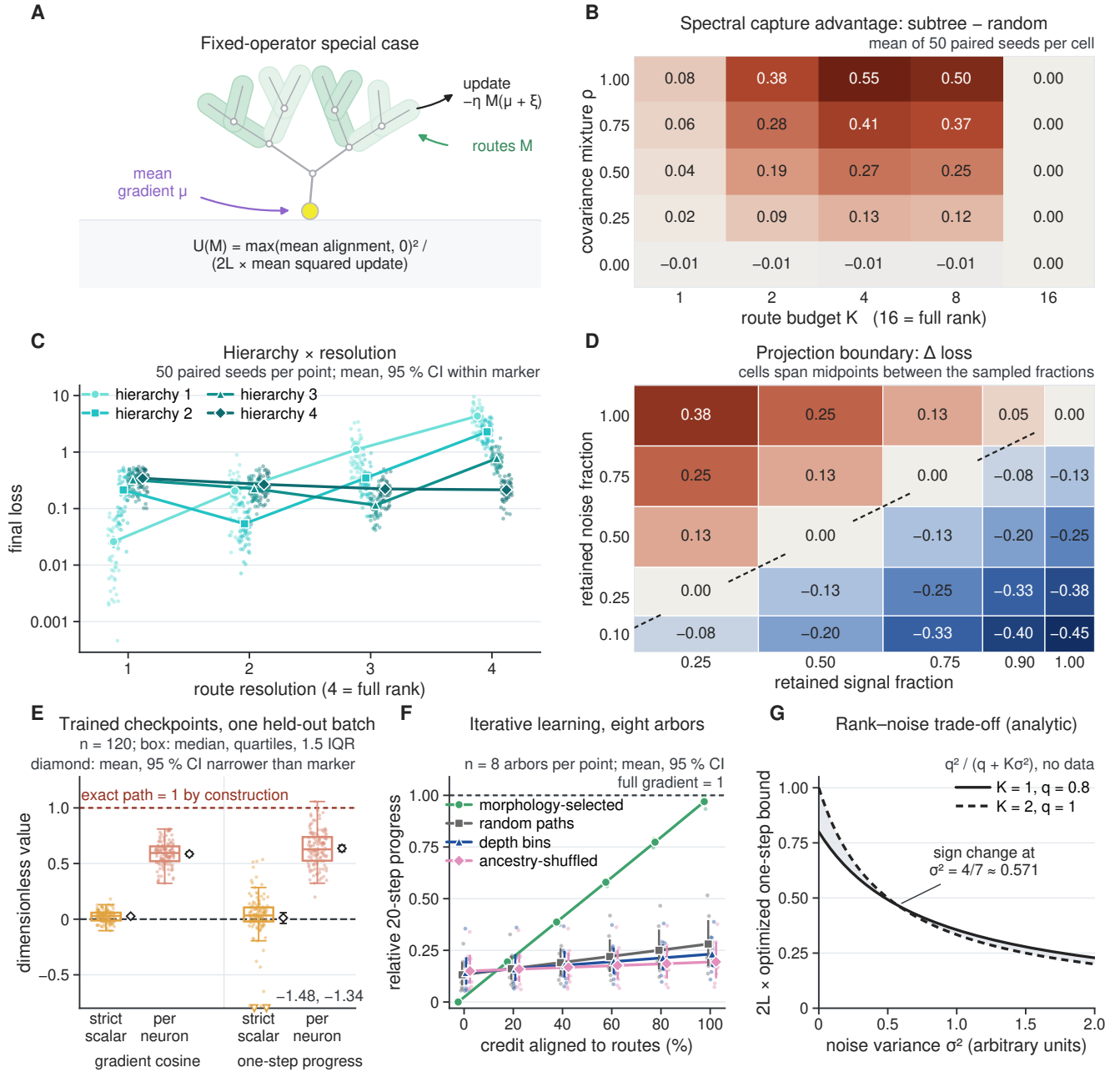


Figure S3: Restricted routes trade signal retention against admitted noise, and the one-step quantity is measurable at trained states. Throughout, small dots are individual units and large symbols their mean with 95% bootstrap whiskers (seeds, **C**; checkpoints, **E**; cells then one Monte Carlo stream per cell, **F**), the whiskers hidden by the marker in **C,E**; **B,D** print each cell (red positive, blue negative, tint by magnitude; **B**'s -0.01 control cells are near-neutral). **A**, Fixed-operator one-step bound: the boxed guarantee, printed in words, $U(M) = \max(\mu^\top M \mu, 0)^2 / (2LE\|M(\mu + \xi)\|^2)$ (ξ , zero-mean noise; L , loss smoothness; η , step size), lower-bounds the expected decrease under $M(\mu + \xi)$, swept in **B–D**. **B**, Subtree-minus-random spectral capture over route budget K and covariance mixture ρ , 50 paired seeds per cell; $K = 16$ and $\rho = 0$ are by-construction controls, only the endpoints isospectral. **C**, Final loss against route resolution D_r for hierarchies $H_c = 1–4$ (teal, light to dark; circle, square, triangle, diamond); $D_r = 4$ resolves every leaf ($M = I$); 50 paired seeds per point, series dodged in x . **D**, Exact projection boundary: retaining signal fraction f_{sig} and noise fraction f_{noise} changes normalized loss by $(f_{\text{noise}} - f_{\text{sig}})/2$; cells centre on the sampled fractions the axes print and the dashed line is the exact boundary $f_{\text{noise}} = f_{\text{sig}}$. **E**, Gradient cosine and norm-matched one-step progress at 120 trained checkpoints (one held-out batch, relative step 10^{-5}); strict scalar amber, per neuron salmon; two strict-scalar values below the floor are open triangles there, printed. Boxes, medians, quartiles and 1.5-interquartile-range whiskers; open diamonds, means. The dashed dark-red rule is the exact path (1 by construction), the dashed grey rule zero. **F**, Positive control, eight reconstructed arbors: loss reduction over twenty projected steps relative to the full gradient (dashed rule, 1) against credit aligned to the routes (green circles, morphology-selected; grey squares, random paths; blue triangles, depth bins; pink diamonds, ancestry-shuffled). **G**, Analytic rank–noise trade-off, $2L$ times the optimized bound $q^2/(q + K\sigma^2)$ for $(K, q) = (1, 0.8)$ (solid) and $(2, 1)$ (dashed), both black, crossing at the annotated $\sigma^2 = 4/7$; grey shading between the curves. Imposed alignment and oracle projections drive **E,F**, which measure neither biological credit nor a long-horizon selector; the bound is local, and endpoint selection fails prospectively (Fig. S34).

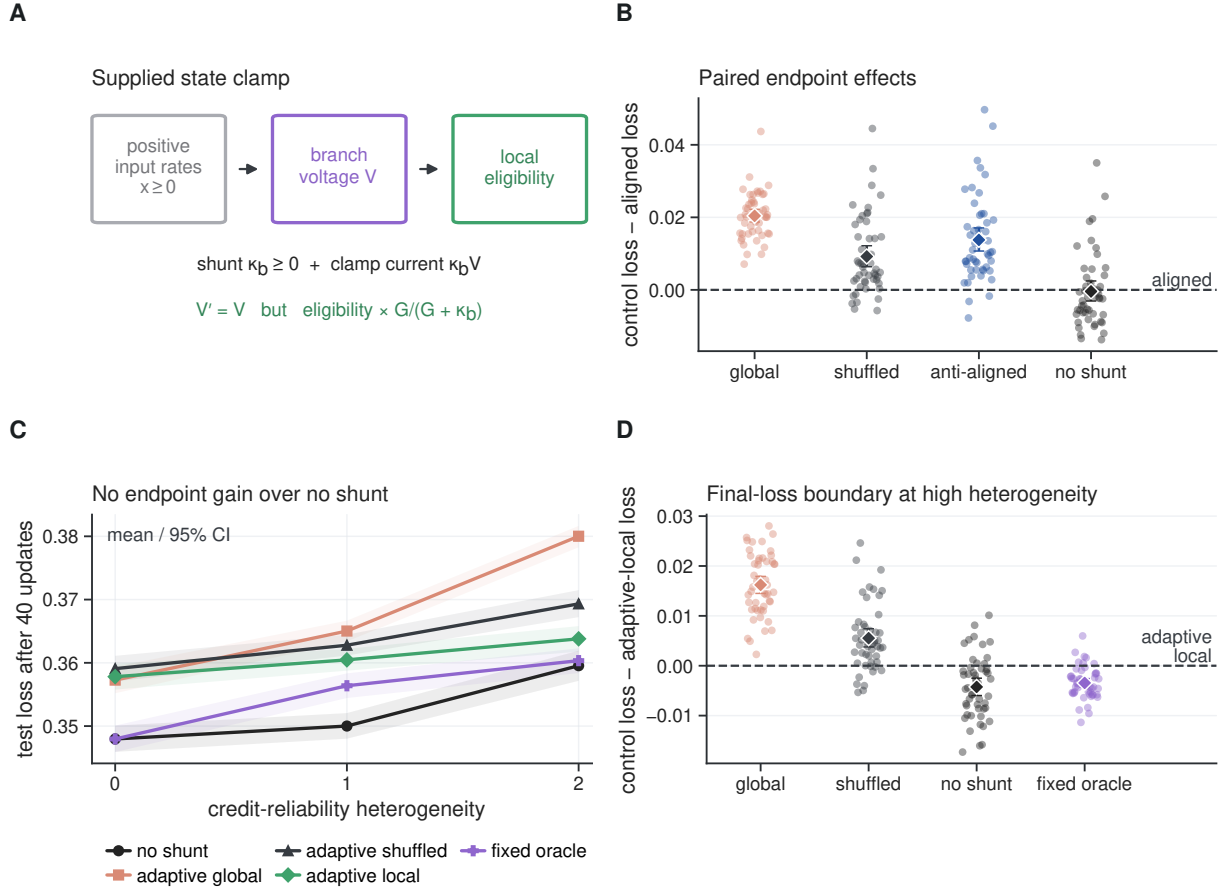


Figure S4: Fixed and estimated reliability gains have bounded learning benefits. All panels use positive-rate synthetic tasks with eight parallel, nonserial branch blocks. **A**, A supplied compensating current preserves branch voltage while a positive shunt reduces input resistance and eligibility. **B**, Paired control-minus-aligned final-loss contrasts at maximal heterogeneity for the best global, shuffled, anti-aligned and unshunted controls; fixed aligned shunting has no reliable final advantage over unshunted noisy learning. **C,D**, Adaptive-rule final loss and paired control-minus-adaptive contrasts. Adaptive local gain beats global and shuffled attenuation but loses to no shunt and the fixed oracle. In **B,D** the dashed line is the aligned or adaptive-local reference, translucent dots are the 50 individual paired seeds, and diamonds with bars are means and 95% paired-seed bootstrap intervals; an interval narrower than its diamond is hidden by it. The point implementation given identical gains reproduces the conductance final loss exactly in all 50 seeds of each study (paired difference 0, 50/50 ties). Curves and shading in **C** are means and 95% paired-seed bootstrap intervals across the same 50 seeds. The supplied compensating current in **A** and the paired gradient observations from which the adaptive rule in **C,D** estimates its gains are explicit information resources.

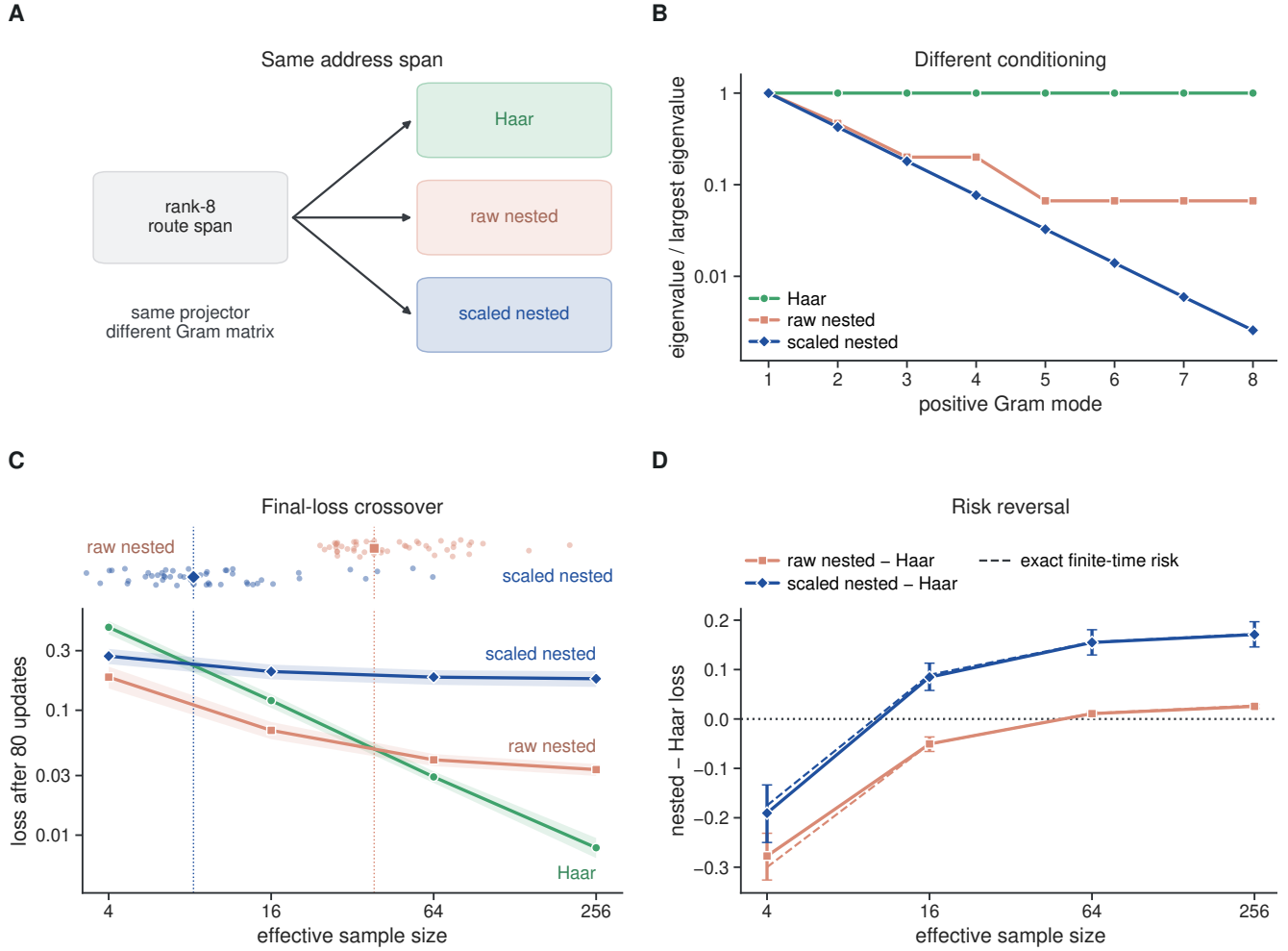


Figure S5: Identical route spans can learn differently through conditioning. **A**, Haar, raw nested (redundant) and scaled nested (statically scaled) coordinates parameterize the same rank-eight span in sixteen abstract coefficients. **B**, Positive Gram eigenvalues; condition numbers are 1, 15 and 388.52. **C**, Population loss after eighty updates across effective sample size on a logarithmic axis; curves and bands are means and 95% seed-bootstrap intervals across fifty seeds. The strip above the axis shows, for every seed, the predicted effective sample size at which the exact finite-time risks of the nested and Haar coordinates are equal (raw nested: median 38.5, interquartile range 31.1–63.7; scaled nested: median 8.2, interquartile range 6.3–12.6); large symbols and dotted guides mark the medians. **D**, Paired nested-minus-Haar contrasts; negative values favor nested coordinates. Points and bars are means and 95% paired-seed bootstrap intervals across fifty seeds ($n = 50$ pairs); dashed curves are the exact finite-time risk. Slow modes reduce variance at low data and retain bias at high data. Gram-preconditioned controls agree to 1.15×10^{-15} . This is a coefficient-learning comparison within one span, not a simulation of forward dendritic morphology or a biological implementation of the preconditioner.

S2 Image benchmarks and feedback controls

2.1 Shared architecture, data and local optimization

The MNIST core contains one dendritic population of 128 neurons, with no separate inhibitory-cell population. Each neuron has a soma, three proximal compartments and nine distal compartments, denoted $[3, 3]$. Separate TopK arrays select 40 excitatory and 20 inhibitory contacts from the 784 pixels at each nonsomatic compartment. There are no somatic input synapses: 92,160 active contacts are selected from 2,408,448 candidate synaptic parameters. Inactive local weights are not updated. Softplus transforms enforce nonnegative synaptic and coupling parameters. A linear 128×10 decoder gives ten class scores. The complete network has 2,414,602 parameter slots, including 1,290 decoder parameters.

Pixels are divided by 255 and flattened without additional normalization or augmentation. Each seed divides the 60,000 official training images into 48,000 fitting and 12,000 validation examples; the official 10,000-image test set is held out. The shared MNIST recipe uses three-factor local eligibility, batches of 256, 180 epochs and Adam with $(\beta_1, \beta_2, \epsilon) = (0.9, 0.999, 10^{-8})$, without weight decay or mixed precision. After minibatch averaging, input, coupling, reactivation and decoder gradients are clipped elementwise to $[-5, 5]$. Their original learning rates are 1.5×10^{-3} , 7×10^{-4} , 5×10^{-4} and 1.5×10^{-3} , respectively. Early stopping is disabled and minimum validation loss selects the retained checkpoint. Deterministic PyTorch settings disable TF32 and cuDNN benchmarking and use the highest float32 matrix-multiplication precision.

Every compartment uses the learned output nonlinearity

$$f_n(V) = \frac{\tanh[m_n(V - b_n)] + 1}{2}, \quad f'_n(V) = \frac{m_n}{2} \{1 - \tanh^2[m_n(V - b_n)]\}, \quad (65)$$

where m_n and b_n are slope and midpoint. Shunting trees use analytical initialization. Raw-additive trees use the prespecified occupancy-quantile policy: empirical voltage quantiles 0.1 and 0.9 map to outputs 0.1 and 0.9, with minimum quantile width 10^{-3} , maximum slope 50 and reversion on an invalid fit. The comparisons therefore retain the architecture-specific initialization policies, not identical initial voltage scales.

2.2 Additive controls and gradient coordinates

The raw-additive alias `dendritic_additive` disables both shunting and additive normalization. Its current sums and local derivatives are

$$J_n^{\text{E,add}} = \sum_{i \in \mathcal{I}_n^{\text{E}}} g_i^{\text{E}} x_i^{\text{E}}, \quad J_n^{\text{I,add}} = \sum_{j \in \mathcal{I}_n^{\text{I}}} g_j^{\text{I}} x_j^{\text{I}}, \quad (66)$$

$$J_n^{\text{child,add}} = \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} a_c, \quad V_n^{\text{add}} = J_n^{\text{E,add}} - J_n^{\text{I,add}} + J_n^{\text{child,add}}, \quad (67)$$

with $a_n = f_n(V_n^{\text{add}})$, and

$$\frac{\partial V_n^{\text{add}}}{\partial g_i^{\text{E}}} = x_i^{\text{E}}, \quad \frac{\partial V_n^{\text{add}}}{\partial g_j^{\text{I}}} = -x_j^{\text{I}}, \quad \frac{\partial V_n^{\text{add}}}{\partial g_{c \rightarrow n}^{\text{den}}} = a_c, \quad \delta_c^V = \delta_n^V g_{c \rightarrow n}^{\text{den}} f'_c(V_c). \quad (68)$$

Inhibition contributes a subtractive current, not a negative conductance. Raw-parameter gradients include the soft-plus derivative. The inherited normalized-additive control instead standardizes each example across a stage's output coordinates. With $Z_{sn} = J_{sn}^{\text{E,add}} - J_{sn}^{\text{I,add}} + J_{sn}^{\text{child,add}}$, it uses

$$V_{sn}^{\text{norm}} = \frac{Z_{sn} - \langle Z_s \rangle_n}{\text{sd}_n(Z_s) + \epsilon}, \quad a_{sn} = \frac{\tanh(0.1 V_{sn}^{\text{norm}}) + 1}{2}. \quad (69)$$

Its fixed slope and midpoint are 0.1 and zero. Because this changes forward computation, its outcomes are not pooled with raw-additive results.

The fresh dictionary experiment routes errors of compartment output activation; f'_n remains in local eligibility, including gate-parameter derivatives. Neuron-specific feedback repeats each exact somatic error within that tree. The projected $K = 1$ rule averages exact activation errors over all twelve nonsomatic compartments; $K = 3$ averages them separately within each proximal compartment and its three children. Both acquire coefficients from the same exact field and preserve its common mean at identical weights, but need not preserve equal field norms. Their somatic error

remains exact. Strict scalar instead reduces the neuronal error vector at every core stage, including the soma, using Eq. 25. Exact transport retains every compartment error. The linear decoder receives exact gradients in all arms. Decoder-only learning suppresses core and gate gradient writeback; initial/final core hashes verify the intended freeze. These projected comparisons test delivery capacity with oracle coefficients.

2.3 Fresh dictionary controls and retained image cohorts

The six-rule MNIST experiment compares common rate multipliers 0.3, 1, 3. Three development seeds per architecture, rule and multiplier give 108 fits. A separately frozen addendum specified projected $K = 1$ before fresh outcomes, with the same tuning procedure. Mean minimum validation loss selects each rule’s multiplier. Ten new paired seeds per architecture and rule then test the selected rate and, when different, the original multiplier one. Coincident conditions reuse a fit, giving 190 unique fresh fits; twelve three-epoch canaries are excluded. Pairing covers initialization, contacts, splits and minibatches. Descriptive intervals use 50,000 whole-seed bootstrap draws. Full configurations and seed endpoints are in `source_data/image_ladder_controls/`.

Capture was measured at initial and exact-trained checkpoints on the first 2,048 official test images. For each example/neuron, the field comprises twelve nonsomatic errors. The primary endpoint averages nonzero field-wise squared-energy ratios; aggregate energy capture is also retained. Activation-space capture matches the trained projections, while voltage-space capture preserves the earlier diagnostic coordinate. Two historical checkpoints reproduced their recorded voltage captures to numerical precision; instrumentation left outputs unchanged. In the earlier fifteen-seed cohort, moving from one to three profiles increased mean voltage-credit capture from 0.480 to 0.625 in shunting trees and from 0.519 to 0.718 in additive trees. These values describe a separate cohort and coordinate from the activation-error capture in main Fig. 1G.

The original MNIST ladder uses seeds 42–56: strict scalar, neuron-specific and exact credit in both architectures give 90 fits; matched-width scalar fallback adds 30. The latter differs from strict scalar in a small near-somatic parameter group, as quantified in Section 1.8. These consistently executed runs replace the superseded mixed-source set; the replacement decision preceded outcome inspection. The independent direct-feedback-alignment factorial uses fifteen paired seeds per dynamics and 120 new fits with frozen exact-readout references. A seed-fixed Gaussian output-to-neuron matrix, scaled by inverse square root of output dimension, replaces the decoder derivative. Ownership controls permute neuronal recipients without changing values or bandwidth.

Fashion-MNIST retains the MNIST architecture, contacts, optimizer, initialization and 180-epoch budget without retuning. Seeds 10200–10209 pair 48,000/12,000 fitting/validation images and the 10,000-image official test set. Sixty fits cross two architectures with matched-width scalar fallback, neuron-specific and exact fields. Neuron-specific-minus-fallback accuracy is 4.46 percentage points (95% interval 4.09–4.88) for shunting and 3.78 (3.43–4.14) for additive trees; all ten differences are positive in each. Exact-minus-neuron-specific differences are -0.11 (-0.21 to -0.01 ; 2/10 positive) and 0.13 (-0.08 to 0.35 ; 8/10). The prespecified neuronal-identity criterion required at least 8/10 positive differences and an interval above zero in each architecture. All sixty fits were complete and finite without feedback fallbacks. Full means and tests remain in `source_data/fashion_feedback_ladder/`.

2.4 CIFAR-10 and exact-gradient validation

The raw-additive CIFAR-10 cohort uses flattened images, no augmentation or normalization, a 90:10 training/validation split, one 20-soma $[3, 3, 3, 3]$ population, 25 contacts of each sign per branch and a 20–32–16–10 decoder. The realized `input_mode=1` model has direct excitatory/inhibitory inputs and no inhibitory-cell population. A validation-only BP screen jointly selected adaptive initialization and empirical stage calibration from three batches; it cannot attribute their benefit separately. A provisional configuration with mean BP accuracy 38.98% was excluded for an inadequate operating point. The selected three-seed calibration reached 49.960% validation and 49.723% test accuracy. A separate five-seed check reproduced normalized-additive and shunting reference endpoints to within 0.026 percentage points.

All four fresh rules use Adam at 10^{-3} , batch size 256, at most 200 epochs, patience 40 and the validation-best checkpoint. Weight decay is zero; the separate 0.01 configuration field controls sparse-weight maintenance. Seeds 10800–10819 give eighty finite, complete fits. Mean accuracies are 0.34472 (strict scalar), 0.50865 (neuron-specific), 0.50006 (exact path) and 0.50122 (matched BP). The exact-minus-neuron-specific difference is -0.00859 (95% interval -0.01171 to -0.00547 ; 2/20 positive), so the exact-path superiority criteria fail. Exact-minus-BP is -0.00115 (-0.00457 to 0.00226), with two one-sided tests supporting equivalence within ± 0.01 (90% interval -0.00398 to 0.00167 ; $P = 1.58 \times 10^{-5}$). This is a compact transfer assay, not a competitive vision benchmark.

A separate five-seed exact-transport factorial (42–46) crosses three-/five-factor rules with local/BP decoder updates over 200 epochs and a ReLU core output. Local rates for input, coupling, reactivation and decoder are 1.5×10^{-3} , 7×10^{-4} , 5×10^{-4} , 1.5×10^{-3} ; matched BP uses 10^{-3} , 5×10^{-4} , 5×10^{-4} , 10^{-3} . Analytic and automatic gradients were compared in 100 diagnostic runs: median relative error 7.43×10^{-8} , maximum 1.88×10^{-7} , with parameter tensors as diagnostic units. The architectures and method-specific rates above distinguish these protocols.

2.5 Depth, ownership and input-map controls

The original depth-by-feedback factorial crosses two tasks, two cores, stages D1–D4, three local fields plus BP and ten seeds: 640 planned fits. Completeness requires the specified seed/configuration, checkpoint and full curves, expected stages, nondegenerate couplings, finite metrics and no transport-shape fallback. All completed, but a later outcome-independent validity check excludes 160 synthetic-noise shunting fits with signed conductance drive. The retained 480 fits contain both cores on MNIST and additive noise-task models. Depth changes compartment and parameter count. The ownership comparison at D2/D4 retains 120/160 fits after the same exclusion. A 400-fit inhibitory-dose family is entirely excluded from inference because its primary shunting contrast uses invalid signed input, although configurations and exclusion records remain available.

The clean exact/BP rerun instead uses valid inputs: two tasks, two cores, four depths, two methods and seeds 42–51 give 320 fits. Shunting uses explicit ReLU transfer; additive noise inputs remain signed. Source commits are 74792ca9 for implementation and 4f3612a5 for the nested generator. All 320 fits are finite and complete; 42 use streaming evaluation of a terminal metric. Failed attempts with missing generator code, insufficient memory or rejected signed shunting have no included endpoint. Re-executed configurations retain their scientific settings. Comparisons first average four depths within each seed, then use 20,000 paired bootstrap draws and descriptive Wilcoxon tests. Across all 160 method pairs, mean exact-minus-BP accuracy is -0.0545 points; the largest absolute pair difference is 5.90 points. This does not establish pointwise equality.

The spatial-input control uses depth four, 21 contacts per distal branch and 320 planned fits. Spatial branches recursively partition the 28×28 feature plane into sixteen disjoint regions; random branches sample all 784 coordinates. The 240 conductance-valid fits cross maps, methods, cores and tasks; the noise task retains only additive models. Persistence under BP and random task projection identifies a forward input-coverage effect. A separate 320-fit noise comparison holds sixteen terminal branches and 960, 960, 968 or 960 active contacts per soma across D1–D4; it retains 160 additive fits. Compartments, couplings and reactivation parameters still increase with depth.

At 120 valid BP checkpoints, a fixed held-out batch compares scalar fallback, neuron-specific and exact fields using nonsomatic stages and trainable branch parameters only. Outcomes are optimally scaled field capture, eligibility-weighted gradient capture/cosine and norm-matched loss change. Relative step 10^{-5} is primary; 10^{-6} , 10^{-4} , 10^{-3} are sensitivity values. Checkpoints are neither updated nor selected by this diagnostic, and bootstrap resampling preserves paired fields. Complete controls and validity accounting accompany the figure-linked Source Data (Supplementary Fig. S3).

2.6 Identity of the historical noise tasks

The key `noise_resilience` named two generators. The clean rerun’s recovered source, commit 4f3612a5, draws class y uniformly from $\{0, 1, 2\}$ and constructs a square image L_y : blank, a uniformly selected horizontal row of ones, or a uniformly selected vertical column of ones. It returns

$$x = \text{vec}(L_y + 0.2\varepsilon), \quad \varepsilon_{ab} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (70)$$

The executed dispatcher’s default side length is 784, giving 614,656 inputs, independently confirmed by checkpoint sizes. Its 10,000 unnormalized, unclipped examples split into 8,000/1,000/1,000 fitting/validation/test samples using run-seeded generators. Projected-noise fields in launch configurations were ignored. Thus these rows test implementation agreement on three classes, not corrupted-MNIST robustness.

The documented prospective depth/routing/dose/fixed-budget protocol instead perturbs MNIST digits as

$$\tilde{x} = \text{clip}_{[0,1]}(x + 1.5Pz), \quad z \sim \mathcal{N}(0, I_{50}), \quad (71)$$

where $x \in [0, 1]^{784}$, $P \in \mathbb{R}^{784 \times 50}$ has independent normal entries and unit-norm columns, and digit labels are unchanged. Projection seed is zero; training/validation/test latent-noise seeds are 1/2/3. The intended splits are 48,000/12,000/10,000 without further gain or normalization. Frozen configurations and model sizes support this intended protocol, but no hash

683 of the executed nested generator survives. The fixed-budget outcome/validity join also lacks raw resolved configurations
684 and does not close that gap.

685 Inherited aggregate feedback/gradient panels lack a complete run-to-generator join. Twenty-six historical shunting
686 points remain unresolved and are excluded from conductance-valid figures and inference; neither generator is assigned
687 to them retrospectively. The fixed-budget audit retains 160 additive and excludes 160 signed-shunting outcomes. The
688 release exposes `legacy_noisy_lines` and `projected_noise_mnist` separately and rejects the ambiguous key without
689 explicit choice. The latter is a frozen implementation of the documented protocol, not recovered historical execution. See
690 `source_data/release_task_identity/task_identity.json` and `source_data/review_curve_lineage/`.

691 Inherited regime sweeps use five seeds per condition; Gaussian broadcast noise is scaled relative to the original
692 teaching signal, and depth varies at fixed somatic width. Their normalized-additive, raw-additive and shunting variants
693 remain separate. They test operating regimes, not isolated backward effects of inhibition.

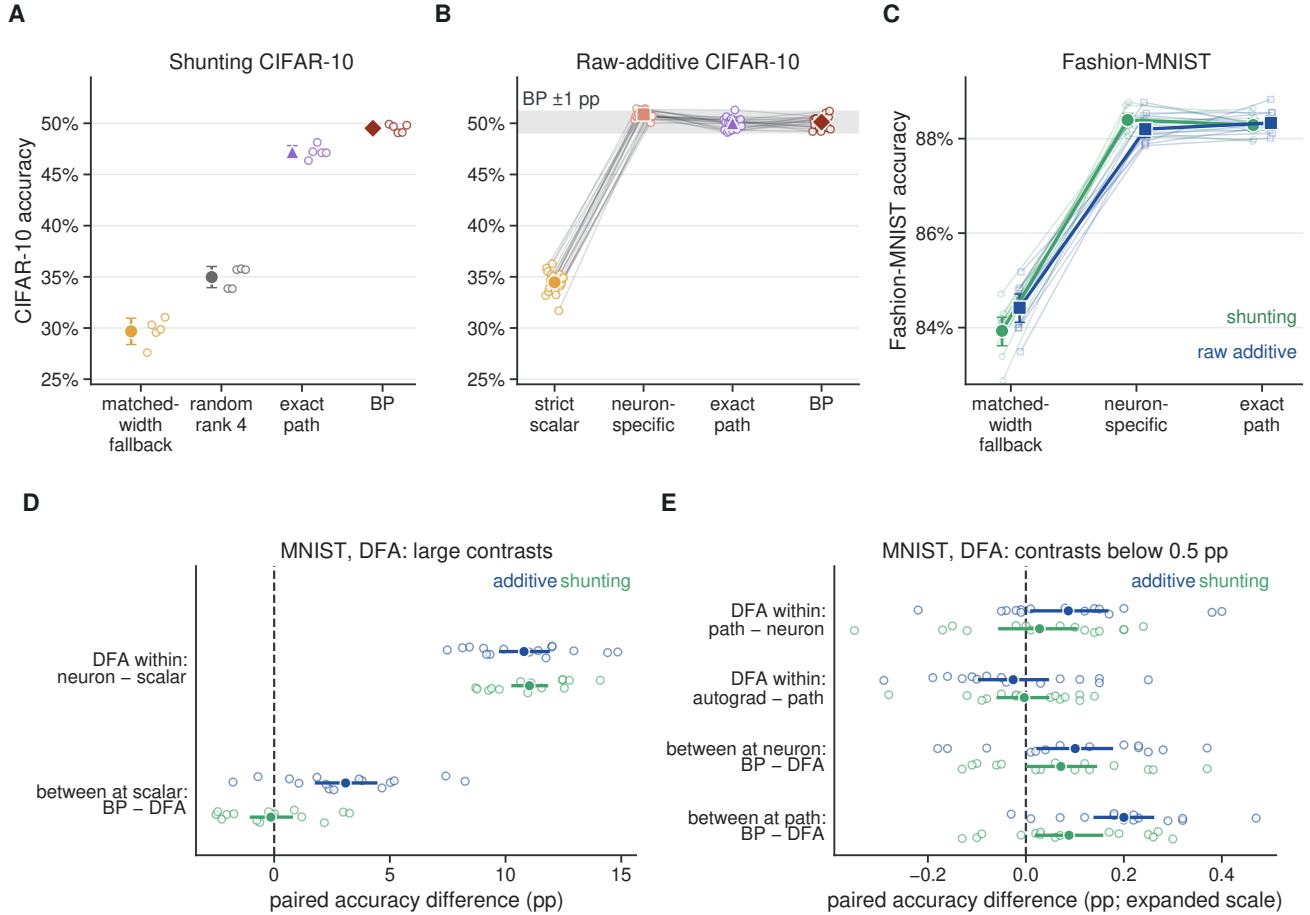


Figure S6: The neuron-identity bottleneck generalizes across image and feedback controls. **A**, Flattened CIFAR-10 in the shunting architecture, with matched-width fallback, random rank-four, exact-path and backpropagation rules (five seeds; bars are ± 1 s.d.). **A** and **B** share one accuracy axis. **B**, Raw-additive CIFAR-10 with strict scalar, per-neuron, exact-path and backpropagation feedback (twenty paired seeds; 95% Student-*t* intervals, at most 0.5 percentage points and mostly hidden by the mean symbols; grey band, the prespecified ± 1 percentage-point backpropagation equivalence margin). The strict scalar of **B**, **D** and **E** is a separately trained control distinct from the matched-width fallback of **A** and **C**. Exact path is 0.859 percentage points below per-neuron feedback and equivalent to backpropagation within the margin. **C**, Fashion-MNIST in shunting and raw-additive trees (ten paired seeds; 95% bootstrap intervals); path resolution adds no reliable benefit. In **A–C**, filled symbols are cohort means and small open symbols individual seeds. **D,E**, Paired accuracy contrasts in the MNIST between-by-within factorial under fixed random soma feedback (DFA): contrasts of order one percentage point or more in **D** and contrasts below half a point in **E**, whose axis is expanded about twentyfold. Open symbols are the fifteen paired seed differences, filled symbols their means and bars 95% paired-seed bootstrap intervals; blue, raw additive; green, shunting. The top rows of **D** and **E** are the MNIST-DFA estimates of main Fig. 1E,F at supplement scale. The per-neuron-minus-strict-scalar contrast of about eleven percentage points is drawn in **D**. DFA preserves the neuron-versus-scalar benefit, but does not add a second dendritic layer or increase the dimensionality of the ten-class readout error.

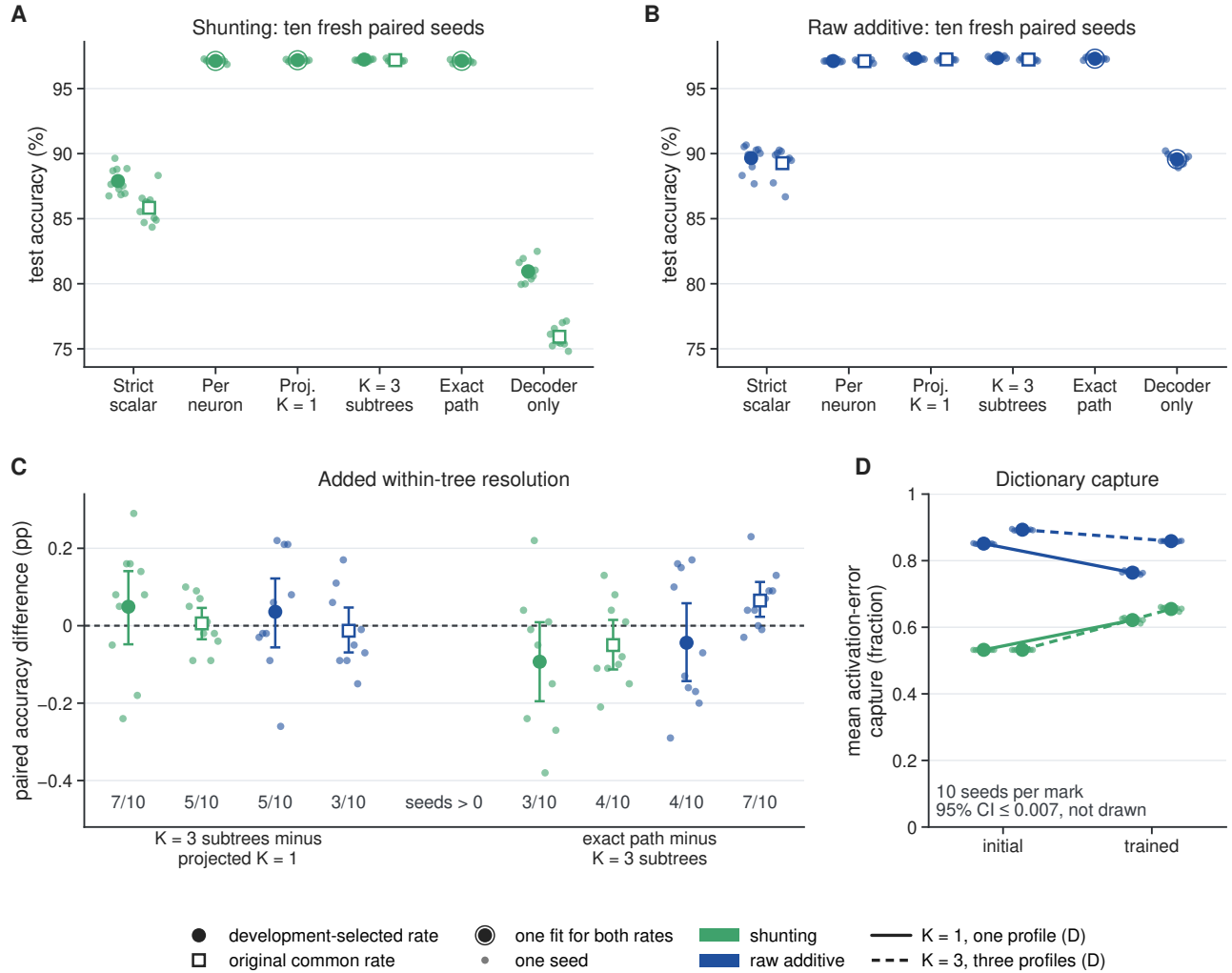


Figure S7: Intermediate dictionaries capture more image-task credit without improving accuracy. **A,B**, Six-rule MNIST ladder in shunting and raw-additive networks; ten fresh paired seeds per architecture. Throughout the sheet colour names the architecture (green, shunting; blue, raw additive) and the rule is named at each tick (Proj. $K = 1$ is the projected one-profile rule). Small translucent dots are the ten individual seeds; filled circles are means at the development-selected rate and open squares means at the original common rate, drawn left and right of the tick where the two rates differ. Where the selected multiplier equals the original common rate (shunting per neuron, projected $K = 1$ and exact path; additive exact path and decoder only) the ten runs are one fit for both rates and are drawn once, as a filled circle inside a thin ring. The accuracy axis is shared by **A** and **B** and spans the data (74–98%); the four dendritic arms lie within 0.25 percentage points of one another in each architecture, below the symbol size at this scale, and their paired within-tree contrasts are resolved in **C**. Decoder-only learning freezes the initialized core. **C**, Paired accuracy differences for three profiles versus a projected common profile ($K = 3$ subtrees minus projected $K = 1$) and for exact versus three-profile credit (exact path minus $K = 3$ subtrees). Within each group the four marks are shunting selected, shunting common, additive selected and additive common, filled and open as in **A,B**, each with its ten per-seed paired differences drawn as small dots behind the mean and a descriptive 95% seed-bootstrap interval; the dashed rule marks zero. The count printed under each mark is the number of seeds (of ten) whose difference is positive: 7, 5, 5 and 3 for $K = 3$ minus projected $K = 1$ and 3, 4, 4 and 7 for exact minus $K = 3$. Note the scale: the full y range of **C** is 0.86 percentage points, about one twenty-ninth of the 25-point range of **A** and **B**. **D**, Initial and trained mean activation-error capture (fraction) by uniform $K = 1$ (solid lines, marks left of each checkpoint) or subtree $K = 3$ (dashed lines, marks right of each checkpoint) profiles, averaged over 2,048 held-out images within each seed; filled circles are the ten-seed means and small dots the ten per-seed values, which lie within the symbol at initialization. At shunting initialization the two profiles coincide (0.532 for both). Projection coefficients use the exact activation-error field of this ten-seed cohort, so **D** reproduces the $K = 1$ and $K = 3$ points of main Fig. 1G for this cohort; voltage-space capture and the aggregate energy capture of the same fits are retained in Source Data and are not the plotted quantity. **A** and **B** show the ten per-seed values with the mean and no interval bars; **C** draws the ten per-seed differences and the mean with a descriptive 95% seed-bootstrap interval; the intervals in **D** are at most 0.007 capture units, below the symbol size, and are not drawn, as the panel states. Greater capture did not yield an image accuracy benefit. All image cohorts have one dendritic layer and a linear readout.

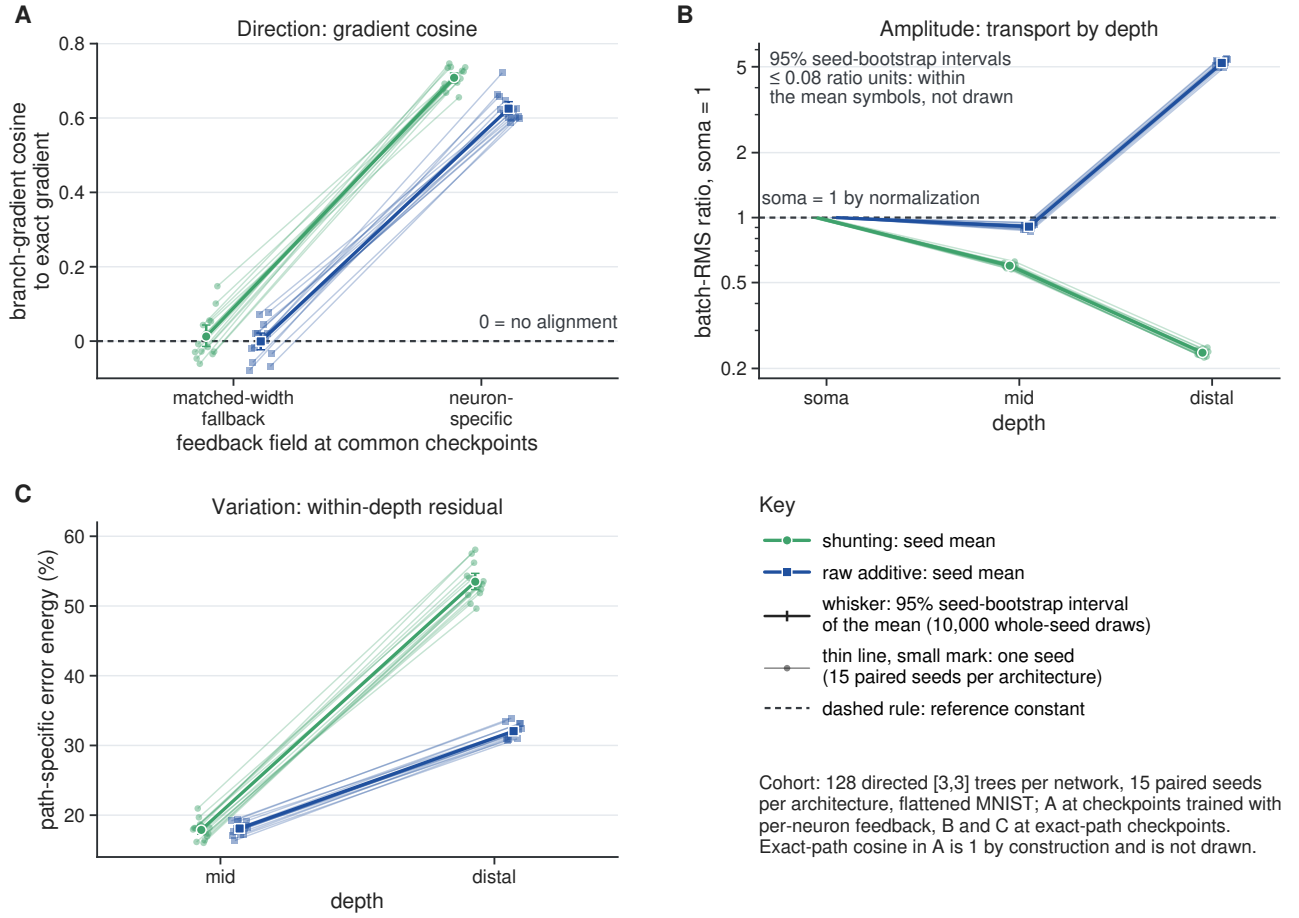


Figure S8: Error-field geometry separates direction, amplitude and spatial variation. Independent fifteen-seed flattened-MNIST cohort of 128 directed [3,3] trees; green circles denote shunting and blue squares raw-additive trees, offset sideways from each other at every category; panel titles name the property each panel isolates (direction, amplitude, spatial variation). **A**, Branch-gradient cosine with the exact gradient at common fixed checkpoints trained with per-neuron feedback; matched-width scalar-fallback and per-neuron fields are evaluated at identical weights and states, and the exact-path cosine is one by construction and is not drawn. **B**, Exact voltage-error magnitude by depth at exact-path checkpoints: the batch root-mean-square ratio $\text{RMS}(\delta_{n,u}^V)/\text{RMS}(\delta_{0,u}^V)$, averaged over neurons and paths, on a logarithmic axis, with somatic value one by normalization, marked by the dashed grey rule at one from which every per-seed profile starts; no symbol is drawn at the soma. **C**, Within-depth path-specific residual error energy at the exact-path checkpoints of **B**, after subtracting the mean across paths within depth d : distal shunting attenuates magnitude but retains the larger path-specific fraction, rising from 17.9% (shunting) and 18.1% (raw additive) at mid to 53% and 32% at distal. In every panel the thin lines with small marks are the fifteen paired seeds per architecture and the large symbols joined by the heavy line are seed means. In **A** and **C**, whiskers in the series colour are 95% seed-bootstrap intervals of the mean (10,000 whole-seed draws), visible where they protrude beyond the symbol and otherwise within it (half-widths 0.014–0.031 in **A**, 0.44–1.2 percentage points in **C**); in **B** every interval is at most 0.08 ratio units, within the symbol, and none is drawn. The dashed grey rule in **A** marks cosine zero; the key on the sheet names each glyph. No additional models are trained here; greater path-specific variation did not yield an image accuracy benefit, and voltage- and activation-space capture are different quantities.

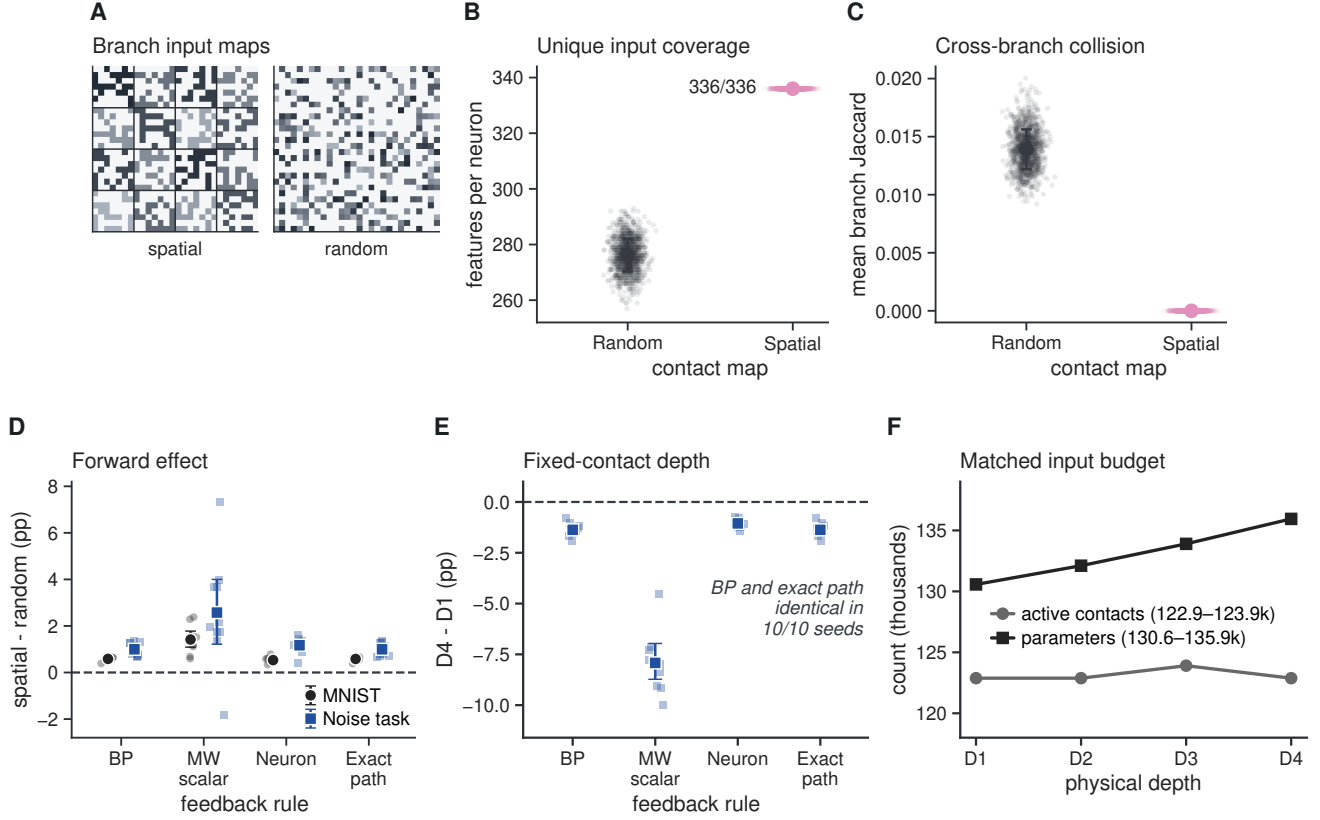


Figure S9: Input coverage and ordinary-task depth are distinct from credit resolution. **A–D**, Spatial versus randomly assigned contacts in depth-four $[2, 2, 2, 2]$ trees. **A**, The sixteen branches of one example neuron on the 28×28 input plane under the spatial sampler (left; grid lines mark its sixteen disjoint regions) and under the matched random sampler of the same seed (right); shade identifies the branch and pale pixels are uncontacted (57%). **B,C**, Unique input coverage and cross-branch overlap for all 1,280 owner maps per condition (128 neurons $\times$ ten seeds; mean $\pm$ s.d.): contact count is matched, but spatial maps reach all 336 contacted inputs (336/336) and avoid collisions. **D**, Paired spatial-minus-random test accuracy under backpropagation (BP), the matched-width scalar field (MW scalar), the shared per-soma field (Neuron) and exact path transport (Exact path). Small symbols are the ten paired seeds; large symbols and bars are means and 95% paired-seed bootstrap intervals, and a bar narrower than its symbol is hidden by it. Black circles are MNIST, each seed averaging the shunting and raw-additive cores; blue squares are the randomly projected noise task, raw-additive core only, in which BP and exact path coincide by construction. The accuracy benefit persists under backpropagation and therefore reflects a forward sparsity prior. **E,F**, Separate raw-additive historical noise cohort with sixteen terminals (blue squares; small symbols are the ten paired seeds): all four D4-minus-D1 accuracy contrasts are negative in 10/10 paired seeds, with BP and exact path identical in every seed because exact path transport reproduces backpropagation in an additive tree; active contact counts stay within 122.9–123.9 thousand across D1–D4 while parameter counts rise from 130.6 to 135.9 thousand. The noise-task executed generator remains unresolved and its shunting runs fail the positive-conductance input criterion. These retained additive outcomes are a depth boundary, not evidence for shunting; exclusions are enumerated in Table S5.

S3 Branch conflict, ancestry routing and coefficient learning

3.1 Context-gated Fashion-MNIST protocol

The context-gated task used Fashion-MNIST classes 0 and 6. Inputs were average-pooled to 7×7 , standardized using training data and augmented with a constant coordinate. Each seed used 8,192 training and 2,000 test draws with replacement from the official binary splits, balanced across context and label. Branch counts were $B = 2, 4, 8$; conflict doses were 0, 0.25, 0.5, 4/7, 0.6, 2/3, 0.75, 1. Thresholding one underlying uniform mask made conflict nested across doses. Correct, shared, cyclically deranged, BP and gated-point rules reused inputs and initialization. Full-batch descent used 250 updates at step 0.04 and an initial weight standard deviation (SD) of 0.01. Three exploratory seeds (100–102) selected a stable setting; an additional implementation-check seed was also excluded. Analysis seeds 11000–11019 were fixed before outcomes, with no outcome-dependent stopping.

The primary estimand was each seed’s least-squares slope of correct-minus-shared accuracy on conflict, separately for each B . Prespecified criteria required positive slopes in at least 16/20 seeds, at least ten percentage points of separation at full conflict, less than two points at zero conflict, correct routing above derangement and exact BP/gated-point agreement. All criteria passed. The maximum analytic-versus- autodifferentiation discrepancy was 3.43×10^{-7} ; equivalent-rule accuracy differences were zero. Intervals used 20,000 paired-seed bootstrap draws; tests were two-sided Wilcoxon tests. Mean-field assumptions and empirical chance crossings are distinguished below.

The same twenty seeds were replayed for post hoc gradient diagnostics at epochs 0, 1, 10, 50, 100 and 250, comparing both each method’s own state and a common exact-rule state. Official input-file checksums were verified; the input manifest and all 28,800 rows are retained in `source_data/review_branch_trajectories/`. Endpoint drift was below 5×10^{-9} in loss. At conflict 0.5, common-state shared-gradient cosine fell from 0.987 to -0.519 for four branches and from 0.952 to -0.701 for eight branches by epoch 250 (Supplementary Fig. S10). The initialization boundary below was retained without refitting; trained gradient direction and trained accuracy crossing are separate endpoints.

3.2 Mean-field boundary for context-gated branch selection

The Fashion-MNIST conflict experiment selects one branch c uniformly from B branches. Only $\mathbf{w}_c^\top \mathbf{x}_c$ enters the logit, so the exact derivative of every unselected branch is zero. Shared feedback omits that selector and updates every active eligibility. For the centered mean-field prediction, assume balanced class signs $s \in \{-1, 1\}$, class means $\mathbb{E}[\mathbf{x} | s] = s\mathbf{v}$, and zero logits, so the logistic error is $\delta = -s/2$. An unselected input copies the selected image with probability $1 - \chi$ and is independently drawn from the opposite class with probability χ . Its mean class evidence relative to the selected view is therefore $1 - 2\chi$. The resulting compatibility matrix is

$$C_\chi = (1 - 2\chi)\mathbf{1}\mathbf{1}^\top + 2\chi I.$$

Its shared-mode eigenvalue is $B - 2\chi(B - 1)$, which vanishes at $\chi_c = B/[2(B - 1)]$. Under these population and initialization assumptions, the expected shared update loses positive alignment with the exact update at that boundary. Logistic derivatives and class-conditional moments need not preserve this relation as weights change. The empirical trained chance crossing is consequently a separate descriptive endpoint, not a theorem about the entire optimization trajectory. The missing branch selector may be supplied in feedback or in local eligibility; it is not evidence that the unselected branches require opposite exact errors.

3.3 Two-branch subtree-address experiment

The minimal two-sibling experiment used ten paired seeds (4100–4109), following an excluded numerical-check seed 4099. The check verified nondegenerate shared capture, exact-route capture one and analytic exact/BP agreement. The subsequent eight-stream factorial varied route budget.

For seed-specific unit vectors $\mathbf{u}_A$ and $\mathbf{u}_B$, binary label sign $y \in \{-1, 1\}$ and context $c \in \{0, 1\}$, the two input means were

$$\mathbb{E}[\mathbf{x}_A | y, c] = y [\mathbb{I}(c = 0) - 0.75\mathbb{I}(c = 1)]\mathbf{u}_A, \quad (72)$$

$$\mathbb{E}[\mathbf{x}_B | y, c] = y [\mathbb{I}(c = 1) - 0.75\mathbb{I}(c = 0)]\mathbf{u}_B. \quad (73)$$

Each stream had 12 coordinates and independent Gaussian noise with standard deviation 1.15. The 2,048-example train and test splits contained exactly 512 examples in each context–label cell. The somatic logit selected branch A when $c = 0$

and branch B when $c = 1$. Both inputs nevertheless remained active, preventing the one-coordinate rule from acquiring branch specificity through a zero eligibility on the distractor.

For somatic logit z and logistic sigmoid σ , let $\delta = \sigma(z) - \mathbb{I}(y = 1)$ and $\mathbf{r}(c) = (\mathbb{I}[c = 0], \mathbb{I}[c = 1])$. Exact and correct-subtree coefficients were $\delta \mathbf{r}(c)$; within-neuron derangement reversed the two entries, giving $\delta \mathbf{r}(c)_{::-1}$; the shared neuronal and same-depth controls used $\delta(1, 1)$; and the random rank-two control used $\delta \mathbf{r}(c) \mathbf{R}_s^T$ for a seed-fixed orthogonal rotation $\mathbf{R}_s$. The gated point emulation used one somatic coefficient together with the context-gated features $(\mathbb{I}[c = 0] \mathbf{x}_A, \mathbb{I}[c = 1] \mathbf{x}_B)$, making its alternative address resource explicit. All conditions had 24 trainable weights and the same paired initialization and batch order. The random dense control had four routing nonzeros; the sparse two-route conditions had two.

Training used mini-batch stochastic gradient descent (SGD), learning rate 0.3, batch size 64 and 60 epochs. The switch endpoint applied another 12 epochs to 1,024 context-1 examples with balanced labels and measured the change in context-0 accuracy. For gradient geometry, exact and candidate per-example gradients were concatenated over both branches. Scaled capture was squared cosine after optimal scalar rescaling. The one-step candidate was normalized to the exact batch-gradient norm and evaluated at step 0.25. The Source Data indexed in Table S12 retain every tested condition; no condition was removed on the basis of its outcome.

All controls, including poor deranged and variable random-map outcomes, are retained. Correct routing, exact transport, BP and gated-point emulation coincide because the exact path coefficient is the binary selector. Thus this experiment isolates the supplied address resource; the gated-point control reproduces it.

3.4 Subtree-address bandwidth and architecture factorial

To vary feedback bandwidth, the complete factorial extended the two-sibling task to eight context-selected streams arranged in a balanced binary hierarchy. Context and binary label were exactly balanced in train and test sets, every stream remained active on every example, and the sign of the task signal reversed outside the selected context. Thus a shared coefficient could not recover the required address simply because off-route eligibilities were zero. The prespecified feedback budgets were $K \in \{1, 2, 4, 8\}$ nested routes. Twenty paired analysis seeds (4200–4219) crossed five representations and 27 method–budget combinations, giving 135 fits per seed and 2,700 fits in total.

The dendritic tree, explicitly gated point model, flat-compartment model and grouped-point model were equivalent controls: each contained 64 trainable weights, 64 active input contacts and one output nonlinearity, and each received the same routed coefficient field. The fifth representation was a degree- and depth-matched rewired tree. Its resource counts were unchanged, but its ancestry groups were permuted relative to the task hierarchy. Feedback families comprised correct ancestry, fixed within-neuron derangement, depth-interleaved groups, random sparse groups, a seed-fixed random rank- K map, and a task-fitted rank- K upper bound. Neuron-shared, exact-transport and analytic-backpropagation references completed the design. Route fields were unit-normalized per example. Data, initialization and every stochastic control map were keyed to seed, method family and K , so matched representations saw identical stochastic realizations.

Pre-analysis checks required balanced labels and contexts, nonzero activity in every stream, shared capture below 0.2, exact capture one, analytic exact/BP equality and no fallback. An initial 2,700-fit cohort used stochastic maps unpaired across representation labels and was excluded. Only the complete repeat with paired maps and cross-representation equivalence checks enters inference.

The primary bandwidth contrasts are retained in the Source Data indexed in Table S12. Correct nested-subtree routing strongly exceeded route derangement at $K = 2$ and $K = 4$, but its comparison with the best matched non-anatomical control was deliberately more stringent: ancestry lost at $K = 1$ and $K = 2$, won by 0.0127 accuracy at $K = 4$ (95% paired seed-bootstrap interval 0.00593–0.01946; 15/20 seeds; paired Wilcoxon signed-rank Holm-adjusted $P = 0.0101$ across the four budgets; the Source Data retain the paired mean-sign-flip sensitivity test), and tied at full rank. Correct routing on the task-matched tree exceeded the rewired tree at $K = 2$ and $K = 4$ in all 20 seeds. The four matched representations were exactly equal in every condition (maximum held-out-accuracy difference zero). The experiment therefore supports an intermediate-bandwidth advantage for a task-aligned nested address, while showing that an explicitly routed point implementation can reproduce it.

3.5 Trained partition-residual reconstruction

Using the saved configurations and outputs, we quantified irreducible address error in a post hoc diagnostic of the complete factorial. For example b , let c_b denote context, δ_b the somatic error, $e_{c_b,i}$ the indicator selecting leaf i , and $\mathbf{x}_{bi}$ its input vector. We defined the exact coefficient field $c_{bi} = \delta_b e_{c_b,i}$, coordinate weight $w_{bi} = \|\mathbf{x}_{bi}\|_2^2$, and active route field

785 ϕ_{bi} . Allowing the best single coefficient for that route on each example gives

$$z_b^* = \frac{\sum_i w_{bi} \phi_{bi} c_{bi}}{\sum_i w_{bi} \phi_{bi}^2}, \quad \mathcal{E}_{\text{part}} = 1 - \frac{\sum_{b,i} w_{bi} (c_{bi} - z_b^* \phi_{bi})^2}{\sum_{b,i} w_{bi} c_{bi}^2}. \quad (74)$$

786 This is the normalized complement of the shared-coefficient residual in Eq. 53. It grants an optimal route coefficient and
787 therefore isolates address/support error from the coefficient scale actually used during training.

788 All 20 seeds and 135 conditions per seed were reconstructed at initialization and after 80 training epochs. Recon-
789 structed losses agreed with the recorded values within 4.97×10^{-9} and accuracies within 5.01×10^{-11} , the precision of
790 the saved decimal tables; the address-plus-coefficient Pythagorean decomposition closed within 7.28×10^{-12} . On the
791 dendritic-tree representation, trained correct-ancestry capture was 0.162, 0.315, 0.547 and 1.000 at $K = 1, 2, 4, 8$. Fixed
792 coordinate-to-arbor derangement had the common scalar value at $K = 1$ and zero capture at $K = 2, 4, 8$, because its active
793 support excluded the target leaf (Supplementary Fig. S11A).

794 For each seed we averaged implementation-equivalent representations within feedback family and budget, then
795 correlated trained capture with held-out accuracy across conditions. The mean within-seed Spearman correlation was
796 0.824 (95% seed-bootstrap interval 0.810–0.838; 20/20 positive; two-sided Wilcoxon $P = 1.91 \times 10^{-6}$). Restricting the
797 analysis to the six route families and $K < 8$ removed the exact full-rank boundary and gave 0.639 (0.610–0.665; 20/20
798 positive; $P = 1.91 \times 10^{-6}$; Source Data, Table S12). This association is descriptive and post hoc. In particular, ancestry,
799 depth-interleaved and random sparse partitions had similar address capture at matched K , so the residual does not explain
800 the small trained ancestry advantage at $K = 4$.

801 3.6 Learning coefficients from an explicit local context cue

802 To test whether coefficients could be learned from a supplied cue, we used a four-route dictionary on fresh versions of
803 the balanced eight-context task. We retained its 64 forward weights, paired initialization, 2,048-example train and test
804 sets, and 80 full-batch task-training updates. The dictionary was fixed before task training: $\Phi_{bk} = \mathbf{1}[k = \lfloor b/2 \rfloor] / \sqrt{2}$
805 for zero-indexed blocks $b = 0, \dots, 7$ and routes $k = 0, \dots, 3$. For trial i with context c_i , the supplied local cue was
806 $\mathbf{h}_i = \mathbf{e}_{c_i} + \sigma_h \boldsymbol{\varepsilon}_i$, with $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(0, I_8)$. Here $\mathbf{e}_{c_i}$ is the one-hot context vector and σ_h is cue-noise SD. The cue is an
807 additional information resource; the experiment does not infer a biological signal.

808 Separate balanced calibration trials supplied the target route-activation vector $\mathbf{e}_{\lfloor c_i/2 \rfloor}$. A four-output softmax encoder
809 with an intercept learned this mapping by minibatch delta-rule updates. Weights started at zero; calibration used 30
810 epochs, batch size 16, step 0.2 and weight decay 0.001 on non-bias weights. Calibration sets of 16, 64 and 256 trials
811 were nested within each cue-noise condition. Because epoch count was fixed, increasing sample count also increased the
812 number of encoder updates. This axis therefore changes both calibration data and computation. No task labels, exact task
813 gradients or held-out task outcomes entered encoder training. Its parameters were frozen before the classification weights
814 were trained.

815 At cue delays $d = 0, 1, 4$ trials, the encoder received the cue array circularly shifted by d . Predicted coefficients $\mathbf{c}_i$
816 produced the normalized field $\Phi \mathbf{c}_i / \|\Phi \mathbf{c}_i\|$. Delay is a mismatch between the current eligibility and an earlier independent
817 context; it is not a calibrated physiological time constant. Controls used exact one-hot route selection, a uniform frozen
818 profile, or a cyclic permutation of the learned coefficient coordinates. Exact and frozen controls repeated across nuisance
819 conditions are algebraic repetitions rather than independent evidence. Task updates used the original factor-eight context
820 rescaling and step 0.3.

821 Twenty analysis seeds (52000–52019) followed the excluded implementation-check seed 51999. Crossing three
822 calibration sizes, cue-noise SDs 0, 0.5, 1, delays and four rules produced 2,160 task fits and 10,800 checkpoint rows.
823 At task epochs 0, 1, 10, 40 and 80, we recorded held-out accuracy/loss and delivered-versus- exact gradient cosine
824 at each method's own state. Held-out cue classification, coefficient mean squared error (MSE) and route-norm errors
825 supplied separate encoder diagnostics. The protocol, fixed before new outcomes but not publicly preregistered, is
826 `configs/review_completion/coefficient_encoder.json`.

827 The primary condition used 256 calibration trials, cue-noise SD 0.5 and no delay. Learned coefficients improved
828 accuracy over the frozen profile by 7.16 percentage points (95% paired-seed bootstrap interval 6.26–8.07; 20/20 positive)
829 and over mismatched coefficients by 7.10 points (6.19–7.99; 20/20). They remained 54.47 points below exact route
830 selection (53.54–55.45; 20/20 lower). Two-sided Wilcoxon tests restored the exact 1/2048 accuracy-count grid before
831 ranking and used Holm correction across these three primary contrasts. Adjusted P values were 5.72×10^{-6} versus the
832 frozen rule and 1.77×10^{-4} versus each other control. Intervals used 20,000 paired-seed draws; other grid contrasts are

descriptive sensitivities. All fits were finite. Learning from the supplied cue improved performance relative to fixed or misassigned coefficients, but left a large gap to the oracle. The experiment does not establish autonomous discovery of credit coordinates in a biological neuron. The complete outputs are in `source_data/review_coefficient_encoder/`.

An exploratory follow-up was specified after observing the soft-encoder outcomes. The same trained encoder, original paired seeds and task data were retained, but its maximum-probability route was delivered as a one-hot vector. No encoder parameter was refitted and no hyperparameter was selected from the task outcomes. This hard readout is a paired sensitivity, not a new independent replication or a replacement for the primary soft rule (Supplementary Fig. S11). At the primary noise SD 0.5, mean accuracy was 25.825% for soft coefficients, 64.031% for hard selection, 80.291% for exact selection and 18.669% for the frozen profile. Hard-minus-oracle accuracy remained -16.26 percentage points (95% paired-seed interval -17.83 to -14.83), while hard-minus- frozen was 45.36 points (43.69–46.92). At zero noise and 256 calibration trials, hard selection equaled the oracle numerically, whereas the soft rule reached 79.978%. With only 16 noise-free calibration trials, the soft encoder already identified every context’s route correctly, yet its graded output gave 19.392% task accuracy; selecting its maximum-probability route gave the oracle’s 80.291%. This sensitivity isolates leakage into incorrect routes from failure to identify the correct route. All hard-readout intervals and unadjusted tests are descriptive; these reused conditions do not define a new primary family. Both learned readouts failed when cues were delayed across independent contexts. Thus accurate route identification does not guarantee that graded coefficients are suitable for the learning rule: readout and timing are separate constraints. All exploratory fits and unchanged encoder settings are in `source_data/review_coefficient_hard_readout/`.

3.7 Route overlap and task interference

We finally examined how overlap between addressed regions and leakage outside an intended route determine interference with an already learned task. Branch-specific plasticity during motor learning suggests that different tasks can recruit different dendritic regions [4]. Let $\mathbf{w}_A$ be an optimum of a quadratic Task-A objective with Hessian H_A , and let an approximate Task-B teaching field produce update $-\eta M_B \mathbf{g}_B$, where η is the step size, $\mathbf{g}_B$ is the Task-B gradient and M_B is its routing operator. Expansion about the Task-A optimum removes the first-order term, leaving the exact loss increase

$$\mathcal{L}_A(\mathbf{w}_A - \eta M_B \mathbf{g}_B) - \mathcal{L}_A(\mathbf{w}_A) = \frac{\eta^2}{2} \mathbf{g}_B^\top M_B^\top H_A M_B \mathbf{g}_B. \quad (75)$$

In the equal-route example, opposing updates on the shared fraction ω have magnitude two, whereas leakage onto the Task-A-only fraction has magnitude λ . Dividing by route size gives $\eta^2[4\omega + \lambda^2(1 - \omega)]/2$. Numerical vector updates matched this identity to 4.2×10^{-17} absolute error (Eq. 75). Separated routes therefore protect a learned task only when both anatomical overlap and teaching-field leakage are small. The published somatostatin (SST) perturbation is consistent with the resulting prediction that loss of branch-selective inhibition increases effective leakage and cross-task interference [4], but it does not uniquely identify the credit-routing mechanism. This static identity does not model dendritic calcium, spikes, eligibility timing or the SST intervention.

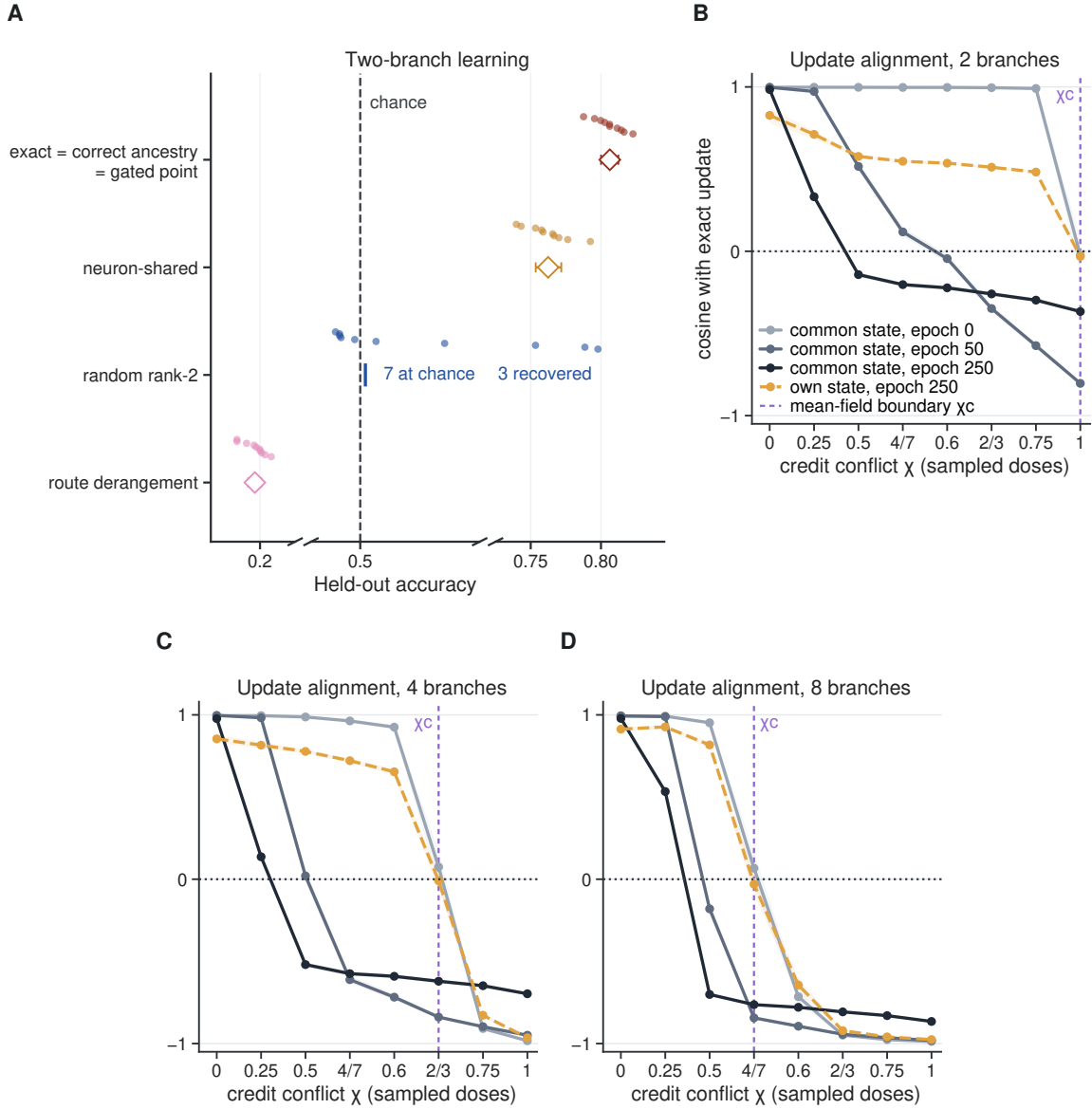


Figure S10: Branch-selective credit limits interference and preserves update direction. **A**, Two-branch learning in ten paired seeds: held-out accuracy on the balanced binary task, with chance at 0.5 (dashed rule) and the axis broken over the two spans that hold no seed. Points show seeds; open diamonds and bars are means and 95% paired-seed bootstrap intervals, which are narrower than the diamond where no bar is visible. The exact-path, correct-ancestry and gated-point rules (and backpropagation) give bitwise-identical accuracy in every seed and share one row. The random rank-2 control is bimodal across seeds (seven at chance, three recovered), so its row shows the median as a vertical bar and the two counts instead of a mean. The context-0 forgetting of the same cohort is drawn at the main-text type scale in main Fig. 2E. **B–D**, Shared-versus-exact gradient cosine in the separate branch-conflict experiment with two (**B**), four (**C**) or eight (**D**) branches, drawn against the eight sampled conflict doses at equal spacing (tick labels give χ ; the odd doses $4/7$ and $2/3$ bracket the boundaries). Solid curves evaluate common exact-learning states at epochs 0, 50 and 250; the dashed amber curve evaluates the shared learner’s own final state. The violet dashed rule is the mean-field boundary $\chi_c = B/[2(B - 1)]$, at 1, $2/3$ and $4/7$. Curves use twenty paired seeds; the 95% bootstrap bands are narrower than the line stroke except near the boundary at epoch 0. Per-seed zero-alignment doses are tabulated in Source Data. An exact quadratic interference calculation is given in the accompanying derivation. Learned-state geometry can depart from the initialization mean-field approximation.

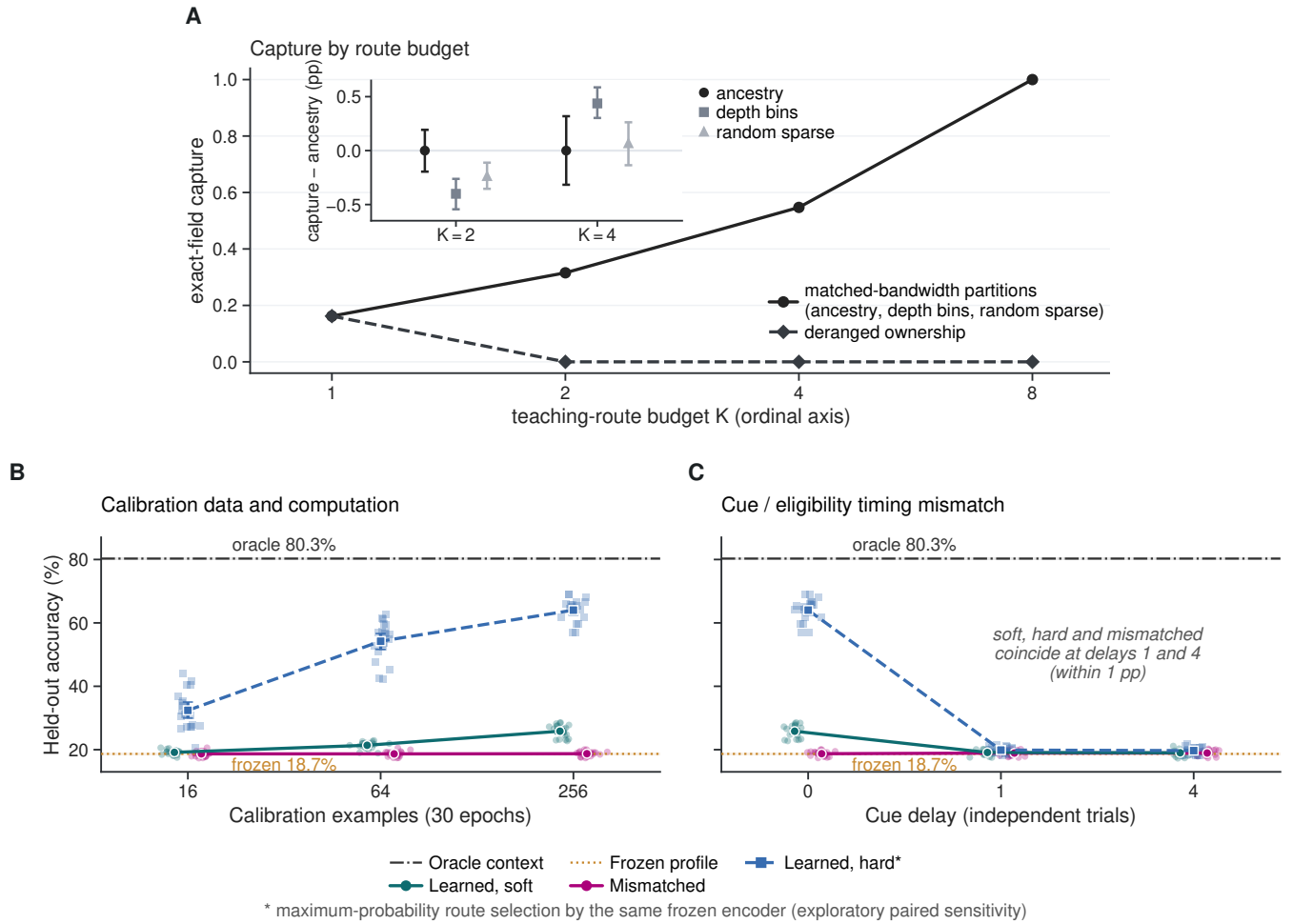


Figure S11: Available route span and learned route coefficients are different constraints. **A**, Trained field capture by route budget in the eight-context factorial, with K on an ordinal axis. The ancestry, depth-bin and random-sparse partitions have coincident capture at matched bandwidth (within 0.005 at every budget; one stroke, drawn at the ancestry values), and derangement removes the target coordinate. The inset resolves the three partitions at $K = 2$ and $K = 4$ as capture minus the ancestry mean, in percentage points of capture, with each family's own 95% interval; $n = 20$ seeds throughout, and on the main axes the intervals are narrower than the symbols. Capture therefore does not explain the small ancestry-specific advantage, and capture alone does not predict trained utility; the within-seed Spearman correlation between capture and utility is about 0.82 over all budgets and 0.64 for $K < 8$, and is retained numerically in Source Data. **B**, Held-out accuracy versus calibration set size (16, 64 and 256 examples, ordinal axis) at cue-noise standard deviation 0.5 and zero delay; 30 epochs are fixed, so both data and computation increase. **C**, Cue delay (0, 1 and 4 trials, ordinal axis, sharing the accuracy axis of **B**) at 256 calibration trials and noise SD 0.5; delays mismatch independent contexts and are measured in trials, not physiological time. In **B,C** the oracle-context and frozen-profile controls do not depend on the panel factor and are drawn as labelled reference lines (dash-dot and dotted) with their 95% seed intervals as bands; the mismatched-encoder control coincides with the frozen profile in **B**, and the soft, hard and mismatched conditions coincide within one percentage point at delays 1 and 4 in **C**, where the three series are offset slightly in x . The dashed, square-marked curves marked with an asterisk use maximum-probability route selection by the same frozen encoder, an exploratory paired sensitivity specified after the soft outcomes with no encoder refit. Small symbols are the twenty fresh analysis seeds; curves, circles, squares and bars are means and 95% paired-seed bootstrap intervals, and a bar narrower than its symbol is hidden by it. The cue-noise sensitivity of the same encoder is drawn at the main-text type scale in main Fig. 3F. The calibration cue and activation targets are supplied resources. No serial forward dendritic computation is present.

S4 Interaction structure, compatible trees and spatial credit

4.1 Interaction structure at fixed input-sensitivity rank

A tree constrains representation only if its units have restricted input access. A dense trainable leaf-input map can compensate for generic route rotations: if $W = \Phi V$, with orthonormal K -column routes Φ , and $Q_r^T H_T \Phi$ has row rank r , a suitable V reproduces every noiseless context in the linear task of Section 10.1. The algebraic study below instead fixes each input to one named leaf. Its exhaustive capacity analysis was formulated after the prospective-selector outcomes and is a population diagnostic.

To impose local input access, we placed eight independent Rademacher inputs (each equally likely to be -1 or $+1$) at named leaves of a binary tree. For internal node v , let u_L and u_R be its child outputs and let a_v, b_v, c_v, d_v be its four trainable coefficients. Each of the seven internal nodes computes

$$u_v = a_v + b_v u_L + c_v u_R + d_v u_L u_R. \quad (76)$$

Only the root is read out. Every candidate has 28 free coefficients and 14 edges, with no dense input mixing, intermediate readout or access to inputs outside a node's subtree. The fixed pool combines three shapes—a balanced tree, a comb and a tree with a $3+5$ root split—with the same four predetermined leaf permutations, giving twelve candidates. Equal edge and coefficient counts do not imply equal cable length, latency or biological energy cost. These signed algebraic nodes are not conductance compartments.

The primary target census contains all 105 perfect matchings M and all 35 unordered four-plus-four partitions A of the eight inputs:

$$f_M(\mathbf{x}) = \frac{1}{2} \sum_{(i,j) \in M} x_i x_j, \quad f_A(\mathbf{x}) = \frac{1}{2} \left(\prod_{i \in A} x_i + \prod_{i \notin A} x_i \right). \quad (77)$$

The targets are enumerated independently of the candidate trees. Evaluating all $2^8 = 256$ equally weighted inputs gives target variances 1 and $1/2$, respectively. For the continuous derivative of each target's unique multilinear extension,

$$\mathbb{E}[\nabla_{\mathbf{x}} f \nabla_{\mathbf{x}} f^T] = I_8/4. \quad (78)$$

Thus both algebraic and participation-ratio effective rank are eight, and the complete input-sensitivity spectrum agrees. This is a property of the target, not equality of changing model parameter-gradient or synaptic-credit moments. Normalized MSE (NMSE) divides population squared error by the stated target variance.

Interaction-tensor descriptions have previously been used to analyze structured neural function classes [5]. The centered constant-feature criterion below is derived for the specific local model in Eq. 76.

4.2 A necessary and sufficient centered-cut condition

For each subset U of input indices, define the Walsh function $\chi_U(\mathbf{x}) = \prod_{i \in U} x_i$, with $\chi_\emptyset = 1$. These functions form an orthonormal basis under the uniform input distribution. Writing c_U for the coefficient of χ_U gives $f = \sum_U c_U \chi_U$. If $\mathbf{c}_f$ and $\mathbf{c}_g$ collect the coefficients of two functions f and g , Parseval's identity gives $\mathbb{E}[(f - g)^2] = \|\mathbf{c}_f - \mathbf{c}_g\|^2$. For the leaf set S of an actual subtree, reshape the target coefficients into C_S , with row index a subset of S and column index a subset of its complement $\bar{S}$. Let C_S° denote this matrix after removing the row for the empty subset of S .

Proposition 8 (Exact representation by a scalar multi-affine tree). *For a fixed binary tree with one named Rademacher input at each leaf, a target is represented exactly by Eq. 76 and a root-only output if and only if $\text{rank}(C_S^\circ) \leq 1$ at every proper subtree cut.*

Proof. Every ancestor is affine in a particular child's output, and sibling inputs are disjoint. Consequently a represented function can be written $g = A(\mathbf{x}_{\bar{S}}) + h_S(\mathbf{x}_S)B(\mathbf{x}_{\bar{S}})$. Its full cut matrix has rank at most two; removing the empty- S row leaves rank at most one. This proves necessity.

For sufficiency, call a cut active when $C_S^\circ \neq 0$. At each active cut, choose a centered function h_S whose Fourier coefficients span the column space of C_S° , and define $V_S = \text{span}\{1, h_S\}$. If $C_S^\circ = 0$, take $h_S = 0$ and $V_S = \text{span}\{1\}$. At a leaf use $V_i = \text{span}\{1, x_i\}$. The condition implies $f \in V_S \otimes \mathcal{H}_{\bar{S}}$, where $\mathcal{H}$ is the complete finite-dimensional function space on the indicated inputs.

For sibling leaf sets L and R , the orthogonal projections onto V_L and V_R commute because they act on disjoint input factors. Hence $f \in V_L \otimes V_R \otimes \mathcal{H}_{\overline{L \cup R}}$. For a non-root parent $P = L \cup R$ with nonzero centered component, write

906 $f = 1_P \otimes A + h_P \otimes B$, with $B \neq 0$. Contracting the outside factor against a linear functional that maps B to one gives
 907 $h_P + c1_P \in V_L \otimes V_R$, where c is a scalar contributed by A . Since the constant function also belongs to this product space,
 908 so does h_P . It is therefore a linear combination of $1, h_L, h_R, h_L h_R$, exactly the four allowed features. Inactive parent
 909 features are zero and cause no exception. At the root, the complete target itself belongs to the product of its children's
 910 spaces. Induction supplies all local coefficients and proves sufficiency. $\square$

911 Retaining the constant feature in every subtree makes this criterion stronger than an uncentered rank-two constraint.
 912 Let $\mathcal{T}_{\text{proper}}$ be the leaf sets of the tree's proper subtrees, and let $\sigma_j(C_S^\circ)$ be the singular values in decreasing order. The
 913 optimal low-rank approximation identity and Parseval then give

$$\text{MSE}(f, g) \geq \max_{S \in \mathcal{T}_{\text{proper}}} \sum_{j \geq 2} \sigma_j(C_S^\circ)^2. \quad (79)$$

914 Taking the maximum preserves a lower bound. Summing tails from overlapping cuts instead gives a heuristic score,
 915 without the same guarantee. A positive maximum-cut bound need not equal the true optimum. The condition applies to
 916 actual subtree cuts; the complement of a subtree need not itself be a subtree. For the matching (12)(34)(56)(78), an
 917 even-odd root split has centered tail 0.75, whereas the full rank-two tail is 0.5. For the two-quartet target with supports
 918 1234 and 5678, a five-leaf subtree 12345 has centered tail 0.25 despite a zero full rank-two tail. Supplementary Fig. S12
 919 compares both bounds with fitted population error.

920 The proof requires ancestors to be affine in each child's output. A general nonlinear scalar ancestor can generate
 921 higher cut rank from one scalar feature. Neither the necessity condition nor Eq. 79 therefore applies automatically to a
 922 tanh tree or a conductance model.

923 4.3 Population fitting, complete-tree construction and depth

924 All twelve candidates were fitted to every primary target by 32 alternating least-squares (ALS) sweeps with four fixed
 925 random restarts. Conditional on the other nodes, the design for one node is $q(\mathbf{x})[1, u_L, u_R, u_L u_R]$, where q is the derivative
 926 of the root with respect to that node's output. The offset is the current root output minus qu_v . Sweeps alternate postorder
 927 and preorder. Hidden mean/scale gauges are normalized with exact parent compensation. Every restart and checkpoints
 928 at 0, 2, 8 and 32 sweeps are retained: 6,720 fits and 26,880 checkpoint rows. The endpoint is the smallest achieved NMSE
 929 among the four restarts. The difference between this endpoint and Eq. 79 can reflect both incomplete optimization and a
 930 loose bound, so it is not a certified optimization gap. Full-domain fitting has no held-out generalization interpretation.

931 For score-based selection among the twelve fixed candidates, values within 10^{-10} are treated as ties and resolved by
 932 lexicographic candidate identifier. A separately retained summary applies this tolerance to the raw floating-point minima
 933 without changing any fit. The two-sweep pilot fits every candidate; the best-fixed baseline chooses a single candidate
 934 per family in hindsight. The uniform-random baseline is its exact candidate expectation. Family averages and paired
 935 shape/assignment counts are descriptive censuses, with error comparisons at tolerance 10^{-7} and no sampling confidence
 936 intervals over the enumerated tasks. The lower bound is attained within this tolerance in 1,407/1,680 endpoints. Although
 937 some restarts develop large cancelling coefficients, every case selected by a deterministic policy has an existing equivalent
 938 restart within 10^{-7} NMSE whose coefficient norm is at most 2.645752. All original fits, including the remaining poorly
 939 conditioned case, remain included.

940 To search beyond the fixed pool, we used an exploratory dynamic program over all input subsets. Let $c(S)$ be the
 941 normalized cut tail from Eq. 79, with zero cost at the root and leaves. Let $b(S)$ be the smallest achievable maximum cut
 942 cost for a subtree on leaf set S , with base case $b(\{i\}) = 0$. Then

$$b(S) = \max \left\{ c(S), \min_{L \cup R = S} \max[b(L), b(R)] \right\}. \quad (80)$$

943 Every binary tree decomposes into one root split and two independent children, so this recurrence covers all trees. A
 944 second pass restricts every cut to the globally minimal threshold, then minimizes total tail, depth and deterministic split
 945 ties. Costs are rounded to 12 decimals for numerical ties. This corrected two-pass version is retained separately from
 946 the first exploratory construction and agrees with exhaustive enumeration of all 945 six-input trees on four random cost
 947 tables. At a zero threshold, the theorem makes the minimum-depth solution the shallowest exact tree in this class; for a
 948 positive threshold only the cut score, not fitted error, is optimized.

Dominant left singular vectors of the centered cut matrices define local scalar features, and four-feature least squares recovers each node’s coefficients. All 105 matching and 35 quartic targets reconstruct at minimum depth three. A separate positive control uses 24 unique seeded permutations π of the nested target

$$f_{\pi}(\mathbf{x}) = \frac{1}{2} \sum_{k \in \{2,4,6,8\}} \prod_{j=1}^k x_{\pi(j)}. \quad (81)$$

Its rank is eight but its spectrum $(1/4, 1/4, 1/2, 1/2, 3/4, 3/4, 1, 1)$ differs from the primary families; it is constant only within this control family. Every nested target requires depth four. An independent depth-constrained recurrence gives minimum centered-cut NMSE bound 0.146446609407 at depth at most three and zero at depth four. All 164 constructions have NMSE below 7×10^{-30} and maximum absolute coefficient one up to rounding (Supplementary Fig. S12). The algorithm uses the complete target Fourier tensor: these are exact-target capacity certificates, not predictions from limited calibration data.

4.4 Boolean interactions, scalar tree structure and conditional credit

We used four-input Boolean functions to make the structural and credit questions explicit. The complete domain comprises sixteen equally weighted patterns. Writing $b_i = (x_i + 1)/2$ for Rademacher inputs $x_i \in \{-1, 1\}$, the raw labels $y \in \{0, 1\}$ are defined below. The symbols $\wedge$, $\vee$ and $\oplus$ denote AND, OR and exclusive OR (XOR), respectively:

$$y_{\text{AND}} = b_1 \wedge b_2 \wedge b_3 \wedge b_4, \quad y_{\text{OR}} = b_1 \vee b_2 \vee b_3 \vee b_4, \quad (82)$$

$$y_{\text{parity}} = b_1 \oplus b_2 \oplus b_3 \oplus b_4, \quad (83)$$

$$y_{\text{OR(AND)}} = (b_1 \wedge b_2) \vee (b_3 \wedge b_4), \quad (84)$$

$$y_{\text{XOR(AND)}} = (b_1 \wedge b_2) \oplus (b_3 \wedge b_4), \quad (85)$$

$$y_{\text{AND(XOR)}} = (b_1 \oplus b_2) \wedge (b_3 \oplus b_4), \quad (86)$$

$$y_{\text{nested}} = b_1 \wedge [b_2 \vee (b_3 \wedge b_4)]. \quad (87)$$

These are specified mechanistic examples rather than a representative sample of natural tasks. The associative AND, OR and parity families control for claims that greater depth or a particular input grouping is inherently necessary. Task permutations preserve a family’s truth-table distribution; comparisons between families need not preserve its input-gradient spectrum.

Every candidate has three scalar internal units, twelve trainable coefficients and six edges. For child outputs ℓ, r and trainable coefficients a, b, c, d , each unit computes $u = a + b\ell + cr + d\ell r$. Only the root supplies the output. All local coefficients can change during learning. There is no additional dense decoder or access to inputs outside a unit’s descendant subtree. This is the multi-affine algebraic class of Proposition 8, not a positive-conductance neuron. An independent enumeration of all fifteen unordered, rooted, full binary trees on four labeled inputs supplies the structural analysis: three balanced trees have depth two, and twelve comb trees have depth three. An actual subtree cut is evaluated by removing the Fourier row for the constant function inside that subtree and retaining the singular-value tail beyond rank one. The maximum cut tail is a lower bound on regression MSE for the entire tree; zero tails on every actual cut are necessary and sufficient for exact representation in this class. These capacity statements do not by themselves establish convergence of a particular optimizer or attainability under an additional parameter box.

We distinguish exact truth-value regression from threshold classification. Let $\mu_y = \mathbb{E}[y]$ and $\sigma_y = \sqrt{\text{Var}(y)}$ be the mean and standard deviation over all sixteen patterns. Training targets are $t = (y - \mu_y)/\sigma_y$. Mean squared error on t therefore equals raw-output MSE divided by $\text{Var}(y)$. A predicted normalized output $\hat{t}$ corresponds to the raw Boolean decision $\hat{t} \geq (0.5 - \mu_y)/\sigma_y$. High classification accuracy does not imply small regression error, and a positive regression lower bound does not establish a classification impossibility. AND and OR are imbalanced: a constant majority prediction has accuracy 15/16. Balanced accuracy averages the true-positive rate (TPR) and true-negative rate (TNR), $(\text{TPR} + \text{TNR})/2$, and distinguishes this trivial result from accurate decisions on both classes. All sixteen truth patterns contribute to each reported population endpoint; they are not independent biological replicates.

Proposition 9 (An explicit Boolean grouping obstruction). *Let $f = (b_1 \wedge b_2) \text{ XOR } (b_3 \wedge b_4)$ on uniform independent bits, and let $z = (f - 3/8)/\sqrt{15/64}$. A scalar multi-affine balanced tree grouped as $(b_1, b_3) \mid (b_2, b_4)$ or $(b_1, b_4) \mid (b_2, b_3)$ has population NMSE at least 8/15 for every choice of its coefficients. The aligned grouping $(b_1, b_2) \mid (b_3, b_4)$ represents z exactly with coefficients in $[-2, 2]$.*

985 *Proof.* For the crossed cut $b_1b_3 \mid b_2b_4$, the raw-target centered Walsh matrix, with nonconstant inside rows ordered as
 986 x_1, x_3, x_1x_3 , is

$$C = \frac{1}{8} \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ -1 & -1 & -1 & -1 \end{pmatrix}.$$

987 The columns correspond to $1, x_2, x_4, x_2x_4$. Its three orthogonal rows each have squared norm $1/16$, so the rank-one tail is
 988 exactly $1/8$. Dividing by $\text{Var}(f) = 15/64$ gives $8/15$; the other crossed pairing is symmetric. For the aligned tree, set

$$u = 2b_1b_2 - 1 = \frac{-1 + x_1 + x_2 + x_1x_2}{2}, \quad v = 2b_3b_4 - 1 = \frac{-1 + x_3 + x_4 + x_3x_4}{2}, \quad z = \frac{1 - 4uv}{\sqrt{15}}.$$

989 The largest absolute coefficient is $4/\sqrt{15} < 2$. □

990 Exact rational Walsh coefficients were checked for all 105 family–tree pairs. All 49 zero-bound cases were
 991 reconstructed by signed Boolean slices and four-corner interpolation: AND, OR and parity admit all fifteen trees; each
 992 mixed-gate target admits one. Every normalized construction lies within $[-2, 2]$, with maximum absolute coefficient
 993 $4/\sqrt{15}$. This verifies existence within the experimental box, independently of optimization.

994 The OR-of-AND target and the nested target each use two AND gates and one OR gate, yet their minimum
 995 exact depths are two and three. For the nested function, the smallest lower bound among all three balanced trees is
 996 $(22 - 2\sqrt{57})/55 \simeq 0.12546$ NMSE; the stated comb tree is exact. All seven functions have input-gradient rank four, but
 997 their full spectra generally differ. This is a rank-versus-interaction example, not an isospectral comparison between the
 998 mixed-gate families. The maximum over cuts remains a lower bound; we do not assume every incompatible whole tree
 999 attains it.

Gate coordinates illustrate conditional derivatives. The elementary gates can also illustrate how another branch affects a local derivative. In raw Boolean branch-output coordinates $u, v \in \{0, 1\}$, their multilinear extensions are

$$F_{\text{AND}}(u, v) = uv, \quad \partial_u F_{\text{AND}} = v, \quad (88)$$

$$F_{\text{OR}}(u, v) = u + v - uv, \quad \partial_u F_{\text{OR}} = 1 - v, \quad (89)$$

$$F_{\text{XOR}}(u, v) = u + v - 2uv, \quad \partial_u F_{\text{XOR}} = 1 - 2v. \quad (90)$$

1000 The other branch thus gates a derivative’s magnitude or sign in these canonical coordinates. The XOR-of-AND diagram
 1001 illustrates this dependence and a specific input-grouping constraint. Its gate labels are explanatory; the learned units
 1002 are not constrained to remain Boolean gates or to preserve that representation. A broadcast learner can change internal
 1003 coordinates, so the derivative diagram is not a proof that broadcast cannot learn the task. Positive affine normalization of
 1004 the raw target changes derivative scale but not these signs. Actual learned update fields and finite-training outcomes
 1005 require the separate experimental comparisons below.

1006 **Frozen learning comparisons.** Four candidate trees were tested: the three balanced pairings and the comb $b_1 \mid [b_2 \mid$
 1007 $(b_3, b_4)]$. These are four of the fifteen trees in the independent capacity analysis. Each seed applied one common physical
 1008 input permutation to both the semantic task variables and candidate leaves. Per-family streams supplied initialization, a
 1009 256-observation training cache, 1,024 independent test observations and minibatch indices. All paired tree, optimizer,
 1010 credit and rate conditions shared these streams within family. Data were drawn uniformly with replacement from the
 1011 sixteen patterns; training and test labels received independent Gaussian noise of SD 0.15 in normalized-target units.
 1012 Independent draws could repeat the same input pattern.

1013 Every coefficient began from an independent mean-zero Gaussian with SD 0.5, except zero biases; no target gates,
 1014 coefficients or exact-construction weights were used. Initial weights and every subsequent update were clipped to $[-2, 2]$.
 1015 Training used 2,048 updates, minibatches of 32, and gradient norm clipping at ten. Exact credit multiplied each local
 1016 eligibility by the current complete path derivative. Broadcast set the two nonroot internal sensitivities to one; root
 1017 sensitivity remained one under both rules. All twelve coefficients were trainable. Stochastic gradient descent (SGD) and
 1018 Adam were tested separately; Adam used $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\varepsilon = 10^{-8}$.

1019 Five development seeds (2026110–2026114) selected a rate separately for each optimizer and rule from 0.003, 0.01
 1020 and 0.03, minimizing mean final clean population NMSE over all seven families and four trees. Ties within 10^{-12}
 1021 selected the smaller rate. Adam selected 0.003 for exact credit and 0.01 for broadcast; SGD selected 0.03 and 0.01,

respectively. These choices were recorded with cryptographic hashes before twenty new seeds (2026210–2026229), which retained all three rates. The resulting 6,720 fresh fits supplied 53,760 records at updates 0, 1, 16, 64, 256, 512, 1,024 and 2,048. An implementation check using seed 2026000 exercised all 336 conditions and was excluded from inference. These seeds vary initialization, permutation and sampled noisy data for seven fixed logical templates; they do not sample new Boolean function families.

The primary endpoint was clean population NMSE evaluated on all sixteen patterns. No sampled test variance or observed class-frequency denominator entered this endpoint. Noisy-test NMSE, raw threshold accuracy and balanced accuracy were secondary. Gradient cosines compare the delivered and exact aggregate clean-population gradients at each learner’s own trained state, before clipping or optimizer transformation; they are not mean per-example cosines or common-state comparisons between separately trained rules.

Two primary Adam contrasts concerned XOR-of-AND: the mean of the two crossed balanced-tree errors minus aligned-tree exact-credit error, and broadcast minus exact error on the aligned tree. The latter compares the prespecified development-selected recipes, not identical learning rates. Ten thousand resamples of twenty paired seed blocks supplied two-sided Bonferroni-adjusted 97.5% intervals for these two contrasts. A positive adjusted lower limit and mean improvement of at least 0.01 NMSE were both required. Other task, optimizer and common-rate contrasts have descriptive pointwise 95% intervals.

The grouping improvement was 0.56317 NMSE [0.54914, 0.58275], satisfying both primary criteria. The credit difference was 0.002507 [0.001780, 0.003318]: its adjusted interval excluded zero, but it did not meet the prespecified 0.01-NMSE mean-effect margin. Both recipes learned the aligned XOR-of-AND task well and classified all sixteen patterns correctly in every fresh seed. All three same-rate Adam credit differences were also below 0.003 NMSE. The low-rate SGD contrast had a wide interval and is retained. The larger parity credit differences are descriptive outcomes of a prespecified control task, not primary discoveries. Neither the gate illustration nor these observations establish an impossibility theorem for broadcast learning, or show that a particular change of internal coordinates caused its success. Full-rate outcomes, constant-mean and random-candidate references, clipping records, weights and source/selection hashes are retained under `source_data/boolean_morphology`.

Post hoc diagnostic of local target information. After examining the fresh learning outcomes, we asked how much target information was available within each proper input subset S . In this diagnostic, f denotes a raw Boolean target, $z = (f - \mathbb{E}[f]) / \sqrt{\text{Var}(f)}$ its normalized version, and $\hat{f}_U$ its Walsh coefficient on input subset U . The squared L^2 norm below averages over the uniform input distribution. We defined

$$P_S = \|\mathbb{E}[z \mid x_S]\|_{L^2}^2 = \frac{1}{\text{Var}(f)} \sum_{\emptyset \neq U \subseteq S} \hat{f}_U^2.$$

We cross-checked all ninety-eight task–subset values by exact rational evaluation of conditional means on the full truth table. Parity4 has $P_S = 0$ for every proper subset. XOR-of-ANDs instead has $P_S = 1/5$ for every two-input subset, including compatible and crossed pairs. Thus this diagnostic quantifies accessible lower-order target information rather than grouping compatibility.

Let F denote the model’s normalized prediction. At fixed parameters, a local eligibility $e_S(x_S)$ at a nonroot subtree is a function only of that subtree’s inputs. Conditional expectation gives $\mathbb{E}[ze_S] = \mathbb{E}[\mathbb{E}[z \mid x_S]e_S]$; when $P_S = 0$, the direct target contribution to the population broadcast update $\mathbb{E}[(F - z)e_S]$ vanishes. This identity does not imply an inability to learn: label-dependent changes in root coefficients can subsequently affect F , and finite training samples and stochastic updates need not exhibit exact population cancellation. The diagnostic was added after observing the learning results and did not change the training protocol or its predeclared criteria.

4.5 Learning compatible trees with exact and restricted credit

To separate credit assignment from uncertainty in structure estimation, this experiment supplied a tree compatible with the full target. Three development seeds and twenty fresh seeds generated independent signed and permuted matching, quartet and nested-interaction targets. The complete target Fourier tensor supplied a compatible tree through the preceding dynamic program; no target-derived coefficients initialize learning. An independent permutation of leaf labels preserved the exact shape, 28 coefficients and 14 edges. Chance compatible permutations were retained. The primary matching and quartet families retained identical input-sensitivity spectra, while the nested family had the previously stated anisotropic spectrum.

1069 All seven internal units initialized their non-bias coefficients independently from $\mathcal{N}(0, 0.5^2)$, with zero biases. The
1070 same initial coefficients, training cache and minibatch draws were used across paired assignments, credit rules and
1071 optimizers. Local features were $(1, u_L, u_R, u_L u_R)$ and the exact path derivative $q_v = \partial u_{\text{root}} / \partial u_v$ supplied credit. Exact
1072 gradients agreed with central finite differences for every coefficient in all three families. Root sensitivity was one under
1073 every rule. The six nonroot internal units received either the exact field, a unit root broadcast, the orthogonal projection
1074 onto a single constant profile, or the projection onto two proximal-subtree indicator profiles. A size-matched permutation
1075 of the latter profiles served as a control. All projected coefficients were computed from the current exact field and are
1076 therefore oracle coefficients; the experiment did not learn a circuit estimator for them.

1077 Training used 2,048 independently sampled Rademacher examples with output noise SD 0.15 and 1,024 minibatch
1078 updates of size 64. The gradient corresponded to half-MSE divided by the known clean target variance. Stochastic
1079 gradient descent (SGD) and Adam ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$) each tested learning rates 0.003, 0.01 and 0.03.
1080 Update gradients were clipped at norm 10 and parameters projected to $[-2, 2]$, which contains all exact constructions.
1081 Every clipping/projection count was recorded. These are constrained finite-horizon training results, not estimates of
1082 unrestricted optimization optima.

1083 A separate learning rate for each optimizer and rule minimized mean development population NMSE over all families
1084 and both assignments. A common rate per optimizer was also selected on development. All choices were recorded with
1085 cryptographic hashes before any fresh fit; all three rates remain in the released outcomes. Every SGD rule selected the
1086 upper tested rate, so the grid does not establish its best attainable optimizer setting. The independent test set contained
1087 4,096 examples with fresh SD 0.15 output noise. Test NMSE, normalized by the clean target variance, was primary;
1088 noiseless population NMSE on all 256 inputs was secondary. Evaluation at steps 0, 1, 16, 64, 256 and 1,024 retained
1089 21,600 checkpoint rows from 3,600 fresh fits. The endpoints at the development-selected rates contained 1,200 fits. No
1090 run was excluded. Ten thousand bootstrap samples of twenty whole paired seed blocks were drawn for pointwise 95%
1091 intervals, averaging families within seed for pooled contrasts. Multiple subgroup intervals are descriptive, not separately
1092 multiplicity-adjusted discoveries.

1093 Exact credit strongly improves learning of compatible quartet and nested targets under both frozen optimizer recipes.
1094 Broadcast is already effective for matching targets and outperforms exact credit there under SGD. Restricted two-subtree
1095 projections help in some Adam conditions but do not consistently beat shuffled profiles, and have essentially no pooled
1096 advantage over broadcast under SGD. For compatible trees trained with exact credit and Adam, mean test NMSE was
1097 0.0329; permuting the input labels increased it to 0.6080. At the common Adam rate 0.01, the compatible exact/broadcast
1098 averages remain 0.0329/0.5973. Supplementary Fig. S13 reports the full family, optimizer, profile, assignment and rate
1099 controls. Numerical outcomes and source/selection hashes are in `source_data/morphology_credit/`.

1100 4.6 Matched fixed-profile learning and credit-field geometry

1101 This experiment separates forward compatibility, spatial profile calibration and learning. This matched cohort retains the
1102 seven-unit model, 28 coefficients and compatible constructions above, specializing hierarchical tensor representations to
1103 scalar messages [6, 7]. Pairwise and quartic targets share a balanced tree and seed-specific leaf permutation. Independent
1104 coefficient signs have magnitude one-half; both input moments equal $I_8/4$, while target variances are one and one-half.
1105 The nested target uses its separate compatible tree and anisotropic spectrum.

1106 Let $q_n(x) = \partial u_0 / \partial u_n$ be the exact path derivative for unit n , and let $\delta(x)$ be the scalar derivative of normalized
1107 half-squared error with respect to root output. Local coefficient eligibilities are $1, u_L, u_R, u_L u_R$. Exact credit multiplies
1108 these by δq_n . Unit broadcast replaces q_n by one. An initially calibrated profile uses $a_n = 256^{-1} \sum_x q_n(x)$ over 256
1109 unlabeled initialization examples, then keeps a_n fixed. The sign control retains only $\text{sign}(a_n)$. Profiles require no target
1110 labels. Learning-rate choices are separate from the profile calculation. Fixed positive coordinate scaling largely cancels
1111 under Adam, so calibrated and sign-only Adam trajectories are expected to be nearly identical; they are not independent
1112 mechanistic confirmations.

1113 The protocol preceded three disjoint development seeds, with the same rate grid for each rule and optimizer.
1114 Separate rule-specific and common rates minimized final validation NMSE across families and were sealed before
1115 twenty paired fresh seeds. Matching/quartic conditions share initialization, calibration, examples, noise and minibatches.
1116 The Adam primary contrast is the quartic-minus-pairwise difference in initial-profile-minus-exact held-out error. All
1117 selected/common-rate unions are retained: 432 development, 1,200 fresh fits, and 48 excluded short checks.

1118 Algebraic train/validation/test sizes are 2,048/1,024/4,096; minibatches have 64 examples. Nonbias coefficients
1119 initialize independently with Gaussian SD 0.5 and biases zero. Both SGD and Adam test 0.003/0.01/0.03; Adam uses
1120 $(0.9, 0.999)$ and $\epsilon = 10^{-8}$. Fits retain norm clipping ten, box $[-2, 2]$, all contacts and steps 0, 1, 16, 64, 256, 512 and

1,024. Label-noise SD 0.15 gives NMSE floors 0.0225 for matching/nested and 0.045 for quartic; normalized half-MSE uses each family’s clean variance. All poorly fitted exact endpoints remain. Pointwise intervals use 10,000 whole-seed draws.

The separate positive-conductance comparison uses the compatible sixteen-parameter teacher of Section 5.3, not the algebraic quartic target. Data sizes are 1,024/1,024/4,096 without label noise, box $[-5, 5]$, SGD rates 0.001/0.003/0.01 and Adam rates 0.003/0.01/0.03.

For each saved state, let Q have one row per evaluation example and six columns containing q_n . For a dictionary A , energy capture is $\|QP_A\|_F^2/\|Q\|_F^2$, where P_A is its orthogonal projector. The best fixed rank-one profile is the leading right singular vector of Q , with capture $\sigma_1(Q)^2/\sum_j \sigma_j(Q)^2$. Its profile and per-example amplitudes are oracle fits. The uncentered second moment is $C = Q^T Q/N$, and participation-ratio effective rank is $(\text{tr} C)^2/\text{tr}(C^2)$. We additionally evaluated unit broadcast, the fixed initial profile, ancestry spans, full-rank capture, and loss-error-weighted fields $\text{diag}(\delta)Q$. The latter has its own optimal profile; applying the path-optimal profile after error weighting answers a different question.

Eligibility-weighted reconstruction multiplies exact and projected path fields by their local coefficient features before comparing parameter updates. Site-normalized spectra divide each path coordinate by its root-mean-square magnitude, exposing sensitivity to fixed coordinate scaling. Raw capture energies are not invariant to arbitrary hidden-unit rescaling. At identical weights and inputs, Q is independent of the target; a scalar residual also cancels from normalized per-example projection capture whenever it is nonzero. Eighty common-state relabeling controls verify that identity. These diagnostics characterize learned-state geometry, rather than infer a task-only credit rank at initialization.

An independent saved-state audit checked 20,160 metrics and 120 matching–quartic pairing identities. For the earlier cohort whose states had not been stored, replay reconstructed 3,600 checkpoints and matched 28,800 archived metrics within 9×10^{-16} . These are reconstructed states, distinguished from the directly archived matched-cohort checkpoints. Source Data retain the complete numerical audit.

The positive-conductance comparison uses the prior compatible teacher across three paired input permutations and the same credit definitions. It does not implement the algebraic quartic target. All four rules learn it accurately with Adam; exact credit has mean NMSE 9.39×10^{-6} , compared with 1.35×10^{-6} for the fixed-profile controls. Positive path derivatives make sign and unit broadcasting identical. Both models’ local updates were independently checked by finite differences. Protocols, selections, checkpoints, full trajectories, numerical audits and scripts are supplied in `source_data/credit_rule_bridge/`, `source_data/credit_resolution_bridge/` and their corresponding script directories.

4.7 Balanced extension of the matched credit-rule comparison

After examining the released 1,024-update outcomes, we froze and committed a longer-budget protocol before running any extended trajectories. This is a protocol-led extension of previously observed seed blocks, not a fresh confirmatory cohort. All 720 algebraic trajectories were retained: twenty seed blocks, three task families, four delivery rules, Adam and SGD, and the union of the original development-selected and common rates. Matching and quartic targets retain the same balanced tree, initialization and example streams; nested targets retain their separately compatible trees. No rates were retuned.

Every trajectory was replayed from initialization and run for exactly 16,384 updates, preserving examples, mini-batches, local rules, gradient clipping and coefficient bounds $[-2, 2]$. All original checkpoint arrays, numerical curves and diagnostics through 1,024 updates reproduced. Additional observations occurred every 256 updates after 1,024. Terminal outcomes at 1,024, 4,096, 8,192 and 16,384 updates preserve the original endpoint definition. Secondary states minimized validation NMSE over the declared observations within each budget, with ties assigned to the earlier checkpoint. Test and full-domain population errors were used only for reporting. Independent recomputation of all 5,760 budget-outcome rows from saved weights reproduced test, validation and population NMSE exactly; all 2,880 validation-only selections were checked.

All twenty selected-rate Adam exact quartic fits reached the test noise floor by 4,096 updates, resolving three slow 1,024-update outcomes. By that budget exact pairwise credit also had slightly lower mean error than the initial profile, with eighteen of twenty exact wins. The original pairwise mean ordering reflected slow exact seeds, despite exact credit winning fifteen of twenty initial pairs. Compact endpoints are in Supplementary Table S8.

At 16,384 updates, the selected-rate calibrated-minus-exact quartic-minus-pairwise interaction was 0.9516 (95% paired interval, 0.8829–1.0123) for terminal states and 0.7903 (0.6915–0.8797) for validation-selected states. Corresponding common-rate effects were 0.9028 and 0.7952. All four contrasts were positive in every seed. Intervals use

1173 10,000 whole-seed bootstrap draws and are descriptive because the original outcomes were already known.
1174 The comparisons remain constrained by time, parameter bounds and historical rates. Every selected-rate Adam
1175 initial-profile trajectory had contacted a coefficient bound by 16,384, whereas none of the corresponding exact trajectories
1176 had. Validation-selected initial-profile states still had substantial quartic and nested error, and the full record retains
1177 clipping and bound events. Extended exact-trained path fields remained almost one-dimensional for pairwise targets but
1178 multidirectional for quartic and nested targets; the original main-figure capture analysis is retained unchanged. These
1179 results strengthen the fixed-budget contrast and explain the slow exact runs, without proving that a bounded fixed profile
1180 fails under every optimizer or that full compartment resolution is uniquely necessary.

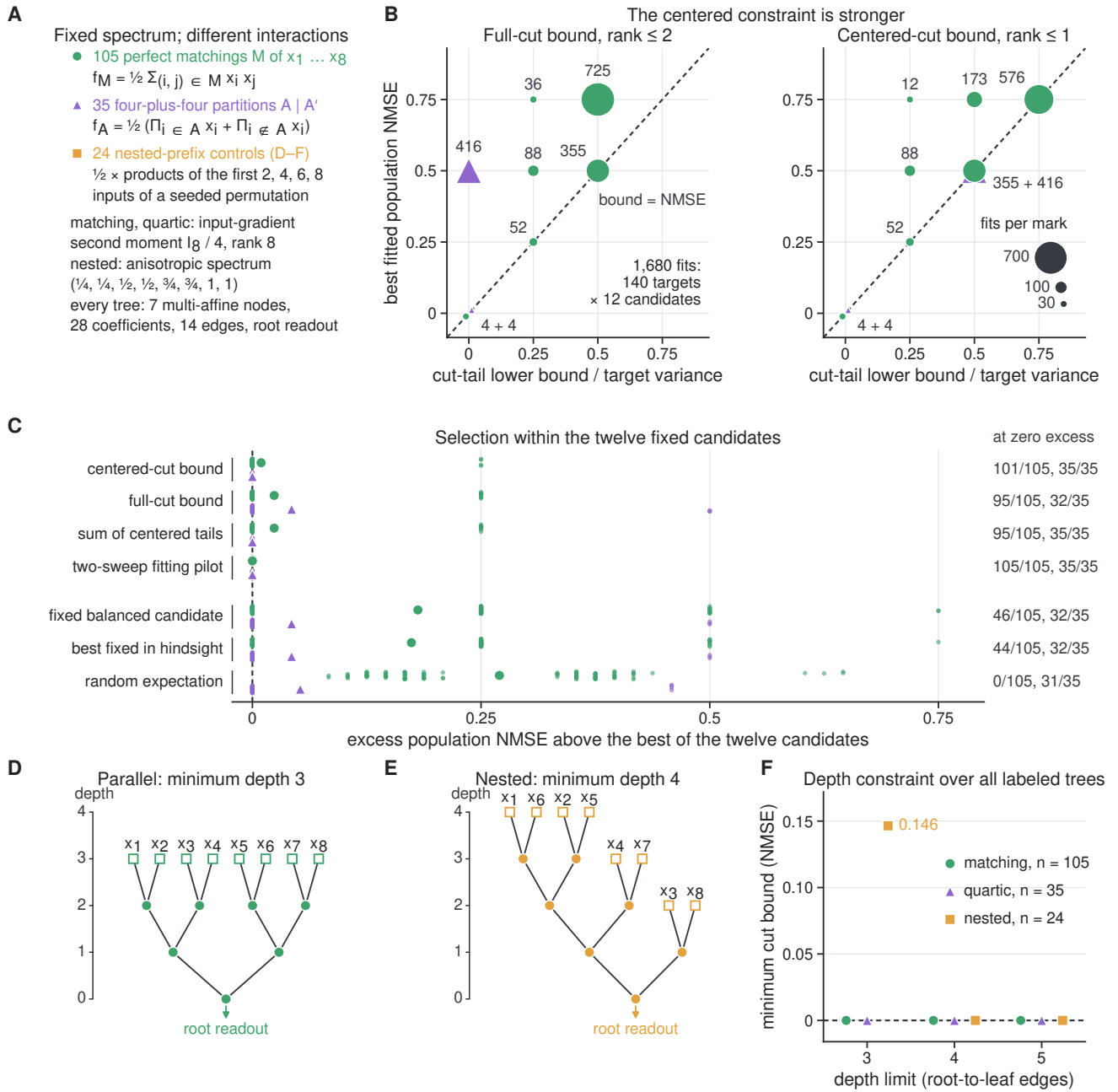


Figure S12: Interaction structure constrains scalar trees at matched input spectra and resources. **A**, Exhaustive pairwise matching and two-quartic target families on eight independent binary inputs and the 24 nested-prefix controls (one anisotropic spectrum); the panel is the key of the sheet: green filled circles, the 105 matching targets; purple filled triangles, the 35 quartic targets; amber filled squares, the 24 nested controls; grey dashed rules are references (equality in **B**, zero in **C** and **F**). **B**, Full rank-two and centered rank-one cut-tail lower bounds versus achieved population NMSE; mark area is proportional to the number of fits at that point, printed beside each mark (1,680 fits, 4–725 per point; marks of at most 18 fits sit at a 3.2 pt floor; area key, 30, 100 and 700 fits). Marks of both families at one point are superposed, the smaller on top, with one label for both counts (355 + 416 at bound 0.5, centered); the two four-fit marks at the origin touch along the equality rule. **C**, Excess NMSE above the best fitted candidate in the fixed pool of twelve: large marks, family means over the 105 matching (upper sub-row) or 35 quartic (lower sub-row) targets; small translucent dots, the per-target values, jittered vertically; the right-hand column counts targets at exactly zero excess. Above the gap, the four bound and pilot selectors; below it, the fixed balanced-shape candidate (mean excess 0.181 matching, 0.043 quartic), best-fixed selection in hindsight and the uniform random expectation. Under the centered-cut bound 101 of 105 matching and all 35 quartic targets have zero excess, so the matching mean 0.0095 rests on four targets at 0.25. **D**, Tree returned for the first enumerated matching, a depth-three parallel construction. **E**, First seeded nested-prefix control tree, of minimum depth four. Inputs enter at the top and the root is read out at the foot, as in main Fig. 4A: open squares, input leaves; filled discs, multi-affine nodes; both trees at one level pitch against the shared depth ruler. **F**, Minimum possible maximum centered-cut bound over all labeled binary trees at depth limits of 3, 4 or 5 edges (ordinal axis; families side by side at each limit), identical within each family ($n = 105, 35$ and 24): zero at every limit for matching and quartic targets; 0.146 at depth limit 3 and zero at 4 and 5 for nested controls; the dashed rule marks zero.

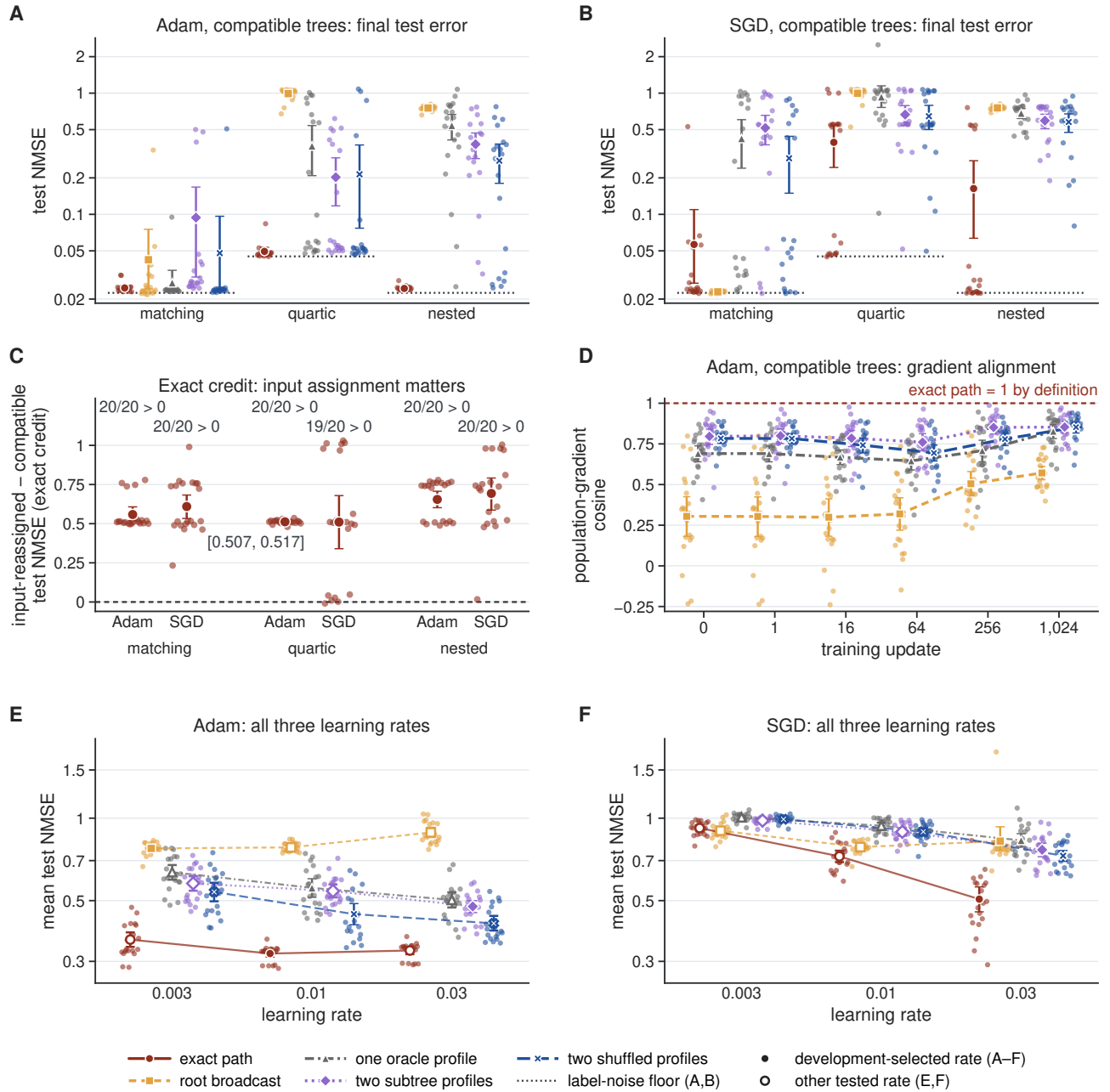


Figure S13: Task-dependent credit requirements in bounded multi-affine trees. **A,B**, Final test normalized mean squared error (NMSE) on oracle-compatible trees under Adam and stochastic gradient descent (SGD), respectively. Exact path (red-brown, circle, solid line), root broadcast (amber, square, dashed), one oracle profile (grey, triangle, dash-dot), two subtree profiles (violet, diamond, dotted) and two shuffled profiles (blue, cross, long dash). The NMSE axis is logarithmic; small translucent points are the twenty individual seeds behind each mean, and dotted rules spanning only the family they apply to mark the known label-noise floors: 0.0225 for matching and nested tasks and 0.045 for quartic tasks. **C**, Paired exact-credit NMSE increase after reassigning the leaf inputs (not the shuffled-profiles rule) while preserving tree shape and parameter count, for Adam and SGD (axis labels); every mark is the exact-path red-brown, the twenty paired seed differences are drawn behind each mean, the number of positive differences out of twenty is printed above each strip, the dashed rule is zero, and the Adam quartic interval (0.507–0.517), narrower than its marker, is printed beside the point. **D**, Adam gradient cosine between the exact and delivered aggregate clean-population updates, evaluated at each rule’s own trained state after 0, 1, 16, 64, 256 and 1,024 updates (equally spaced ordinal axis), each seed averaging the three task families before aggregation; the exact-path cosine is 1 by definition and is the dashed red-brown reference, not a series; the fan behind each mean is the twenty per-seed values. Points and whiskers in **A–F** are means and pointwise 95% intervals from 10,000 whole-seed bootstrap draws, twenty fresh seeds per family. **E,F**, Final NMSE at the three tested learning rates 0.003, 0.01 and 0.03 (categorical positions), averaged over both compatible and reassigned input assignments and all three families, on one common logarithmic NMSE axis; each point carries its twenty per-seed means and whisker, and thin connectors in each rule’s dash pattern join the three rates. Filled markers, the development-selected rate of each rule and optimizer (used in **A–D**); open markers, the other tested rates. The Adam strips of **C** are main Fig. 4H’s contrast (“matching” is “Pairwise” there).

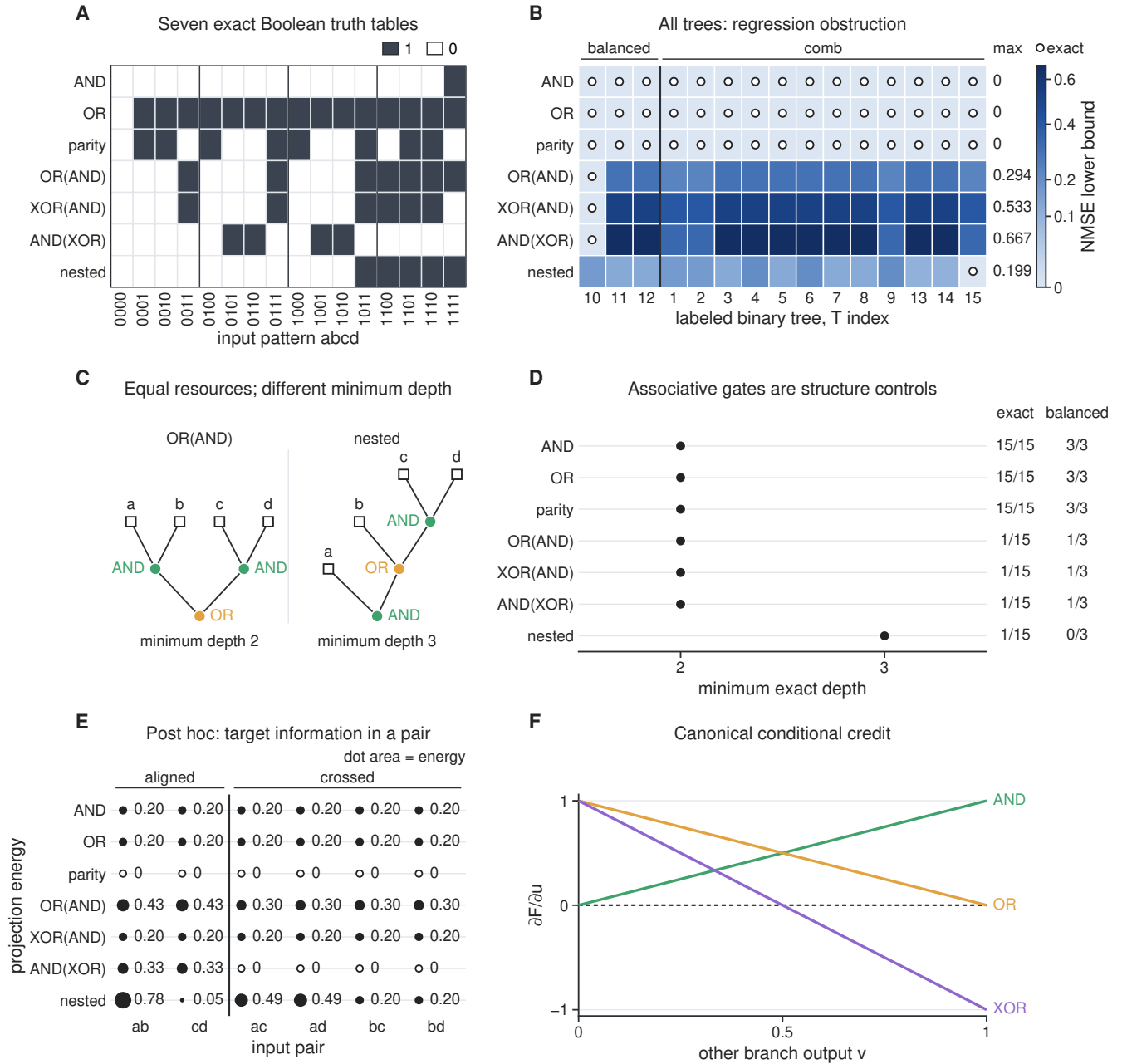


Figure S14: Boolean interactions distinguish structure, depth and canonical credit. **A**, Truth tables of seven four-input templates on the sixteen equally weighted patterns (white, 0; slate, 1); columns are $abcd$ in numeric order, a the most significant bit, hairlines between groups of four sharing a and b . Mixed targets: $(a \wedge b) \vee (c \wedge d)$, $(a \wedge b) \text{ XOR } (c \wedge d)$, $(a \text{ XOR } b) \wedge (c \text{ XOR } d)$ and nested $a \wedge [b \vee (c \wedge d)]$. **B**, Centered-cut normalized-mean-squared-error (NMSE) regression lower bounds for all fifteen labeled binary trees (twelve coefficients, six edges); the rule separates the balanced (depth-two) trees T10–T12 from the twelve comb trees. The colour ramp is a power mapping (exponent 0.6) of the bound from a faint tint (zero cells palest); the nested row is positive on every tree but T15 (0.105–0.199); row maxima are printed at right. Open rings (a marker, not a bound) mark the 49 exactly representable family–tree pairs, certified with coefficients in $[-2, 2]$ (largest $4/\sqrt{15}$; T indices and cut sets in Source Data). **C**, OR-of-AND and nested each use two AND gates and one OR gate (node colours as in **F**; inputs above, output below) but need minimum exact depths two and three. **D**, Minimum exact depth (ordinal axis, 2 or 3); the columns count exactly compatible trees among all fifteen (the ring counts of **B**) and among the three balanced: 15/15 and 3/3 for AND, OR and parity, 1/15 and 1/3 per mixed-gate target, 1/15 and 0/3 for nested. **E**, Post hoc target-projection energy $\|\mathbb{E}[z | x_S]\|^2$ of all seven families on the six two-input subsets, normalized target z : dot area is proportional to the energy, values printed beside the dots, exact zeros are open rings, and the rule separates aligned pairs (ab, cd) from crossed. XOR-of-AND has 1/5 on every pair; parity has zero on every proper subset (all fourteen per family in Source Data). **F**, Canonical derivatives $\partial F / \partial u$ for left-branch output u against right-branch output v : AND, v ; OR, $1 - v$; XOR, $1 - 2v$; dashed rule, zero. Green, amber and violet in **C,F** name the gates AND, OR and XOR, not bound values; **D,E** are drawn in ink only. Panels are exhaustive analytic diagnostics (no sampling error bars): positive bounds concern truth-value regression, not threshold classification; **E** is local target information, predicting neither grouping compatibility nor broadcast-learning failure; **F**'s raw derivatives need not survive training and imply no learning outcome; assumptions: Supplementary Note 4.4.

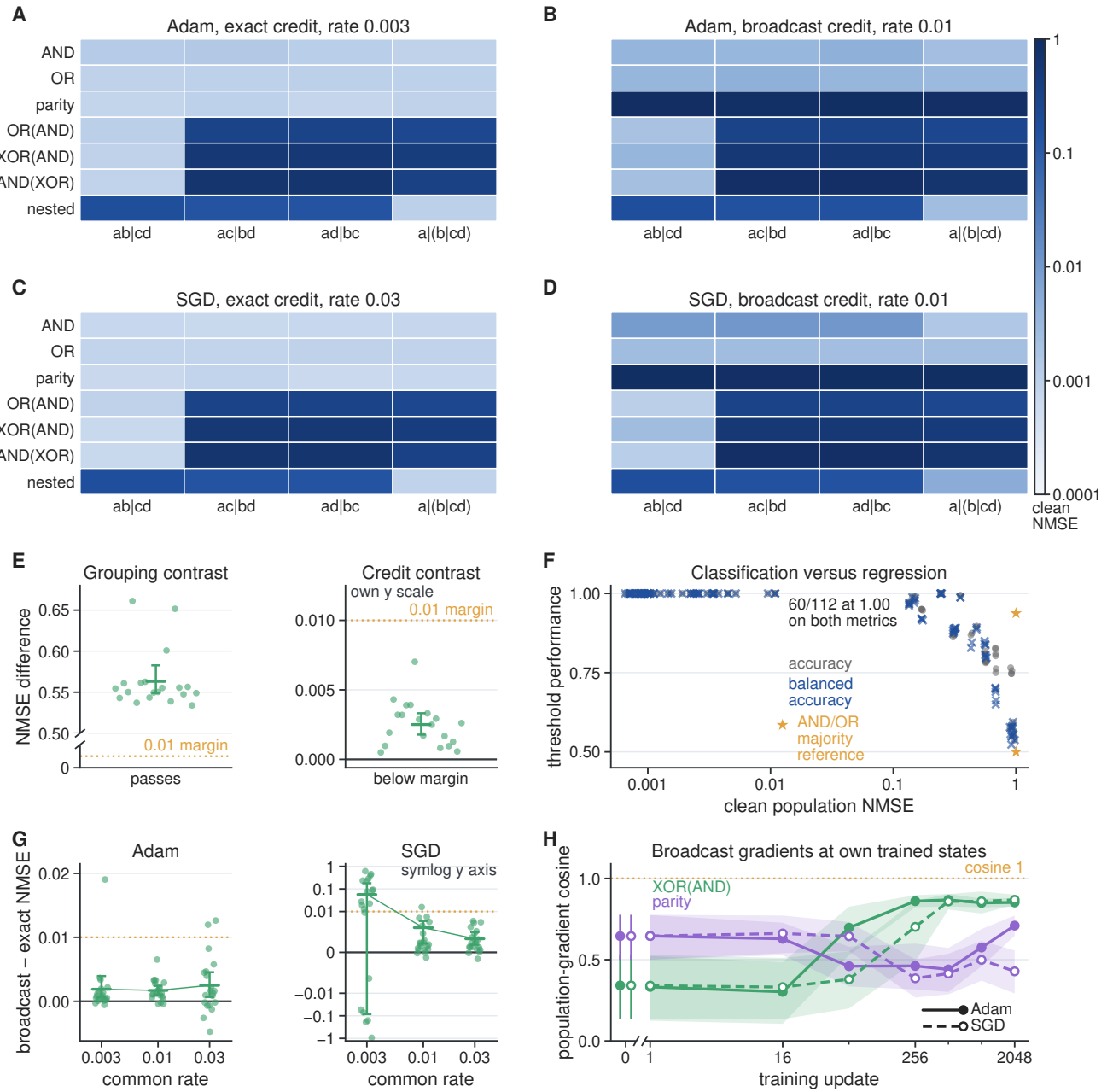


Figure S15: Boolean grouping benefit and a small primary credit effect under frozen recipes. **A–D**, Clean population normalized mean squared error (NMSE), twenty-seed means, for seven target families (rows, labelled at **A,C**) and four trees (three balanced groupings and the comb $a|(b|cd)$) under Adam (**A,B**) and stochastic gradient descent (SGD; **C,D**) with exact (**A,C**) or broadcast (**B,D**) credit, on one logarithmic colour scale (key at right). Rates (0.003, 0.01, 0.03 and 0.01 in **A–D**) were selected per optimizer and rule on five development seeds; per-cell 95% intervals are in Source Data. **E**, Prespecified Adam XOR-of-AND contrasts: grouping, the crossed-tree mean ($ac|bd$, $ad|bc$) minus the aligned $ab|cd$ under exact credit; credit, broadcast minus exact on $ab|cd$. Dots (green, as XOR-of-AND in **G,H**), twenty paired seed differences; rules and whiskers, means and Bonferroni-adjusted 97.5% intervals (grouping 0.549–0.583); amber dotted, the 0.01-NMSE mean-effect margin. The grouping axis is broken, zero and the margin in its foot strip; the credit subpanel has its own y range and a solid zero rule. Grouping passes both criteria, credit does not; both rules classify all sixteen patterns perfectly in every seed. **F**, Accuracy (gray circles) and balanced accuracy (blue crosses) versus NMSE for the 112 conditions at raw-output threshold 0.5; sixty overprint at 1.00 on both. Amber stars, the AND/OR constant-majority references (NMSE one, accuracy 15/16, balanced accuracy 1/2). **G**, Same-rate broadcast-minus-exact contrasts on the aligned XOR-of-AND tree, Adam (left) and SGD (right): dots, twenty paired seed differences per common rate (logarithmic axis); rules and whiskers, joined across rates, means with descriptive 95% intervals; amber dotted, the 0.01 margin; dark solid, zero. The SGD y axis is symmetric-log (linear within ± 0.002) to keep the rate-0.003 seed differences (-0.94 to 0.61) in view. **H**, Broadcast gradient cosine on the aligned balanced tree for XOR-of-AND (green) and parity (violet); solid lines, filled markers Adam; dashed lines, open markers SGD; on all sixteen clean patterns at each learner's own state before clipping or optimizer transformation; eight checkpoint markers on a logarithmic update axis from 1, and update 0 (shared initial state, one whisker per optimizer) left of an axis break; amber dotted, cosine one. Bands and whiskers, pointwise 95% intervals. The large parity credit effect is descriptive.

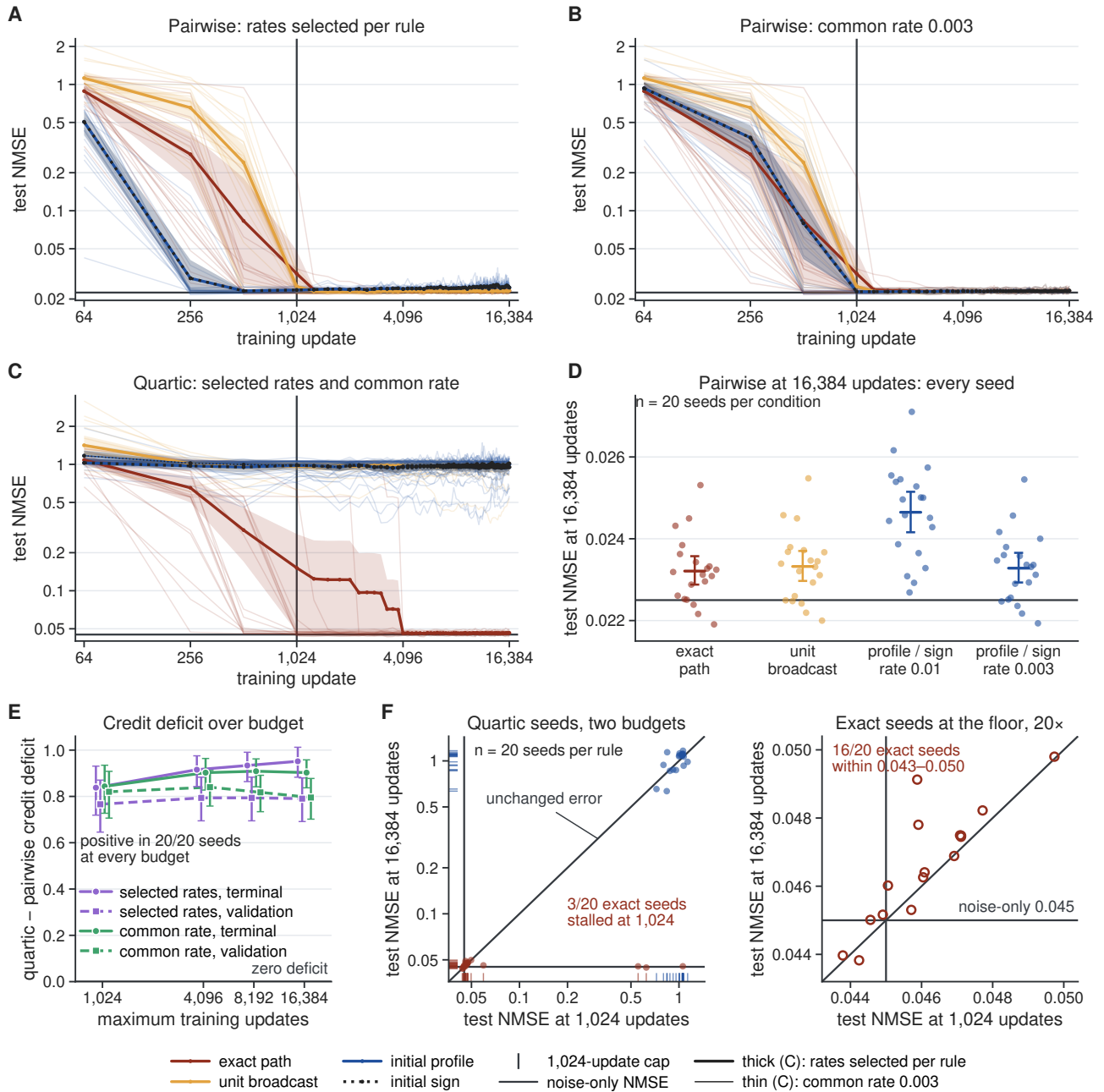


Figure S16: Longer matched budgets retain the task-dependent fixed-profile deficit. **A,B**, Pairwise targets under Adam at the rates selected per rule (0.003 for exact path and unit broadcast, 0.01 for initial profile and initial sign) and at the common rate 0.003. In **A–D** and **F** colour names the credit rule (key below the sheet): dark red, exact path; amber, unit broadcast; blue, initial profile; black dotted, initial sign. Both axes of **A–C** are logarithmic; thick curves are means with a dot at each of the 64 measured checkpoints, tinted bands descriptive 95% whole-seed bootstrap intervals, faint hairlines the twenty seed trajectories of each rule (no separate initial-sign fan: it coincides with the initial-profile fan where the rules share a rate); thin solid grey rules mark the 1,024-update cap (vertical) and the noise-only NMSE (horizontal; 0.0225 pairwise, 0.045 quartic). **C**, Quartic targets under both rate choices: exact-path and unit-broadcast selected rates equal the common rate, so those curves are drawn once; initial profile and initial sign are thick at the selected rate (0.01) and thin at the common rate (0.003). **D**, Test NMSE at 16,384 updates for all twenty pairwise seeds (dots, $n = 20$ per condition), the mean (short rule) and its 95% whole-seed bootstrap interval (whisker) for exact path, unit broadcast and the coinciding initial-profile/initial-sign pair at rate 0.01 and at 0.003, against the 0.0225 floor (thin grey rule). **E**, Quartic-minus-pairwise difference in initial-profile-minus-exact NMSE at budgets 1,024, 4,096, 8,192 and 16,384: circles/solid, terminal states; squares/dashed, validation-selected states; here colour names the rate view: violet, rates selected per rule (the series of main Fig. 4F); green, the common rate 0.003. Whiskers, 95% paired whole-seed bootstrap intervals, offset slightly at each budget; every difference is positive in 20/20 seeds; the thin grey rule marks zero deficit. **F**, All twenty exact-path and twenty initial-profile quartic seeds at 1,024 and 16,384 updates (left, logarithmic axes; dots with a rug tick per seed on each axis); the thin grey diagonal marks unchanged error and the thin grey guides the quartic noise-only NMSE 0.045. The three exact seeds stalled at 1,024 updates (0.55–1.06; main Fig. 4D) are at the floor by 16,384; the right-hand panel enlarges the 0.043–0.050 window twenty-fold, where the other sixteen exact seeds (open circles) lie on both axes.

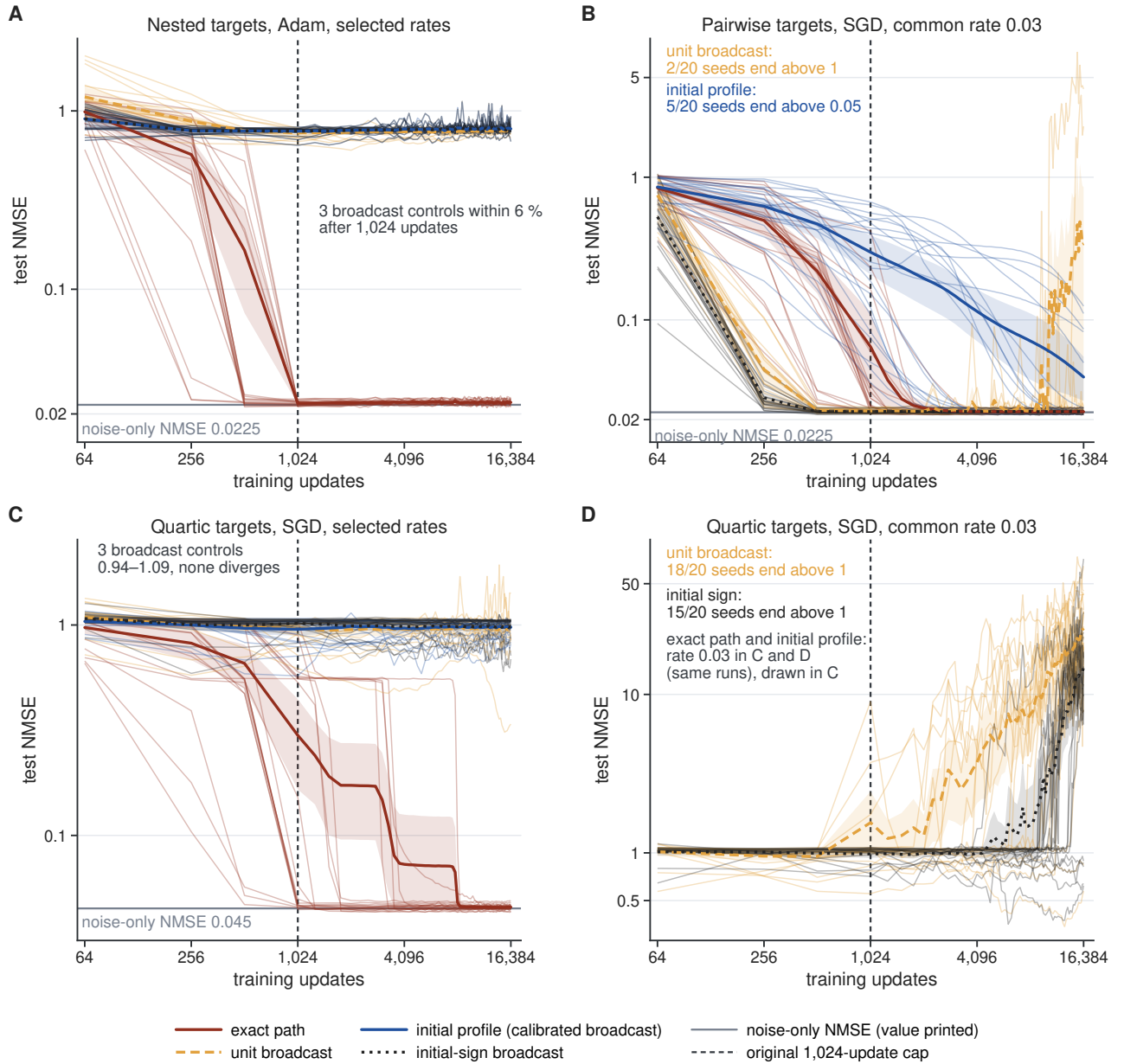


Figure S17: Nested targets and SGD controls in the balanced budget extension. **A**, The same condition as main Fig. 4E, repeated as the reference for the SGD panels: nested targets under Adam at the selected rates, 0.003 for exact-path and unit-broadcast credit and 0.01 for initial-profile and initial-sign credit. **B**, Pairwise targets under SGD at the common rate 0.03 for all four rules. **C,D**, Quartic targets under SGD at the selected rates (**C**: 0.03 for exact-path and initial-profile credit, 0.01 for unit and initial-sign broadcast) and at the common rate 0.03 (**D**); exact-path and initial-profile credit use rate 0.03 in both, so those runs are drawn once, in **C**, and **D** shows only the unit and initial-sign broadcast arms. Colours follow the main figures and line style also names the rule: exact path dark red solid, unit broadcast amber dashed, initial-profile broadcast (the calibrated broadcast of main Fig. 4) blue solid and initial-sign broadcast black dotted. Heavy curves show mean test NMSE from update 64 onward over the 64 saved checkpoints, thin lines are the 20 individual seeds of each rule, and pale bands give descriptive 95% whole-seed bootstrap intervals of the mean (10,000 draws). A mean can sit where no seed does: in **B** the unit-broadcast mean rises to 0.34 at 16,384 updates because 2 of 20 seeds diverge (to 2.3 and 4.2) while 18 end at the floor, and 5 of 20 initial-profile seeds end at 0.050–0.125; in **D**, 18 of 20 unit-broadcast and 15 of 20 initial-sign seeds exceed NMSE 1. In **A** the three broadcast controls coincide to within 6% after 1,024 updates, and in **C** all three stay between 0.94 and 1.09 without diverging, so their curves overprint and are told apart by line style. Vertical dashed lines mark the original 1,024-update cap; light grey solid rules mark the noise-only NMSE, printed under the left end of each rule (0.0225 in **A,B**; 0.045 in **C**); **D** lies entirely above its floor of 0.045. In **B** the initial-sign and unit-broadcast means reach the floor by update 512 and the exact-path mean by 2,304, overprinting on it thereafter apart from the unit-broadcast excursions after 9,472 updates (2 of 20 seeds). **A** and **B** are single conditions, not a matched pair; only **C** and **D** contrast the two rate views at fixed task and optimizer. The update axis is logarithmic in every panel, and each vertical axis is logarithmic and tight to its own panel's seeds: 0.014–2.6 in **A**, 0.014–9.4 in **B**, 0.032–2.6 in **C** and 0.28–95 in **D**, which retains every divergent broadcast seed (up to 74).

1181 S5 Credit delivery in conductance-tree learning

1182 5.1 Conductance-tree tasks with aligned or opposed feature tuning

1183 We tested whether context-dependent branch selection can create a learning advantage for spatially resolved credit in a
 1184 conductance tree. The model contains four terminal compartments, two proximal compartments and a soma. It uses
 1185 DendriNet’s directed steady-state excitatory/inhibitory balance with identity reactivation; it is not a reciprocal cable
 1186 model. All 24 positive conductances are trainable, including inhibitory gains and daughter and somatic couplings.

1187 For each example, two independent latent variables $z_1, z_2 \sim \mathcal{U}[-2, 2]$ generate positive features $x_j^+ = \exp(z_j)$ and
 1188 $x_j^- = \exp(-z_j)$. Both subtrees receive the same latent features: terminals 0 and 2 receive z_1 , and terminals 1 and 3
 1189 receive z_2 . Each terminal has two excitatory and two inhibitory conductances. Its effective excitatory and inhibitory
 1190 conductances and voltage are

$$E_j = g_{E+,j}x_j^+ + g_{E-,j}x_j^-, \quad I_j = g_{I+,j}x_j^+ + g_{I-,j}x_j^-, \quad v_j = \frac{E_j}{1 + E_j + I_j}. \quad (91)$$

1191 This normalization sets leak conductance to 1 and excitatory, inhibitory and leak reversal potentials to 1, 0 and 0. An
 1192 independent binary context c is sampled with equal probability. Proximal compartment $p \in \{0, 1\}$ receives inhibitory
 1193 activity $a_p = A\mathbf{1}[c \neq p]$, with $A = 10$ for the primary task pair. Its voltage and the somatic voltage are

$$u_p = \frac{\sum_{j \in \text{ch}(p)} g_{pj}v_j}{1 + \sum_{j \in \text{ch}(p)} g_{pj} + g_{I,p}a_p}, \quad v_s = \frac{g_{s0}u_0 + g_{s1}u_1}{1 + g_{s0} + g_{s1}}, \quad (92)$$

1194 where $\text{ch}(p)$ denotes the two terminal daughters of proximal compartment p . Here $u_p = v_{4+p}$ denotes a proximal voltage;
 1195 thus v_j , $j = 0, \dots, 5$, indexes all six nonsomatic compartment voltages. All conductances are parameterized by their
 1196 logarithms.

1197 Labels are the noiseless somatic responses of a teacher with this exact model, establishing a constructive zero forward-
 1198 error floor. In the aligned task every teacher terminal has nominal gains (8, 0.25, 0.25, 8), ordered as $(E+, E-, I+, I-)$.
 1199 In the opposed task terminals 2 and 3 instead have (0.25, 8, 8, 0.25). Teacher log gains receive independent Gaussian
 1200 perturbations of standard deviation 0.15, shared across the paired tasks. Parent inhibitory gains are nominally 15, daughter
 1201 couplings are (20, 0.2) in each subtree, and somatic gains are 4. Students start with terminal gains 1 and these same other
 1202 nominal gains, with independent Gaussian perturbations of standard deviation 0.4 applied to all 24 log gains. No exact
 1203 symmetry is imposed. Paired tasks have identical student initializations, input samples, contexts and minibatches; their
 1204 teacher tuning differs. Identical input samples do not imply identical task input-sensitivity spectra.

1205 Training minimizes half the mean squared error divided by training-target variance. Writing $\delta = (v_s - y)/\text{Var}_{\text{train}}(y)$,
 1206 the exact update uses the path factor $q_j = \partial v_s / \partial v_j$ and local eligibility $\partial v_j / \partial \theta_j$. Here θ_j is a local log conductance.
 1207 Excitatory eligibilities retain the factor $(1 - v_j)/D_j$, inhibitory eligibilities retain $-v_j/D_j$, and daughter-coupling
 1208 eligibilities retain $(v_{\text{child}} - v_{\text{parent}})/D_{\text{parent}}$, together with their corresponding input and conductance factors; D is the local
 1209 conductance denominator. Thus changing the credit rule leaves all local voltage derivatives intact. Somatic couplings
 1210 receive the exact somatic factor 1 under every rule.

1211 The six non-somatic path factors are delivered exactly, replaced by 1, replaced by a fixed initial profile, or projected
 1212 into three fixed spatial patterns. The initial profile is the mean initial exact path field on 512 independently generated,
 1213 unlabeled calibration examples. The three-pattern dictionary retains this profile on terminals (0, 1), terminals (2, 3)
 1214 and proximal compartments (4, 5), respectively, with zero entries elsewhere. If A is the resulting six-by-three matrix,
 1215 the oracle rule delivers $A(A^\top A)^{-1}A^\top \mathbf{q}$ for the current example. Its coefficients use the current exact field. It tests a
 1216 restricted spatial dictionary with an oracle encoder; it is not a locally estimated routing rule or evidence that three
 1217 independent external errors are necessary. At fixed parameters the binary context generates at most two path profiles, so
 1218 this experiment concerns branch selection rather than a six-dimensional or quartic-interaction requirement.

1219 Each task/seed has 2,048 training, 1,024 validation, 4,096 test and 1,024 diagnostic examples, independent of the
 1220 calibration sample and of one another. No noise is added to labels. Across the 20 fresh teachers, mean test-target variance
 1221 is 0.024108 for aligned tuning and 0.014691 for opposed tuning. Test NMSE divides mean squared error by each test
 1222 set’s clean target variance.

1223 Development comprised three seeds, four regimes, four credit rules, two optimizers and three rates, for 288 fits.
 1224 Every outcome is retained. The regimes were aligned strong gating, opposed strong gating, opposed moderate gating
 1225 ($A = 1$), and opposed ungated ($A = 0$). Before development outcomes, the primary strong-opposed regime had to satisfy

mean teacher rank-one path capture at most 0.90 and mean exact-Adam validation NMSE at most 0.02, with a declared fallback order. Selection did not use the between-rule gap. Strong opposed tuning qualified, and a separate selection freeze preceded the 20 fresh paired blocks. Adam rates were 0.003, 0.01, 0.03, and SGD rates were 0.03, 0.1, 0.3. Each task/rule/optimizer selected the rate with lowest mean development best-validation NMSE; a common rate minimized the mean over rules. In the fresh task pair all selected rates equal their common rates: 0.03 for Adam and 0.3 for SGD. The common-rate results therefore reuse the same fits. The fresh cohort comprises $20 \times 2 \times 4 \times 2 = 320$ fits.

Minibatches contain 128 examples sampled with replacement. Adam uses $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 10^{-8}$; SGD has no momentum. The original budget is 4,096 updates, with global gradient norm capped at 10 and log conductances projected into $[-7, 7]$. Validation checkpoints are updates 0, 64, 256, 1,024, 2,048 and 4,096; the lowest-validation-loss checkpoint supplies the primary test endpoint. Every fresh fit continues, with its original optimizer and minibatch state, to 16,384 updates, adding validation checkpoints 8,192, 12,288 and 16,384. This extension was declared before fresh outcomes and retains earlier checkpoints when selecting its endpoint. These are fixed optimization budgets, not claims of convergence. No fresh or extended fit undergoes gradient-norm clipping. Exact and oracle fits never reach a parameter bound, whereas broadcast fits can do so.

The primary interaction is the calibrated-minus-exact NMSE gap for opposed tuning minus the corresponding gap for aligned tuning. Confidence intervals are paired percentile bootstrap intervals from 20,000 resamples of the 20 teacher/initialization seed blocks. Full rule-specific, seed-specific and checkpoint-specific outcomes are supplied. Post-fit diagnostics report squared-energy capture for the path field and the loss-weighted field, their best rank-one projections, and projection into the frozen initial profile after weighting by squared local eligibility, with or without squared somatic error. All are oracle diagnostics rather than additional training rules.

To identify cancellation across contexts, each rule is evaluated at the same initial, 256-update, primary-best and extended-best states of each selected Adam fit. For a given state, the cancellation ratio is the norm of the probability-weighted sum of context-conditional distal gradients divided by the probability-weighted sum of their norms. Lower values indicate stronger cancellation. All 2,560 audited mean gradients reproduce the training adapter exactly. Since path factors and fixed profiles are positive, per-example approximate gradient estimates have nonnegative inner products with exact per-example gradients; the failure being tested concerns averaging conflicting examples.

A separately timestamped posthoc sensitivity restarts every fresh condition from its original initialization for 16,384 updates with wider log bounds $[-20, 20]$ and unchanged rates. It also replays every original-bound trajectory to record first contact with each parameter bound. These 640 trajectories reuse the same 20 seed blocks. All 320 original-bound replays reproduce every final parameter exactly. The wider-bound interaction remains positive in all 20 seeds for both optimizers. At the last saved checkpoint before each calibrated Adam fit’s first original bound contact, its opposed-task gap is already positive in all 20 seeds. Some broadcast parameters also reach the wider bounds, so this control establishes robustness across a much wider conductance range rather than unbounded convergence. The retained moderate-gating development regime also shows a substantial broadcast deficit despite teacher rank-one path capture near 0.98. Raw path-energy capture is therefore not a sufficient criterion for learning: a small spatial component can distinguish the gradient contributions of competing contexts.

5.2 Local inhibitory gating in the conductance task

This study was specified after exposure to a referee panel’s exploratory gate results. Those runs are excluded here. The protocol and numerical adapter were committed before execution on twenty new seed blocks, 2026090801–2026090820. Within each seed, aligned and opposed targets and all nine rules shared teacher perturbations, initial students, features, contexts and minibatch streams. The 24-conductance model, nominal gains, input streams and independent perturbations are defined in Section 5.1; every conductance remains plastic.

Let $\delta = (V_s - y) / \text{Var}_{\text{train}}(y)$ denote the somatic error and $a_p \in \{0, 10\}$ the inhibitory activity at parent p . The hard local rule multiplies terminal eligibilities by $\delta \mathbf{1}[a_p = 0]$, while proximal and somatic eligibilities receive δ . It uses neither exact paths nor exact-field projections during learning. A secondary continuous gate uses $\delta / (1 + g_p^I a_p)$ at terminals, again retaining unit proximal and somatic multipliers. Both receive an imposed inhibitory context and an output error; neither generates those signals autonomously.

Exact, uniform, calibrated and three-profile oracle rules retain the previous comparators. Calibration averages initial exact path derivatives on 512 unlabeled examples. Two weighted leaf-subtree profiles with oracle projection amplitudes and unit proximal credit test the context distinction separately from proximal provision. The earlier three-profile oracle also includes a weighted proximal-pair column. The new control therefore has two distal profiles plus a fixed proximal rule, rather than an entirely two-dimensional six-site dictionary. Further controls reverse the distal gate or apply it at the

proximal compartments as well. In the latter, inhibitory eligibility is nonzero only when inhibition is present, while its credit multiplier is then zero; both proximal inhibitory gradients vanish identically.

Data sizes, half-NMSE objective, Adam parameters, batch size 128, clipping threshold ten and log bounds $[-7, 7]$ were unchanged. The primary rate was 0.03; 0.01 and 0.1 were symmetric robustness conditions for all rules. All $20 \times 2 \times 9 \times 3 = 1,080$ trajectories continued to 16,384 updates. Minimum validation NMSE selected a state within the primary 4,096-update and extended windows, while fixed endpoints and every saved observation remain available separately.

Two prespecified paired contrasts used exact two-sided sign-flip tests: opposed-task broadcast-minus-local-gate NMSE, and the difference in this gap between opposed and aligned tasks. Both were positive in all twenty seeds, with Holm-adjusted $P = 3.81 \times 10^{-6}$ in the two-test family. Intervals use 20,000 whole-seed bootstrap draws. Other rates, budgets and contrasts are descriptive. The local-minus-exact opposed-task difference was 1.49×10^{-6} (95% interval, -1.60×10^{-6} to 4.86×10^{-6}), and the maximum local-gate seed error was 5.47×10^{-5} . These satisfy the frozen practical precision reference without establishing asymptotic equivalence.

At the primary rate, exact, proper local gates and both oracle dictionaries contacted no bounds; opposed broadcast contacted bounds in 18/20 fits by 4,096 and 20/20 by 16,384. No fit clipped gradients. Ten mechanism tests verify the derivatives, absence of exact-path access in the hard local update, and the proximal eligibility failure. Excluded historical-seed replays reproduced prior exact, unit and calibrated trajectories bit for bit and the independently projected dictionary to floating-point association error. Protocols, fresh-seed outcomes, states, curves and validation records are in `source_data/conductance_local_gate/`.

Retained first conductance study. A separately frozen first study used the same seven-compartment topology but one excitatory and one inhibitory feature per terminal, giving 16 trainable conductances. An independent context selected one subtree. Its two terminal latent inputs were $z_1, z_2 \sim \mathcal{U}[-2, 2]$; the inactive subtree received $-\rho z_j + \sqrt{1 - \rho^2} u_j$, where u_j were independent draws from the same distribution and $\rho \in \{0, 1\}$ set independent or opposite distractors. Each terminal received $\exp(z)$ through its excitatory contact and $\exp(-z)$ through its inhibitory contact, with the corresponding selected or inactive latent value. The four development regimes were ungated/independent, strong-gated/independent, strong-gated/opposite and moderate-gated/opposite, using inhibitory activities $A = 0, 10, 10, 1$, respectively. Nominal teacher excitatory gains were (0.25, 8, 0.25, 8) and inhibitory gains (8, 0.25, 8, 0.25); nominal student gains were 2 and 0.7, respectively. Parent inhibition, daughter couplings and somatic gains were 15, (20, 0.2) per subtree and 4, as above. Teacher and student log gains had independent Gaussian perturbations of standard deviations 0.15 and 0.4. Labels were noiseless outputs of the same teacher circuit. Dataset sizes, calibration, optimizers, learning-rate grids, batch size, validation cadence and the two update budgets matched those specified above. Three development seeds, four regimes, three rules, two optimizers and three rates gave 216 development fits. The declared teacher-geometry and exact-learnability criteria selected strong-gated/opposite inputs, paired with the ungated/independent control; this comparison changes both gating and distractor dependence. Twenty fresh seeds gave 240 fits, all continued to the larger budget. All outcomes are retained. After 16,384 updates, its gated-task calibrated-minus-exact gap was 0.000502 [0.000267, 0.000772] NMSE, positive in all 20 seeds; under Adam the mean broadcast error remained below 0.0006 NMSE. This small precision effect motivated the separately declared opponent-tuning model above. The first result and all unsuccessful development regimes remain part of the evidence.

Expanded learning-rate screen. Because the original development selection lay at the upper edge of its rate grid, we performed a separately specified post hoc screen using the original three development seeds (201–203). Each of the four credit rules received the same six-rate search on both the aligned and opposed tasks: $\{0.003, 0.01, 0.03, 0.1, 0.3, 1\}$ for Adam and $\{0.03, 0.1, 0.3, 1, 3, 10\}$ for SGD. We reran the original rates alongside the added rates, giving every fit 16,384 updates and selecting its minimum validation NMSE among checkpoints at 0, 64, 256, 1,024, 2,048, 4,096, 8,192, 12,288 and 16,384 updates. All other task, initialization, optimizer and conductance-bound settings remained those of the original opponent study. The screen comprises 288 retained fits; no test outcome entered rate selection.

Before inspecting these outcomes, we specified that an improvement of at least 0.01 NMSE and at least 10% in mean opposed-task validation error for either broadcast rule would trigger a follow-up on the 20 previously used fresh seed blocks. The first criterion compared the best added rate with the best original-grid rate at the same update budget; a pre-outcome addendum also compared the best rate across the whole grid with the originally selected rate (0.03 for Adam and 0.3 for SGD). Any triggered follow-up would retune all four rules on both tasks using development validation alone and would be reported as reuse of existing blocks. Neither criterion was met. No added rate improved

the opposed-task broadcast minimum; within the original grid, reducing Adam’s rate to 0.003 lowered mean validation NMSE by approximately 0.00647 (0.89%), and reducing the unit-broadcast SGD rate to 0.1 lowered it by 0.03348 (4.46%). Calibrated-broadcast SGD retained rate 0.3. We therefore performed no additional fresh-block fits. These results address a finite grid at the stated update budget and do not establish optimizer-independent convergence.

5.3 Input grouping and restricted credit in directed conductance trees

We used a minimal directed shunting model to test input grouping while holding the branching shape and parameter counts fixed. Four terminal compartments fed two proximal compartments and a soma. Every model had seven compartments, six child couplings, four excitatory contacts, six inhibitory contacts and sixteen trainable conductances. Leak conductance was one; excitatory and inhibitory reversal potentials were one and zero. All compartment activations were the identity, and the soma had no direct synaptic inputs or additional decoder. Let $i \in \{0, 1, 2, 3\}$ index terminal compartments and $j \in \{1, 2\}$ proximal compartments. Write T_i , P_j and S for terminal, proximal and somatic voltages. The pair $\mathcal{S}_j$ contains the terminals assigned to proximal compartment j . Terminal i has synaptic conductances g_i^E and g_i^I , and b_{ji} couples it to proximal compartment j . The maximal proximal shunt conductance is h_j , and a_j couples that compartment to the soma. For terminal activities x_i^E, x_i^I and proximal shunt input z_j ,

$$T_i = \frac{g_i^E x_i^E}{1 + g_i^E x_i^E + g_i^I x_i^I}, \quad (93)$$

$$P_j = \frac{\sum_{i \in \mathcal{S}_j} b_{ji} T_i}{1 + \sum_{i \in \mathcal{S}_j} b_{ji} + h_j z_j}, \quad (94)$$

$$S = \frac{a_1 P_1 + a_2 P_2}{1 + a_1 + a_2}. \quad (95)$$

The positive local input z_j controls shunt conductance; it carries no teaching signal. Conductances were parameterized as $g = \exp(\theta)$, with $\theta \in [-5, 5]$. The matched resources were compartment, contact, coupling and parameter counts and the admissible conductance range; total learned conductance was not constrained to be equal. No fresh run reached a parameter bound. The NumPy implementation was checked compartment by compartment against the repository’s production `forward_branch_dynamics` function in its identity-activation, zero-numerical-offset configuration. All sixteen log-conductance derivatives and ten input derivatives agreed with autograd to the numerical tolerances reported in the accompanying tests. This is a directed steady-state model, not a reciprocal cable or spiking model.

The three groupings were (01 | 23), (02 | 13) and (03 | 12). A planted teacher used the same conductance equations, with independent Gaussian perturbations of standard deviation 0.25 in log conductance around $g^E = 2$, $g^I = 0.7$, $h = 3$, $b = 3$ and $a = 4$. Independent student perturbations had standard deviation 0.4 around these centers and were shared across paired candidate groupings, credit rules and optimizer settings. Student initialization did not use teacher coefficients. Teacher tasks permuted terminal excitatory/inhibitory input pairs together; the proximal inputs retained their identities. Paired task inputs were inversely permuted so that the realized target values were identical across the three tasks. Thus each task had the same input distribution and the same spectrum of the input-gradient second moment. The latter was verified from 2,048 paired probes per seed; its matrix entries could differ by the corresponding coordinate permutation. Compatibility meant that the student’s input grouping matched the teacher’s grouping. These tasks are planted mechanistic positive controls within one model class, rather than tests of transfer to an unseen class of biological computations.

Each input was $\exp(\sigma Z - \sigma^2/2)$ for independent standard-normal Z . We used $\sigma = 0.7$ for terminal excitatory and inhibitory inputs and $\sigma = 1.2$ for proximal shunt inputs. All activities and conductances were strictly positive, all voltages lay between the reversal potentials, and there was no label noise. Each task used 1,024 training, 1,024 validation, 4,096 test and 256 calibration examples from independent random streams. Within a task, every candidate and credit rule received the same examples and minibatch indices. Training minimized half-MSE divided by the training target variance. Reported NMSE is MSE divided by the target variance of the evaluation split; it is dimensionless and is not a percentage.

Four feedback conditions shared the forward model. Let V_n be the voltage of compartment n and L the training loss. Write $\rho_n = \partial S / \partial V_n$ for its current exact path sensitivity and $\delta_0 = \partial L / \partial S$ for the somatic loss derivative. Exact transport supplied $\delta_0 \rho_n$. A fixed calibrated broadcast supplied $\delta_0 \gamma_n$, where γ_n was the mean initial path sensitivity of that student’s own forward states on the label-free calibration inputs. The profile was frozen during training. A one-profile projection used the same γ and supplied its best Euclidean fit to the current nonsomatic field. The two-subtree condition restricted γ

1363 to each proximal compartment and its two terminal descendants, giving disjoint columns $\mathbf{A}_j$. Its coefficients were

$$c_j = \frac{\mathbf{A}_j^\top \boldsymbol{\rho}}{\mathbf{A}_j^\top \mathbf{A}_j}. \quad (96)$$

1364 The one-profile projection used the analogous coefficient with $\mathbf{A} = \boldsymbol{\gamma}$. These per-trial coefficients had access to
 1365 the current exact student path field; both projection rules are oracle diagnostics of restricted feedback, not learned
 1366 coefficient estimators. Soma error remained exact in every condition. Each conductance update multiplied the delivered
 1367 compartment error by its local conductance eligibility, including the chain derivative of the exponential parameterization.
 1368 For a child coupling, the local conductance derivative was $(V_{\text{child}} - V_{\text{parent}})/G_{\text{parent}}^{\text{tot}}$, where $G_{\text{parent}}^{\text{tot}}$ is the parent's total
 1369 open conductance. Credit geometry was evaluated at the corresponding exact-learning model's common checkpoint state;
 1370 the reported path-capture normalization excluded the soma.

1371 We used 1,000 minibatch updates of size 128, with checkpoints at 0, 10, 100, 300 and 1,000. Stochastic gradient
 1372 descent (SGD) and Adam were evaluated separately; Adam used $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\varepsilon = 10^{-8}$. One learning
 1373 rate per optimizer was chosen by averaging final validation NMSE over five development seeds (15300–15304), all
 1374 task/student groupings and all four credit rules. The original SGD grid was 0.001, 0.003, 0.01; because its optimum was
 1375 at the upper boundary, an additional development-only bracket tested 0.03, 0.1, 0.3, 1. Every result, including unstable
 1376 high-rate trajectories, was retained. Adam tested 0.003, 0.01, 0.03. The selected rates were 0.3 for SGD and 0.01 for
 1377 Adam. These settings were frozen before the twenty fresh seeds (15400–15419), producing $20 \times 3 \times 3 \times 4 \times 2 = 1,440$
 1378 fits and 7,200 checkpoint outcomes. No learning rate, horizon or credit rule was selected using fresh training outcomes.
 1379 The two primary contrasts were the Adam exact-gradient grouping effect and the Adam calibrated-broadcast-minus-exact
 1380 effect within compatible groupings. Other contrasts are descriptive optimizer and feedback-resolution sensitivities.

1381 Inference used the seed as the independent unit. Each seed's compatible mean averaged its three teacher-task permu-
 1382 tations; its incompatible mean additionally averaged the two mismatched student groupings per task. Confidence intervals
 1383 used 10,000 bootstrap resamples of these paired seed summaries. Intervals are pointwise rather than simultaneous across
 1384 all reported subgroup comparisons. All sources, rates, states, physical-domain checks and numerical tests are retained
 1385 under `source_data/morphology_conductance`; execution code is under `scripts/morphology_conductance`.

1386 **An interaction lower bound for this conductance model.** Let R_i denote a teacher terminal response, $A_j = (1 +$
 1387 $\sum_i b_{ji} + h_j z_j)^{-1}$ its proximal attenuation, and $\beta_j = a_j/(1 + a_1 + a_2)$. The sum in A_j runs over the teacher terminals
 1388 assigned to proximal compartment j . The teacher output is a sum of terms $\beta_j b_{ji} R_i A_j$. Let $p_{\text{teacher}}(i)$ and $p_{\text{student}}(i)$ denote
 1389 the proximal parent assigned to terminal i in the teacher and student, respectively, and write $p(i) = p_{\text{teacher}}(i)$. For
 1390 terminal i , set $j = p(i)$. Because terminal inputs and proximal shunt inputs are independent, a term whose terminal and
 1391 teacher shunt belong to different *student* proximal input blocks has the centered interaction

$$H_i = \beta_j b_{ji} (R_i - \mathbb{E}R_i)(A_j - \mathbb{E}A_j). \quad (97)$$

1392 Every admissible student output is additive across its two proximal input blocks. Each such H_i is orthogonal to all
 1393 functions additive across those blocks, and the omitted H_i are mutually orthogonal. Therefore, for every choice of the
 1394 student's conductances,

$$\mathbb{E}[(S_{\text{teacher}} - S_{\text{student}})^2] \geq \sum_{i: p_{\text{student}}(i) \neq p_{\text{teacher}}(i)} (\beta_{p(i)} b_{p(i)i})^2 \text{Var}(R_i) \text{Var}(A_{p(i)}). \quad (98)$$

1395 The bound is zero for compatible groupings. It is a lower bound for the larger class of arbitrary functions additive
 1396 across the student's proximal blocks, not a claim that the constrained rational student attains it. Its validity relies on the
 1397 specified independent inputs, identity somatic activation and absence of direct somatic synaptic inputs; it does not extend
 1398 automatically to arbitrary conductance architectures or nonlinear readouts. An equivalent obstruction is the absence of a
 1399 mixed derivative between a terminal input and a shunt input assigned to the other student subtree.

1400 We specified this diagnostic before the fresh training outcomes and retained the learner and rate-selection procedure
 1401 unchanged. Population lognormal moments were evaluated with 32-, 64- and 128-node Gauss–Hermite quadrature. The
 1402 primary bound used 128 nodes; the maximum change from 64 to 128 nodes was below 8×10^{-17} in normalized-MSE
 1403 units. This is a converged numerical evaluation of an analytic bound, not a formally certified quadrature error. Across the
 1404 fresh teacher draws, the mean incompatible population-NMSE lower bound was 0.013222, with a minimum of 0.005367
 1405 across task/candidate pairs. These are population lower bounds, distinct from held-out sample NMSE.

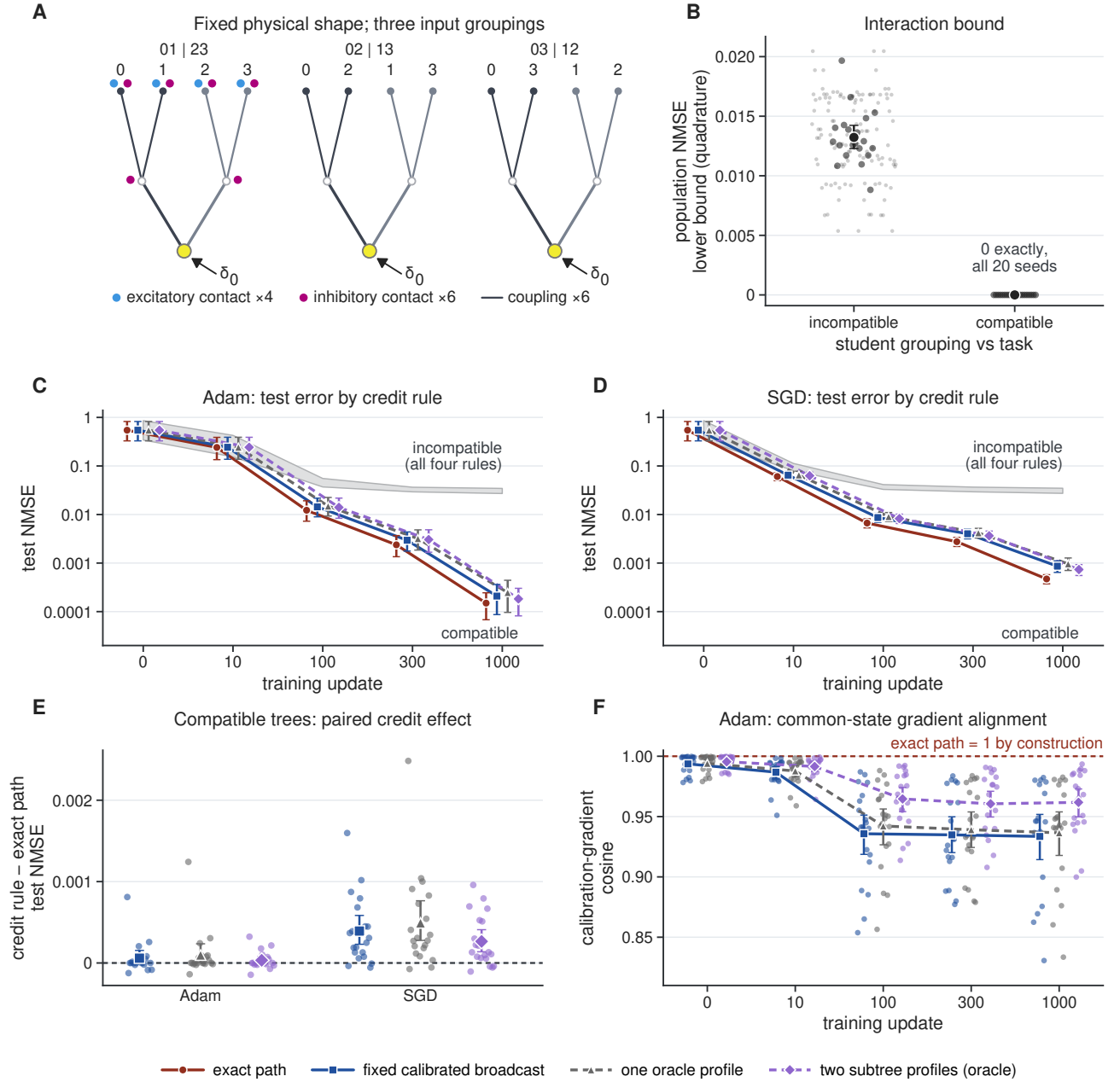


Figure S18: Input grouping and restricted feedback in positive-conductance trees. **A**, Seven-compartment directed shunting model (one physical shape; sixteen positive conductances: four excitatory and six inhibitory contacts, six couplings; soma-only readout) under the groupings 01|23, 02|13 and 03|12. The first tree marks the contacts (per leaf one excitatory, blue, and one inhibitory, carmine; one inhibitory per proximal compartment); its six edges are the couplings, the proximal subtrees two greys, the soma yellow with the somatic-error arrow δ_0 . Teacher-task permutations preserve the input-gradient spectrum. **B**, Analytic population lower bound from interactions crossing the student's two proximal input blocks (converged quadrature, not a certified bound): small pale dots, the 120 incompatible task/grouping pairs (six per seed, two values each); larger dots, the twenty seed means; mean bound 0.013222 population normalized mean squared error (NMSE); compatible groupings are exactly zero in all twenty seeds (superimposed). **C,D**, Held-out sample NMSE at checkpoints 0, 10, 100, 300 and 1,000 updates (ordinal axis, straight segments) under Adam and stochastic gradient descent (SGD) for exact path (dark red circles), fixed calibrated broadcast (blue squares), one oracle profile (grey triangles, dashed) and two subtree profiles (purple diamonds, dashed; main-text colours): coloured curves, compatible means over three task permutations within seed, with 95% whiskers offset per checkpoint; grey band, the union of the four rules' 95% intervals for the incompatible groupings (two per task; final means 0.029–0.032 NMSE under both optimizers); all rules share the initialization error. **E**, Paired compatible-tree NMSE differences from exact path at 1,000 updates for the three other rules under Adam and SGD: dots, the twenty independent seed means; dashed rule, zero; the Adam fixed-broadcast interval includes zero; the small SGD effect implies no universal need for precise credit. **F**, Aggregate gradient cosine on 256 calibration examples, compatible Adam models, same checkpoints: dots, the twenty seed means; symbols and whiskers, their mean and 95% interval (individual records reach 0.79); exact path is 1 by construction, drawn as the dashed reference; fixed broadcast and one oracle profile differ by at most 0.0064. Means and whiskers in **B–F**: 10,000 whole-seed bootstrap draws, pointwise 95% intervals.

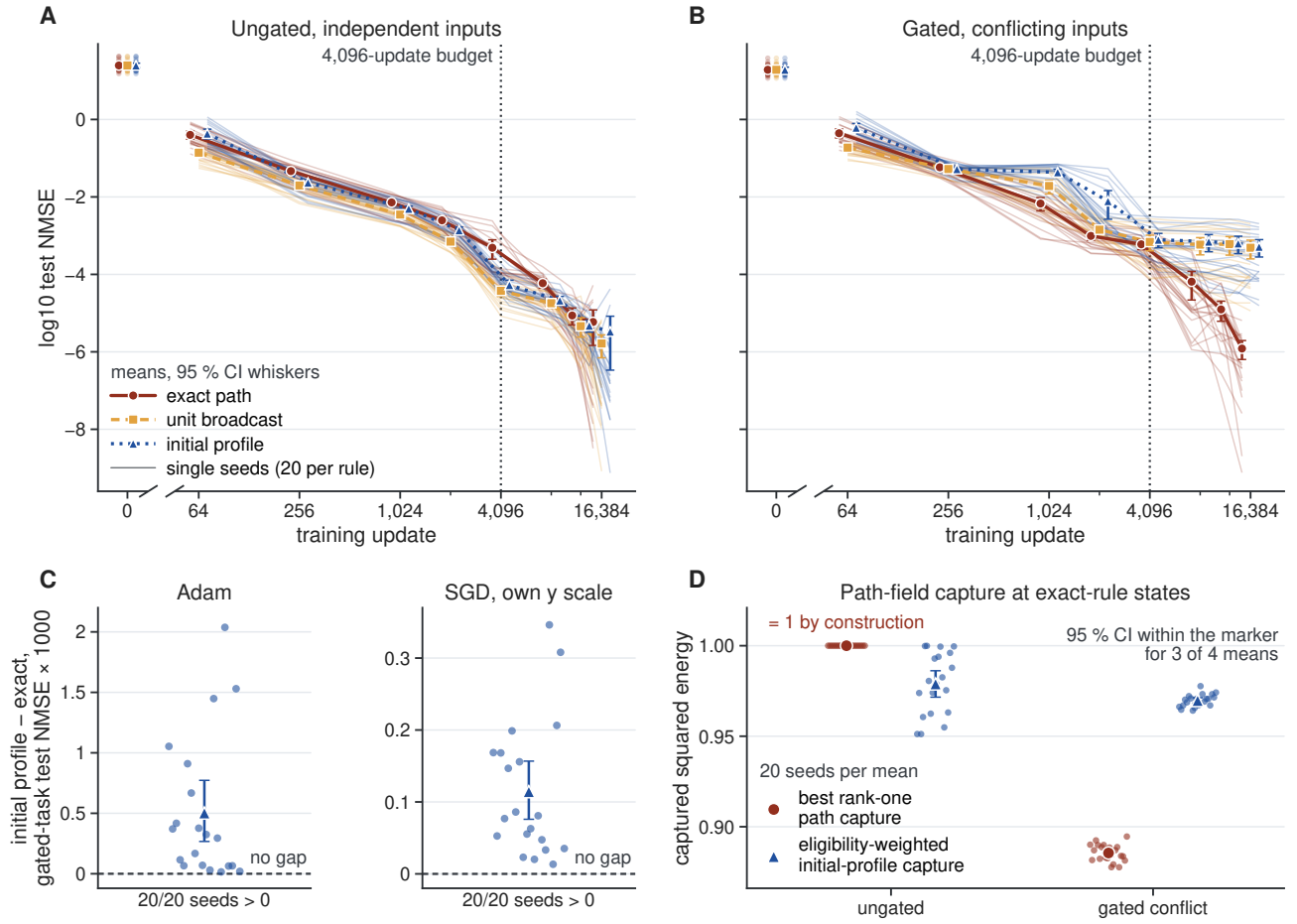


Figure S19: A first conductance task produces a small precision benefit from exact credit. **A,B**, Base-ten logarithm of test normalized mean squared error (NMSE) under Adam at the nine saved checkpoints (0, 64, 256, 1,024, 2,048, 4,096, 8,192, 12,288 and 16,384 updates) for ungated independent inputs and gated conflicting inputs in the 16-conductance model, with exact-path credit (dark red, solid, circles), unit broadcast (amber, dashed, squares) and the initial profile (the fixed calibrated broadcast; blue, dotted, triangles). The update axis is logarithmic from 64 with a marker at every checkpoint (minor ticks at 2,048, 8,192 and 12,288); update 0 is drawn left of the axis break. Symbols are means over 20 fresh seeds joined by straight segments, whiskers are 95% paired-seed bootstrap intervals, and the thin lines behind them are the 20 individual seed trajectories of each rule; the vertical dotted line marks the original 4,096-update budget. All three rules share each seed's initialization error (mean 24.8 in **A**, 19.2 in **B**), drawn at update 0. The key in **A** applies to **B**, which shares its y axis. All fits continue to 16,384 updates with unchanged optimizer and minibatch state. On the ungated task the three rules are indistinguishable at 16,384 updates (exact 5.9×10^{-6} [1.5×10^{-6} , 1.2×10^{-5}], unit broadcast 1.6×10^{-6} [7.0×10^{-7} , 2.7×10^{-6}]); the precision benefit of exact credit is a gated-task effect. Unit-broadcast and initial-profile curves differ here because global gradient clipping can break Adam's invariance to fixed positive coordinate scaling. **C**, Gated-task NMSE differences, initial profile minus exact credit (in thousandths of NMSE), at each fit's best validation checkpoint in the extended 16,384-update window, under Adam (left) and under SGD (right, on its own y scale), the latter at its separately selected rate with the same seeds and window. Blue points are the 20 paired seed differences; the blue triangle and whisker show their mean and 95% bootstrap interval, and the dashed line marks no gap. The Adam gap is 0.000502 [0.000267, 0.000772] and the SGD gap is 0.000114 [0.000076, 0.000157], each positive in all 20 seeds; under Adam the mean gap thus stays below 6×10^{-4} NMSE. **D**, Diagnostics at the exact-rule best-validation states distinguish the best rank-one capture of the six-compartment path field from capture by each student's fixed initial profile after weighting by local parameter eligibility. Both are squared-energy projection ratios with one freely fitted amplitude per example. Small points are the 20 seeds behind each mark; marks are means with 95% bootstrap intervals of the mean, narrower than the marker for three of the four means; ungated path capture is 1 by construction (effective rank 1.000 in every seed). High eligibility-weighted capture helps explain the small learning difference. Colours on this sheet: dark red circles are exact-path credit (**A,B**) and the best rank-one capture of its path field (**D**); amber squares are unit broadcast; blue triangles are the initial profile (**A,B**), its gap against exact credit (**C**) and its eligibility-weighted capture (**D**); line style also separates the three rules in **A,B** (solid, dashed and dotted, in that order). The gated and ungated tasks change both inhibition and distractor dependence, so this first study is a task-family comparison. All 216 development fits, 240 fresh fits and their continuations are retained in Source Data.

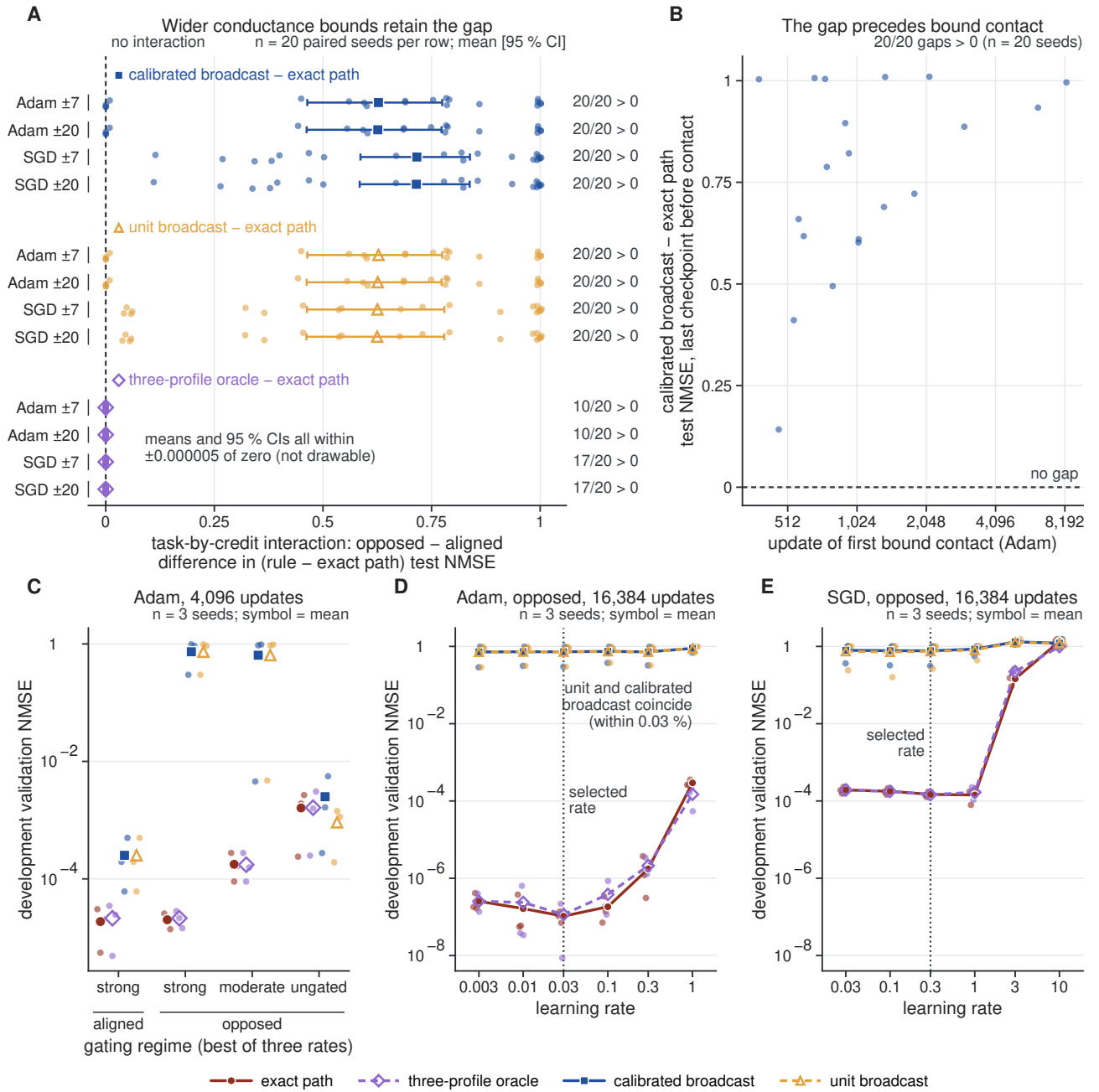


Figure S20: Parameter-range, development and rate controls retain the opposed-task broadcast deficit. **A**, Task-by-credit interaction (opposed minus aligned difference in rule-minus-exact-path test NMSE) for the calibrated broadcast, unit broadcast and three-profile oracle rules, each under original $[-7, 7]$ and wider $[-20, 20]$ log-conductance bounds with Adam and SGD. Small dots are the twenty paired seeds of each row, the large symbol their mean and the bar its 95% paired bootstrap interval; the dashed rule is zero, and each row prints its positive-seed count out of twenty. All 160 broadcast seed values are positive; the oracle means and intervals lie within $\pm 5 \times 10^{-6}$ of zero (too narrow to draw), with 10/20 (Adam) and 17/20 (SGD) seeds positive; unit-broadcast means are within 13% of the calibrated-broadcast means. **B**, The calibrated-broadcast gap at each seed's last saved checkpoint before its fit first reaches an original bound, against the update of that contact (Adam; logarithmic axis; one dot per seed, $n = 20$, no interval): all twenty gaps are positive, the dashed rule is zero, and the seeds with first contact at 1,039 and 1,041 updates nearly coincide. **C**, All four development regimes (aligned strong; opposed strong, moderate and ungated) under Adam at the original 4,096-update budget, each rule at its best of three rates (0.03 for all); small dots are the three development seeds, symbols their means; exact and oracle means agree within 13%, and the two broadcast means coincide except when ungated. **D,E**, Six-rate opposed-tuning sweeps under Adam and SGD at a 16,384-update budget; the dotted vertical line marks the originally selected rate (Adam 0.03, SGD 0.3). Small dots are the three development seeds behind every point and lines join the three-seed means; seed ranges (up to 25-fold) stay far below the three-decade rule separation at rates up to 1. Unit and calibrated broadcast coincide within 0.03% under Adam and 7.4% under SGD, so the open amber triangle sits on the filled blue square wherever they agree. One key serves **A** and **C–E**: dark red filled circles, solid lines, exact path; blue filled squares, solid lines, calibrated broadcast; amber open triangles, dashed lines, unit broadcast; purple open diamonds, dashed lines, three-profile oracle; in **A** the hue names the rule whose deficit is drawn, and the dots of **B** are the calibrated-broadcast blue.

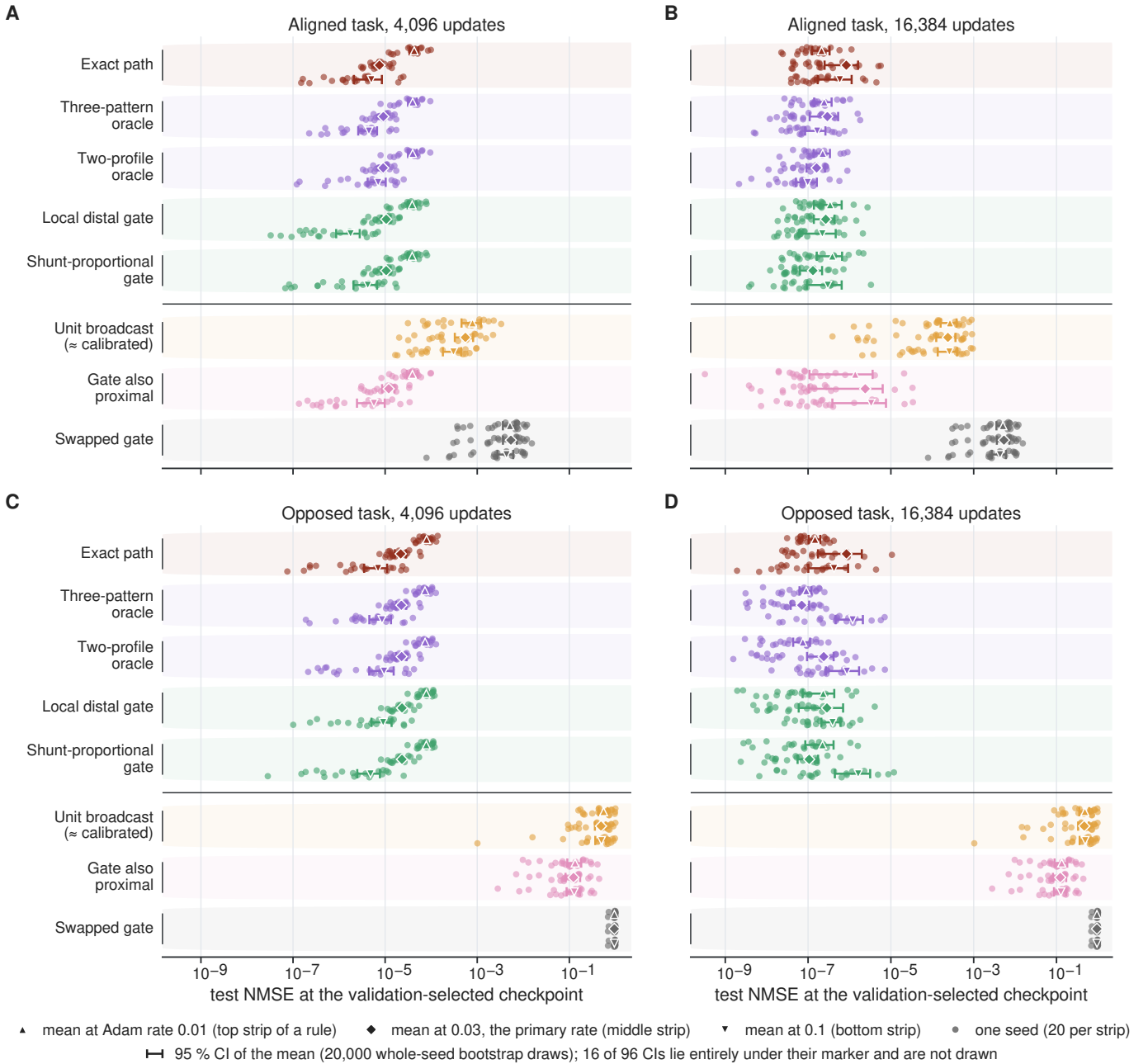


Figure S21: Local conductance gating is robust to rate choice but depends on where it is applied. **A,B**, Aligned tuning at 4,096 and 16,384 updates. **C,D**, Opposed tuning at the same budgets. Each row is one credit rule and holds three strips, top to bottom the Adam rates 0.01 (triangle up), 0.03 (diamond; the primary rate) and 0.1 (triangle down). In every strip the small dots are the test NMSE at the validation-selected checkpoint within the budget for each of $n = 20$ new paired seeds, the marker their arithmetic mean and the whisker its 95% percentile interval from 20,000 whole-seed bootstrap draws; the x axis is one logarithmic scale shared by all four panels, labelled on **C** and **D**. Sixteen of the 96 intervals are narrower than 0.18 decade and lie entirely under their mean marker, so they are not drawn: all six swapped-gate strips in **C** and **D**, the 0.03 strips of the exact path and the two oracles in **A**, and the 0.01 strips of the exact path, both oracles and both local gates plus the 0.03 strips of both local gates in **C**. Colour is the rule family, as in main Fig. 5: dark red exact path, purple oracle projections, green local gates, amber broadcast, rose gate also proximal, grey swapped gate; the pale band behind each row is that row's colour and the thin rule separates the five rules that learn accurately on both tasks (above) from the three that fail on at least one task (below). Row labels follow main Fig. 5: 'Two-profile oracle' is the two leaf patterns with unit proximal credit, 'Local distal gate' the hard distal gate and 'Shunt-proportional gate' its continuous form. The 'Unit broadcast ($\approx$ calibrated)' row draws the unit-broadcast rule only: the calibrated broadcast has a mean equal to it to two significant figures in all twelve task, budget and rate cells, so its row is not repeated; individual seeds differ between the two rules by up to 3.1-fold. In **B** the gate-also-proximal means ($1.4\text{--}3.3 \times 10^{-6}$) sit 17–25-fold above their medians ($0.8\text{--}2.0 \times 10^{-7}$), matching the accurate rules' $0.5\text{--}1.9 \times 10^{-7}$ because of one or two outlier seeds, the isolated dots right of that row's cloud: an outlier effect, not an aligned-task deficit, so the row sits in the lower block for its opposed-task failure alone. About tenfold outlier inflation likewise moves the two-profile oracle mean at 0.03 and the shunt-proportional gate mean at 0.1 in **D**.

1406 S6 Serial physical computation and training budgets

1407 6.1 Generator, architecture and primary comparison

1408 The hierarchical_gain_load task supplies 64 positive excitatory and 64 positive inhibitory rates. Fine, coarse and
1409 global Gaussian node fields generate three mean-corrected lognormal gain factors (hierarchy H3). The product of these
1410 factors modulates distal class-bearing excitation; separate inhibitory sensor blocks report each factor. For a signal-bearing
1411 coordinate,

$$x_E = \max\{b_E + y\Delta + \varepsilon_E, 10^{-4}\} \exp\left(\sum_{\ell=1}^3 \sigma_\ell z_\ell - \frac{1}{2} \sum_{\ell=1}^3 \sigma_\ell^2\right), \quad (99)$$

1412 where $y \in \{0, 1\}$ is balanced class, $\Delta = 0.80$, $\sigma_\ell = 0.25$, z_ℓ are independent standard Gaussians and ε_E is independent
1413 Gaussian noise of SD 0.08. The same noise-floor-gain order applies to inhibitory rates. Class-independent mid-
1414 dle/proximal excitation has baseline 0.02, so flooring makes its effective rate noise nonzero-mean. Each dataset contains
1415 10,000 balanced examples.

1416 Sensor alignment α changes correspondence without changing marginal Gaussian variance:

$$s_\ell(\alpha) = \alpha z_\ell + \sqrt{1 - \alpha^2} \xi_\ell, \quad \xi_\ell \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad (100)$$

1417 with sensor noise independent of latent gains. Alignment uses $\alpha = 1$; independent sensors use zero. A nonzero cyclic
1418 trial shift instead removes trial-specific access while preserving the entire inhibitory field. Reversed placement exchanges
1419 fine/global feature ranges across inventory tiers, preserving their 21/22/21 widths. An earlier three-cycle assignment
1420 changed candidate slots and was excluded before inference.

1421 Branch factors run soma to distal. For factors $[b_1, \dots, b_D]$, the nonsomatic count is $\sum_{j=1}^D \prod_{r=1}^j b_r$. D1 [8], D2 [2, 3]
1422 and D3 [2, 1, 2] each contain eight units per soma; D3's unary middle stage preserves that inventory. Structured masks
1423 place all tiers in D1, fine/coarse distally and global proximally in D2, and one tier per stage in D3. Every architecture has
1424 66,178 trainable parameters, 14,336 active synapses, 21,760 candidate slots and 2,944 persistent state scalars per sample.
1425 Compartments use the directed positive-conductance division, identity reactivation and softplus-positive synapses.

1426 Excluded calibration seeds 10100–10141 tested signal contrast, gain shift and child conductance. Contrast 0.24 with
1427 train/test gain SD 0.25/1.2 remained at chance despite improved training fits at depth. Coupling and contrast sweeps
1428 selected child conductance 16, contrast 0.80 and matched SD 0.25; contrast 0.72 missed the predefined D1 accessibility
1429 threshold (two-seed mean test accuracy at least 0.60 at every depth; D1 reached 0.597 at contrast 0.72 versus 0.609 at
1430 contrast 0.80). All failed calibration settings remain in Source Data.

1431 The primary 270 fits crossed D1–D3, ten paired seeds 10200–10209 and nine conditions: aligned shunting BP;
1432 independent-sensor, shuffled-sensor or reversed-placement BP; aligned raw-additive BP; and aligned/reversed LocalCA
1433 with shared-soma or exact-path credit. The protocol required at least 9/10 aligned-BP seeds above 0.55 at every depth
1434 and every mean below 0.98. All conditions passed completeness, finite-metric, resource and no-fallback checks. Training
1435 used a 180-epoch cap and validation-loss early stopping with patience thirty. The BP recipe used Adam at 0.003;
1436 LocalCA retained split parameter groups with nominal rate 0.0008. Group rates, boosting, decay and update options are
1437 retained in the resolved configurations.

1438 At that budget, aligned-BP D3 minus D1 was 30.86 percentage points; alignment-minus-control depth interactions
1439 exceeded 30 points. Aligned additive BP declined with depth. These contrasts separate forward composition from a
1440 credit-rule effect, whose budget dependence is evaluated below. A post hoc common-checkpoint audit found exact-path
1441 LocalCA gradients collinear with autograd (minimum cosine 0.99999999999994), while shared-soma cosine declined
1442 from 0.946 at D1 to 0.767 at D3. This gradient diagnostic does not establish a converged learning requirement.

1443 6.2 Nonserial and point-model controls

1444 An all-active grouped star retains branch layers, masks and parameters but lets each nonsomatic layer read external
1445 inputs independently; child-aggregation parameters contribute only once at the soma. D1 logits match exactly. A
1446 literal grouped-point control additionally replaces child aggregation by direct projections to the soma, copying the
1447 corresponding initial scalar conductances. These controls retain resource counts while removing serial inheritance of
1448 branch states.

1449 The star/point/credit-coordinate study comprised 200 fits on the same ten H3 seeds: 140 initial controls and sixty
1450 optimizer-matched soma-broadcast fits. Serial exceeded star by 30.81 percentage points at aligned D3 (95% paired

interval, 30.41–31.26), but fell below it by 0.43 points after reversal. Two unconstrained point MLPs matched active parameters (15,055) or nearest realizable total parameters (66,200 versus 66,178). Their mean accuracies, 0.9861 and 0.9902, exceeded the tree. Thus the serial advantage is relative to matched branch resources, not arbitrary point-network expressivity.

Soma-broadcast autograd preserves local synaptic derivatives but replaces nonsomatic loss derivatives by each unit's somatic derivative. Full BP exceeded it by 0.64 percentage points under the BP optimizer. Under the LocalCA optimizer, broadcast autograd and shared-soma LocalCA differed by only -0.06 points (-0.23 to 0.09), whereas changing the broadcast optimizer cost 13.47 points at 180 epochs. The optimizer-matched comparison was specified after discovering the recipe difference. It isolates that implementation contrast; autograd is not a biological mechanism. Full configurations and outcomes are in `source_data/point_dendrite_credit_controls/`.

Sixty additional literal grouped-point H3 fits used aligned/reversed placement and D1–D3 on the same seeds. Grouped-point means remained near 0.61 across depth. Serial-minus-grouped at aligned D3 was 30.87 points, reversing to -0.44 under placement reversal. These observations agree with the star control without retaining its child-aggregation modules.

6.3 Alignment, hierarchy and task-family sensitivities

The alignment interpolation was specified after endpoint outcomes but before intermediate fits. Ninety runs crossed $\alpha = 0.25, 0.50, 0.75$, three depths and the original ten seeds. D3-minus-D1 effects at $\alpha = 0, 0.25, 0.50, 0.75, 1$ were 0.019, 0.207, 0.820, 3.126 and 30.855 percentage points. The prespecified within-seed linear summary had mean slope 25.84 points per unit alignment, but the final interval contributed 27.73 points: the response is strongly nonlinear. Software records identify an unused optional-rule edit; all unique conditions passed resource checks (`source_data/physical_alignment_dose/`).

The H2 follow-up fixed two 32-feature tiers, inventory 4/4 and D1 [8]/D2 [4, 1] before outcomes. Ten new seeds 10300–10309 crossed aligned/reversed placement, serial or literal grouped-point BP, and shared-soma or exact-path LocalCA (160 fits). No inherited signal, conductance or optimization setting was retuned. Aligned serial BP gained 30.84 points from D1 to D2; the placement interaction was 32.40 points. Grouped-point depth effects remained near zero. This is replication within the same synthetic mechanism family. Together with the sixty H3 grouped-point fits, all 220 outcomes are in `source_data/remaining_physical_experiments/`.

H4 used four 16-feature tiers, inventory [4, 2, 1, 1] and D1–D4, adding D4 [1, 1, 2, 2]. All depths retain the same eight-unit resources. The 360 fits crossed aligned/reversed placement, serial/grouped-point BP, both LocalCA rules and aligned raw-additive BP over seeds 10400–10409. An automatic D3 partition error stopped ninety configurations before outcomes; explicit grouping [4], [2], [1 + 1] restored the intended counts, and only failed configurations were repeated. The main contrasts, 50,000-draw intervals, exact sign-flip tests and BH family were specified before analysis, with positive claims additionally requiring 8/10 concordant signs. Aligned BP accuracy peaked at D3: D4 minus D3 was -1.46 points (-1.74 to -1.16), although D4 minus D1 remained 25.63 points. This falsified the predefined simple depth-equals-hierarchy crossover. Exact-path LocalCA's 0.60-point D4–D3 gain failed the direction criterion (7/10); the shunting-minus-additive increment was also inconclusive. All reversed and null endpoints remain in `source_data/physical_depth_h4_factorial/`.

A same-seed computational repeat under source commit a99c3a7 covered all 220 H2/grouped-point fits and 210 principal H3 fits. Median absolute accuracy drift was 0.030 percentage points, mean drift 0.154 and maximum drift 15.01 in a raw-additive control; 357/430 pairs were within 0.1 points. Principal shunting and local-credit directions persisted. Supplementary Table S9 retains this unfavorable reproducibility range; these reused seeds are not fresh replication.

The fixed-D3 task-family study used seeds 10500–10509 and crossed three generators, $\alpha = 0, 0.5, 1$, serial and grouped-point composition and BP or exact-path LocalCA (360 fits). Nested factors use Eq. 99. Flat factors retain the same product and marginal variances but assign each independent factor to the same deterministic eight-bin spatial partition. Local-ratio inputs give excitation and inhibition at each level correlated cumulative hierarchical fields, making divisive information locally available. All families retain 10,000 balanced examples, distal-only class signal and identical model resources. Primary architecture-by-alignment and between-family contrasts used 50,000 paired bootstrap draws and an 8/10 sign criterion. BP alignment interactions were 30.41, 22.26 and 1.30 percentage points for nested, flat and local-ratio tasks; LocalCA effects were 27.70, 12.37 and -11.68 . The local-ratio reversal is retained in `source_data/task_family_alignment/`.

1501 6.4 Longer training and metric dependence

1502 All fifty aligned D3 fits in five relevant credit/optimizer conditions reached the original 180-epoch cap with decreasing
1503 validation loss. A protocol-led extension retained those conditions and ten exact-BP D1 references, using the same ten
1504 seeds, configurations, patience and validation-based selection. Only the cap (600 epochs) and output names changed.
1505 Historical files lacked optimizer and random states, so runs restarted from their original seeds with a99c3a7; they were
1506 not exact checkpoint continuations. Four seed blocks ran on GPU and six on CPU, held fixed across all six conditions
1507 within each seed.

1508 Current and best weights, optimizer and all random states were saved at 180, 300 and 600 epochs or earlier termination.
1509 Final weights exactly matched validation-selected states. Budget comparisons use the best state within the first 180
1510 epochs and within the full new trajectory; test accuracy and cross-entropy never select states. Stopped curves are carried
1511 forward to preserve ten seeds per mean and are marked separately. Source Data retain 34,159 observed epochs, 120
1512 original/extension configurations and sixty complete run records.

1513 At 600 epochs, exact-BP D3 minus D1 was 38.17 percentage points (95% interval, 37.98–38.39). LocalCA exact
1514 minus shared changed from +10.86 points at 180 to −1.52 (−1.83 to −1.23), reversing in all ten seeds. Exact LocalCA
1515 nevertheless retained lower test cross-entropy, 0.2544 versus 0.2758. The BP-recipe exact-minus-broadcast accuracy
1516 difference was 0.48 points; the broadcast BP-minus-LocalCA-recipe difference was 1.54. Intervals use 10,000 paired
1517 bootstrap draws and are descriptive.

1518 All fifty D3 fits and two D1 fits still reached the 600-epoch cap; D3 validation loss continued decreasing over the final
1519 thirty epochs. Relative to historical trajectories, maximum validation-loss drift was 0.005885; median and maximum
1520 selected-accuracy drift were 0.070 and 1.830 percentage points. Comparisons therefore use windows of the same new
1521 run. The extension establishes budget and metric dependence, not convergence or fresh-seed confirmation.

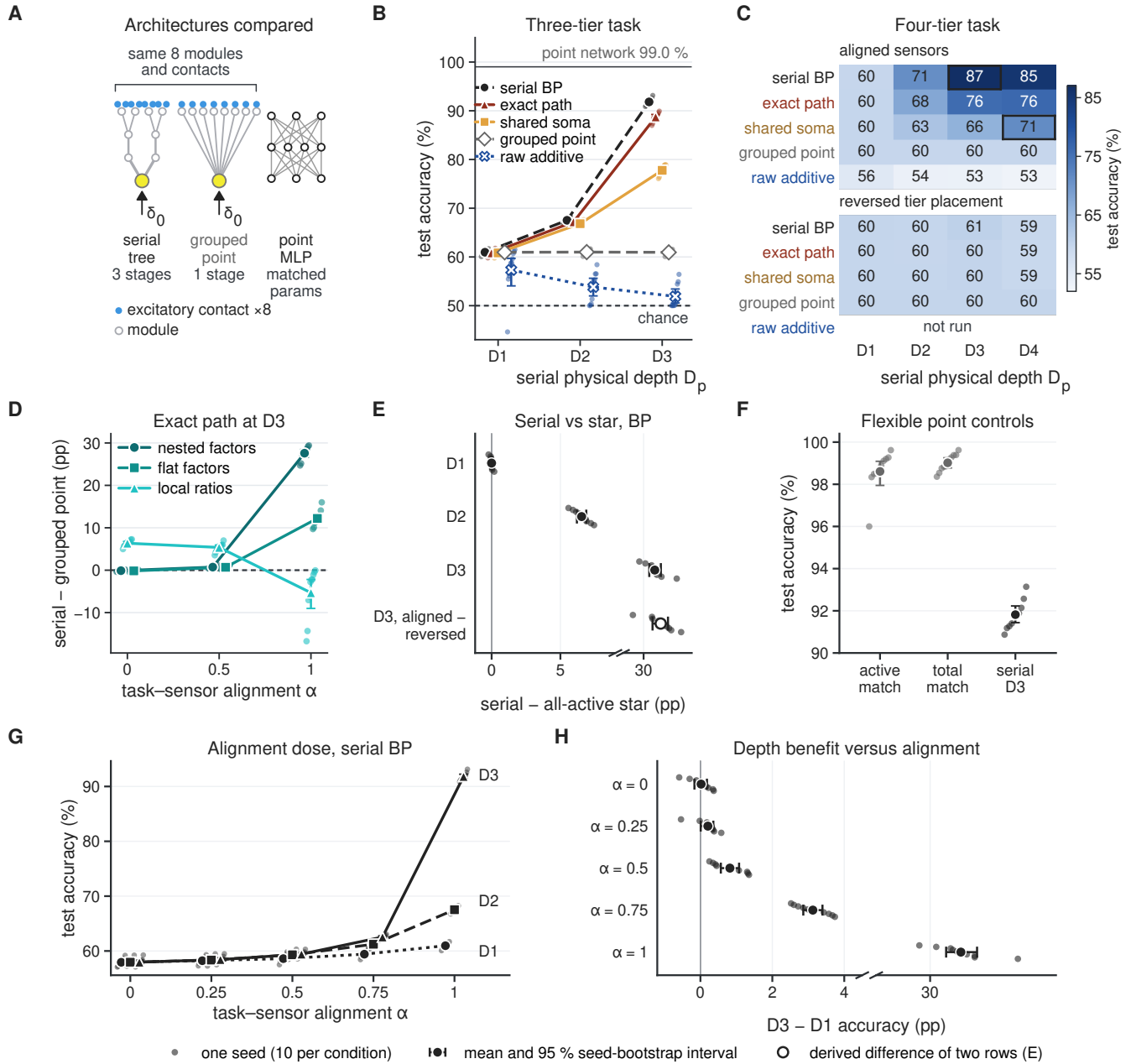


Figure S22: Task-matched serial computation differs from grouped and flexible point controls. Throughout, small dots are the ten seeds per condition, large markers their mean with a 95% seed-bootstrap interval (paired for differences), series dodged in x , and most intervals in **B,G** narrower than the marker. **A**, Serial D3 tree ([2, 1, 2], eight modules in three stages), resource-identical grouped point (same modules, one stage) and parameter-matched flexible point MLP; blue dots, the eight excitatory class-bearing contacts; open rings, modules; yellow disc, soma; black arrow, somatic error δ_0 . **B**, Original-budget three-tier aligned task, D1–D3, five arms in the main Fig. 6D palette: black dashed circles, serial (exact) BP; dark red triangles, exact-path LocalCA; amber squares, shared-soma LocalCA; grey dashed open diamonds, the grouped point; blue dotted open crosses, raw-additive BP. Dark rule and grey band, the total-parameter-matched ceiling (99.0%, 95% interval); dashed rule, chance (50%). **C**, Four-tier task including D4 under aligned sensors or reversed tiers; cells print mean accuracy (%), one ramp spanning the observed 52–87%, row labels in the arm hues of **B**; the raw-additive arm ran aligned only (its reversed row, “not run”). An outlined aligned cell marks a row’s best depth, 95% interval clearing the runner-up’s; the reversed block is flat by design (largest lead 0.8 points, serial BP D3 over D1). **D**, Serial-minus-grouped-point accuracy at D3 for nested-factor (dark teal circles), flat-factor (teal squares) and local-ratio (light teal triangles) families under exact-path LocalCA. **E**, Serial-minus-star accuracy at D1–D3 under BP, the star the resource-identical all-active grouped control (Source Data all_active_star); open marker, derived aligned-minus-reversed D3 difference; axis break, empty 9–28.5 pp span. **F**, Flexible point networks matched to active or total parameter count (grey) beside the serial D3 run (black), the exact-BP run of **B**. **G**, Serial-BP accuracy against task-sensor alignment α for D1 (dotted, circles), D2 (dashed, squares) and D3 (solid, triangles), all black, labelled at $\alpha = 1$. **H**, The D3-minus-D1 advantage at each α , axis broken between 4.5 and 28.5 pp; from $\alpha = 0.25$ to 0.75 it grows by 2.92 points (2.65–3.17), within-seed linear slope 25.8 points per unit α (25.4–26.3). Main Fig. 6A,D re-render **A,B** from the same seeds; the two-tier task is Supplementary Fig. S24B.

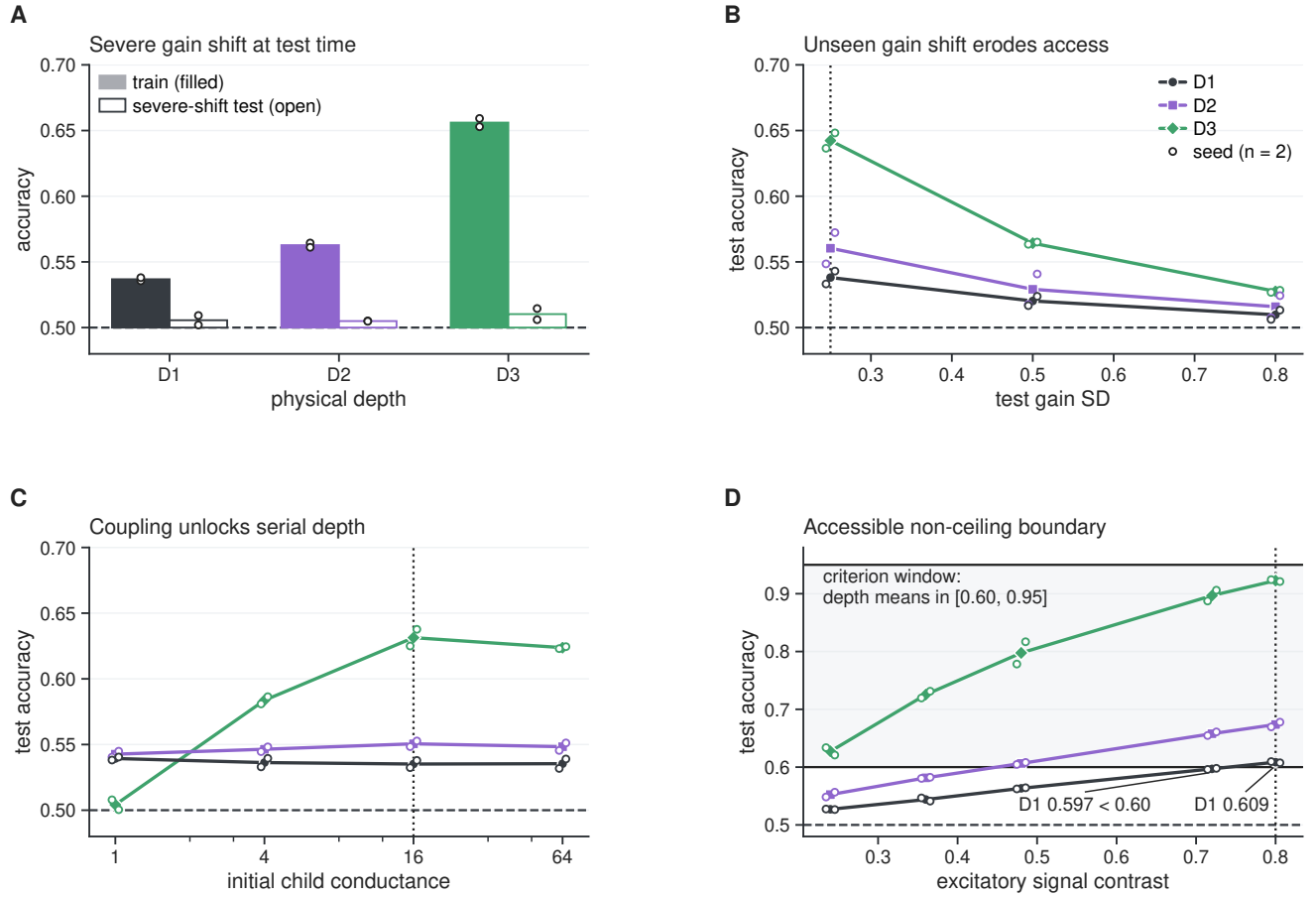


Figure S23: Transparent calibration of the nonlinear physical-depth operating point. All panels use the three-level hierarchical gain-load task and serial shunting morphologies D1 [8], D2 [2,3] and D3 [2,1,2], each with eight nonsomatic compartments per soma. They test the prerequisite for the main hypothesis: whether the mechanism-matched divisive task is accessible at every depth before comparing depth benefits. **A**, The original aligned shunting-backpropagation (BP) pilot at signal contrast 0.24 fit the training distribution increasingly with depth but remained at chance under the prespecified severe gain shift (test gain standard deviation (SD) 1.2). Bars grow from chance (0.5) and show two-seed means; colour encodes depth as in **B–D**, filled bars are the training split, open bars the severe-shift test split, and open circles the individual exploratory seeds. **B**, With training gain SD fixed at 0.25, increasing unseen test gain variation eroded accessibility at every depth. The dotted line marks the matched train/test condition. **C**, At matched gain and signal contrast 0.24, stronger child coupling selectively made serial D3 computation accessible; the dotted line marks child conductance 16 used subsequently. **D**, At matched gain and child conductance 16, increasing excitatory signal contrast moved D1–D3 away from chance without saturating D1 or D2. The shaded window marks the accessibility criterion frozen before the ladder was run: every depth’s two-seed mean test accuracy in [0.60, 0.95], each seed above 0.57 and D3 minus D1 at least 0.02 in both seeds. Contrast 0.72 failed it by 0.003 (D1 mean 0.597); the separately run 0.80 boundary (dotted line; D1 mean 0.609) met it and selected the operating point used for the analysis seeds. Lines and filled symbols show two-seed means and small open circles the two exploratory seeds, dodged slightly in x ; no confidence interval is drawn. Panels **A–C** share an expanded ordinate (0.48–0.70); **D** uses the full range. All pilot, ladder and boundary seeds were excluded from the confirmatory contrasts.

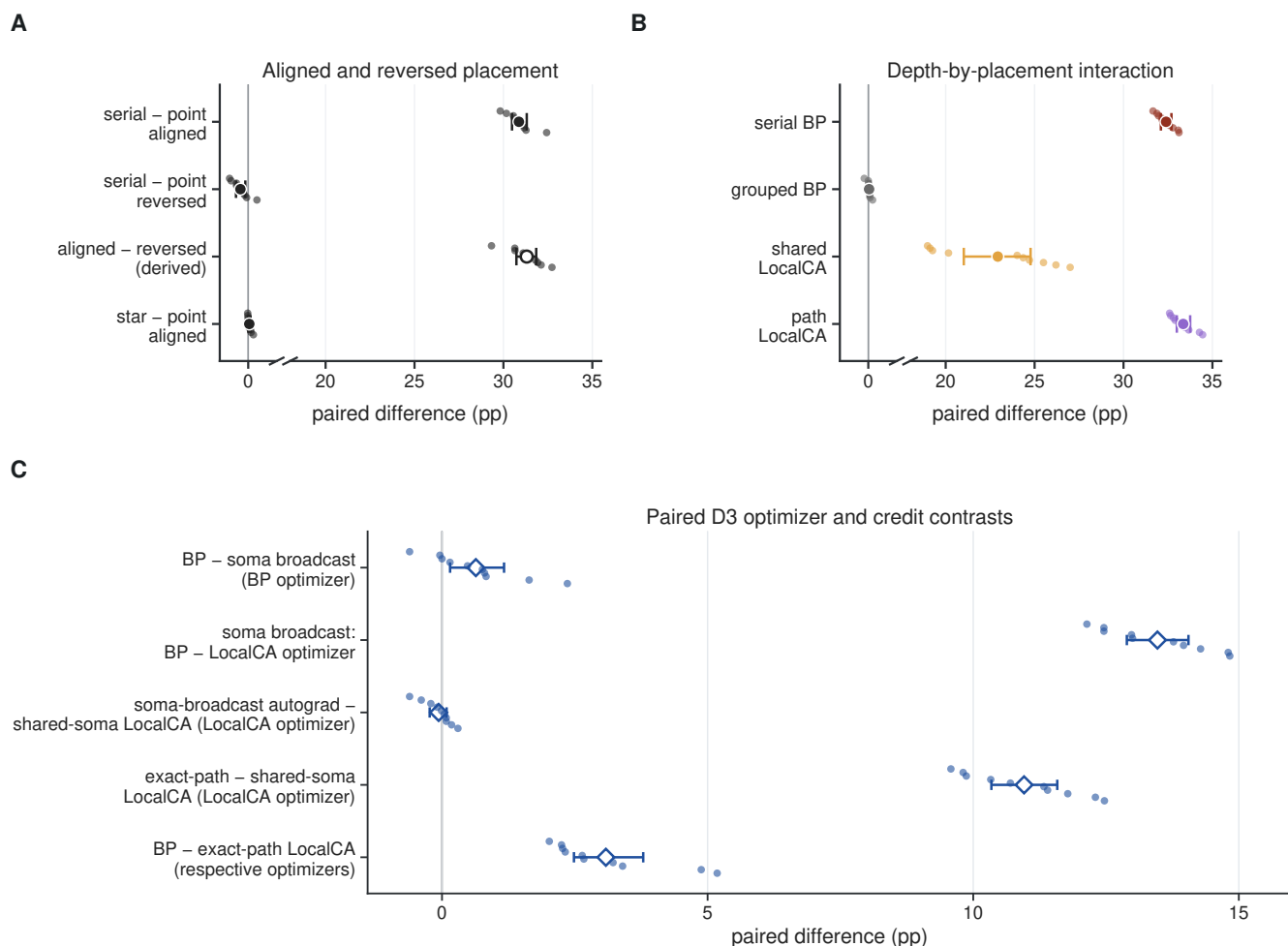


Figure S24: Physical-depth conclusions depend on sensor fidelity, task hierarchy and optimization. **A**, Serial-minus-grouped-point accuracy at D3 under matched (aligned) and reversed sensor placement, the derived aligned-minus-reversed difference (open marker) and the star-minus-point architecture control, in percentage points. **B**, Separate independently seeded two-level task with D1 [8] and D2 [4, 1]: the depth-by-placement interaction (aligned D2-minus-D1 gain minus the reversed gain) for serial and grouped-point backpropagation and for shared or exact-path LocalCA; arm colours follow Supplementary Fig. S22B. **A,B** share one broken axis with 5 pp ticks (the empty 1.6–18 pp span is removed) and draw the ten paired seeds behind every mean. **C**, Original-budget D3 optimizer/credit-coordinate contrasts in the serial conductance morphology [2, 1, 2], in drawn order: BP minus soma broadcast under the BP optimizer, 0.64 points (95% interval 0.15–1.17); soma broadcast, BP minus LocalCA optimizer, 13.47 (12.89–14.05); soma-broadcast autograd minus shared-soma LocalCA, –0.06 (–0.23 to 0.09); exact-path minus shared-soma LocalCA, 10.96 (10.34–11.58); BP minus exact-path LocalCA, 3.08 (2.48–3.79). Means and intervals use ten paired seeds and 95% bootstrap intervals; panel **C** is generated from `source_data/point_dendrite_credit_controls/` and is registered in the Source Data manifest. These earlier recipe-specific effects do not imply a converged credit advantage: at 600 epochs the LocalCA accuracy difference reverses, while exact credit retains lower cross-entropy (main Fig. 6). All D3 continuations hit the cap; D1 usually stops early.

S7 Anatomical dictionaries and controlled field alignment

7.1 Reconstruction, cohorts and electrical normalization

The initial cohort comprises eight layer 2–5 intratelencephalic cells from MICrONS [8]. Stable nuclei were resolved in minnie65_public materialization 1822; skeletons came from the MICrONS skeleton service and incoming contacts from synapses_pni_2. Degree-two chains were collapsed to branch-to-branch cables. Contacts were assigned to the nearest dendritic node and retained within $5\mu\text{m}$. Across cells, 28,669 E/I-classified contacts passed, with mean mapping fraction 0.980. Root traversal checked acyclicity and a unique path to the soma. Cell identifiers, contact counts and source assets are indexed in Supplementary Table S12.

Presynaptic coarse type supplied E/I identity where available. Otherwise the in-development synapse_target_predictions_ssa_v2 table classified high-confidence spine targets as excitatory and shaft/soma targets as inhibitory. This proxy table lacks a stable public release identifier. Direct/proxy agreement was 73.01% among 2,012 jointly labeled contacts (E/E 489, E/I 84, I/E 459, I/I 980). Principal comparisons were repeated on the 2,077 directly typed contacts. Coverage and error asymmetry are observed properties, not missing-at-random assumptions.

A non-overlapping robustness cohort was defined from a 55-cell convenience export before outcomes, resolved in public v661 and excluding the original eight by stable nucleus. Its 47 retained cells are from the same mouse and an earlier reconstruction release. Direct presynaptic labels were available for 11,024 incoming contacts (4.71%) before mapping; 10,516 mapped contacts entered analysis. This layer-imbalanced cohort is not a population or independent-animal replication.

The second-animal cohort used published Pinky v185 mesh, soma-valence and dense-synapse data [9, 10]. Before mesh or selected-cell contact counts were inspected, 362 excitatory soma rows were ordered along the second volume coordinate and split into twelve equal-count strata. Each stratum contributed the root nearest the global first/third-coordinate medians, with root-ID tie-breaking. This deterministic rule spans the volume while reducing boundary truncation; it is not random sampling. Mesh processing retained the largest face-plus-link-edge component and used meshparty TEASAR with $7.5\mu\text{m}$ soma collapse and $12\mu\text{m}$ invalidation. Radius was half the opposite-surface ray distance at mesh anchors (embreeex 2.17.7.post7). Only presynaptic roots labeled in the published soma-valence table supplied E/I identities.

Pinky inherited v661 compression, mapping, normalization, dictionary ranking and 500-shuffle settings. Inclusion required at least twenty mapped typed inputs and three inhibitory inputs. All twelve cells processed; ten qualified, with excluded cells having nineteen and thirteen inputs. Their route-generated $K = 4$ ancestry advantages over random routes, depth bins and shuffling agreed in direction with the other mouse. Cell-bootstrap intervals describe stability within each volume; two mice do not support a population-level animal test. In the equal-broadcast analysis below, only eight Pinky cells qualify at $K = 8$ and none at $K = 16$; lower-budget results retain the other eligible cells.

Raw axial coupling scales as radius squared over length, divided by its within-cell positive median and clipped to 0.05–20. Leak scales as radius times the larger of length and radius, similarly normalized and clipped to 0.1–10. Synaptic areas are divided by the median nonzero total area. These are normalized passive sensitivity weights, not fitted membrane conductances. The reconstructed model uses reciprocal coupling and a symmetric conductance matrix; the directed path product of Eq. 10 applies only to the artificial network.

For excitatory-bearing segment i and inhibitory-bearing route site k , define

$$\Gamma_{ik} = A_{ik}\beta_k, \quad (101)$$

where A_{ik} indicates that k lies on the path from i to soma and β_k is its normalized inhibitory fraction. This ancestry kernel approximates route weighting; it is not the reciprocal Green's-function derivative. Excitatory-area-weighted singular values s_j give participation rank $(\sum_j s_j^2)^2 / \sum_j s_j^4$.

7.2 Mapping, labels and cable compression

The eight-cell uncertainty analysis crosses mapping thresholds 2, 5 and $10\mu\text{m}$ with direct-only or hybrid labels (proxy probability at least 0.8). Radius factors are $\exp(\sigma_r z - \sigma_r^2/2)$ for independent standard Gaussian z , with $\sigma_r = 0, 0.25, 0.5$ and five paired draws at each nonzero dose. Every condition reselects up to four routes by the same leverage. Jaccard overlap measures route changes; capture uses the fixed nominal field, sites and weights, isolating dictionary sensitivity rather than changing both sides of the comparison.

1569 Compression preserves total length and length–radius product to rounding error, but mean-radius resistance $\ell/\bar{r}^2$
1570 differs from the series sum $\sum_e \ell_e/r_e^2$. Median relative error was -0.00200 , with minimum -0.99717 . A second
1571 normalization uses inverse series resistance, retaining leak, synapses and clipping; radius perturbations rescale re-
1572 sistance inversely with squared radius. All 1,056 conditions remain. Nominal mean capture was 0.3293 under
1573 mean-radius and 0.3375 under series normalization; small mean changes do not negate substantial local resistance
1574 errors. Draws were averaged within cell and intervals used 5,000 cell resamples. Audits and cache hashes are in
1575 `source_data/review_morphology_uncertainty/`.

1576 7.3 Route-generated capacity and spatial scales

1577 Route-generated fields multiply independent amplitudes by Eq. 101. Dictionaries comprise a dense PCA oracle, selected
1578 real ancestry routes, random nonempty routes, cable-depth bins and independently shuffled ancestry columns preserving
1579 values and sparsity. Real routes rank by inhibitory fraction times descendant excitatory-area fraction. Fields and
1580 dictionaries are both transformed by square-root excitatory area before least squares (Eq. 33). Twenty Monte Carlo
1581 streams each contain 384 fields and 64 randomized dictionaries per cell and budget; streams are averaged within cell
1582 before inference.

1583 At $K = 8$, original ancestry capture was 0.487 with 6.92% dense wiring. The same model-matched construction
1584 favored ancestry in the direct-label subset and all 47 non-overlapping cells. These positive controls measure internal
1585 compressibility of the route family. The older replication reports one minus residual norm, which differs from squared-
1586 energy capture; Source Data retain each endpoint’s definition.

1587 A deterministic unbalanced tree-Haar basis tests route scales on all 55 cells, with the 47-cell cohort primary. A region
1588 R is split at the descendant subtree closest to half its excitatory weight, giving $A \cup B = R$. For site weights w_i and region
1589 sums W_A, W_B, W_R ,

$$h_{R,i} = \begin{cases} \sqrt{w_i} \sqrt{W_B/(W_A W_R)}, & i \in A, \\ -\sqrt{w_i} \sqrt{W_A/(W_B W_R)}, & i \in B, \\ 0, & i \notin R. \end{cases} \quad (102)$$

1590 With $h_{0,i} = \sqrt{w_i/\sum_j w_j}$ these contrasts form a complete orthonormal basis [11]. All bases were full rank, with maximum
1591 orthonormality error 4.44×10^{-16} . For $W = \text{diag}(w_i)$,

$$\tilde{\Gamma} = (I - h_0 h_0^T) W^{1/2} \Gamma, \quad p_r = \frac{\|h_r^T \tilde{\Gamma}\|^2}{\|\tilde{\Gamma}\|_F^2}. \quad (103)$$

1592 Coarse supports exceed one quarter of total weight; intermediate supports exceed one sixteenth but not one quarter;
1593 others are fine. A scale with m_s of $n - 1$ nonconstant dimensions has isotropic reference $m_s/(n - 1)$. The anatomy null
1594 independently permutes raw route columns before weighting and centering, using 256 draws per cell.

1595 Coarse modes contained 0.4065 of nonconstant energy but 0.1923 of dimensions in the 47-cell cohort. Enrichment
1596 over isotropic noise was 2.351 (95% interval, 1.995–2.880). Actual-minus-permuted coarse energy was 0.0345 (-0.0025
1597 to 0.0713), and intermediate/fine permutation contrasts were also null. The original eight-cell positive permutation
1598 contrast did not reproduce in the larger cohort. Coarse enrichment therefore does not establish special fine-layout
1599 organization.

1600 7.4 Passive response fields and equal broadcast budgets

1601 Separately generated fields use focal shunts at inhibitory-bearing segments of the reciprocal cable. Each dose is 0.001
1602 times local leak-plus-synaptic conductance; a solved compensating current restores baseline somatic voltage. The
1603 response matrix H contains finite-difference changes in log absolute excitatory-conductance gradients. The inherited
1604 normalized model uses E/I scales 0.35 and reversals 1 and -0.2 . This generator is separate from Eq. 101, but remains
1605 ancestry-dependent through the passive identity in Section S8.

1606 The original dictionaries lacked a common field in selected subtrees while depth bins included it. The correction
1607 gives every family one all-site broadcast and $K - 1$ spatial columns at $K = 1, 2, 4, 8, 16$ when permitted. The initial eight
1608 cells served as development data; the protocol and inputs were frozen before new passive-field results on the existing
1609 47-cell and Pinky cohorts. This is a frozen follow-up on previously used anatomy, not new animal sampling.

1610 Ancestry retains its outcome-independent leverage order. Random routes sample without replacement; site shuffles
 1611 preserve each column's values/nonzeros and leave broadcast intact. Two hundred surrogate trees per cell preserve every
 1612 node depth and parent out-degree by reassigning children among parent slots, keeping attributes with children, then
 1613 recomputing ancestry and ranking. The depth dictionary uses broadcast plus the first $K - 1$ inherited bins, preserving the
 1614 original K -bin span. Equal column counts may have unequal numerical ranks or nonzero counts, which are reported.

1615 For $Y = W^{1/2}H$ and normalized weighted constant q , let $R = (I - qq^T)Y$. Total capture is $\|P_D Y\|_F^2 / \|Y\|_F^2$ and residual
 1616 capture is $\|P_D R\|_F^2 / \|R\|_F^2$. Because every dictionary contains q , total capture equals broadcast capture plus one minus
 1617 broadcast capture times residual capture. The constrained oracle adds the leading $K - 1$ left singular vectors of R .

1618 Randomized controls are averaged within cell before 20,000-draw bootstrap inference. Holm corrections cover
 1619 four controls separately for each endpoint/cohort at primary $K = 8$; other budgets are descriptive. The 65 operators
 1620 and 149,331 dictionary evaluations remain in `source_data/anatomy_commonmode/`, including 148,200 randomized
 1621 realizations. Validation checks weighted identities, ceilings, depth-span equivalence, ranks, common-mode coverage and
 1622 exact ancestry/shuffle nonzero matching. Small Pinky domains explain near-saturation at eight columns.

1623 7.5 Observed inhibitory contacts and descendant domains

1624 The inhibitory census uses the complete stable-nucleus intersection of v661 and v795: twenty targets, with no outcome
 1625 selection. An affine raw-to-slanted transform reproduced paired published locations within $10^{-6} \mu\text{m}$; five cross-release
 1626 root changes were joined by stable nucleus. Mapping within $5 \mu\text{m}$ retained 114,775 of 116,401 inputs and 2,637 of 2,789
 1627 known inhibitory contacts, from 114 of 116 inhibitory cells. No unreported proofreading status was inferred.

1628 One largest contact per published axonal clump avoids counting adjacent contacts as independent sites. Within-
 1629 target/compartment candidate pools match log soma distance. For each site, a descendant-input indicator multiplied
 1630 by square-root synapse size and normalized gives r_k ; $r_k^T r_l$ measures weighted domain overlap, not inhibitory activity
 1631 correlation. The 402 connections with at least two clumps are nested within targets.

1632 Primary target-level spatial associations were retested by averaging within 92 presynaptic axons and by jointly
 1633 matching 3D and path distance over 500 assignments. Coarse tree-distance association survived axon aggregation;
 1634 shared-path and descendant overlap did not. No endpoint survived joint geometry matching. Thus the census establishes
 1635 spatial co-targeting, not geometry-independent credit organization. Capacity columns use actual inhibitory-bearing
 1636 segments weighted by contact size, with 200 random/shuffled controls. Their target is the full actual-route matrix, making
 1637 this another compressibility diagnostic. All primary and sensitivity endpoints are indexed in Supplementary Table S12.

1638 7.6 Optimization with explicitly imposed alignment

1639 Eight-cell kernels supply eight leverage-selected routes. For their weighted orthonormal basis Q , paired unit directions
 1640 parallel and orthogonal to its span define

$$g(a) = \sqrt{a} \mathbf{u}_{\parallel} + \sqrt{1-a} \mathbf{u}_{\perp}, \quad a = 0, 0.2, 0.4, 0.6, 0.8, 1. \quad (104)$$

1641 Capture in the chosen span is therefore exactly a by construction. Eight random paths, depth bins and column-wise row
 1642 shuffles are paired across alignment levels within each of forty streams per cell, each containing 256 fields. A positive
 1643 diagonal quadratic uses curvature 0.5–1.5, permuted per stream. One-step updates match norm at 0.1 times exact-gradient
 1644 norm; iterative projected descent uses step 0.25 for twenty steps. Progress is normalized to full-gradient progress. The
 1645 median within-cell capture/progress correlation was 0.990. This establishes usefulness conditional on imposed alignment
 1646 and oracle coefficients, not observed biological alignment or autonomous coefficient learning.

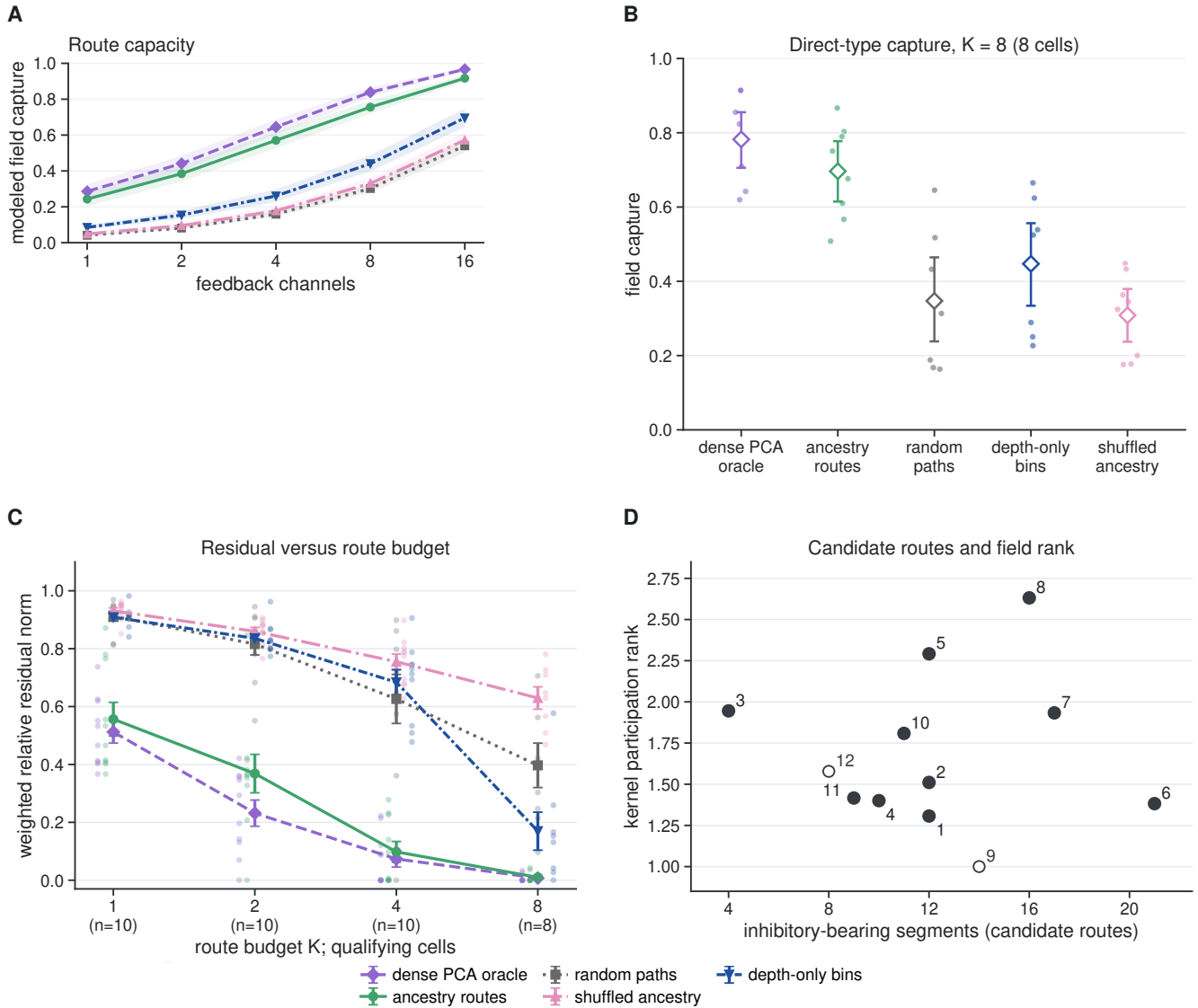
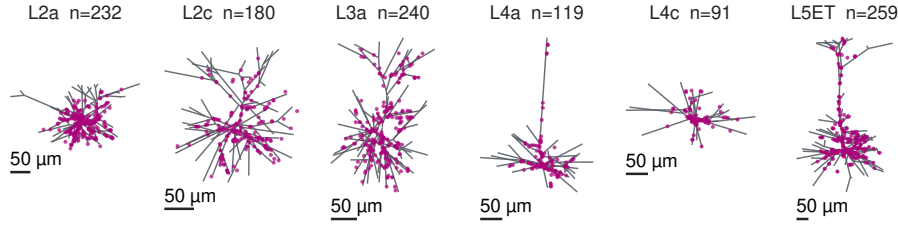


Figure S25: Cohort, label and arbor-size controls bound modeled anatomical capacity. **A**, Route-generated capture of constructed target fields versus budget in 47 disjoint same-mouse cells (46 at 16 channels); morphology routes use 8.1% of dense-feedback wiring at eight channels (wiring density per cell and arm is tabulated in Source Data). The key below the sheet applies to **A** and **C**. Twenty modeled streams are averaged within each cell; bands are 95% cell-bootstrap intervals of the mean, several times narrower than the cell-to-cell spread, and the two null arms (random paths, shuffled ancestry) nearly coincide. **B**, Initial eight-cell cohort: squared-energy capture at $K = 8$ restricted to directly typed presynaptic contacts, every cell drawn, with the dense PCA oracle as the ceiling. **C,D**, Second-mouse Pinky cohort: relative residual norm versus route budget against the dense principal-component ceiling, every qualifying cell drawn faintly behind its arm mean, and the number of inhibitory-bearing segments (candidate routes) versus effective field rank for the ten of twelve selected cells that passed the typed-input criterion (open markers: the two excluded cells; 78–114 reconstructed segments per tree). Eligibility changes from ten cells at $K \leq 4$ to eight at $K = 8$. Panel **C** uses mean $\pm$ s.e.m., and its residual norm is not squared-energy capture. All fields are model generated. Main Fig. 7 equalizes a common broadcast and leads with actual-tree surrogate controls; these route-generated comparisons are structural positive controls, not measured biological teaching signals.

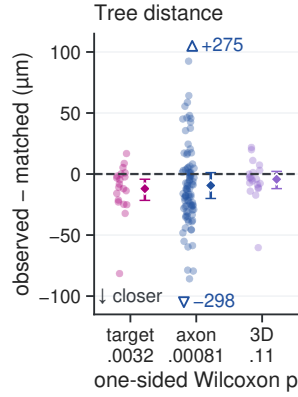
A

Inhibitory contacts on reconstructed arbors

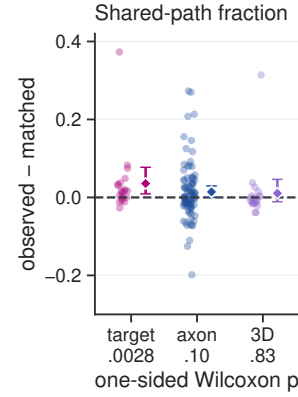
— arbor skeleton ● inhibitory contact



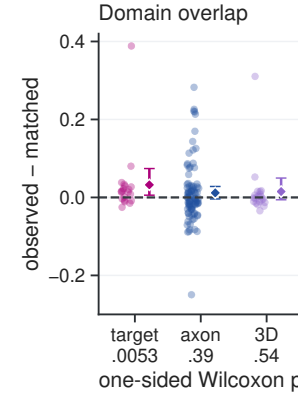
B



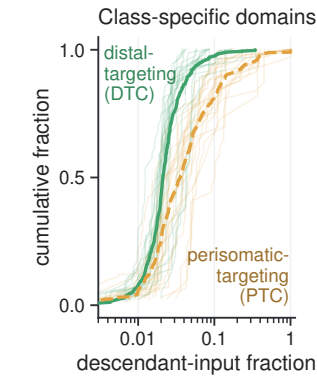
C



D



E



F

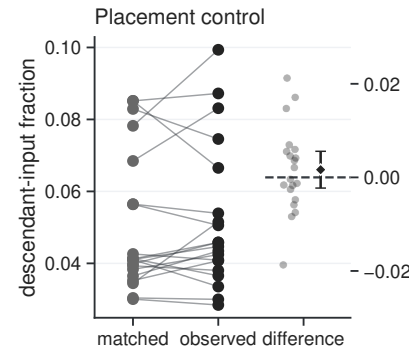


Figure S26: Spatial matching limits apparent inhibitory targeting of ancestry domains. **A**, Six layer-diverse reconstructions with mapped inhibitory contacts in x - y projection, pia up. Grey lines are the compressed dendritic skeleton (segment end points joined by straight edges); each cell is scaled independently and its own bar marks $50\ \mu\text{m}$. **B–D**, Observed-minus-matched tree distance, shared-path fraction and weighted descendant-domain overlap (“domain overlap”) in the twenty-target analysis, after aggregation within 92 presynaptic axons, and after joint path-distance/Euclidean matching (“3D”). Points are the individual units; diamonds and whiskers are means and 95% bootstrap intervals over the displayed inferential unit; the value beneath each unit is the one-sided Wilcoxon signed-rank p (alternative: closer in **B**, larger in **C,D**). In **B**, two axon values beyond $\pm 105\ \mu\text{m}$ are drawn as open triangles at the frame with their values; they remain in every statistic. Persistence is judged by the Wilcoxon test: only tree distance persists after axon aggregation ($p = 0.0008$, 62 of 92 axons closer, versus $p = 0.10$ and 0.39 for the two overlap measures), although its axon-level bootstrap interval of the mean spans zero because of the two off-scale axons; no endpoint of the **B–D** triplet persists after joint three-dimensional matching ($p \geq 0.11$). Shared-path fraction and domain overlap are nearly collinear (Pearson $r = 0.98$, 0.83 and 0.99 over the target, axon and 3D units), so **D** is a weighted restatement of **C**. **E**, Cumulative descendant-input fraction of contacts from distal-dendrite-targeting (DTC, $n = 1,142$) versus perisomatic-targeting (PTC, $n = 840$) interneurons, retaining one contact per axonal clump; faint curves are the twenty individual targets and bold curves the pooled contacts. The per-target median DTC-minus-PTC difference is -0.016 (95% bootstrap interval -0.023 to -0.011 ; negative in 20 of 20 targets). **F**, Descendant-input fraction at observed inhibitory contacts versus matched locations in each of the twenty targets; the difference column shows the per-target observed-minus-matched values on the right axis with their mean and 95% bootstrap interval ($+0.002$, -0.002 to $+0.006$; two-sided Wilcoxon $p = 0.55$). No reliable overall localization advantage is detected. Cells derive from one mouse. These structural comparisons do not show that contacts carried teaching signals.

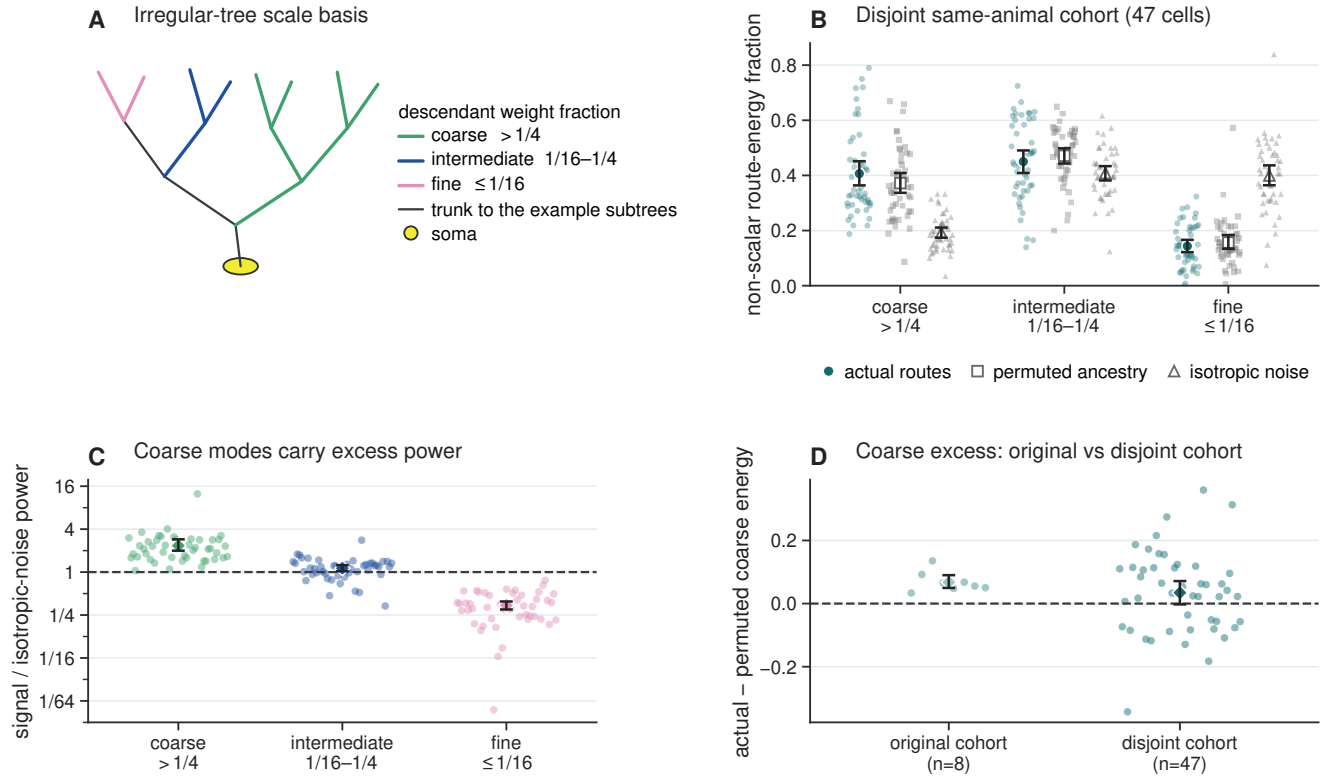


Figure S27: Irregular reconstructed trees concentrate modeled route-field energy coarsely, but the anatomy-specific excess does not replicate. **A**, Weighted unbalanced tree-Haar contrasts recursively partition each reconstructed arbor into coarse, intermediate and fine supports, defined by descendant excitatory-weight fractions $> 1/4$, $1/16-1/4$ and $\leq 1/16$, respectively; the schematic colours one example subtree per bin (key), with the grey trunk leading to them and the soma drawn as the yellow ellipse. Green, blue and rose mark scale bins throughout the figure. **B**, Non-scalar conductance-weighted ancestry-route energy fraction in the disjoint 47-cell cohort (teal, filled circles), compared with column-permuted ancestry (grey, open squares) and dimension-matched isotropic noise (grey, open triangles). Small points are individual cells; large marks are cell means, and the black whiskers drawn over them are 95% cell-bootstrap intervals for the actual-route series and for both nulls. **C**, Signal-to-isotropic-noise power by scale for all 47 cells (points); diamonds show means, with 95% cell-bootstrap intervals drawn over them as black whiskers (the intermediate-scale interval is only slightly wider than the diamond), and the dashed line marks equality with isotropic noise. **D**, Actual-minus-permuted coarse energy per cell (points) in the original 8-cell cohort (light teal) and the disjoint 47-cell cohort (teal); diamonds show cohort means with 95% cell-bootstrap intervals as black whiskers, and the dashed line marks zero. The positive original-cohort contrast is not reliable in the 47-cell cohort. Cells are the inferential units; all reconstructions derive from one mouse. Modeled route fields are not measured teaching signals.

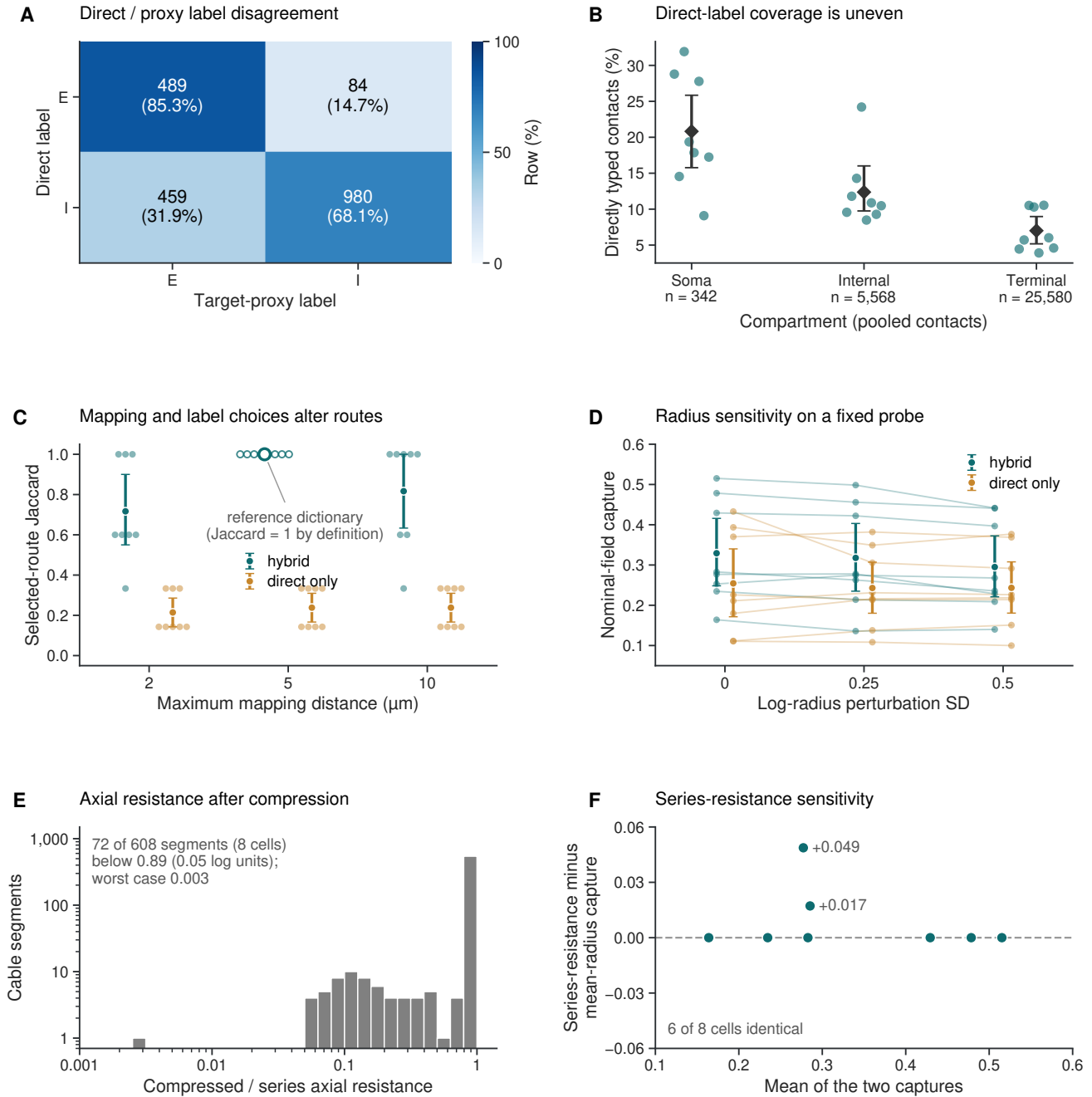


Figure S28: Label coverage and geometric preprocessing affect anatomical route dictionaries. All analyses use eight initial reconstructed cells from one mouse. **A**, Direct-presynaptic versus target-proxy excitatory/inhibitory (E/I) labels for 2,012 jointly labeled mapped contacts; cells print counts and row percentages and are shaded by row percentage (colour bar). **B**, Direct-label fraction by compartment: individual cells and cell means with 95% cell-bootstrap intervals. Denominators differ (22–66 contacts per cell at the soma, 342 pooled; 157–1,541 internal, 5,568 pooled; 1,148–5,697 terminal, 25,580 pooled), so per-cell points carry unequal weight. **C**, Jaccard overlap of the four selected routes with the nominal hybrid-label, $5\mu\text{m}$ dictionary across mapping thresholds (ordinal, equally spaced) and label choices; the open symbol at hybrid, $5\mu\text{m}$ is the reference compared with itself (Jaccard = 1). With hybrid labels, 5 of 8 cells change at least one route at $2\mu\text{m}$ (mean 0.72) and 3 of 8 at $10\mu\text{m}$ (mean 0.82); with direct-only labels all eight differ at every threshold (means 0.21–0.24). **D**, Capture of the fixed nominal field under heterogeneous log-radius perturbations indexed by their standard deviation (SD), for the hybrid and direct-only dictionaries as in **C**; thin lines join the eight cells across SD. The paired within-cell change from SD 0 to 0.5 is -0.034 with hybrid labels (-0.050 to -0.018 ; 7 of 8 cells decline) and -0.011 with direct-only labels (-0.053 to 0.019). Large symbols and bars in **C,D** show cell means and 95% cell-bootstrap intervals; small symbols are individual cells, tied values spread side by side. **E**, Distribution of the mean-radius compressed axial resistance divided by the exact series sum over raw edges (0.1-log-unit bins). No segment exceeds its series sum; 72 of 608 segments (8 cells) fall below 0.89 (0.05 log units), the worst to 0.003 (2.5 log units). **F**, Series-resistance minus mean-radius capture of the nominal field against the mean of the two, at the nominal condition (hybrid labels, $5\mu\text{m}$, SD 0), one point per cell; the dashed line marks equality. Six of eight cells are identical and two differ by $+0.049$ and $+0.017$.

1647 S8 Focal shunting and spatial transport

1648 8.1 Passive focal shunts attenuate a positive adjoint

1649 **Proposition 10** (No sign reversal in a passive grounded tree). *Let G be a symmetric positive-definite nonsingular M -*
 1650 *matrix, with nonpositive off-diagonal entries. For a nonnegative adjoint source $\mathbf{s}$, define $\mathbf{q} = G^{-1}\mathbf{s}$. Adding a nonnegative*
 1651 *shunt κ at compartment k gives $G' = G + \kappa \mathbf{e}_k \mathbf{e}_k^T$ and*

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa q_k}{1 + \kappa(G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (105)$$

1652 *Both G^{-1} and $(G')^{-1}$ are elementwise nonnegative, hence $0 \leq \mathbf{q}' \leq \mathbf{q}$ componentwise. A focal passive shunt attenuates*
 1653 *but cannot reverse a positive adjoint field.*

1654 *Proof.* Equation 105 is the Sherman–Morrison identity. A nonsingular M -matrix has a nonnegative inverse; adding a
 1655 nonnegative diagonal term preserves the M -matrix property [2]. The subtracted vector is nonnegative, and $\mathbf{q}' = (G')^{-1}\mathbf{s} \geq$
 1656 0 . $\square$

1657 This ordering does not by itself guarantee descendant selectivity relative to a matched current injection; that spatial
 1658 comparison depends on the Green's function, focal site and electrotonic regime and is evaluated numerically in the focal
 1659 experiments.

1660 **Corollary 2** (A focal shunt applies gains to its ancestry partition). *Assume additionally that the nonzero off-diagonal*
 1661 *entries of G form a connected tree rooted at the soma s , and that the adjoint source is $\ell \mathbf{e}_s$ for a fixed scalar somatic loss*
 1662 *derivative ℓ . Write $R = G^{-1}$ and $\mathbf{h} = R \mathbf{e}_s > 0$. For a shunt at k , let $\mathcal{A}_k$ be the vertices on the path from s to k , including*
 1663 *both endpoints. For each $a \in \mathcal{A}_k$, define the disjoint block $P_a = \{i : \text{LCA}(i, k) = a\}$, where LCA is the lowest common*
 1664 *ancestor. Then*

$$q'_i = \eta_a q_i \quad (i \in P_a), \quad \eta_a = 1 - \frac{\kappa R_{ks} R_{ak}}{(1 + \kappa R_{kk}) R_{as}}, \quad \eta_k = \frac{1}{1 + \kappa R_{kk}}. \quad (106)$$

1665 *For finite $\kappa \geq 0$, every gain satisfies $0 < \eta_a \leq 1$. Let B_k have columns $\mathbf{1}_{P_a}$ and define the baseline-weighted partition*
 1666 *dictionary $D_k = \text{diag}(\mathbf{h}) B_k$. With $\Gamma_k = \text{diag}(\eta_a)$,*

$$\mathbf{q} = \ell D_k \mathbf{1}, \quad \mathbf{q}' = \ell D_k \Gamma_k \mathbf{1}. \quad (107)$$

1667 *Thus the adjoint effect is exactly a diagonal gain in this partition basis.*

1668 *Proof.* Removing a vertex a separates a tree into components. If a lies on the path from i to j , solving the source-free
 1669 equations in the component containing i , with boundary voltage at a , gives $R_{ij} = R_{ia} R_{aj} / R_{aa}$. For $a = \text{LCA}(i, k)$, apply
 1670 this identity to (i, k) and (i, s) ; it also holds when an endpoint equals a . Hence $R_{ik} / R_{is} = R_{ak} / R_{as}$. Substitution into
 1671 Eq. 105 gives Eq. 106. Positivity of both tree inverse matrices gives positive ratios, and the preceding proposition bounds
 1672 them by one. The formula also holds when $\ell = 0$, when both adjoints vanish. The disjoint blocks partition all vertices,
 1673 yielding Eq. 107. $\square$

1674 The block P_k is the whole subtree below the shunt. For an ancestor a above k , P_a contains a and its sister subtrees off
 1675 the path to k . If c is the next vertex toward k , its indicator is $\mathbf{1}_{P_a} = \mathbf{1}_{\text{Desc}(a)} - \mathbf{1}_{\text{Desc}(c)}$, whereas $\mathbf{1}_{P_k} = \mathbf{1}_{\text{Desc}(k)}$. The gain
 1676 field therefore lies in the span of nested ancestry indicators. The diagonal form requires the disjoint partition basis; an
 1677 arbitrary selected set of overlapping routes need not have diagonal gains. Even for the ancestry chain, a change from
 1678 partition to nested-subtree coordinates generally makes the gain matrix non-diagonal, while preserving that full span.

1679 For the synaptic gradient, the local driving force remains a separate factor: $\gamma'_i = \eta_a q_i (E_E - V_i')$ for $i \in P_a$. Its
 1680 within-block variation is therefore not constrained to a single gain relative to $\gamma_i = q_i (E_E - V_i)$. The corollary describes
 1681 somatic-source adjoint transport on a passive tree, rather than the complete synaptic update, a distributed-error source or
 1682 a nonlinear finite-dose response. It connects the morphology-defined dictionary to a conductance gain without implying
 1683 a learning advantage.

1684 **Corollary 3** (Spectral size of a focal passive perturbation). *Under the preceding assumptions, let $\mathcal{R} = G^{-1}$ and*
 1685 *$\mathcal{R}' = (G + \kappa \mathbf{e}_k \mathbf{e}_k^T)^{-1}$. The inverse-operator change has rank at most one, with rank one for $\kappa > 0$, and*

$$\|\mathcal{R}' - \mathcal{R}\|_2 = \frac{\kappa \|\mathcal{R} \mathbf{e}_k\|_2^2}{1 + \kappa \mathcal{R}_{kk}} \leq \frac{\kappa \|\mathcal{R}\|_2^2}{1 + \kappa / \lambda_{\max}(G)}. \quad (108)$$

1686 *Consequently, for any adjoint source $\mathbf{s}$, $\|(\mathcal{R}' - \mathcal{R})\mathbf{s}\|_2 \leq \|\mathcal{R}' - \mathcal{R}\|_2 \|\mathbf{s}\|_2$.*

1687 *Proof.* Sherman–Morrison gives $\mathcal{R}' - \mathcal{R} = -\kappa(\mathcal{R}\mathbf{e}_k)(\mathcal{R}\mathbf{e}_k)^\top / (1 + \kappa\mathcal{R}_{kk})$. The norm of this symmetric outer-product
 1688 matrix is its scalar coefficient times $\|\mathcal{R}\mathbf{e}_k\|_2^2$. Use $\|\mathcal{R}\mathbf{e}_k\|_2 \leq \|\mathcal{R}\|_2$ and $\mathcal{R}_{kk} \geq \lambda_{\min}(\mathcal{R}) = 1/\lambda_{\max}(G)$ for the bound. $\square$
 1689 Equation 108 provides only a global magnitude ceiling on the total adjoint change. It does not predict the localization
 1690 contrast or the observed crossover with membrane conductance. Those spatial effects depend on the Green’s-function
 1691 profile and electrotonic length constant of the cable.

1692 8.2 Focal shunting perturbations

1693 Route capacity does not show that conductance can selectively alter transport. We therefore compared a focal shunt with
 1694 a baseline-current-matched injection at the same site of each reciprocal reconstructed cable and quantified the resulting
 1695 change in the exact modeled gradient.

1696 8.3 Adjoint formulation and baseline-current-matched current injection

1697 For passive conductance matrix G and current-source vector $\mathbf{b}$, the voltage vector was $\mathbf{V} = G^{-1}\mathbf{b}$. Let $\mathbf{e}_s$ select the
 1698 somatic coordinate and let w_i denote the excitatory conductance at site i . For a loss depending on somatic voltage, the
 1699 adjoint error and conductance gradient were

$$\mathbf{q} = G^{-1}\mathbf{e}_s \frac{\partial \mathcal{L}}{\partial V_s}, \quad \frac{\partial \mathcal{L}}{\partial w_i} = q_i(E_E - V_i). \quad (109)$$

1700 Adding focal inhibitory conductance κ_k at node k , selected by the coordinate vector $\mathbf{e}_k$, gives

$$G' = G + \kappa_k \mathbf{e}_k \mathbf{e}_k^\top, \quad \mathbf{b}' = \mathbf{b} + \kappa_k E_I \mathbf{e}_k.$$

1701 The shunt therefore changes both conductance loading and the forward source. Holding the somatic loss derivative fixed,
 1702 the Sherman–Morrison identity gives the transport change

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa_k q_k}{1 + \kappa_k (G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (110)$$

1703 The reversal E_I does not appear in Eq. 110 because the adjoint transport depends on G' , not on the forward source $\mathbf{b}'$. It
 1704 remains in the solved voltage and in every inhibitory driving force. Corollary 2 supplies the exact partition-gain form.
 1705 The full synaptic gradient additionally changes through its local driving force.

1706 Let $r_k = (G^{-1})_{kk} > 0$. Then

$$\Delta \mathbf{q} = -a_k G^{-1} \mathbf{e}_k, \quad a_k = \frac{\kappa_k q_k}{1 + \kappa_k r_k}, \quad \frac{\partial |a_k|}{\partial \kappa_k} = \frac{|q_k|}{(1 + \kappa_k r_k)^2} \geq 0. \quad (111)$$

1707 The derivative is strictly positive when $q_k \neq 0$; if $q_k = 0$, the perturbation vanishes. Its amplitude saturates at $|q_k|/r_k$.
 1708 For a nonzero perturbation, the normalized spatial profile is fixed by the Green’s-function column $G^{-1}\mathbf{e}_k$ in this linear
 1709 passive state. For a descendant set $D(k)$ and a prespecified matched comparison set $C(k)$, define

$$S_k(D, C) = \frac{\text{median}_{i \in D(k)} |(G^{-1})_{ik}|}{\text{median}_{j \in C(k)} |(G^{-1})_{jk}|}. \quad (112)$$

1710 For $\kappa_k > 0$, $q_k \neq 0$ and a positive comparison median, the common amplitude $|a_k|$ cancels. Under these conditions,
 1711 $S_k > 1$ is necessary and sufficient for a positive descendant-minus-comparison contrast in absolute adjoint change
 1712 at fixed k . It is only a transport-level criterion. Relative or logarithmic changes additionally divide by the baseline
 1713 q_i , and the full synaptic gradient also contains the post-perturbation driving force. This is why axial coupling, leak,
 1714 background conductance and local input resistance must be varied explicitly rather than inferred from topology alone.
 1715 The baseline-current-matched current injection applied $\kappa_k(E_I - V_k)$ at the same node without changing G . This equals
 1716 the current carried by the added shunt at the unperturbed voltage; its local voltage effect matches the shunt only to first
 1717 order in dose. An independently solved state-matching compensatory current injection restored baseline somatic voltage
 1718 after both interventions. This is not a continuously adjusted voltage clamp. The somatic scalar error was therefore fixed.
 1719 Shunting changed adjoint transport, whereas the current-injection control left it unchanged. Dendritic voltages were not
 1720 clamped, so both interventions could also change local driving forces.

8.4 Prespecified endpoint and numerical validation

Before running the analysis, we fixed the site-selection rule, primary dose, comparison groups, and localization endpoint. Focal sites were selected by a depth-stratified rule without access to gradient outcomes. A site required at least three excitatory descendants and three comparison sites. At most 16 eligible sites were sampled per cell across quartiles of topological depth. The primary model used excitatory and inhibitory synaptic scales of 0.35 and normalized reversal potentials $E_E = 1$ and $E_I = -0.2$. Synapses were classified as descendants, ancestors, sister-subtree sites or unrelated sites. Unrelated synapses within each template were matched to descendant cable depth. For baseline gradient g_i and perturbed gradient g'_i , we defined $m_i = |\log[(|g'_i| + \epsilon)/(|g_i| + \epsilon)]|$, where $\epsilon = 10^{-15} \max(1, \max_i |g_i|)$. For each site, the localization index was the median m_i among descendants minus the median among depth-matched unrelated synapses. Site effects were first averaged within cell.

The somatic loss was one-half the squared difference between somatic voltage and a target one normalized unit below its baseline value. Conductance doses were 0.25, 0.5, 1, and 2 times the local leak-plus-synaptic conductance. The state-matching current was solved after each perturbation and then treated as fixed when evaluating the post-intervention conductance gradient. The topology control sampled 200 relation templates from other focal sites, stratified by focal-site topological-depth quartile when another site was available; otherwise it sampled another site in the same cell.

At unit dose, 101 sites in eight cells gave mean localization 0.104 for shunting and 0.035 for current injection. Their paired cell-level contrast was 0.069 log units (95% interval, 0.053–0.086), positive in all eight cells. True relations also exceeded the depth-stratified relation control. The contrast grew with relative dose; Source Data retain every site, dose and template.

We separated the two factors in the exact gradient algebraically. Writing $d_i = E_E - V_i$ and $d'_i = E_E - V'_i$, full shunting uses $q'_i d'_i$, adjoint-only substitution uses $q'_i d_i$, and driving-force-only substitution uses $q_i d'_i$. At unit dose, their mean localization indices were 0.104, 0.130 and 0.026, respectively, compared with 0.035 for the current-injection control. Adjoint-only minus driving-force-only localization was 0.104 (95% interval 0.080–0.125), positive in all eight cells.

Full shunting and the driving-force-only state provide a direct comparison of adjoint transport: they share the same post-shunt voltage field and differ only in whether the adjoint field is updated. Their localization difference was 0.079 and was positive in all eight cells. This is the exact adjoint-replacement contrast plotted in main Fig. 8E.

The gradient factors and localization statistic are nonlinear, so these four outcomes do not form an additive decomposition. To allocate the full-shunt-minus-current-injection contrast, we used two-factor Shapley values. For v_{00} equal to current-injection localization, v_{10} adjoint-only, v_{01} driving-force-only and v_{11} full-shunt localization,

$$\phi_q = \frac{1}{2}[(v_{10} - v_{00}) + (v_{11} - v_{01})], \quad (113)$$

$$\phi_d = \frac{1}{2}[(v_{01} - v_{00}) + (v_{11} - v_{10})]. \quad (114)$$

The allocations sum exactly to the full-shunt-minus-current contrast. Mean adjoint and driving-force allocations were 0.087 and -0.017 , respectively: driving-force change partly cancelled the adjoint contribution. This is an algebraic allocation of a nonlinear statistic; the current-matched control is not a literal no-factor corner of a physical factorial. Repeating the allocation with the unperturbed field as its reference is retained in Source Data. Factor substitutions are not independent biological interventions.

All conductance matrices in the primary analysis were positive definite. The somatic state-matching residual was numerically zero. Analytic adjoint gradients agreed with finite differences to a maximum relative error of 1.02×10^{-6} . The Source Data indexed in Table S12 retain prespecified conductance-scale and reversal-potential sensitivity runs. The direction of the shunting-minus-current-injection effect was positive in every condition, with seven or eight positive cells. These results establish the mechanism in a passive model on measured trees; they do not establish endogenous inhibitory teaching signals in vivo.

With proxy-classified contacts removed, the primary shunt-minus-current-injection contrast was 0.101 (0.075–0.129) across 55 sites and was positive in all eight cells. In the independent, non-overlapping v661 sample, 45 cells supplied 235 eligible sites. The same contrast was 0.138 (0.116–0.162), positive in 45/45 cells. Forty cells with at least two eligible sites supplied the reassigned-relation control; the true relation exceeded this control by 0.118 (0.094–0.144), with 39 positive cells and one numerical tie (Supplementary Fig. S30A,B). Cells, rather than sites or relation draws, were the units of inference.

To test sensitivity to electrical calibration, we also computed axial and leak conductances in physical units from the same compressed geometry. For segment radius r , length ℓ , axial resistivity R_a and specific membrane resistance R_m , a cylindrical non-root segment had axial conductance $\pi r^2/(R_a \ell)$ and membrane leak $2\pi r \ell/R_m$; the root was assigned

spherical area $4\pi r^2$. All conductances were converted to nanosiemens (nS). Because MICrONS synapse size is not an electrical conductance, synaptic sizes retained the prespecified relative scaling with the median physical leak as reference. We varied R_a and R_m while retaining the site-selection rule, relative dose, gradient definition and current-injection control. For each condition and intervention, we re-solved the compensating somatic current to restore that condition's baseline somatic voltage. The Source Data indexed in Table S12 retain this electrotonic dependence. The large contrast observed under the normalized proxy was recovered only in low- R_m , high-conductance settings; at $R_a = 150 \Omega \text{cm}$ and $R_m = 15,000 \Omega \text{cm}^2$ it was negligible or small. These are sensitivity settings, not fits to the recorded cells, and the model remains passive and steady-state.

A broader parameter matrix, specified before numerical validation, tested membrane resistance, background conductance and dose definition jointly. It used $R_a = 150 \Omega \text{cm}$ and crossed $R_m \in \{300, 1000, 15,000\} \Omega \text{cm}^2$, distributed leak-like background conductance of 0, 1 or 4 times baseline leak, and three dose definitions. Relative-local doses were 0.25, 1 and 2 times membrane-plus-synaptic conductance; fixed doses were 0.05, 0.5 and 5 nS; input-normalized doses were 0.25, 1 and 4 times $[(G^{-1})_{kk}]^{-1}$. The same 101 sites in eight cells supplied 16,362 site-condition-intervention rows. Every conductance matrix was positive definite (minimum eigenvalue 0.107 nS) and the maximum somatic-state residual was 3.33×10^{-16} .

At fixed absolute dose, the conductance-specific contrast did not follow the raw focal-shunt localization alone. At 0.5 nS, shunt-minus-current-injection localization was 0.0104 (0.0080–0.0126), 0.0073 (0.0044–0.0104), and -0.0039 (-0.0075 to -0.0004) at $R_m = 300, 1,000$ and $15,000$, respectively. At 5 nS the corresponding values were 0.0974, 0.0741 and 0.0061; the last interval included zero. Distributed background conductance restored the input-normalized effect in the high-resistance condition: at unit input-conductance dose, the contrast changed from -0.0004 with no background to 0.1210 and 0.2587 at background multipliers one and four. Across all 8,181 focal-shunt rows, every descendant gradient magnitude was attenuated, none was enhanced and no sign flip occurred. The passive mechanism therefore selectively attenuated gradients; sign reversal would require dynamics or active conductances absent from this phase (Supplementary Fig. S29).

8.5 Weak-channel steady-state linearization check

We checked the local-Jacobian calculation after adding weak voltage-dependent currents to the high-conductance cable. The physical reconstructed-tree system used $R_a = 150 \Omega \text{cm}$, $R_m = 1,000 \Omega \text{cm}^2$ and distributed background conductance equal to baseline leak. Steady-state sodium (Na), potassium (K), calcium (Ca), hyperpolarization-activated cyclic nucleotide-gated (HCN) and N-methyl-D-aspartate receptor (NMDA) current families had fixed reversal potentials, half-activation voltages, slopes and gate powers specified in the predefined parameter file. Global and segment-level maximal conductances were drawn independently from prespecified log-normal distributions. Median maximal Na, K, Ca and HCN conductances were 0.06, 0.10, 0.025 and 0.05 times baseline leak, respectively; NMDA conductance was 0.15 times the excitatory conductance. Gate activation further scales these currents. These prespecified weak-channel draws are not cell-specific density estimates or a test of strongly regenerative operation.

For each draw, a residual-reducing Newton solve found the nonlinear steady state. The predefined acceptance criteria required all voltages to lie in the specified range, maximum absolute state residual below 10^{-10} , and minimum eigenvalue of the symmetrized local Jacobian above 10^{-8} . The implicit-adjoint calculation then used the exact inverse of this baseline Jacobian. For channel family a , let $\bar{g}_a$ be maximal conductance, $h_a(V)$ the steady-state gating factor and E_a the reversal potential. The current $I_a(V) = \bar{g}_a h_a(V)(V - E_a)$ contributed the diagonal term $\bar{g}_a [h_a(V) + h'_a(V)(V - E_a)]$; thus the baseline Jacobian includes voltage-dependent gate derivatives. Focal shunting and a baseline-current-matched current injection were evaluated at 0.25, 1 and 4 times local input conductance within this fixed local linearization. Sherman–Morrison updated the Jacobian inverse for the added focal conductance, and a compensating somatic current restored baseline somatic voltage in that linearized system. The active gates and their derivatives were not re-evaluated at a new nonlinear equilibrium after each finite dose. Up to 256 attempts were permitted to obtain 64 accepted draws per cell. All first 64 attempts were accepted in every cell, giving 512 equilibria, 101 focal sites and 38,784 site-draw-condition rows. All values were finite, the largest equilibrium residual was 9.51×10^{-11} and the smallest Jacobian eigenvalue was 1.084.

Channel draws and focal sites were averaged within each of the eight cells before inference. At doses 0.25, 1 and 4, mean shunt localization was 0.159, 0.529 and 1.339, respectively (Table S12). At unit dose, shunting exceeded the current-injection control by 0.382 (95% cell-bootstrap interval 0.356–0.409), positive in 8/8 cells (two-sided Wilcoxon $P = 0.0078$). Every modeled descendant gradient was attenuated, none was enhanced or changed sign, and descendant gradient energy fell to 0.330 of baseline. The dose response closely followed the passive response at the same electrical

1819 calibration. This agreement checks the weak-channel linearization; it does not establish robustness to strongly active
1820 dendrites, channel kinetics or spiking.

1821 **8.6 Driving-force cancellation and the ancestry dependence of passive fields**

1822 We reanalyzed the eight original cable operators used for the common-broadcast comparison. For each focal site, the
1823 perturbation was the original dose 0.001 with somatic voltage restored. The finite log-gradient difference divided by dose
1824 separates into an adjoint term and a driving-force term. Energy uses the original excitatory-area site weights and sums
1825 over focal perturbations. The full focal ancestry partition is defined separately for each shunt and is distinct from the
1826 seven selected routes in the budgeted anatomical dictionary.

1827 Driving-force energy was 0.990–3.496 times full-field energy across cells; these are ratios of squared magnitudes,
1828 not amplitudes. Weighted centered correlations with the adjoint term were -0.979 to -0.913 . The negative cross term
1829 explains why large components yield a smaller full response. The full response retained 0.9711–0.9993 of its energy
1830 in the focal partition. Reconstruction agrees with the original operators to 10^{-8} ; numerical decomposition closes to
1831 rounding error. Cellwise energies, correlations and partition captures are supplied in Source Data.

1832 The passive corollary depends on a tree Green's function, a somatic adjoint source and a diagonal conductance change.
1833 Choosing other passive perturbations under the same assumptions does not make the anatomical field ancestry-neutral.
1834 The budgeted dictionary comparison measures compression within this response family. In the disjoint 47-cell cohort,
1835 actual ancestry exceeds the mean degree–depth surrogate in 38 cells, but in eleven cells more than half of 200 surrogates
1836 match or exceed the actual tree. This descriptive count uses tolerance 10^{-12} and retains the full within-cell distribution.
1837 A different error source, active dynamics or measured fields would be required to test beyond the present structural
1838 dependence.

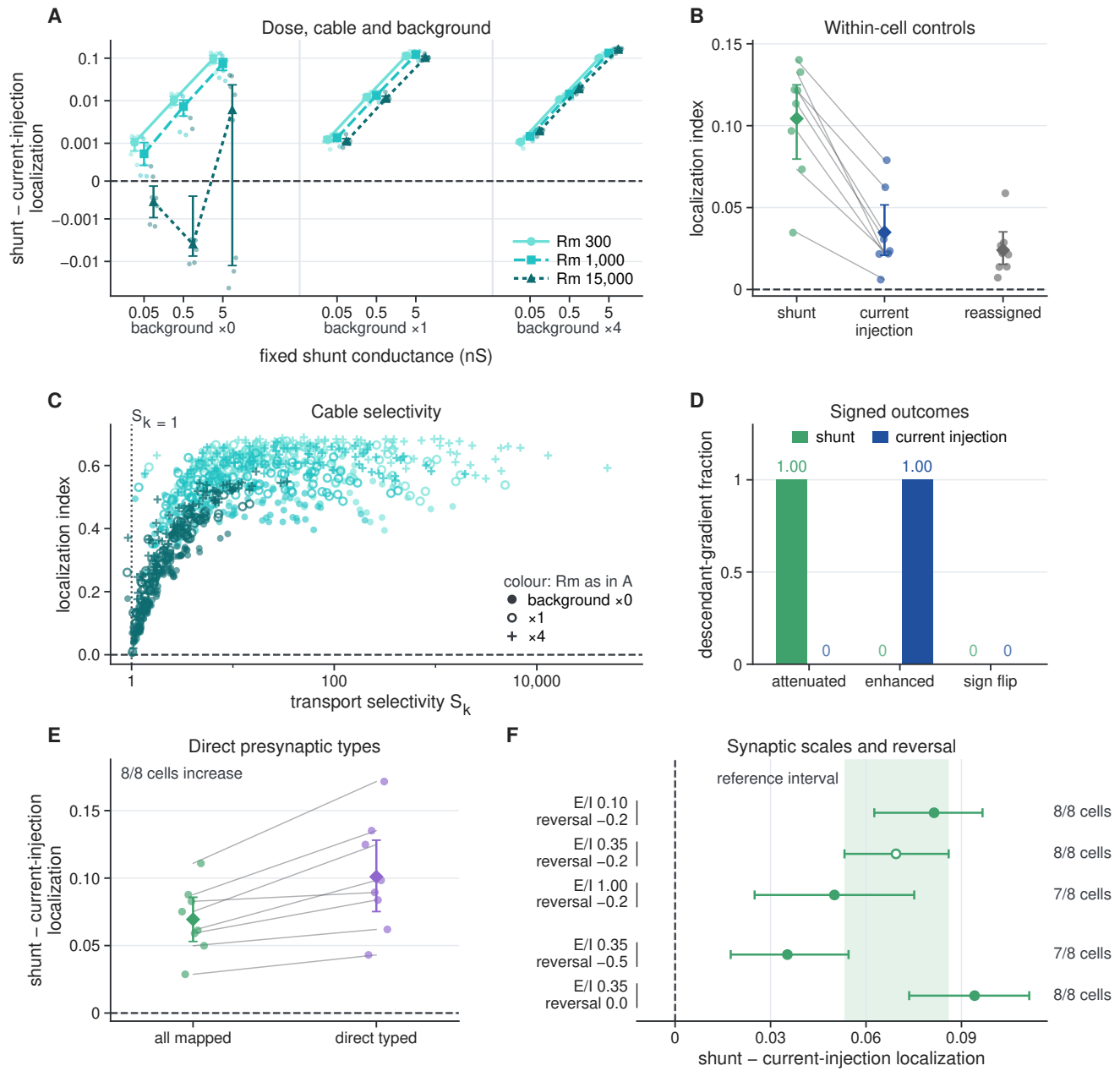


Figure S29: Dose, cable parameters and input mapping determine focal-shunt selectivity. **A**, Shunt-minus-current-injection localization across fixed absolute doses at three membrane resistances (R_m of 300, 1,000 and 15,000 Ωcm^2 : one lightness step, marker and dash pattern each) and three background-conductance multipliers (column groups), on a symmetric-log axis that is linear within ± 0.001 . The contrast is close to proportional to dose at every setting, and the R_m ordering reverses with background: with no background the $R_m = 15,000$ contrast stays near zero (slightly negative at 0.05 and 0.5 nS, an interval spanning zero at 5 nS), whereas at background multiplier four it is the largest of the three at every dose, with 8/8 cells positive. **B**, Shunt and matched-injection localization (grey lines pair cells) and the true relation against a reassigned foreign template (unpaired); positive favors the shunt or the true relation. **C**, Transport selectivity S_k against full synaptic-gradient localization at unit input-conductance-normalized dose, 101 sites; descriptive site-regime points, R_m colours as in **A**, background multipliers 0, 1 and 4 as filled, open and plus markers; dotted line, $S_k = 1$. **D**, Signed census of the same sites at $R_m = 1,000\Omega\text{cm}^2$, background multiplier one, unit dose: fraction of descendant gradients attenuated, enhanced or sign reversed by shunt (attenuates only) and matched injection (enhances only), as in all 81 passive conditions. **E**, All mapped versus directly typed contacts (grey lines pair cells); the contrast increases in 8/8 cells. **F**, E/I scale and inhibitory-reversal sensitivity by row; open marker and band, the reference condition (E/I 0.35, reversal -0.2) and its interval; right column, cells with a positive contrast. **A,B,E,F**, means and 95% cell-bootstrap intervals over eight cells, every cell drawn: in **A** the eight per-cell contrasts are the small dots behind each mean, and the three R_m series are offset within each dose so their fans and intervals stay apart.

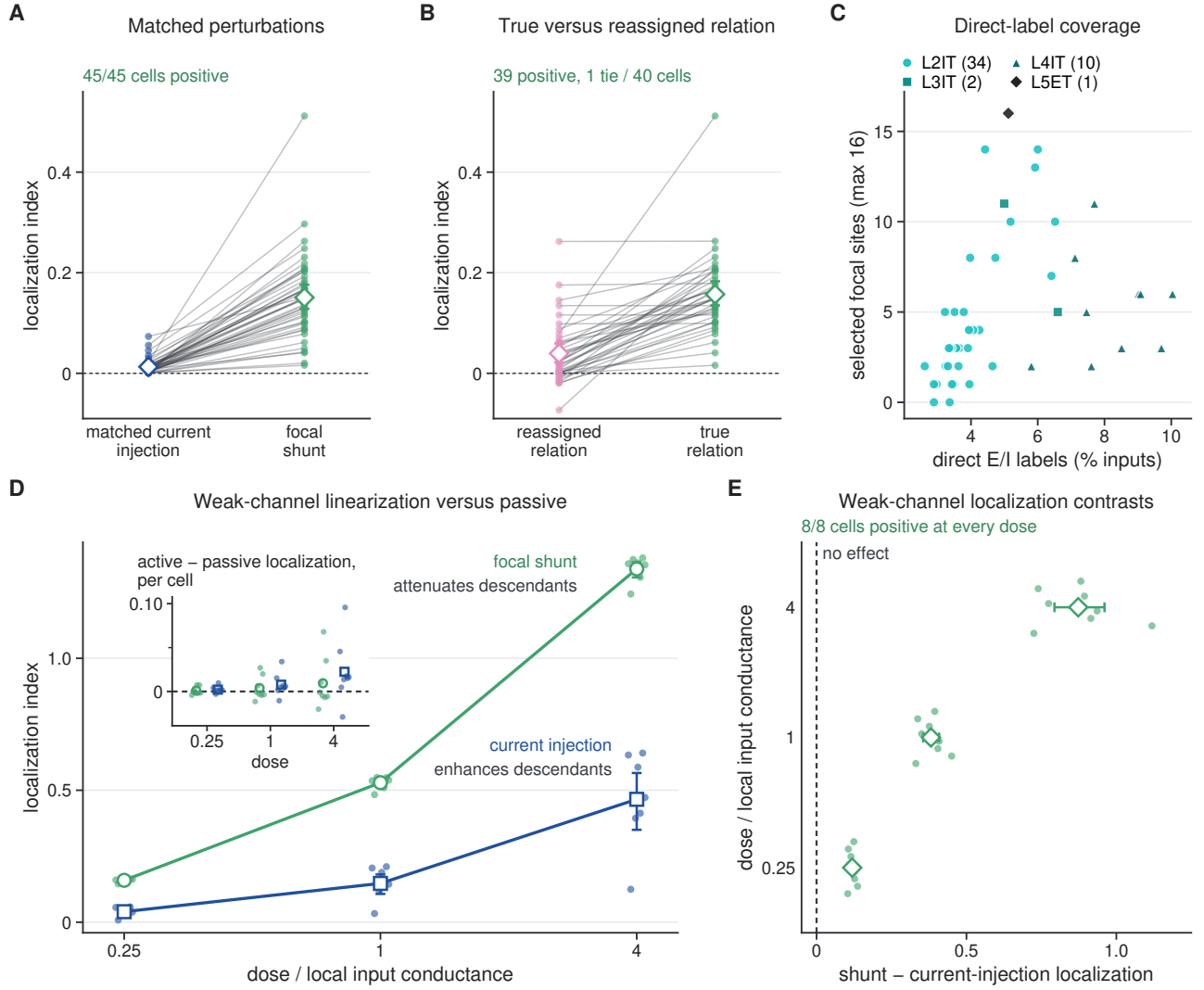


Figure S30: Focal-shunt controls replicate structurally and under weak-channel linearization. **A**, Focal shunt (green) versus baseline-current-matched injection (blue) in 45 disjoint same-mouse cells (235 sites): small dots are cell means joined by a hairline per cell, open diamonds and whiskers the mean and 95% cell-bootstrap interval (20,000 draws), and the dashed rule is zero; the injection mean (0.013) lies on the zero rule with its interval narrower than the diamond. **B**, True (green) versus reassigned (rose) descendant relation templates in forty cells (230 sites), drawn as in **A**; the true-relation column repeats the focal-shunt values of **A** for the forty cells that have a reassigned-relation partner. The count above **A** and **B** is the number of cells whose contrast is positive. **C**, Direct E/I label coverage and the number of selected focal sites per cell, capped at 16 (one L5ET cell had 47 eligible sites); cell classes are the teal ordinal ramp with distinct markers, L2IT circles (34), L3IT squares (2), L4IT triangles (10) and the one L5ET cell a black diamond. **D,E**, Separate initial eight-cell weak-channel ensemble with a fixed steady-state Jacobian around each solved weak-channel state. **D**, Localization versus dose (dose divided by local input conductance) for the focal shunt (green, open circles) and the matched current injection (blue, open squares): small dots are the eight cell means at each dose, markers and whiskers the mean and 95% cell-bootstrap interval, narrower than the marker except for the shunt at dose 4 and the injection at doses 1 and 4. Shunting attenuates descendants whereas matched current injection enhances them (signed localization negative in 8/8 cells for the shunt and positive in 8/8 for the injection at every dose). Inset, the same eight cells' active-minus-passive localization at each dose (passive model at the same calibration, $R_m = 1,000 \Omega \text{cm}^2$, background conductance equal to leak): small dots per cell, open markers the mean difference, dashed rule zero; the mean difference is at most 0.010 localization units for the shunt and 0.023 for the injection. **E**, Paired shunt-minus-current localization by dose row: small dots are the eight cells, open diamonds and whiskers the mean and 95% cell-bootstrap interval (narrower than the diamond at dose 0.25), the dashed rule zero, and the contrast is positive in 8/8 cells at every dose (two-sided Wilcoxon $p = 0.0078$). Sixty-four channel draws and sites are averaged within each cell. Weak conductances test local linearization near the passive response and do not establish robustness to regenerative dynamics or channel kinetics.

S9 Measured responses, model diagnostics and detection sensitivity

9.1 Functional topology and task-derived credit

Anatomical routes can be available without aligning with the credit required by a measured task. We therefore used recorded visual responses to define a prediction objective on reconstructed target cells. Within each model, we changed the backward field while holding the forward morphology fixed.

9.2 Direct anatomy–function join

Functionally imaged excitatory partners were linked to anatomical contacts through MICrONS/CAVE and DANDI mappings. The analysis contained 69 connected presynaptic partners across seven postsynaptic target cells. Automatic matches came from `coregistration_auto_phase3_fwd_apl_vess_combined_v2` in materialization 1822. We retained residual at most 10 and score at least zero, then deduplicated each presynaptic root by ascending residual and descending score. All mapped contacts from a retained presynaptic root contributed to its partner-level contact summary. For the one-site-per-partner model, contacts were first grouped by postsynaptic segment. The partner was assigned to the segment with greatest summed contact size, breaking ties by synapse count, and its major branch was the first segment below the soma on that segment’s ancestry chain. Thus a partner with contacts on several segments supplied one activity feature and one modeled learning site; its total contact size and synapse count still included all retained contacts.

The target universe was the initial eight-cell morphology convenience sample. The analysis plan, specified before outcome inspection, included every target in that sample with a DANDI scan and directly connected, functionally imaged excitatory partners; seven targets qualified. Predefined inclusion criteria required exactly one postsynaptic target in an extract, agreement between partner and contact counts, at least five mapped partners for the topology screen, at least two sites after any requested task filter, and at least one nonempty inhibitory nested-subtree route. The primary task analysis applied neither a manual-match restriction nor a repeat-reliability threshold. The exact target/session/scan pairs are retained in the source records indexed in Table S12. The original analysis plan did not record a deterministic rule for choosing among multiple eligible scans of the same target. We therefore treat these scan choices as part of the initial selected-scan sample, not as the output of a coverage-ranking procedure. The omitted structural target, root 864691136043380566 (nucleus 292648), had one source-data row, for session 8/scan 4, but `has_dandi_asset=False` and no DANDI asset identifier or path. No additional exclusion reason is inferred.

A deterministic sensitivity retained every automatic-conservative target–session–scan cohort with at least five imaged connected presynaptic roots: thirteen scans across the same seven targets. Statistics were computed within scan, averaged within target and inferred over targets. The mean partial shared-path correlation was -0.0485 (95% target-bootstrap interval, -0.224 to 0.118), positive in three targets. The selected-scan and scan-complete analyses therefore both provide no evidence of preferential ancestry alignment. Full topology endpoints are indexed in Supplementary Table S12.

Each target had 464 trials and 280 unique condition hashes, of which 136 were repeated. Trial families comprised 384 Clip, 40 Monet2 and 40 Trippy presentations. For each trial and region of interest, the response was the mean of the supplied `RoiResponseSeries` fluorescence trace over samples whose timestamps fell from trial start through trial stop, including both endpoints. Start and stop offsets were zero. These are the supplied fluorescence traces; our analysis did not deconvolve or apply additional raw-fluorescence preprocessing to them. No trials were dropped; every response used here was finite. The records indexed in Table S12 give target-level asset identifiers and sample counts.

Functional similarity between two partners was the Pearson correlation of their vectors of condition-mean fluorescence over the 136 condition identities with at least two trials. Repeat reliability was computed separately for each region of interest: successive occurrences of every repeated condition were assigned alternately to two groups, responses were averaged within each group, and the two vectors of condition means were compared by Spearman correlation. No reliability correction was applied, and the primary cohort retained all partners regardless of this diagnostic; thresholds of 0.1 and 0.2 were used only in sensitivity analyses.

The controlled shared-ancestry statistic was computed separately within each target. Functional similarity and shared-path fraction were rank transformed. The two controls were $\log(1 + d_{\text{Euclidean}})$ and $\log(1 + |d_{\text{soma},i} - d_{\text{soma},j}|)$, where the second term is the absolute difference in soma path length; each control was also rank transformed. We regressed each ranked variable on an intercept and the two ranked controls by ordinary least squares, then took the Pearson correlation between the two residual vectors. The mean target-level partial-rank effect was -0.069 with a target-bootstrap 95% interval of -0.251 to 0.106 ; three of seven targets had positive effects. Pairwise partner comparisons were not treated as independent replicates.

Several features limit the interpretation of this cohort. MICrONS calcium imaging measured excitatory neurons, the observed partners cover only a small fraction of each target’s complete inputs, and response reliability is modest. The volume comes from one animal, and the recordings were not made during learning.

9.3 Task-derived branch credit

A branch model predicted the standardized postsynaptic response from the standardized responses of its connected presynaptic partners. Let x_i be the response of partner/site i , w_i its input weight, and $b(i)$ the dominant major branch receiving its contacts. Branch b had summed input d_b , activity h_b and readout weight v_b ; c was an output bias. The fitted prediction $\hat{y}$ was

$$d_b = \sum_{i:b(i)=b} x_i w_i, \quad h_b = \tanh(d_b), \quad \hat{y} = \sum_b v_b h_b + c. \quad (115)$$

For N training trials with target responses y_t , the objective was

$$\frac{1}{2N} \sum_{t=1}^N (\hat{y}_t - y_t)^2 + \frac{\lambda}{2} (\|\mathbf{w}\|_2^2 + \|\mathbf{v}\|_2^2), \quad \lambda = 10^{-3}, \quad (116)$$

with no penalty on c . The exact voltage-space error at site i was $(\hat{y} - y)v_{b(i)}[1 - \tanh^2(d_{b(i)})]$.

Whole condition-hash stimulus identities, not trials of the same identity, were split: 20% of the 280 identities and all of their trials were held out. Input columns and the target were standardized separately using training-set means and sample standard deviations (ddof = 1); invalid or zero scales were replaced by one and remaining nonfinite standardized values by zero. A stable SeedSequence combined base seed 20260721, target root, stream and replicate. There were ten fixed split and initialization replicates per target.

With n_{site} input sites and n_{branch} occupied branches, input weights were initialized as $\mathcal{N}(0, 0.12^2/n_{\text{site}})$, branch readout weights as $\mathcal{N}(0, 0.25^2/n_{\text{branch}})$, and $c = 0$. Each method used 1,200 full-batch Adam steps with learning rate 0.015, $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. The exact model was trained first. Its exact site-error fields on training trials defined the dictionaries and its held-out exact field defined reported capture. All projected-learning methods then restarted from the identical initial state. Only the input-weight error was projected; $\mathbf{v}$ and c always received their exact gradients.

The primary budget requested four channels, reduced only when a target had fewer sites or nonempty candidate routes. Ancestry candidates were columns of ancestry multiplied by local inhibitory fraction g_i/g_{tot} ; routes were ranked by this fraction times their descendant-site fraction. Random paths sampled nonempty candidates uniformly without replacement. Each ancestry-shuffle column was permuted independently over sites. Depth controls were path-length quantile bins, scalar feedback was an all-ones column, and the dense oracle used the leading right singular vectors of the exact training error. Projectors used relative singular-value tolerance 10^{-10} . No held-out response was used to fit a dictionary.

Held-out normalized MSE divided a method’s held-out MSE by the MSE of predicting the training mean response. Replicates were averaged within target before inference; the seven target cells were the units. Intervals used 20,000 target bootstrap draws. Paired method comparisons used two-sided Wilcoxon tests, and the capture–utility analysis used method-label permutations stratified within target.

The four-channel primary outcomes are retained in the Source Data indexed in Table S12. The dense four-dimensional oracle nearly matched exact learning, establishing that feedback dimension alone was not limiting in this surrogate. Ancestry routes did not reliably outperform ancestry shuffling in capture or learning. The within-target association between capture and normalized error was $\rho = -0.150$ with a target-stratified method-label permutation $P = 0.419$. The target-bootstrap 95% interval for ρ was -0.449 to 0.144 . The coefficients are least-squares oracles and therefore upper bounds on the capacity of each feedback family, not biologically implemented encoders.

Sensitivity analyses at one, two and eight channels, with direct manual matches, with repeat-reliability thresholds, and with a linear branch model did not establish a reliable morphology-specific learning benefit. Some subsets improved morphology-specific capture without improving learning. The separation between capture and learning motivates the distinction between anatomical route capacity and task alignment.

1927 9.4 Complete reconstructed-tree learning across all eligible scans

1928 To retain the complete spatial state, we replaced the major-branch reduction with a segment-resolved model. Each
 1929 complete compressed reconstruction was instantiated as a reciprocal passive conductance system. Each imaged partner
 1930 supplied a trainable non-negative excitatory conductance at its dominant mapped segment. For trial-specific soma
 1931 prediction error, the exact input gradient factored into the local term $x_i(E_E - V_i)$ and the exact soma-to-segment adjoint
 1932 from the inverse conductance operator. Readout and bias parameters received exact gradients under every method.

1933 Restricted rules received scalar output error, local eligibility and one fixed site profile lying in a four-route span.
 1934 That profile was calibrated before training from a conductance-weighted topology dictionary, an equal-rank random
 1935 selection of nonempty anatomical routes, or a column-value-preserving site shuffle. No target response entered this
 1936 calibration. Each trial scaled the same profile by the scalar error; the rule did not infer four independent trial-dependent
 1937 coefficients. The comparison therefore tests this frozen-profile rule within each dictionary, rather than all estimators
 1938 its four-route span could support. Ten seed-fixed splits per scan held out complete stimulus identities. Runs were
 1939 averaged within scan and then within target. The resulting design contained four methods, 13 scans and ten splits, for
 1940 520 fits. The exclusion criteria were missing routes, rank mismatch, nonfinite values, incomplete scan coverage or failed
 1941 finite-difference validation. None of the runs met an exclusion criterion. One exact conductance coordinate per run
 1942 agreed with finite differences to maximum relative error 3.05×10^{-7} .

1943 Let δ_i^{out} be the scalar output error on trial i , $\mathbf{v}_0$ the baseline soma-to-site transfer profile and P_D the projector into
 1944 dictionary D . The delivered nonsomatic field is

$$\hat{\delta}_{i,D}^V = \delta_i^{\text{out}} P_D \mathbf{v}_0. \quad (117)$$

1945 This scalar-times-fixed-profile rule does not estimate four trial-dependent coefficients. Its prescribed updates U_D , with
 1946 rows corresponding to trials and columns to trainable input coordinates, are compared with exact updates U_{exact} by

$$\mathcal{M}_D = 1 - \frac{\|U_{\text{exact}} - U_D\|_F^2}{\|U_{\text{exact}}\|_F^2}. \quad (118)$$

1947 Because these coefficients are prescribed, $\mathcal{M}_D$ is normalized update match rather than an optimal projection-capacity
 1948 bound. Table S6 retains all primary learning and geometry outcomes. Topology did not reliably improve learning over
 1949 random or shuffled routes.

1950 A retrospective geometry check separated this coefficient restriction from the capacity of the fixed route spans. Full
 1951 checkpoint states were not archived, so we reexecuted the 130 exact-learning fits using their resolved settings, stimulus
 1952 splits and recorded random-route identities. At each resulting checkpoint we compared the frozen profile, a trialwise
 1953 exact-field projection, and an eligibility-weighted update oracle. For fixed dictionary Φ and trial eligibility vector ℓ_i ,
 1954 define $E_i = \text{diag}(\ell_i)$. The update oracle fitted

$$\mathbf{c}_i^* = \arg \min_{\mathbf{c}} \|\mathbf{u}_i - E_i \Phi \mathbf{c}\|_2^2, \quad \hat{\mathbf{u}}_i = E_i \Phi \mathbf{c}_i^*.$$

1955 This is the best attainable per-trial update inside that dictionary's eligibility-weighted span. Its normalized update match
 1956 equals projected energy capture. Split-level values were averaged within scan and then within target, with 20,000
 1957 bootstrap draws over the seven targets.

1958 The update oracle increased ancestry match from 0.2322 to 0.2436, compared with oracle matches 0.3847 for
 1959 random routes and 0.4677 for shuffling. The unrestricted baseline transfer profile reached 0.9756 (95% target interval,
 1960 0.9641–0.9858). Thus most exact-update geometry was already carried by one nearly fixed profile; restricting it to the
 1961 selected ancestry supports removed useful coverage. Selected routes covered about 48% of observed input coordinates,
 1962 averaging 1.31 inputs per route, with six of thirteen scan dictionaries containing only single-input selected routes.
 1963 Multiple observed inputs may map to the same physical segment. These geometric restrictions limit what the learning
 1964 comparison can reveal about anatomical alignment.

1965 Three of 130 replayed exact fits differed from archived NMSE by more than 10^{-5} (maximum 5.74×10^{-4}); maximum
 1966 frozen-profile match drift was 2.57×10^{-4} . All fits are retained as reexecuted states, not asserted to reproduce every
 1967 historical checkpoint exactly. `source_data/fulltree_within_span_oracle/` contains every geometry estimate and
 1968 replay discrepancy. These common-state projections were not trained and do not establish eventual oracle-learning
 1969 accuracy.

1970 9.5 Nested linear baselines and observed-input sensitivity

1971 To establish the predictive difficulty of the measured-response task, we fitted an intercept-only training-mean predictor,
1972 ordinary least squares and ridge regression on the same 130 outer stimulus-identity splits as the complete-tree analysis. All
1973 13 scans and seven targets were retained. Three inner folds partitioned training stimulus identities; no outer test identity
1974 entered preprocessing, reliability filtering or penalty selection. The primary linear models used the training-quantile input
1975 transform from the complete-tree analysis (10th and 90th percentiles, clipped at three), followed by column centering
1976 and standardization using training data. These quantities were refitted inside each inner fold. Raw standardized inputs
1977 supplied a secondary ridge reference.

1978 Ridge regression minimized mean squared training error plus $\lambda \|\mathbf{w}\|^2$, with an unpenalized intercept. The penalty
1979 grid was $\{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$; the minimum mean inner-fold validation error selected the penalty, with
1980 ties resolved toward the earlier, smaller penalty. An eigendecomposition solved each fit. The selected primary penalties
1981 were 0.1 in 90 splits, 1 in 36 and zero in four. Held-out normalized MSE used the same training-mean denominator as
1982 the nonlinear models. Archived nonlinear outcomes were read unchanged and paired to the new linear fits; they were not
1983 refitted under the sensitivity masks.

1984 Observed-input sensitivity used three explicit restrictions. First, the manual-only mask retained partners with archived
1985 manual-match provenance. Second, repeat reliability was the Spearman correlation between condition means formed
1986 from alternating repeats; it was recomputed using only inner training identities during tuning and only outer training
1987 identities for the final fit. Partners passing thresholds 0, 0.1 or 0.2 were retained. Third, five independent partner orderings
1988 per split defined nested masks retaining 25%, 50% or 75% of observed inputs, with at least one input. Empty manual or
1989 reliability masks used the intercept-only model and were retained. A separate measurement-noise series added Gaussian
1990 predictor noise with SD 0.25, 0.5 or 1 in units of the training-standardized input, in both training and test examples.
1991 The same underlying noise draws were paired across doses. Noise was applied after the quantile transform and training
1992 standardization; the intercept was refitted after perturbation.

1993 The secondary baseline protocol was fixed before its outcomes but was not publicly preregistered. Its 3,380
1994 evaluations are in `source_data/review_response_baselines/`. Masks were averaged within split, ten splits within
1995 scan and scans within target; intervals use 20,000 resamples of seven targets. Input hashes and all reconstructed split
1996 counts matched retained records. A held-out-data substitution audit confirmed that tuning, masks and training reliability
1997 were unchanged. Historical split arrays were reconstructed with the unchanged helper and seed rule. These are descriptive
1998 sensitivities within one mouse.

1999 Mean normalized MSE was 1 for the training mean, 0.807 for least squares and 0.803 for nested ridge (Supplementary
2000 Fig. S32). Exact nonlinear learning reached 0.788, a paired nonlinear-minus-ridge difference of -0.0152 (95% target-
2001 bootstrap interval -0.0271 to -0.0042 ; six of seven targets lower). The nonlinear model therefore provided a small
2002 predictive improvement over this linear baseline on the measured task. The corresponding subtree-minus-ridge difference
2003 was 0.0293 (-0.0130 to 0.0736), which does not establish a benefit for the frozen subtree rule. Manual-only inputs
2004 raised ridge MSE to 0.912, with 3.18 retained partners on average compared with 8.98 for all inputs. Retaining only
2005 25% raised MSE to 0.917; adding unit-SD predictor noise raised it to 0.869. The strictest reliability mask retained 5.63
2006 partners and reached MSE 0.837. These manipulations measure sensitivity to degradation of the observed inputs. They
2007 neither recover unobserved partners nor estimate the power of a complete-population experiment, and do not establish
2008 that missing inputs explain the absence of preferential ancestry alignment.

2009 9.6 Sensitivity of the measured ancestry-response test

2010 We estimated detection sensitivity conditional on the observed sampling, using a Gaussian response model frozen before
2011 simulation. The analysis retained 125 partner observations of 102 presynaptic roots across thirteen scans and seven
2012 targets, including twenty target-partner combinations repeated across scans and two roots shared between targets. Each
2013 scan contained 136 repeated stimuli: 130 presented twice and six ten times. Original condition hashes identified 316
2014 distinct repeated stimuli across scans. Pair tables, contact geometry, recording identities and the required observed
2015 responses are included in Source Data.

2016 For each target arbor, an ancestry covariance was defined by the shared soma-to-common-ancestor path length
2017 divided by the geometric mean of the two soma-to-contact lengths. This is a positive-semidefinite Brownian path kernel;
2018 zero-depth contacts contributed independent diagonal components. The seven kernels were embedded in common
2019 presynaptic-root coordinates, summed and diagonal-normalized, so each shared root had one consistent latent response.
2020 Reliable tuning mixed independent Gaussian unit-specific and ancestry-correlated components with variance fractions

2021 $1 - \lambda$ and λ , respectively. We tested $\lambda = 0, 0.05, \dots, 1$, retaining the actual partner sets and stimulus identities.
2022 Measurement noise was calibrated to each recording's odd/even split-half Spearman reliability. Values were clipped
2023 to $[0, 1]$ and converted to the Gaussian Pearson parameter $r_p = 2 \sin(\pi r_s / 6)$; the two negative estimates remained archived
2024 and contributed zero reliable variance. With h the mean inverse half-repeat count, trial responses had reliable variance
2025 r_p and noise variance $(1 - r_p)/h$. Condition-mean noise was divided by the actual repeat count. This matches the
2026 pooled Pearson second moment; its Spearman calibration is approximate because repeat counts vary. An independent
2027 1,000-dataset audit found mean and maximum absolute Spearman discrepancies of 0.0020 and 0.0069, without retuning.
2028 A prespecified perfect-reliability comparison removed measurement noise.
2029 In 4,000 simulated complete datasets, all pairwise similarities were computed from joint response matrices. The
2030 original partial-rank statistic controlled ranked Euclidean separation and soma-to-contact path-length difference; scan
2031 effects were averaged within target. Positive detection required a positive mean effect and an exact two-sided signed-rank
2032 $P \leq 0.05$ over the seven targets. All 128 sign assignments were enumerated. For seven nonzero untied effects, the
2033 minimum two-sided P is 0.015625, and six of 128 null sign assignments reject at 0.05. This discreteness does not imply
2034 a universal power ceiling below one. Gaussian draws were paired across signal fractions and reliability scenarios; Wilson
2035 intervals describe Monte Carlo uncertainty.
2036 With measured reliability, the first sustained tested 80% crossing was $\lambda = 0.60$, bracketed by 0.55–0.60. Descriptive
2037 interpolation gave $\lambda = 0.589$ and a mean simulated target-level partial-rank effect of 0.249. Detection at $\lambda = 0.60$ was
2038 80.85% (95% Monte Carlo interval, 79.60–82.04%); the lower interval first exceeded 80% at $\lambda = 0.65$. At maximal
2039 specified signal, detection was 96.23% (95.59–96.77%). The zero-signal two-sided rejection rate was 4.95% (4.32–
2040 5.67%). Perfect reliability reduced the interpolated variance fraction to 0.206, with a similar mean partial-rank effect of
2041 0.248 at detection. These sensitivity estimates assume the specified Gaussian ancestry covariance, reliability calibration
2042 and known shared-input dependence; they do not exclude weaker effects, other covariance patterns, unknown biological
2043 dependence or alignment involving unobserved partners.

2044 9.7 External signed-credit analysis and task-interference identity

2045 At the neuronal level, the framework gives a simpler prediction: when a causal task assigns opposite learning signs
2046 to two neurons, their measured plasticity-related contrasts should separate along the corresponding signed coordi-
2047 nate. The following reanalysis tests that prediction but does not address dendritic morphology. The Francioni et al.
2048 workbook was downloaded from the source-data link of the published article; file identity was verified by SHA-256
2049 be1b87481d0f1d1bffc85d8a4cabfd37b7055d640306afab5b3622de0ced8264. The animal-level contrasts in sheet
2050 “Ext 13E” were used directly. The endpoint was the difference in soma–dendrite residual, in z-score units, between
2051 error-reduction and error-increase epochs. P+ and P- denote populations assigned opposite causal signs in the source
2052 brain–computer-interface mapping. For the source-defined P+ and P- groups, we required the table's signed-separation
2053 column to equal P+ minus P- to numerical precision. The orthonormal common and signed projections sum exactly to
2054 total two-coordinate energy. Sheet “5E” supplied the four neuron-level distributions in the display but was not used to
2055 increase the sample size of the animal-level test.

2056 For $n = 6$ animals, the one-sided exact random-sign test enumerated all 2^6 sign assignments. Bootstrap intervals
2057 sampled animals with replacement. The directions were fixed by the experimental mapping: P+ positive, P- negative,
2058 and P+ minus P- positive. This is a retrospective test because the prediction was formulated after publication of the
2059 experiment, although it was evaluated from the numerical source table rather than the displayed figure.

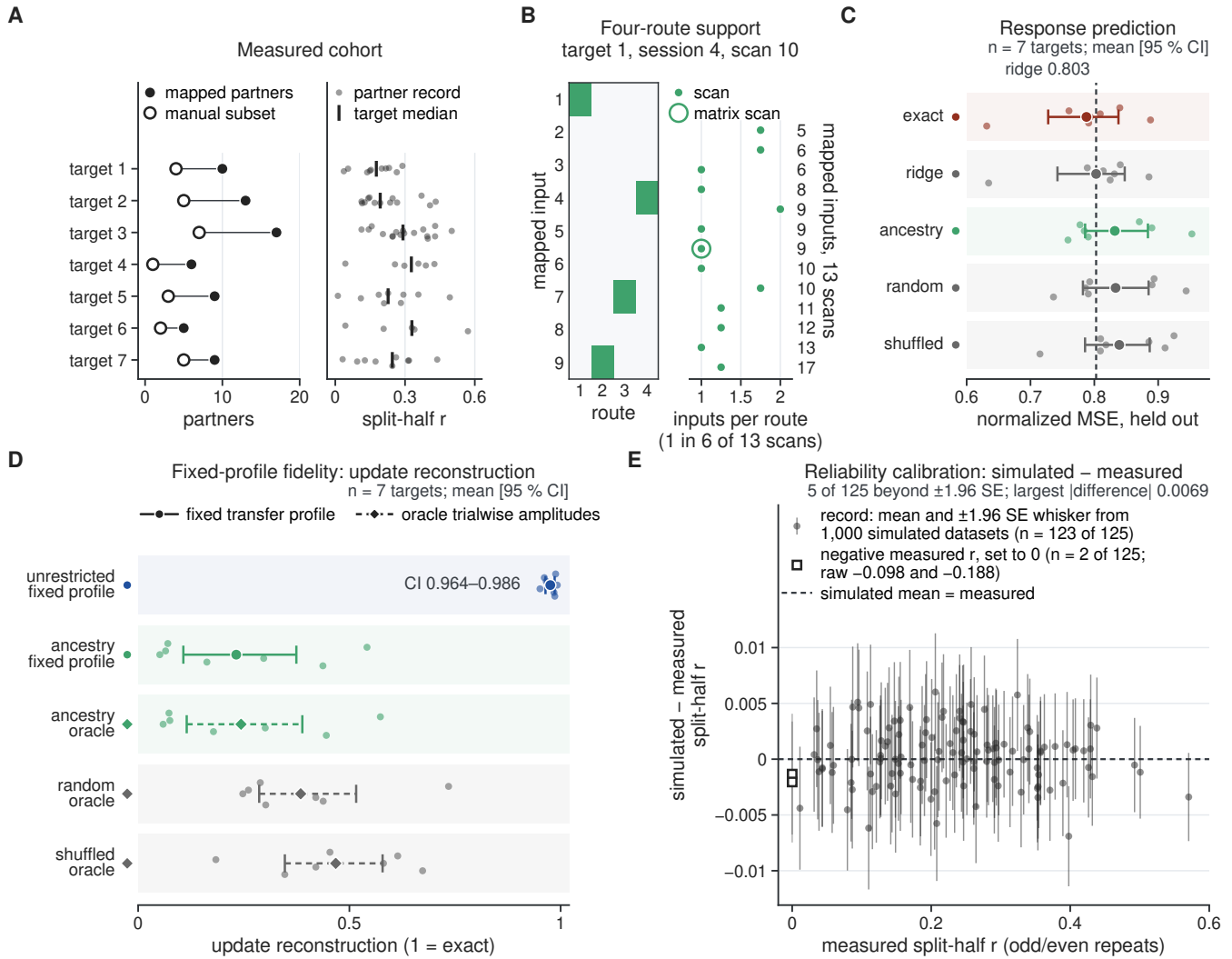


Figure S31: The measured-response learning comparison is limited by mapped coverage and transfer geometry. **A**, The seven targets (one mouse; one selected scan per target): mapped functional presynaptic partners (filled circles), the manually curated subset (open circles) and the split-half repeat reliability of each partner's responses (odd/even repeats, Spearman r ; small dots, 69 records) with the target median (bar). **B**, Left, the four-route support of the scan selected by median mapped-input count with identifier tie breaking (target 1, session 4, scan 10; nine mapped inputs): rows are input coordinates, columns selected routes, and green cells the four occupied coordinates, one per route (the matrix of main Fig. 9E). Right, input coordinates per route for all thirteen scans (one dot per scan; rows ordered by mapped-input count, which labels each row; the ring marks the scan drawn at left): 6 of 13 supports place one coordinate per route (mean 1.31). **C**, Response-prediction normalized MSE (held out) for the exact compartment-error fit (dark red), treeless ridge (grey), the ancestry-restricted dictionary (green) and the random-anatomical and site-shuffled surrogate dictionaries (grey); small dots are the seven target means, circles and bars their mean and 95% target-bootstrap interval (20,000 draws), and the dashed rule marks ridge (0.803). **D**, Common-checkpoint update reconstruction (1 = exact) by the unrestricted fixed transfer profile (blue; 0.976, interval 0.964–0.986, its bar covered by the marker), the ancestry-restricted fixed profile (green circle, solid bar; 0.232) and oracle trialwise amplitudes (diamonds, dashed bars) on ancestry (green; 0.244), random-anatomical (grey; 0.385) and site-shuffled routes (grey; 0.468); dots, means and intervals as in **C**. The unrestricted fixed profile is nearly sufficient because somatic error mainly scales a stable transfer vector, and both surrogates with oracle amplitudes reconstruct the update better on average than the ancestry restriction, although their target-bootstrap intervals overlap it. **E**, Calibration check: mean split-half (odd/even repeat) Spearman reliability in 1,000 independent calibration datasets minus the measured value, against the measured value, for 125 partner records (dots; whiskers, ± 1.96 Monte Carlo standard errors, SE 0.0020–0.0028); the two records with negative measured reliability (open squares; raw -0.098 and -0.188) are assigned zero reliable variance and drawn at 0, and the dashed rule is zero difference, not a fit. 5 of 125 simulated means differ from the measured value by more than 1.96 standard errors (largest difference 0.0069). n is seven targets in **C,D** and 125 partner-scan records in **E**; **A** and **B** are descriptive. Target-level associations for all four topology measures are in main Fig. 9C, and coverage across all thirteen scans is in main Fig. 9F.

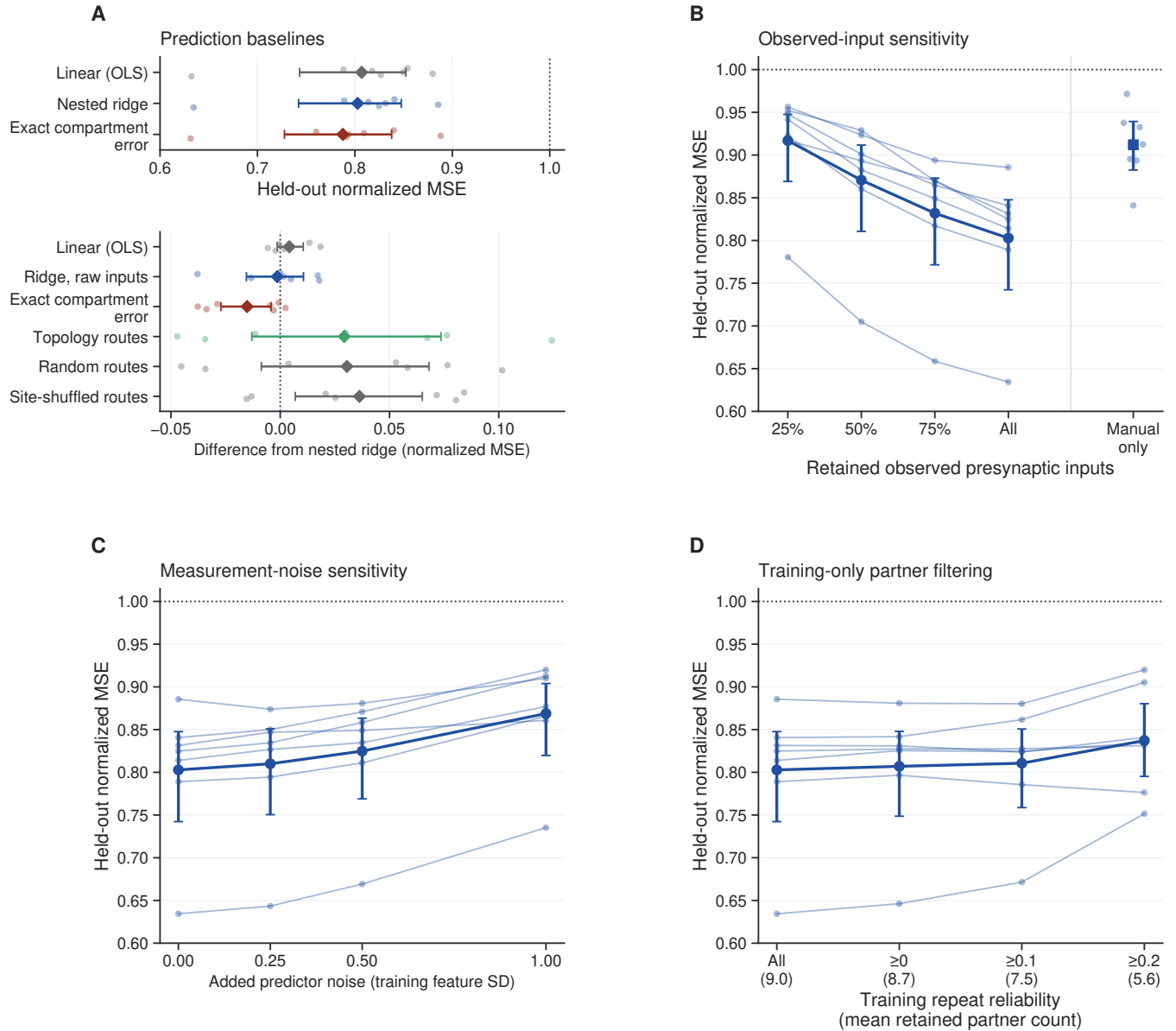


Figure S32: Linear baselines explain most measured-response prediction, and degrading observed inputs worsens performance. Thirteen scans retain the complete-tree analysis's 130 outer stimulus-identity splits. All preprocessing, reliability masks and ridge tuning use training identities only. **A**, Ordinary least-squares (OLS) and nested ridge-regression baselines against the archived exact compartment-error fit; the dotted line is the training-mean predictor (normalized MSE = 1 by construction). Lower normalized MSE is better. Below, paired differences from nested ridge (dotted line, zero difference) for each comparator: OLS, ridge on untransformed inputs, the exact compartment-error fit, and the archived route-restricted fits with topology-matched (ancestry), random anatomical and site-shuffled routes, whose absolute errors are in Supplementary Fig. S31C. The exact fit gives a small additional nonlinear benefit, mean -0.0152 (95% interval -0.0271 to -0.0042), whereas topology-matched routes (0.029 ; -0.013 to 0.074) are matched by random routes (0.031 ; -0.009 to 0.068). **B**, Ridge prediction when retaining nested random fractions of observed presynaptic inputs; five masks per split are averaged before inference. Manual-only partners, right of the separator, are the same ridge model under a provenance restriction, not a random retention dose. Empty masks use the training mean and remain included. **C**, Added Gaussian predictor noise in both training and test data, in units of training input standard deviation (SD) after the quantile transform. **D**, Reliability filtering based only on training repeats; parentheses give mean retained partner counts. In **B–D** the dotted horizontal line marks the training-mean predictor (normalized MSE = 1). Thin traces and small dots show seven target means after averaging splits within scan and scans within target. Larger symbols and bars are means and 95% target-bootstrap intervals (20,000 draws). All targets come from one mouse. Only ridge is refitted under the sensitivity conditions. Removing observed inputs does not recover unknown partners, establish population power or explain away the absence of preferential ancestry alignment.

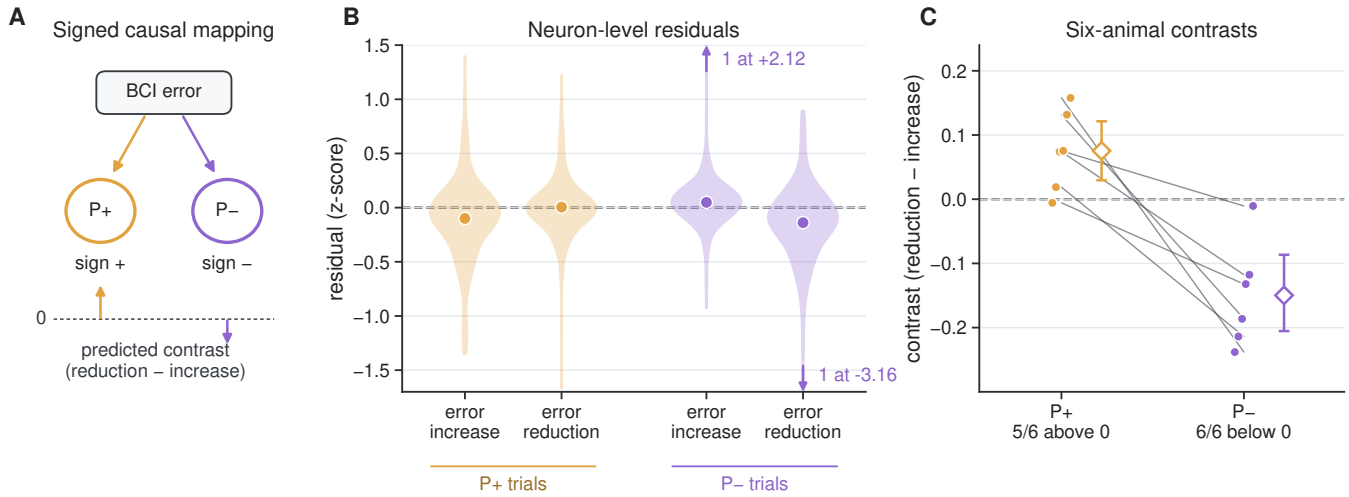


Figure S33: Published animal data are consistent with signed per-neuron teaching coordinates. This reanalysis contains no reconstructed dendritic morphology and measures no within-tree learning signal; it tests only the sign of two neuron-level coordinates. **A**, In the source brain–computer-interface (BCI) experiment, P+ and P- populations had positive and negative causal decoder weights, predicting opposite dendritic residual contrasts. **B**, Descriptive neuron-level residual distributions from published source sheet “5E” for error-increasing and error-reducing trials, in the source z-score normalization; dots show means, and the neuron-level standard errors (0.015–0.024 z) are smaller than the marker. The axis is clipped to -1.7 to 1.5 z; the two neurons beyond it are counted at the arrow-capped ticks. These are descriptive and are not animal-bootstrap intervals. **C**, The six paired animal contrasts (reduction minus increase; lines join the two populations of one animal), with means and animal-bootstrap 95% intervals from Source Data summary.json: P+ 0.075 (0.030–0.121), five of six animals positive; P- -0.150 (-0.205 to -0.087), six of six negative. The mode-energy decomposition uses the same $n = 6$ animals and is reported in Table S7; animals are the inferential units for those tests. This retrospective external comparison is consistent with signed per-neuron information but does not resolve within-tree transport or uniquely identify a conductance mechanism. This reanalysis is restored as a figure so that the six-animal signed-credit comparison described in Methods has drawn support in C; the animal-level contrasts also remain tabulated.

2061 **10.1 Prospective costed selection in tree-constrained linear learning**

2062 These secondary studies test structure selection separately from the main credit comparisons. The first asks whether an
 2063 initialization descent bound can select a model for later learning. Five explicit 15-node trees supplied eight labeled leaves:
 2064 four balanced hierarchies with distinct leaf orders and one contiguous comb. Each supplied orthonormal subtree-cut
 2065 dictionaries at $K = 1, 2, 4, 8$, giving 20 candidates per arm. Every candidate retained 64 trainable leaf-input weights. In
 2066 the feedback-only arm, all candidates used identity forward transfer $H_T = I_8$. In the joint arm, H_T was the leaf block
 2067 of $(I_{15} + \kappa L_T)^{-1}$, where L_T was the full-tree weighted Laplacian, $\kappa = 1$, unit leak was present at every node and edge
 2068 conductance was inverse embedded length. Internal nodes were retained. For input $\mathbf{x} \in \mathbb{R}^8$, trainable weights $W \in \mathbb{R}^{8 \times 8}$
 2069 and an imposed context decoder $\mathbf{a} \in \mathbb{R}^8$, the scalar prediction was $\mathbf{a}^\top H_T W \mathbf{x}$. Here W denotes model weights. This
 2070 passive-tree linear abstraction uses an imposed context readout, not a somatic voltage or the nonlinear physical-depth
 2071 model.

2072 A random orthogonal teacher mapped isotropic Gaussian inputs to labels. Context decoders were sampled uniformly
 2073 from the first r columns of an orthogonal basis Q , with $r \in \{1, 2, 4, 8\}$ specifying the reference credit rank. Observation-
 2074 noise standard deviation (SD) was $\sigma_y = 0$ or 0.75. At zero initialization, the exact reference decoder-space credit second
 2075 moment was

$$S_0 = (1 + \sigma_y^2) Q \text{diag}(\underbrace{1/r, \dots, 1/r}_r, 0, \dots, 0) Q^\top.$$

2076 Rotation preserves this reference spectrum, but the joint arm's transfer can change transported-credit and parameter-
 2077 gradient spectra. Angle and noise comparisons shared underlying random draws within seed and rank. Five development
 2078 seeds (6100–6104) used angles $0, \pi/4, \pi/2$; 20 separate confirmation seeds (7200–7219) used held-out angles $\pi/8, 3\pi/8$.
 2079 The 16 rank–noise–angle conditions within each confirmation seed were paired, giving 320 tasks per arm.

2080 For dictionary projector P , the actual synaptic descent direction was $\mathbf{d}_i = P H_T^\top \mathbf{a}_i \delta_i \mathbf{x}_i^\top$, where δ_i was the scalar
 2081 prediction error on example i . Inner products and covariances use the vectorized 64-coordinate update. Coefficients
 2082 receive the known decoder and transfer, rather than a learned local encoder. Each candidate had 64 fixed task-decoder
 2083 coefficients, K channels, $8K$ possible context-to-route encoder coefficients, eight nonzero delivery entries, 15 electrical
 2084 nodes and 14 edges. Let ℓ_T be the total cable length in the fixed two-dimensional embedding and $\ell_{\max}$ the largest
 2085 candidate length. The normalized geometric cable cost was $\ell_T/\ell_{\max}$; the resource cost was $C_T = K/8 + 0.1\ell_T/\ell_{\max}$. This
 2086 declared cost proxy does not measure biological expenditure. Candidates with identical projectors but different cable
 2087 costs remain identified as equivalence classes, not replicates.

2088 Independent calibration, training and test sets contained 1,024, 2,048 and 4,096 examples. Calibration evaluated the
 2089 actual routed update mean and empirical single-example covariance, giving $\mathbb{E}\|\mathbf{d}_{\text{batch}}\|^2 = \|\mathbb{E}\mathbf{d}_i\|^2 + \text{trCov}(\mathbf{d}_i)/32$ for
 2090 minibatches of 32 independent and identically distributed (iid) draws. The initialization score was the bound at the actual
 2091 step $\eta = 0.35$,

$$U_T = \eta \mathbf{g}^\top \mathbb{E}\mathbf{d}_i - \frac{\eta^2 L_T^{\text{cal}}}{2} \left(\|\mathbb{E}\mathbf{d}_i\|^2 + \frac{\text{trCov}(\mathbf{d}_i)}{32} \right).$$

2092 Here L_T^{cal} is the largest Hessian eigenvalue of calibration half mean squared error (MSE), global in W . The bound
 2093 concerns calibration loss, not independent-test or population loss. All weights started at zero. Paired candidates received
 2094 the same iid minibatch stream for 256 updates, with checkpoints at 16, 64 and 256. Only the final checkpoint defined the
 2095 primary endpoint, test half-MSE plus $0.08C_T$.

2096 Development's 4,800 fits fixed one nonnegative utility-to-final-loss scale $s \in [0, 256]$ per arm, using within-task
 2097 centered outcomes. The moment policy minimizes $\mathcal{L}_{\text{cal}}(0) - sU_T + 0.08C_T$. Development also fixes the constant,
 2098 maximum-budget and known-rank policies; sampled and exact-expected random choice are references. All 3,200
 2099 confirmation choices were hash-sealed before training without retuning. The prospective protocol retains all 12,800
 2100 confirmation fits and 38,400 checkpoint rows, with no outcome exclusions.

2101 Regret subtracts the retrospectively best candidate's costed loss on each task; this oracle is unavailable to selection.
 2102 Intervals use 10,000 bootstrap draws over 20 whole-seed averages. Moment-policy regret was 0.08055 (95% interval
 2103 0.07262–0.08781) for feedback-only selection and 0.10404 (0.09975–0.10791) for joint selection. Fixed maximum-
 2104 budget regret was 0.04495 and 0.04269; rank-only regret was 0.00831 and 0.00821. The moment selector lost to
 2105 both in every seed average in both arms. Rank-only improved over fixed by 0.03664 (0.03630–0.03697) and 0.03448
 2106 (0.03427–0.03468). The retrospectively best budget equaled the imposed rank on every confirmation task, while the

contiguous balanced tree won 308/320 feedback-only and 303/320 joint tasks. Thus the constructed family supports rank-dependent capacity but little fine tree selection.

Post hoc checks separate this failure from local prediction: the bound held in all 12,800 calibration conditions, and utility correlated 0.984/0.974 with first-update test improvement. Algebraic, covariance, derivative and source checks passed; 120 independent endpoints replayed within 5.55×10^{-17} . Complete matrices, policies and audits are in `source_data/prospective_morphology_selection/`.

10.2 Finite, noisy calibration of interaction-based tree selection

The finite-calibration study uses the eight-input, 28-coefficient model and twelve-candidate pool of Section 4.1, with four equally weighted families: four disjoint pairs, two complementary quartets, nested products of degrees 2/4/6/8, and a random-interaction control selecting four distinct degree-2–8 monomials. Input permutations, coefficient signs and magnitudes $\mathcal{U}[0.5, 1.5]$ vary by task; coefficients are normalized to unit squared norm. Targets have zero mean and unit variance on uniform Rademacher inputs, without matched spectra across families.

The estimator receives input/output examples and all 255 nonconstant Walsh monomials, without family, planted support, true coefficients or derivatives. Lasso fits an unpenalized intercept and uses penalty $c \text{sd}(y) \sqrt{2 \log(256)/n_{\text{fit}}}$, cyclic coordinate descent, tolerance 10^{-8} and at most 20,000 iterations. Five development blocks select c from 0.15/0.35/0.65, fixing 0.15. Calibration budgets 64/256/1,024 cross Gaussian label-noise SD 0/0.5. The first 75% of observations fit the estimator and pilots; the final 25% is a calibration gate. Primary calibration therefore uses 192 fit observations out of 256 queries at SD 0.5.

The selector minimizes the maximum normalized centered-cut tail beyond rank one, breaking 10^{-10} ties by summed tails and candidate identifier. Only true-coefficient maximum tails are bounds; estimated tails and summed-tail tie-breaking are scores. Secondary subset dynamic programming searches beyond the pool at the primary condition.

Development fixes a constant candidate and an estimated-rank rule using the rank that explains 99.5% of estimated input-gradient energy. Both reduce to `balanced_p0`; sparse bins use the constant fallback. Other references are fixed shape representatives, exact uniform-random candidate expectation and a two-sweep ALS pilot. The pilot fits every candidate from one fixed initialization on the calibration-fit subset and chooses by gate MSE, then discards all pilot parameters. True-coefficient selectors and the retrospective best-trained candidate are privileged references.

Twenty new seed blocks supply eighty tasks; calibration inputs, scores, choices and adaptive trees were sealed before fitting. Pre-fit implementation repairs preserved those choices and are recorded in Source Data. Final fitting uses separate 1,024-example training data with noise SD 0.3, 32 ALS sweeps, two paired random restarts and output-preserving gauge normalization. Independent 512-example noisy validation chooses the restart. Twelve fixed candidates, one estimated adaptive tree and one true-coefficient adaptive reference give 2,240 fits; fixed candidate outcomes are reused across calibration conditions.

Primary evaluation uses clean-target MSE on 4,096 independent inputs and unit population variance; noisy-test and all-256-pattern errors are secondary. Streams can repeat patterns with independent observation identities/noise. Regret subtracts the best validation-selected fixed-pool result, retaining negative adaptive regret. Failed/nonfinite fits receive loss 10^6 , with warnings and counts retained.

The primary 256-query/noise-0.5 comparisons against development-best fixed and the two-sweep pilot use 10,000 whole-seed bootstrap draws, averaging families within seed, Bonferroni-adjusted 97.5% intervals and a 0.01-NMSE margin. Other comparisons are descriptive. The exponential Walsh dictionary and ALS fitter constrain scalability and biological interpretation. Full outcomes and costs are in Source Data and Supplementary Fig. S36.

10.3 Direct confirmation from noisy morphology estimates to gradient learning

Twenty independent seed blocks 930910–930929 generate eighty new functions from the same four-family mixture. The fixed Lasso ($c = 0.15$) receives 192 of 256 calibration observations at label-noise SD 0.5; the remaining 64 are reserved, unused. All estimated trees are hash-sealed before learning. The earlier best fixed balanced tree and a post-seal true-coefficient construction are references. Construction coefficients never initialize training.

Each of three trees receives exact path credit or unit root broadcast, giving 480 fits. The seven multi-affine units, 28 coefficients and root-only output are unchanged; broadcast replaces six nonroot sensitivities by one. All conditions share Gaussian nonbias initialization of SD 0.5, zero biases and independent calibration/training/test/minibatch streams. Coefficients remain in $[-2, 2]$ and gradients are clipped at norm ten. Adam uses (0.9, 0.999) and $\epsilon = 10^{-8}$, with inherited rates 0.01 for exact and 0.003 for broadcast. Thus comparisons are between frozen recipes, not common rates.

Learning uses 1,024 updates of 64 examples from a 2,048-example cache with label-noise SD 0.15, minimizing half-MSE divided by known target variance. Separate 4,096-example test data use fresh noise. No validation selection, retuning or continuation from calibration fits occurs. Primary noisy-test NMSE has expected floor 0.0225; clean test and all-pattern errors are secondary. Within-seed family-averaged comparisons use 10,000 bootstrap draws, adjusted 97.5% intervals for the two primary contrasts and the same 0.01 effect margin. Fixed-minus-estimated exact-credit error was 0.45913 [0.38489, 0.53227]; broadcast-minus-exact on estimated trees was 0.60750 [0.56916, 0.64610]. Matching, quartet and nested tasks approached the floor with estimated-tree exact credit, while random interactions retained error and broadcast performed slightly better on matching.

Six full-horizon checks (seed 721000) are excluded; a documented pre-fit repair preserved settings and choices. Independent reconstruction reproduced all eighty trees and endpoints exactly. All 480 trajectories and validation records are in `source_data/morphology_calibration/end_to_end`. The pipeline demonstrates finite-data construction and learning within this class, not endpoint-optimal morphology.

10.4 Finite-horizon forecasts for the contextual linear learner

A finite-horizon predictor tests the failed scalar extrapolation while retaining the candidate pool, $\eta = 0.35$, minibatch size $B = 32$, 256-update horizon and cost. Five original development seeds informed its design; the predictor was frozen before fresh calibration or training. Twenty new seeds (9300–9319) crossed four ranks, two noise levels, two held-out angles, both transfer arms and both training regimes. Every candidate was trained, yielding 25,600 new fits and 76,800 checkpoint outcomes at updates 16, 64 and 256. Non-pilot scores were sealed before each seed’s outcomes without endpoint tuning. The predictor separates mode-dependent dynamics from a scalar initial slope, motivated by deep linear learning and task alignment with kernel eigenfunctions [12, 13].

To forecast learning, we express the weights in route coordinates. For orthonormal routes Φ , the learner stays in $W = \Phi V$. For context j with decoder $\mathbf{a}_j$, define the effective route vector $\mathbf{c}_j = \Phi^\top H_T^\top \mathbf{a}_j$. The scalar prediction is $\mathbf{c}_j^\top V \mathbf{x}$ and the actual update is minibatch descent in V . Let $\mathbf{v} = \text{vec}(V)$ stack the route-coordinate weights. For calibration example i with input $\mathbf{x}_i$, label y_i and effective route vector $\mathbf{c}_i$, define $\mathbf{z}_i = \text{vec}(\mathbf{c}_i \mathbf{x}_i^\top)$. With $\mathbb{E}_i$ denoting an average over the calibration-fit examples, set $H_{\text{emp}} = \mathbb{E}_i[\mathbf{z}_i \mathbf{z}_i^\top]$ and $\mathbf{b}_{\text{emp}} = \mathbb{E}_i[y_i \mathbf{z}_i]$. Write $H_{\text{emp}} = U \text{diag}(\lambda) U^\top$, with orthonormal eigenvectors in U and eigenvalues in λ . After T updates, full-batch descent from zero gives

$$\mathbf{v}_T = U \text{diag}\left(\frac{1 - (1 - \eta\lambda)^T}{\lambda}\right) U^\top \mathbf{b}_{\text{emp}}, \quad (119)$$

The diagonal expression is evaluated separately at each eigenvalue, with continuous value ηT when that eigenvalue is zero. The empirical selector fits the first 512 calibration examples and evaluates on the other 512; Eq. 119 is exact for that full-batch surrogate, not for the original cached-data SGD process.

For fresh inputs $\mathbf{x} \sim \mathcal{N}(0, I_d)$, where d is the input dimension, a smaller recurrence describes stochastic gradient descent (SGD). Let p_j be the context probabilities and $\mathbf{t}_j \in \mathbb{R}^d$ the target coefficient vector in context j . Suppose $y = \mathbf{t}_j^\top \mathbf{x} + \varepsilon$, with independent zero-mean noise of variance σ_j^2 . The exact recurrence closes on the mean weights $M = \mathbb{E}[V] \in \mathbb{R}^{K \times d}$ and the route-space variance matrix $\Gamma = \mathbb{E}[(V - M)(V - M)^\top]$. The latter sums weight covariance over the input coordinates; it is not the full vectorized parameter covariance. Define

$$C = \sum_j p_j \mathbf{c}_j \mathbf{c}_j^\top, \quad D = \sum_j p_j \mathbf{c}_j \mathbf{t}_j^\top, \quad \bar{G} = CM - D, \quad (120)$$

$$R_j = \|M^\top \mathbf{c}_j - \mathbf{t}_j\|^2 + \mathbf{c}_j^\top \Gamma \mathbf{c}_j, \quad A_\eta = I - \eta C, \quad (121)$$

$$J = \sum_j p_j \mathbf{c}_j \mathbf{c}_j^\top [(d+2)R_j + d\sigma_j^2], \quad K_G = C\Gamma C + \bar{G}\bar{G}^\top. \quad (122)$$

Here $\bar{G}$ is the mean gradient, R_j is the expected squared residual excluding observation noise, J is the second moment of the single-example gradient Gram matrix, and K_G is the corresponding second moment of the conditional-mean gradient. For a fixed residual coefficient vector $\mathbf{e}$, Gaussian fourth moments yield $\mathbb{E}[(\mathbf{e}^\top \mathbf{x} - \varepsilon)^2 \|\mathbf{x}\|^2] = (d+2)\|\mathbf{e}\|^2 + d\sigma_j^2$. Conditioning first on the current weights, the Gram of a minibatch-mean gradient is its conditional-mean Gram plus $1/B$

times its conditional variance. Total covariance therefore gives

$$M_{t+1} = M_t - \eta \bar{G}_t, \quad (123)$$

$$\Gamma_{t+1} = A_\eta \Gamma_t A_\eta^\top + \frac{\eta^2}{B} (J_t - K_{G,t}), \quad (124)$$

$$\mathbb{E}[\mathcal{L}_T] = \frac{1}{2} \sum_j p_j [R_j(T) + \sigma_j^2]. \quad (125)$$

The forecast updates noise with the residual and weight variance. It requires Gaussian input fourth moments and fresh independent minibatches; noise needs only the stated first two moments. This closure does not cover nonlinear targets, anisotropic inputs, adaptive optimizers or correlated samples.

The primary plug-in predictor estimates context frequencies, conditional coefficient vectors and residual noise variances using 1,024 independent calibration examples. Coefficients use ordinary least squares without an intercept, and noise variances use residual degrees of freedom. Its recurrence is exact for the fitted Gaussian population and is a plug-in approximation for the generating population. Population-oracle versions instead receive true teacher coefficients and noise moments and are labeled privileged. The 16-step pilot performs real candidate updates and evaluates on calibration observations; its computation is included. An observed-context-count baseline chooses the smallest permitted K covering the number of distinct observed context vectors, followed by the cheapest tree. This and an observed-count-restricted conditioning baseline were frozen after development inspection, before new outcomes. Context count equals generating rank only in this constructed family.

We ran both fresh-example SGD and minibatches resampled from a fixed 2,048-example cache, retaining a common 4,096-example test set. Reusing a cache makes later weights dependent on those examples and violates the fresh recurrence’s independence assumption. Population-oracle prediction minus exact population loss of the realized weights quantifies that discrepancy. All conditions are averaged within seed before 10,000 whole-seed bootstrap draws. Pointwise intervals for secondary comparisons carry no multiplicity-adjusted confirmation claim. Numerically constant cases retain undefined correlations with explicit defined-task counts, without outcome exclusions.

Supplementary Fig. S35 shows endpoint regret and strong baselines, selection timing, and the forecast-versus-outcome comparisons; the joint-arm cache sensitivity is retained in Source Data. The complete timing and prediction outcomes are in Source Data (Table S12). The primary forecast reaches mean fixed-cache regret 0.000064 and 0.000042 in the feedback and joint arms, while the cheap observed-count baseline reaches 0.000134 and 0.000197. The scalar-score improvement therefore does not establish a large fine-selection advantage. Tests verify spectral forecasts against direct descent, stochastic moments against Gaussian quadrature, sealed choices and exclusion of training/test observations from predictors. Complete data, protocols and source hashes are in `source_data/morphology_structure/` and `source_data/morphology_finite_horizon/`.

10.5 Statistical analysis, reproducibility and interpretation

Central comparisons report paired effect sizes, the number of positive cell or seed pairs, and 95% intervals unless a different level is specified for that analysis. Tests were two sided unless a direction had been specified before analysis. Exact Wilcoxon signed-rank tests were used for small paired cell cohorts. Reconstructed-cell intervals resampled cells first and, where relevant, one Monte Carlo stream within cell. Measured-response intervals resampled target cells. Artificial-network summaries used independently initialized training runs. Inhibitory-census outcomes were first averaged within postsynaptic target. Signed-credit intervals and exact sign tests used animals; neuron-level source rows were descriptive only.

Bootstrap intervals used 20,000 draws unless an experiment’s protocol states otherwise; the Fashion-MNIST ladder, the physical-depth, H4 and task-family cohorts and the external animal reanalysis used 50,000 paired draws.

Benjamini–Hochberg correction was applied within each prespecified comparison family—the archived four-budget subtree scan across $K = 1, 2, 4, 8$, the six bandwidth-matched routing contrasts and the H4 primary family. The additional four-budget ancestry-minus-best-control reanalysis enumerated all 2^{20} paired sign assignments for a two-sided mean-difference test and applied Holm correction across the four budgets. Accuracy differences were reconstructed on the exact 1/2048 test-count grid before testing. Initialization-score correlation intervals used 5,000 whole-seed bootstrap draws. The local-context soft-encoder analysis used Holm correction across its three primary contrasts; the hard-readout follow-up remains exploratory. The prospective selection experiment resampled 20 whole seed blocks 10,000 times. The finite-horizon, morphology-calibration, credit and conductance studies also used 10,000 whole-seed draws, retaining

2230 task, candidate and optimizer conditions together within each seed. The two primary comparisons in each of the
2231 finite-calibration and separate end-to-end experiments used Bonferroni-adjusted 97.5% intervals and a prespecified 0.01
2232 improvement margin in normalized MSE (NMSE). Other bridge intervals are pointwise and descriptive. Exhaustively
2233 enumerated structural targets have no biological sampling interpretation. Response-baseline sensitivity retained seven
2234 targets and 20,000 draws. Other parameter sweeps and sensitivity analyses received no multiplicity correction, are
2235 labeled secondary, and are not interpreted as independent confirmation.

2236 Cross-mouse contrasts are summarized within volume; cell-level tests do not provide an animal-level P value from
2237 two mice. The 12-cell Pinky cohort was sized to span 12 equal-count spatial strata while keeping complete mesh
2238 skeletonization and 500-shuffle processing feasible. The task-family factorial used ten paired seeds, matching the sample
2239 size used for the physical-depth experiments; positive directional language requires a paired 95% interval excluding zero
2240 and at least 8/10 seed signs.

2241 Cross-architecture conclusions exclude pooled error-field metrics that mix dendritic stages with a somatic stage
2242 whose reconstruction is exact by construction. They also exclude unweighted path-gain dispersion as evidence of gradient
2243 direction and post hoc comparisons between unmatched training cohorts. Table S10 records the supported interpretation
2244 and boundary of each evidence class.

2245 Each empirical main panel has a machine-readable table with values, independent and nested units, seeds, conditions
2246 and analysis version. Automated checks verify these figure inputs. Authentication tokens, restricted caches, and raw data
2247 already hosted by MICrONS or DANDI are not redistributed.

2248 Source Data retain configurations, outcomes, exclusions, selections and hashes; software contains implementations,
2249 joins and figure builders. Manifests authenticate source identities and path translations without changing numerical
2250 settings. The source index maps compact tables to complete records.

2251 An exact package lock was not archived for the June exact-transport or backpropagation jobs, nor for the measured-
2252 response analysis. These versions are therefore not guessed from the current environment. The complete July feedback
2253 analysis records Python 3.10.13, its Git identity and selected source hashes. Its archived source state nevertheless
2254 includes tracked modifications. Table S3 and the accompanying reproducibility README distinguish exact, partial and
2255 unverifiable provenance fields.

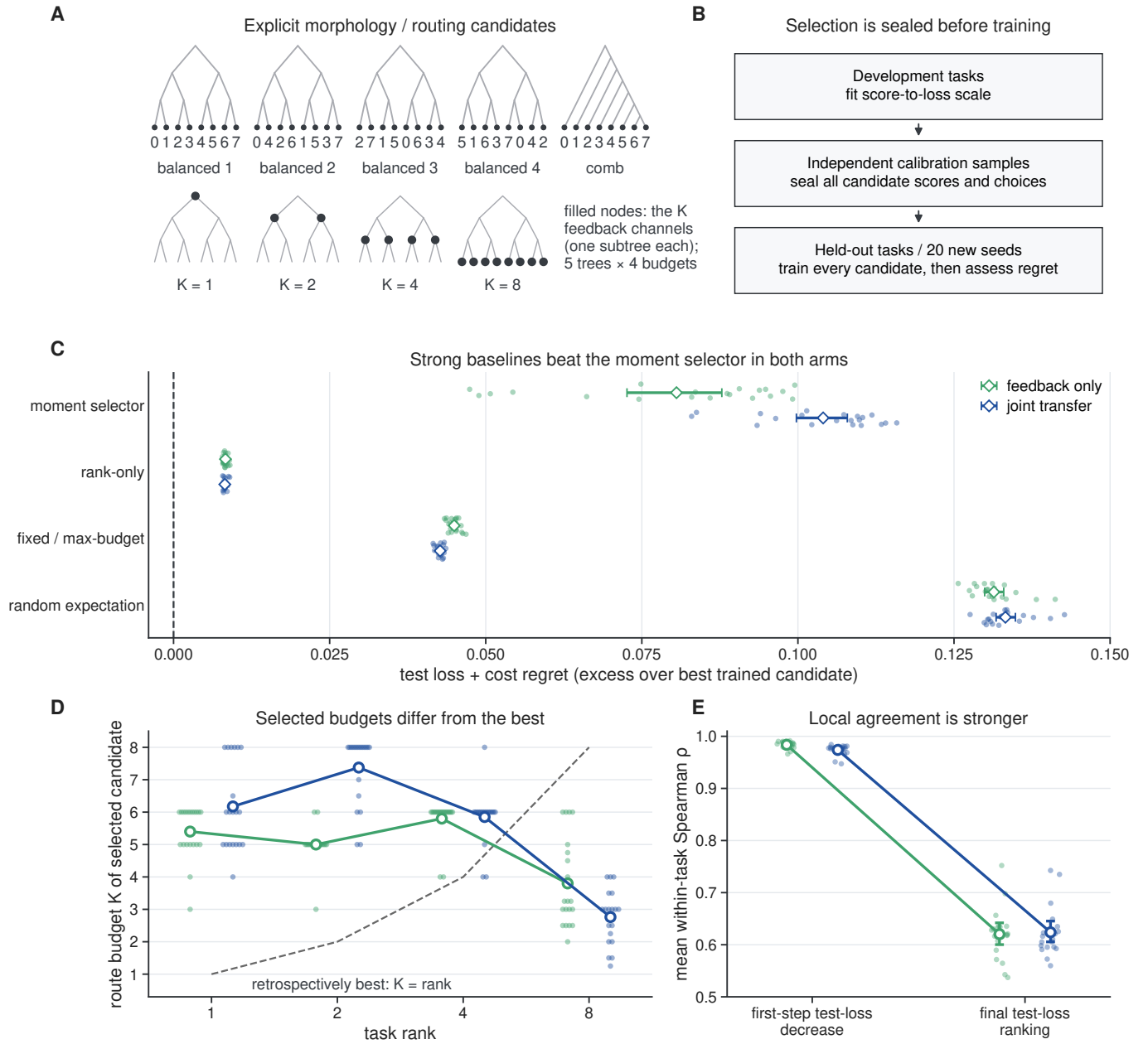


Figure S34: The original scalar initialization score predicts a first step but fails at prospective endpoint selection. **A**, Four balanced leaf assignments (balanced 1–4; the number under each leaf is the input index at that leaf slot) and one comb, each at route budgets $K = 1, 2, 4, 8$: twenty candidates with prescribed cost. Filled nodes mark the K feedback channels, one subtree each. **B**, Development, independent calibration, sealed choices and confirmation training. **C**, Final test half-mean-squared-error (MSE) plus cost, expressed as excess over the best trained candidate, for the feedback-only (green) and joint passive-transfer (blue) arms; the dashed line marks zero regret. Small points are the twenty whole-seed means; open diamonds and whiskers are means and 95% seed-bootstrap intervals (10,000 draws), which are narrower than the diamond in the rank-only, fixed / max-budget and random-expectation rows. The fixed / max-budget baseline (the development choice, which is also the maximum budget) and the privileged rank-only baseline outperform the moment selector; random expectation averages all candidates uniformly. **D**, Route budget of the candidate chosen by the moment selector versus imposed rank; small points are the twenty seed-block means, open circles their mean, and the gray dashed line the retrospectively best candidate's budget, which equals the rank in every task of both arms. **E**, Mean within-task Spearman correlation of initialization utility with the actual first-step test-loss decrease and with the final test-loss ranking; small points are the twenty seed-block means, open circles and whiskers their mean and 95% interval resampling whole seeds ($n = 20$ seed blocks). Colours in **D,E** as in **C**. All 12,800 confirmation candidate fits and 38,400 checkpoints remain included. Every model has a freely trained dense weight matrix; passive transfer changes conditioning and cost but does not necessarily restrict representation. This experiment is distinct from the locally connected multi-affine construction and the separately frozen finite-horizon follow-up.

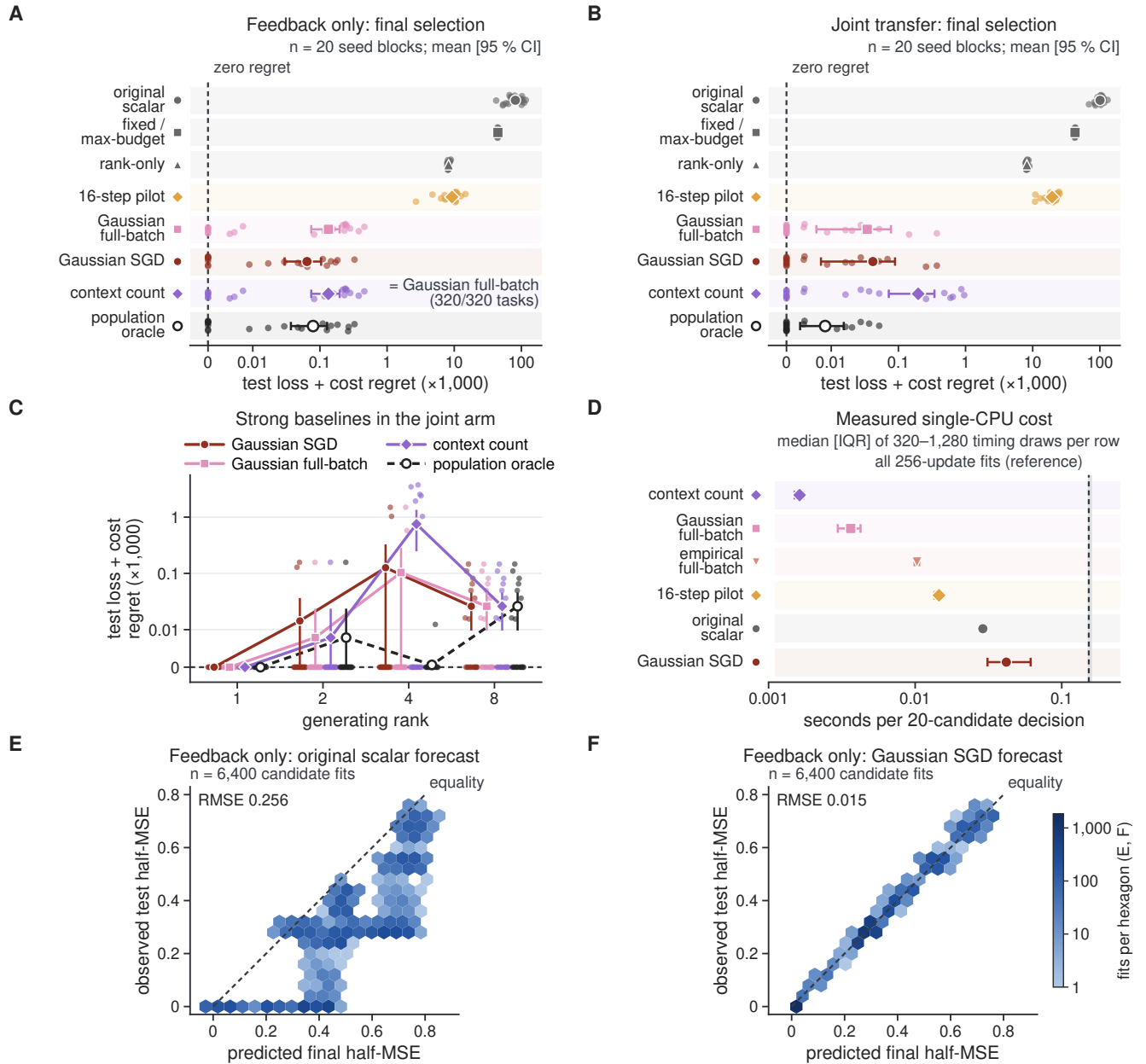


Figure S35: Finite-horizon prediction improves selection but strong simple baselines remain. **A,B**, Final test loss + cost regret ($\times 1,000$) pooled over generating ranks in the feedback-only and joint-transfer fixed-cache learners; regret is excess above the best trained candidate, and the axis is linear from zero to 0.01 and logarithmic above, with the dashed rule at zero regret. Small points in **A–C** are the twenty whole-seed means, markers and whiskers their mean and 95% interval. The top three grey rows are the baselines of Supplementary Fig. S34C,D (moment selector there, original scalar here) on independent fresh seeds; in **A** the Gaussian full-batch and context-count rows coincide because they chose the same candidate in all 320 tasks. **C**, The four strong selectors at each generating rank in the joint arm, on the regret scale of **A,B**: Gaussian SGD (dark red circles, the primary method), Gaussian full-batch (rose squares), observed context count (violet diamonds) and the privileged population oracle (hollow circles, dashed). **D**, Median elapsed time on one central processing unit per twenty-candidate decision, rows sorted by cost on a logarithmic time axis; ‘empirical full-batch’ is the split-calibration forecast, and the dashed rule with its grey band is the median and interquartile range of ‘all 256-update fits’, the cost of training every candidate. Timings include calibration-model preparation but not shared calibration generation. **E**, Original scalar forecast versus final test half-mean-squared-error in the feedback-only arm. **F**, The calibration-derived Gaussian SGD forecast on the same cases and arm. Hexagons (one 21×10 grid, identical limits and scale in both panels) bin the 6,400 fits per panel on the logarithmic colour bar beside **F**, shared by **E,F**; the dashed grey line marks equality and each panel prints its root-mean-square forecast error. Source Data retain the joint arm, which reproduces **E,F**, and the population-oracle forecast bias (exact only under fresh examples). Markers and whiskers in **A–C** are means and pointwise 95% intervals from 10,000 whole-seed bootstrap draws over twenty seed blocks, hidden by the marker in the fixed / max-budget and rank-only rows; **D** shows medians and interquartile ranges of 320 (context count), 640 (forecasts and original scalar) or 1,280 (pilot and full training) timing draws, hidden by the marker where narrower than it. Marker shapes and colours are shared across **A–D**.

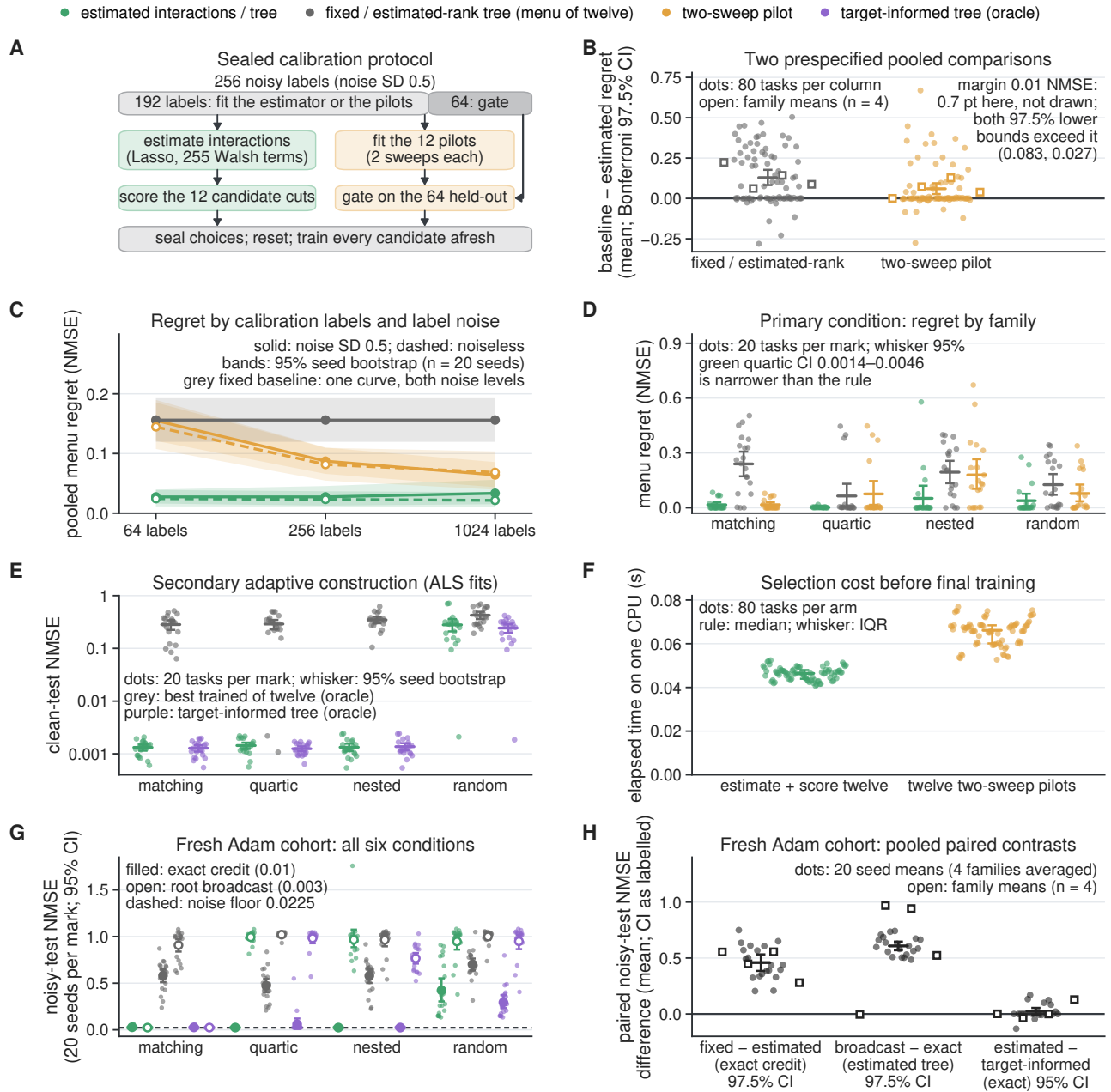


Figure S36: Finite calibration, adaptive construction and direct gradient learning. **A**, Flow chart of the sealed calibration protocol (Section 10.2): green, estimator branch; orange, two-sweep pilot branch; merging in the sealed reset-and-train step. **B**, Pooled fixed-baseline-minus-estimated (left, grey) and pilot-minus-estimated (right, orange) NMSE regret: small dots, 80 per-task paired differences; open squares, family means; short rule, pooled mean; whisker, Bonferroni-adjusted 97.5% paired interval; solid rule, zero; the 0.01-NMSE margin (on the panel, not drawn) lies below both lower bounds (0.083, 0.027). **C**, Pooled menu regret (excess clean-test NMSE above the best trained candidate) against calibration labels: solid curves, filled markers, noise SD 0.5; dashed, open markers, no noise; bands, 95% intervals; the fixed / estimated-rank baseline is one grey curve for both noise levels, and the estimated-cut noise contrast lies within the seed interval (1,024 labels: 0.022 [0.010, 0.035] noiseless, 0.033 [0.015, 0.056] at SD 0.5). **D**, Family-specific primary regret (green, estimated interactions; grey, fixed/estimated-rank baseline; orange, pilot): small dots, 20 per-task regrets; short rule, mean; whisker, 95% interval; the quartic estimated-cut interval (0.0027 [0.0014, 0.0046]) is narrower than the rule. **E**, Secondary adaptive trees (green; ALS fits) against the best trained candidate (grey) and the true-target tree (purple), clean-test NMSE, marks as in **D**. **F**, Selection time on one CPU (eighty primary tasks; axis from zero): dots, per-task times; short rule, median (0.046 s, estimation plus scoring, green; 0.066 s, twelve two-sweep pilots, orange; 1.4-fold); whisker, interquartile range. **G**, Fresh Adam cohort: estimated (green), development-fixed (grey) and target-informed (purple) trees under exact credit (filled) and root broadcast (open), noisy-test NMSE: small dots, 20 per-seed endpoints; whisker, 95% interval; dashed rule, label-noise floor 0.0225. **H**, Pooled paired noisy-test differences in neutral ink: small dots, 20 seed differences; open squares, family means (matching, quartic, nested, random, left to right); short rule, pooled mean; whisker, Bonferroni-adjusted 97.5% interval for the two primary contrasts and pointwise 95% for estimated-minus-target-informed; solid rule, zero. Target-informed construction is not a performance ceiling. Intervals in **C–E** and **G**: pointwise 95%, whole-seed bootstrap, $n = 20$.

Table S1: Core notation. Symbols are defined at first use; this index distinguishes quantities that recur in different coordinate systems.

Symbol	Meaning
$V_n, a_n = f_n(V_n)$	Compartment voltage and transmitted activity.
$g_i, x_i, g_i x_i; E_i^{\text{rev}}$	Maximal conductance, presynaptic activity, open conductance and reversal potential; E/I polarity does not require signed conductances.
$G_n^{\text{tot}}, R_n^{\text{tot}}$	Total open conductance and its reciprocal input resistance.
$\delta_n^V, \delta_n^a; q_n$	Errors in voltage/activity coordinates; root-to-unit output derivative. In reciprocal cables, q denotes the current-source adjoint.
Φ, K, c	Spatial-profile dictionary, column budget and delivered coefficients. Rank can be smaller than column count.
Γ	Route-gain matrix in the delivery equation; the rectangular anatomical kernel of Eq. 101 is separately defined.
$W, P_{\Phi, W}; \mathcal{C}_K$	Positive coordinate weights, weighted projector and explicitly specified per-field or aggregate energy capture.
$\mathbf{g}, \mathbf{d}, \mathbf{m}, V$	Exact gradient, routed-update estimator, its conditional mean and covariance.
M, Σ, U	Fixed routing operator, estimator-noise covariance and one-step moment bound.
$H_c, D_r; H_p, D_p$	Task-credit hierarchy/routed resolution, and latent task hierarchy/physical stage count.
χ, α, a	Branch conflict, physical-task sensor alignment and imposed route-span energy fraction, respectively.
G, R_a, R_m, κ	Reciprocal conductance matrix, axial resistivity, specific membrane resistance and focal shunt dose.

Table S2: Delivery profiles and coefficient sources. A useful span and an implementable coefficient rule are separate requirements.

Comparison	Field, information source and model
Images	Twelve nonsomatic activation coordinates per [3, 3] tree. Neuron-specific error is supplied by the exact linear readout; $K = 1/K = 3$ coefficients project the exact activation field. Local output derivatives remain in eligibility.
Quadratic screens	Sixteen abstract parameter coordinates; exact/noisy projections and supplied or paired-probe reliability estimates. No forward neuron is simulated in the Haar screens.
Context and ancestry	Eight terminal blocks receive $\delta_i \Phi_{c_i, \cdot}$. Context is supplied; the separate cue encoder learns coefficients before task training.
Multi-affine trees	Six nonroot unit errors receive exact paths, a unit or initial signed profile, or oracle-projected coefficients. Best-rank-one capture is a separate common-state diagnostic.
Conductance learning	Six nonsomatic voltages. Three-profile coefficients use current exact paths. Two leaf profiles plus unit proximal credit retain oracle amplitudes. The hard local rule instead gates distal somatic error by parent inhibition, retaining unit proximal credit; all 24 conductances learn.
Reconstructed arbors	Weighted input sites; broadcast plus ancestry supports. Free projection coefficients measure capacity for modeled passive fields. No encoder or biological learning is measured.
Measured-response models	Four-route spans contain one frozen baseline-transfer profile scaled by output error. Trialwise field/update oracles are geometry diagnostics and are not trained.

Table S3: Regular-tree execution records. Missing historical software versions are not inferred from a current environment. Complete configurations, manifests and exclusion ledgers accompany Source Data.

Cohort	Retained provenance
120 MNIST feedback runs	Configurations, checkpoints, logs and hashes; Python 3.10.13; source identities 6c1aaa25/bd5f905e. Scientific signatures and executable sources match across fields within architecture. Heterogeneous NVIDIA devices.
60 gradient-diagnostic runs	Same records; Python 3.10.13, source 4ba38bde with archived tracked modifications and explicit source hashes; H200, 4,006 summed GPU-hours.
640 depth-feedback runs	480 conductance-valid retained; configurations, curves, resource records and the complete validity ledger. Later inactive-source edits were checked for numerical equivalence. Heterogeneous NVIDIA devices, 73.29 summed GPU-hours.
160 diagnostic checkpoints	120 valid; five fields/four step sizes per checkpoint, paired/clustered analysis and complete exclusions. No new training.
320 exact/BP repeats	Unmodified root 74792ca, nested generator 4f3612a5, configurations, seeds, checkpoints and scheduler records; H200. Signed additive and ReLU shunting inputs remain distinct.
20 exact-factorial / five BP references	Configurations, checkpoints, logs and grouped outcomes on A100-SXM4-40GB; 2.953/0.474 summed GPU-hours. Exact commit, Python and package versions were not recorded.

Table S4: Repeated exact-transport/BP outcomes, averaged over four depths within each of ten paired seeds. Accuracy is a percentage and the contrast is percentage points, with a 20,000-draw paired 95% interval. The full seed contrasts and descriptive tests remain in Source Data.

Task and core	Exact	BP	Exact minus BP (95% interval)
MNIST, raw additive	97.013	97.008	0.005 (−0.028 to 0.041)
MNIST, shunting	96.960	96.973	−0.013 (−0.047 to 0.022)
Noisy lines, raw additive	33.378	33.490	−0.112 (−0.465 to 0.247)
Noisy lines, shunting (ReLU)	36.690	36.788	−0.098 (−0.653 to 0.390)

The 320 runs form 160 method pairs. One depth-averaged MNIST additive seed was tied. Exact transport is an information-oracle implementation control; an interval containing zero is not an equivalence test.

Table S5: Outcome-independent conductance-validity accounting for the artificial-tree experiments. “Included” counts retained training runs, or checkpoints in the diagnostic row; these unit types are not added together. The entire inhibitory-dose family is excluded because its prespecified endpoint compares against invalid signed-input shunting cells.

Experiment	Planned units	Invalid shunting	Unmatched controls	Included units
Depth by feedback	640 runs	160	0	480
Matched routing	160 runs	40	0	120
Inhibitory dose	400 runs	200	200	0
Spatial topology	320 runs	80	0	240
Fixed-budget depth	320 runs	160	0	160
Checkpoint diagnostic	160 checkpoints	40	0	120

The exclusion rule is task plus core plus resolved transfer activation; no accuracy or gradient outcome enters it. The complete 1,840-run validity table and all included row tables are supplied as Source Data.

Table S6: Complete reconstructed-tree learning across all eligible scans. Means average ten held-out-stimulus splits within each of 13 scans, then scans within each of seven target cells. Mean squared error (MSE) is normalized to the training-mean predictor. Each restricted learning rule scales one fixed site profile in a four-route span by scalar prediction error. Common-checkpoint update match is one minus normalized squared update error for that prescribed profile, rather than the best attainable projection into the full span.

Teaching field	Normalized held-out MSE	Common-checkpoint update match	Fits
Exact compartment error	0.7877	1.0000	130
Topology routes	0.8321	0.2322	130
Random routes	0.8333	0.3584	130
Site-shuffled routes	0.8391	0.4392	130

Positive MSE improvement favors topology. Topology improved over site shuffling by 0.0070 (95% interval -0.0207 to 0.0321) and over random routes by 0.0013 (-0.0090 to 0.0113). All restricted dictionaries had identical effective rank within run. None of

520 fits was excluded. A separate common-state diagnostic at 130 reexecuted exact-learning checkpoints gave per-trial eligibility-weighted within-span oracle captures of 0.2436, 0.3847 and 0.4677 for ancestry, random and shuffled routes, respectively.

Those oracle values are geometric upper bounds at the evaluated states; no oracle-feedback learning accuracy was measured.

Table S7: Retrospective six-animal signed-credit reanalysis of Francioni et al. source sheet “Ext 13E”. The contrast is error-reduction minus error-increase soma–dendrite residual, in the source z-score units. P+ and P- denote populations with positive and negative causal decoder weights. Directions follow this published mapping; intervals bootstrap paired animal-level contrasts, retaining animals as the inferential unit.

Contrast	Mean	95% interval	Ordered animals
P+	0.0753	[0.0295, 0.1214]	5/6 positive
P-	-0.1498	[-0.2054, -0.0866]	6/6 negative
P+ minus P-	0.2251	[0.1416, 0.3108]	6/6 positive

Exact one-sided random-sign permutation P values were 0.0313, 0.0156, and 0.0156. The signed and common modes contained 83.7% and 16.3% of pooled squared projection energy across the six animal contrast vectors, respectively. The pooled fraction is a ratio of summed energies, not the mean of animal-level fractions. The signed fraction was 45.8–97.1% across individual animals, with an animal-bootstrap 95% interval of 60.6–95.7% for the pooled fraction and a leave-one-animal-out range of 76.5–90.2%.

Animal-level values are provided in Source Data.

Table S8: Terminal outcomes in the balanced longer-budget follow-up. Rates are the original development-selected Adam rates. NMSE values are mean (median) across the same twenty previously examined seed blocks. The final column counts exact/profile fits with clean full-domain population NMSE below 0.02; this descriptive threshold was not used for stopping, selection or inference. Complete seedwise outcomes, validation-selected states, common-rate and SGD controls remain in Source Data.

Updates	Task	Exact NMSE mean (median)	Initial-profile NMSE mean (median)	Clean error < 0.02 exact/profile counts
1,024	Pairwise	0.0317 (0.0231)	0.0236 (0.0236)	19/20; 20/20
1,024	Quartic	0.1516 (0.0465)	0.9813 (1.0277)	17/20; 0/20
1,024	Nested	0.0230 (0.0229)	0.7690 (0.7829)	20/20; 0/20
4,096	Pairwise	0.0232 (0.0231)	0.0240 (0.0241)	20/20; 20/20
4,096	Quartic	0.0461 (0.0458)	0.9624 (1.0205)	20/20; 0/20
4,096	Nested	0.0233 (0.0232)	0.7913 (0.7853)	20/20; 0/20
8,192	Pairwise	0.0231 (0.0231)	0.0243 (0.0243)	20/20; 20/20
8,192	Quartic	0.0463 (0.0463)	0.9811 (1.0353)	20/20; 0/20
8,192	Nested	0.0233 (0.0233)	0.8039 (0.7918)	20/20; 0/20
16,384	Pairwise	0.0232 (0.0232)	0.0246 (0.0248)	20/20; 20/20
16,384	Quartic	0.0464 (0.0461)	0.9994 (1.0769)	20/20; 0/20
16,384	Nested	0.0232 (0.0232)	0.7944 (0.7826)	20/20; 0/20

Table S9: Same-seed physical-depth reproducibility. All 430 repeated fits used versioned source a99c3a7 and passed completeness, finite-metric and resource checks. Accuracy differences are percentage points. This is computational repetition of existing seeds.

Measure	Value
Repeated H2/grouped-point and principal H3 fits	220 + 210
Median absolute original/repeated accuracy difference	0.030
Mean absolute difference	0.154
Maximum absolute difference (raw-additive control)	15.01
Pairs within 0.1 points	357/430
Repeated H2/H3 aligned serial-BP depth effects	30.84 / 30.86
Repeated H2/H3 architecture-by-placement interactions	32.43 / 31.27

Table S10: Claim and evidence index. Each row identifies the question answered by the corresponding controlled comparison; scope and inferential details are stated once in its methods.

Question	Evidence and interpretation
Which spatial distinctions matter?	Image, conflict and ancestry comparisons hold the forward model fixed while changing delivery. Their scope is within-arbor credit with the stated somatic signal.
Can the tree represent the task?	Centered-cut theorem and Boolean constructions characterize the scalar multi-affine class; they do not prove optimizer convergence.
Does a fixed profile learn?	Matched algebraic tasks and longer budgets separate learned credit geometry and finite-training performance. Oracle capture is distinct from a trained rule.
Can a local rule supply selection?	Fresh conductance gates use parent inhibition and somatic error without exact paths. Both signals are supplied; autonomous error generation is unresolved.
Does physical depth help?	Serial/divisive controls support task-dependent forward benefit. Longer training reverses the LocalCA accuracy effect and retains metric dependence.
What does anatomy supply?	Budgeted broadcast-plus-ancestry dictionaries compress modeled passive responses; topology-preserving surrogates test finer structure.
Can conductance change route gain?	The passive partition identity and reconstructed-cable interventions establish modeled transport; the weak-channel ensemble checks a local linearization.
Are the routes used in measured responses?	Seven-target analyses provide no evidence of preferential alignment. Simulated sensitivity is conditional on recorded coverage/reliability and the specified Gaussian model.
Can an initial score choose morphology?	The original scalar selector lost to a fixed tree. Finite-horizon and interaction estimators answer narrower selection questions under their declared priors.

Table S11: Cohort and protocol index. Counts refer to runs within a study, not sums of independent replications. Detailed seed ranges, pairing, stopping and selection are given in the cited sections and source configurations.

Question	Units and retained design	Protocol section
Exact/rule screens	Fifty seeds per screen; fixed/adaptive conductance reliability 2,250/1,050 outcomes; same-span 4,800 checkpoint rows	S1; probes/checkpoints nested within seeds.
Images	Original MNIST/Fashion/CIFAR: 15/10/20 seeds, 90/60/80 fits; random feedback 120 new fits; fresh dictionary 108 development and 190 fresh fits	S2; twelve short dictionary checks excluded.
Depth/input controls	480/640 depth, 120/160 ownership, 240/320 spatial and 160/320 fixed-budget fits valid; separate 320-run exact/BP repeat	S2; Table S5.
Branch/ancestry	Twenty conflict seeds; ten two-branch seeds; twenty eight-context seeds and 2,700 fits; cue encoder 2,160 fits	S3; hard cue readout exploratory.
Interaction/Boolean capacity	Census of 140 primary +24 nested targets; 164 constructions. Boolean: seven templates, fifteen trees and 49 exact constructions	S4; complete-domain analyses.
Algebraic learning	Twenty fresh seed blocks each: 3,600 restricted-credit and 1,200 matched-profile fits; Boolean 6,720 fits. Extension reuses 720 original algebraic trajectories	S4; no extension retuning/new seeds.
Conductance learning	Grouping 1,440 fits; first context study 240 fresh; opposed tuning 320 fresh; wider bounds 640 same-seed trajectories; added-rate development 288 fits	S5; distinct models/cohorts.
Local conductance gate	Twenty new seeds, nine rules, two tasks, three rates: 1,080 trajectories to 16,384 updates	5.2; protocol frozen after prior exploratory exposure.
Physical composition	Ten seeds per cohort; H3 270, point/optimizer 200, dose 90, grouped H3/H2 220, H4 360, task families 360; same-seed repeat 430 and budget extension sixty	S6; finite budgets and metric dependence retained.
Anatomy	Eight initial +47 disjoint cells in one mouse; twelve selected/ten eligible Pinky cells in a second. Equal-broadcast: 65 operators, 149,331 dictionary evaluations	S7; separate cohort inference.
Inhibitory contacts/shunts	Census twenty targets, 92-axon sensitivity. Focal studies: eight initial cells, 45/47 disjoint cells eligible; 64 weak-channel draws per initial cell	7.5, S8; draws/sites nested in cells.
Measured responses	Seven targets/thirteen scans; ten splits, 520 full-tree fits, 130 geometry replays and 3,380 baseline evaluations	S9; target is inferential unit.
Detection sensitivity	4,000 complete simulated datasets across 21 signal levels/two reliability scenarios; 1,000 repeat-calibration datasets	9.6; simulations are not biological replicates.

Question	Units and retained design	Protocol section
Structure selection	Original scalar 12,800 confirmation fits; finite-horizon 25,600 fresh fits; noisy calibration 2,240 and end-to-end 480 fits	S10; sealed choices and independent observations.
Imposed/external controls	Eight arbors with forty streams each; six external animals	7.6, 9.7; imposed and retrospective outcomes.

Table S12: Source index for endpoint grids and metadata. All paths are relative to the released `source_data/` directory. These records retain the complete per-unit outcomes, contrasts and exclusions behind condensed or omitted print tables; figure-source mappings provide the individual panel joins.

Evidence	Machine-readable record
Two-branch controls	<code>trained_subtree_address/condition_summary.csv</code> ; <code>trained_subtree_address/paired_contrasts.csv</code>
Ancestry budgets	<code>trained_subtree_address_full_factorial/condition_summary.csv</code> ; <code>review_evidence_reanalysis/ancestry_control_contrasts.csv</code>
Moment bound/reanalysis	<code>credit_phase_theory/primary_contrasts.csv</code> ; <code>review_evidence_reanalysis/pooled_correlations.csv</code> ; <code>review_evidence_reanalysis/moment_scores.csv</code>
Fixed/adaptive reliability	<code>positive_conductance_reliability_step_consistent/paired_contrasts.csv</code> ; <code>adaptive_conductance_reliability/paired_contrasts.csv</code>
Same-span learning	<code>same_span_coefficient_learning/paired_contrasts.csv</code> ; <code>same_span_coefficient_learning/condition_summary.csv</code>
Original image outcomes	<code>mnist_feedback_ladder/condition_summary.csv</code> ; <code>figure2/feedback_gradient_runs.csv</code> ; <code>figure2/exact_transport_and_backprop_runs.csv</code>
Strict scalar audit	<code>mnist_feedback_ladder/strict_scalar_implementation_audit.csv</code> ; <code>mnist_feedback_ladder/strict_scalar_paired_contrasts.csv</code>
Valid depth/ownership	<code>prospective_input_validity/central_valid_condition_summary.csv</code> ; <code>prospective_input_validity/routing_valid_paired_contrasts.csv</code>
Spatial coverage	<code>spatial_topology_audit/owner_metrics.csv</code> ; <code>spatial_topology_audit/summary.csv</code>
Fixed-checkpoint step	<code>prospective_input_validity/mechanism_feedback_summary_valid.csv</code> ; <code>prospective_input_validity/mechanism_paired_contrasts_valid.csv</code>
Operating-regime grids	<code>regular_tree_regimes/competence_summary.csv</code> ; <code>regular_tree_regimes/depth_scaling_summary.csv</code> ; <code>regular_tree_regimes/broadcast_noise_summary.csv</code>
Morphology inputs/positive controls	<code>figure3/cell_metrics.csv</code> ; <code>figure3/routing_capacity_summary.json</code> ; <code>figure3/typed_only_compression_curves.csv</code>
Passive-field controls	<code>reciprocal_routing/cell_method_capture.csv</code> ; <code>reciprocal_routing/summary.json</code> ; <code>anatomy_commonmode/</code>
Focal scale/reversal sensitivity	<code>figure4/summary.json</code> ; <code>figure4/scale0p1_summary.json</code> ; <code>figure4/scale1p0_summary.json</code> ; <code>figure4/irev0_summary.json</code> ; <code>figure4/irevm0p5_summary.json</code>
Physical/weak-channel shunts	<code>physical_cable_sensitivity/cell_primary_contrasts.csv</code> ; <code>focal_selectivity_active_ensemble/condition_summary.csv</code>
Measured target/scan metadata	<code>figure5/functional_target_metrics.csv</code> ; <code>figure5/functional_summary.json</code> ; <code>functional_topology_all_scans/cell_metrics.csv</code>
Branch-model restricted credit	<code>figure5/task_target_method_means_ch4.csv</code> ; <code>figure5/task_summary_ch4.json</code>

Evidence	Machine-readable record
Inhibitory census controls	microns_inhibitory_routes/target_metrics.csv; microns_inhibitory_routes/inhibitory_axon_metrics.csv; microns_inhibitory_routes/inhibitory_3d_matched_target_metrics.csv
Imposed alignment	alignment_controlled/cell_alignment_metrics.csv; alignment_controlled/cell_paired_contrasts.csv
Physical rerun outliers	physical_depth_clean_source_replication/historical_concordance.csv; physical_depth_clean_source_replication/seed_outcomes.csv
Algebraic budget endpoints	credit_rule_extension/summaries/all_budget_outcomes.csv; credit_rule_extension/summaries/compact_terminal_table.csv

References

- 2258 [1] Houman Safaai, Maceo Richards, and Bernardo L. Sabatini. Shunting inhibition and dendritic branching shape
2259 local credit assignment. *arXiv preprint arXiv:2607.03556*, 2026. URL <https://arxiv.org/abs/2607.03556>.
- 2260 [2] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, 2 edition, 2013. doi:
2261 10.1017/CBO9781139020411.
- 2262 [3] Yurii Nesterov. *Introductory Lectures on Convex Optimization: A Basic Course*, volume 87 of *Applied Optimization*.
2263 Springer, 2004. doi: 10.1007/978-1-4419-8853-9.
- 2264 [4] Joseph Cichon and Wen-Biao Gan. Branch-specific dendritic ca^{2+} spikes cause persistent synaptic plasticity.
2265 *Nature*, 520:180–185, 2015. doi: 10.1038/nature14251.
- 2266 [5] Nadav Cohen, Or Sharir, and Amnon Shashua. On the expressive power of deep learning: A tensor analysis. In
2267 *Proceedings of the 29th Conference on Learning Theory*, volume 49 of *Proceedings of Machine Learning Research*,
2268 pages 698–728, 2016. URL <https://proceedings.mlr.press/v49/cohen16.html>.
- 2269 [6] Wolfgang Hackbusch and Stefan Kühn. A new scheme for the tensor representation. *Journal of Fourier Analysis*
2270 *and Applications*, 15:706–722, 2009. doi: 10.1007/s00041-009-9094-9.
- 2271 [7] Lars Grasedyck. Hierarchical singular value decomposition of tensors. *SIAM Journal on Matrix Analysis and*
2272 *Applications*, 31(4):2029–2054, 2010. doi: 10.1137/090764189.
- 2273 [8] MICrONS Consortium. Functional connectomics spanning multiple areas of mouse visual cortex. *Nature*, 640:
2274 435–447, 2025. doi: 10.1038/s41586-025-08790-w.
- 2275 [9] Nicholas L. Turner, Thomas Macrina, J. Alexander Bae, Runzhe Yang, Alyssa M. Wilson, Casey Schneider-Mizell,
2276 Kisuk Lee, Ran Lu, Jingpeng Wu, Agnes L. Bodor, et al. Reconstruction of neocortex: Organelles, compartments,
2277 cells, circuits, and activity. *Cell*, 185(6):1082–1100.e24, 2022. doi: 10.1016/j.cell.2022.01.023.
- 2278 [10] MICrONS Consortium. MICrONS Layer 2/3 Data Tables, 2020. URL [https://doi.org/10.5281/zenodo.](https://doi.org/10.5281/zenodo.3710459)
2279 3710459.
- 2280 [11] Matan Gavish, Boaz Nadler, and Ronald R. Coifman. Multiscale wavelets on trees, graphs and high dimensional
2281 data: Theory and applications to semi supervised learning. In *Proceedings of the 27th International Conference on*
2282 *Machine Learning*, 2010. URL <https://icml.cc/Conferences/2010/papers/137.pdf>.
- 2283 [12] Andrew K. Lampinen and Surya Ganguli. An analytic theory of generalization dynamics and transfer learning in
2284 deep linear networks. In *International Conference on Learning Representations*, 2019. URL [https://openreview.](https://openreview.net/forum?id=ryfMLoCqtQ)
2285 [net/forum?id=ryfMLoCqtQ](https://openreview.net/forum?id=ryfMLoCqtQ).
- 2286 [13] Abdulkadir Canatar, Blake Bordelon, and Cengiz Pehlevan. Spectral bias and task-model alignment explain
2287 generalization in kernel regression and infinitely wide neural networks. *Nature Communications*, 12:2914, 2021.
2288 doi: 10.1038/s41467-021-23103-1.